MIDDLE YEARS PROGRAMME
YEAR 1 · MATHEMATICS

MYP

MATHEMATICS 1

STUDENT COURSEBOOK

Improved First Edition

A concept-driven, inquiry-led course in number, proportion, algebra, shape, measurement, sets and data.

ENGAGE · EXPLORE · EXPLAIN · PRACTISE · APPLY · REFLECT

Ten chapters · Five parts · Full answers, investigations and criterion tasks

SADIKOU MOUKODDIM
Head of Mathematics Department · Mathematics Teacher · Category 1

- Number systems and place value
- Powers and order of operations
- Factors, multiples and primes
- Fractions and decimals
- Percentages and applications
- Patterns, expressions and equations
- Lines, angles and constructions
- Sets and classification
- Perimeter, area and volume
- Data, tables and graphs

All diagrams original to this edition

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MYP

MATHEMATICS 1

Student Coursebook

Improved First Edition

An inquiry-led introduction to number, place value, proportion, algebra, geometry, measurement, sets and data, built around the concepts, contexts and assessment criteria of the Middle Years Programme.

SADIKOU MOUKODDIM
Head of Mathematics Department · Mathematics Teacher · Category 1

About this edition

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Contents

FRONT MATTER —
Welcome vii
How to use this book viii
Visual guide ix
The MYP framework x
Assessment criteria xi
Course map xii
Approaches to learning xiii
Command terms and assessment xiv
Notation and habits xv
PART I · THE GRAMMAR OF NUMBER xvi
1 Number systems and place value 1
1.1 Number systems and the number line 6
1.2 Place value and the role of zero 9
1.3 Expanded form and equivalent representations 12
1.4 Comparing, ordering and rounding 14
1.5 Estimation and sensible accuracy 18
Investigation 1 How many numbers can a set of digit cards make? 20
Application 1 The community noticeboard 21
Task C Explaining place value to a younger student 23
Review Consolidating Chapter 1 23
Review questions Practice across the whole chapter 25
Mixed 1 Fluency, reasoning and problem solving 26
Cumulative practice 1 Questions that mix this chapter with earlier ones 27
2 Powers, expanded form and order of operations 28
2.1 From repeated multiplication to powers 33
2.2 Powers of ten and expanded form 35
2.3 Squares, cubes and roots 38
2.4 Order of operations 40
2.5 Substitution and expressions with powers 42
Investigation 2 Does doubling always stay ahead? 44
Application 2 The school data store 45
Task C Explaining a calculator's answer 46
Review Consolidating Chapter 2 47
Review questions Practice across the whole chapter 48
Mixed 2 Fluency, reasoning and problem solving 49
Cumulative practice 2 Questions that mix this chapter with earlier ones 50
3 Factors, multiples and primes 51
3.1 Factors, multiples and factor pairs 56
3.2 Divisibility tests 58
3.3 Prime and composite numbers 60
3.4 Prime factorisation 63
3.5 Highest common factor and lowest common multiple 65
Investigation 3 The odd number of factors 67
Application 3 The fairness relay 69
Task C Writing the divisibility guide 70
Review Consolidating Chapter 3 70
Review questions Practice across the whole chapter 72
Mixed 3 Fluency, reasoning and problem solving 72
Cumulative practice 3 Questions that mix this chapter with earlier ones 73
PART II · PARTS AND PROPORTIONS 74
4 Fractions and decimals 75
4.1 What a fraction means 80
4.2 Equivalent fractions and simplifying 82
4.3 Comparing and ordering fractions 84
4.4 Fractions and decimals 86
4.5 Calculating with decimals 89

4.6 Adding and subtracting fractions 91
4.7 Multiplying and dividing fractions 93
Investigation 4 Which product is larger? 95
Application 4 Scaling the canteen recipe 96
Task C Explaining the rule for dividing 97
Review Consolidating Chapter 4 97
Review questions Practice across the whole chapter 99
Mixed 4 Fluency, reasoning and problem solving 100
Cumulative practice 4 Questions that mix this chapter with earlier ones 100
5 Percentages and applications 102
5.1 What a percentage is 107
5.2 Finding a percentage of a quantity 109
5.3 One quantity as a percentage of another 111
5.4 Percentage increase and decrease 113
5.5 Discounts, taxes and budgets 115
Investigation 5 Do two discounts add up? 118
Application 5 Choosing a phone plan 119
Task C Interrogating an advertisement 120
Review Consolidating Chapter 5 121
Review questions Practice across the whole chapter 122
Mixed 5 Fluency, reasoning and problem solving 123
Cumulative practice 5 Questions that mix this chapter with earlier ones 124
PART III · PATTERNS AND SYMBOLS 125
6 Patterns, expressions and equations 126
6.1 Patterns and rules 130
6.2 Expressions and substitution 133
6.3 Simplifying expressions 135
6.4 Solving linear equations 137
6.5 Forming and using equations 139
Investigation 6 The matchstick chain 141
Application 6 The school event budget 142
Task C Explaining the balancing method 143
Review Consolidating Chapter 6 144
Review questions Practice across the whole chapter 145
Mixed 6 Fluency, reasoning and problem solving 146
Cumulative practice 6 Questions that mix this chapter with earlier ones 147
PART IV · SHAPE, STRUCTURE AND MEASURE 148
7 Lines, angles, triangles and constructions 149
7.1 Points, lines, angles and notation 154
7.2 Angle relationships 156
7.3 Parallel lines and transversals 159
7.4 Triangles 161
7.5 Polygons and their angles 164
7.6 Constructions with compasses 167
Investigation 7 The triangle inequality 170
Application 7 Laying out the school garden 171
Task C Writing up a construction 172
Review Consolidating Chapter 7 172
Review questions Practice across the whole chapter 174
Mixed 7 Fluency, reasoning and problem solving 175
Cumulative practice 7 Questions that mix this chapter with earlier ones 175
8 Sets and classification 177
8.1 Sets and set notation 182
8.2 Subsets, union, intersection and complement 184
8.3 Venn diagrams and survey problems 186
8.4 Classification and fair decisions 188
8.5 Number sets and the real line 191
Investigation 8 How much overlap is possible? 194
Application 8 The library survey 195
Task C Explaining a misleading headline 196

Review Consolidating Chapter 8 196
Review questions Practice across the whole chapter 198
Mixed 8 Fluency, reasoning and problem solving 198
Cumulative practice 8 Questions that mix this chapter with earlier ones 199
9 Perimeter, area, surface area and volume 200
9.1 Perimeter and area of rectangles and triangles 205
9.2 Area of parallelograms and compound shapes 207
9.3 Circles: circumference and area 209
9.4 Surface area of cuboids and prisms 212
9.5 Volume of cuboids and prisms 214
9.6 Using measurement to make decisions 217
Investigation 9 The packaging problem 219
Application 9 The water tank decision 220
Task C Writing the measurement guide 221
Review Consolidating Chapter 9 221
Review questions Practice across the whole chapter 223
Mixed 9 Fluency, reasoning and problem solving 224
Cumulative practice 9 Questions that mix this chapter with earlier ones 224
PART V · DATA AND DECISIONS 226
10 Data, tables and graphs 227
10.1 Collecting and organising data 231
10.2 Displaying data: bar charts, dot plots and pictograms 234
10.3 Averages and range 237
10.4 Grouped data 239
10.5 Reading data critically 241
Investigation 10 How should data be grouped? 244
Application 10 The canteen survey 245
Task C Writing the graph-checking guide 246
Review Consolidating Chapter 10 246
Review questions Practice across the whole chapter 248
Mixed 10 Fluency, reasoning and problem solving 248
Cumulative practice 10 Questions that mix this chapter with earlier ones 249
REFERENCE 250
Cumulative revision 250
Cumulative revision 1 Chapters 1–3 250
Cumulative revision 2 Chapters 4–5 251
Cumulative revision 3 Chapters 6–7 251
Cumulative revision 4 Chapters 8–9 252
Cumulative revision 5 Chapters 1–10 253
Part review I The grammar of number (Chapters 1–3) 254
Part review II Parts and proportions (Chapters 4–5) 254
Part review III Algebra and shape (Chapters 6–7) 255
Part review IV Sets and measurement (Chapters 8–9) 256
Part review V Data and decisions (Chapter 10, with Chapters 1–9) 256
Extended project Designing a mathematics trail 257
Practice paper A Chapters 1–5 258
Practice paper B Chapters 6–10 259
Practice paper C Mixed paper across all ten chapters 261
Worked solutions Twenty-five problems solved in full 262
Annotated student work Four responses with examiner commentary 265
Revision planner Twelve weeks to the assessment 267
Reference notes Using a calculator well 268
Answers and worked solutions 269
Formula bank 283
Glossary 285
Index 288
References and acknowledgements 290

Welcome to Year 1 mathematics

This is a working book. Expect to write in the margins, test an idea, compare your method with a classmate's and return to a question after your thinking has changed.

Year 1 is where many familiar ideas become precise. You already know how numbers are written, how shapes are named and how data can be displayed. This course asks you to look underneath those familiar forms: why a digit's position changes its value, why equivalent fractions name the same quantity, why an angle rule must hold for every triangle rather than merely for the one drawn on the page.

The book is arranged in five parts. **Part I** builds the grammar of number: place value, powers, factors and the rules that keep calculation reliable. **Part II** treats fractions, decimals and percentages as three ways of describing the same quantity. **Part III** turns pattern into algebra, and algebra into a tool for finding what is unknown. **Part IV** brings sets, shape and measurement together, with constructions that make geometry reproducible. **Part V** examines data, graphs and the claims people make from them.

Every chapter follows the same rhythm. A situation raises a question. An exploration gives you something to notice before anyone names the rule. The explanation then gives names, notation and reasons. Carefully written worked examples make the reasoning visible, practice makes the method dependable, and an investigation, an application and a criterion task ask you to use the mathematics beyond the original example.

The answer section confirms your reasoning rather than replacing it. Attempt the question fully before you look: if your answer differs, find the first line where your reasoning diverges. That line is the most useful part of the page.

NOTE

How the course is organised

Each part opens with a page naming the chapters it contains. Each chapter opens with a full-page invitation and then a unit planner: the concepts, the global context, the statement of inquiry, the inquiry questions and the criteria the chapter develops. The teaching pages follow the engage–explore–explain–practice–apply–reflect rhythm described on the next pages.

How to use this coursebook

Six movements are repeated in every chapter. Knowing what each movement is for helps you use the pages deliberately rather than reading them passively.

01 **Engage**
A situation, puzzle or claim you cannot yet explain. Your first job is to ask questions, not to calculate.

02 **Explore**
A structured investigation with materials, a table or a diagram. Record what you notice before anyone tells you the rule.

03 **Explain**
Definitions, vocabulary, notation and the reasons behind each rule. This is where the mathematics gets its name.

04 **Worked examples**
Given, asked, method, steps, answer and check. Cover the answer, follow the method, then reproduce it unaided.

05 **Practice**
Fluency, reasoning, problem solving and extension, labelled so you can choose an entry point and then move beyond it.

06 **Investigate, apply and reflect**
Investigations, real-life applications, criterion tasks, a chapter review and mixed practice that combines earlier chapters.

EXERCISE TYPES

Practice questions carry a short label so that you can see what kind of thinking is being asked for. A question may belong to more than one type; the label names the dominant demand.

F **Fluency** Short, deliberate practice that makes a method accurate and efficient.

R **Reasoning** Explain, compare, justify, find a counterexample or diagnose an error.

P **Problem solving** Multi-step and unfamiliar contexts that require choosing a strategy.

E **Extension** Deeper structure, generalisation or a more open task within Year 1.

APPROACHES TO LEARNING
Using the answer key well

Attempt the question fully before checking. If your answer differs, locate the first line where your reasoning diverges, mark the correction in a different colour and write one sentence explaining why the alternative was wrong. That sentence is often worth more than the correct number.

Visual guide to the book

Every coloured panel has a job. The label at the top of the panel tells you what kind of information to expect, so that you can decide whether to read it carefully, copy it into your notes or use it as a checklist.

DEFINITION	**Definition** The precise meaning of a term, in the words the course uses.
KEY IDEA	**Key idea** The principle behind a method, so that the method can be adapted.
COMMON ERROR	**Common error** A mistake worth anticipating, with the correction beside it.
APPROACHES TO LEARNING	**ATL skill** An approaches-to-learning skill, and the activity that develops it.
NOTE	**Note** A reminder, a convention or a useful piece of background information.
MATHEMATICAL LANGUAGE	**Mathematical language** Vocabulary to use, spell and explain accurately.
METHOD	**Method** A reliable sequence of steps for a familiar type of question.
INVESTIGATION	**Investigation** A structured exploration with a question to answer.
APPLICATION	**Application** Mathematics used in a realistic setting with a decision to make.
CRITERION TASK	**Criterion task** An assessed task naming the criterion strands it develops.

KEY IDEA

Three habits this book rewards

Estimate first. Predict the size of an answer before calculating; an estimate turns a wrong keypress into a visible mistake.

Say the reason. After each method, write one sentence that begins “This works because…”. If you cannot finish the sentence, the method is not yet yours.

Compare representations. Ask how the diagram, the table, the fraction, the decimal and the formula describe the same quantity. Moving between them is the core skill of Year 1.

The MYP framework in this course

The Middle Years Programme asks students to learn mathematics as a connected, conceptual and applicable discipline. Concepts organise the content, contexts give it purpose, and criteria make the quality of thinking visible. In Year 1 the emphasis is on strong foundations: fluent number sense, careful use of notation, spatial reasoning, and the first deliberate steps from arithmetic towards algebra.

KEY CONCEPTS USED IN THIS BOOK

Form	The shape, structure and organisation of mathematical objects: how a decimal is arranged, how an angle is drawn, how an expression is built. Form helps us notice what is the same beneath different appearances.
Logic	The reasoning that connects claims to conclusions. Logic asks not only “Is it true?” but “How do we know, and for which cases?”
Relationships	Connections between quantities, variables, representations and ideas. A percentage, a formula, an angle pair and a graph are different ways of expressing relationships.
Change	How one quantity responds when another changes — growth, discount, sequence, trend.

GLOBAL CONTEXTS IN THIS COURSE

Identities and relationships	How categories, groups and belonging can be described and questioned; how sets and classification represent inclusion and difference.
Orientation in space and time	How number systems, calendars and measurement developed across cultures; how scale and position structure space.
Personal and cultural expression	How proportion, pattern and geometry shape design, craft and architecture.
Scientific and technical innovation	How measurement, formulas and data displays support invention, engineering and scientific claims.
Fairness and development	How budgets, percentages, comparisons and data can support or distort fair decisions.
Globalization and sustainability	How quantities are compared across communities and environments, and how claims about resources are communicated.

THE INQUIRY PATHWAY

PROVOKE	QUESTION	REPRESENT	GENERALISE	APPLY	REFLECT
Notice a situation, pattern or claim that is not yet explainable.	Turn the noticing into a mathematical question with quantities or objects.	Choose diagrams, tables, symbols or materials that reveal structure.	Move from examples to a rule, testing it against further cases.	Use the rule in a context and judge whether the answer makes sense.	Evaluate the method, the evidence and the limits of the conclusion.

Assessment criteria in practice

MYP mathematics uses four criteria. They describe different qualities of mathematical work, not four separate syllabuses: a single rich task can provide evidence for more than one criterion. Most questions in this book are formative practice. A criterion task is different: it is planned to collect evidence, and it names its audience, product and expectations.

CRITERION A · KNOWING AND UNDERSTANDING

Students:

- Select appropriate mathematics for familiar and unfamiliar problems.
- Apply the selected mathematics successfully.
- Solve problems correctly across a range of contexts.

In this course:

Definitions, place value, operations, fractions, percentages, angle rules, area and volume.

CRITERION B · INVESTIGATING PATTERNS

Students:

- Apply problem-solving techniques to recognise patterns.
- Describe patterns as relationships or general rules consistent with the findings.
- Verify whether a pattern works for other examples.

In this course:

Sequences, factor structures, constructed figures, set patterns and measurement relationships.

CRITERION C · COMMUNICATING

Students:

- Use appropriate mathematical language, notation and symbols.
- Use different forms of representation and move between them.
- Present complete, coherent and concise lines of reasoning.
- Organise information using a logical structure.

In this course:

Explanations, diagrams, tables, graphs, units and justified working.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

Students:

- Identify the relevant elements of authentic situations.
- Select and apply mathematical strategies.
- Justify the degree of accuracy of a result.
- Interpret results in context and evaluate the solution.

In this course:

Budgets, packaging, timetables, surveys, design constraints and public claims.

NOTE

Formative practice is not a formal assessment

Practice questions are designed for learning, feedback and improvement, and may be attempted more than once as understanding develops. The criterion label at the head of a task states which criteria and strands are targeted; it is not a formal IB grade, and the rubrics in this book are classroom tools written for formative use.

Course map and progression

The ten chapters form five parts. Each chapter names what it assumes, what it teaches and what it prepares. Reading the map before you begin helps you see that the course is a sequence of connected ideas rather than ten separate topics.

Part	Chapter	Core content	Builds towards
I · The grammar of number	**1 Number systems and place value**	Number families, decimal place value, expanded form, ordering, rounding, estimation	All later number work; measurement precision
	2 Powers and order of operations	Index notation, powers of ten, squares and roots, order of operations, substitution	Prime factorisation; algebra; standard form
	3 Factors, multiples and primes	Factor pairs, divisibility, primes, prime factorisation, HCF and LCM	Fraction simplification; scheduling problems
II · Parts and proportions	**4 Fractions and decimals**	Equivalence, ordering, four operations, decimal place value, rounding, conversions	Percentages; ratio; algebraic fractions
	5 Percentages and applications	Percent of a quantity, increase and decrease, discounts, taxes, budgets	Proportional reasoning; financial contexts
III · Patterns and symbols	**6 Patterns, expressions and equations**	Sequences, general rules, substitution, like terms, expanding, one- and two-step equations	Linear functions; formula rearrangement
IV · Shape, structure and measure	**7 Lines, angles, triangles and constructions**	Angle facts, parallel lines, triangle and polygon angle sums, ruler-and-compass work	Pythagoras; congruence; circle theorems
	8 Sets and classification	Set notation, subsets, Venn diagrams, number sets, surveys and decisions	Probability; logic; data handling
	9 Perimeter, area, surface area and volume	Perimeter and area, circles, composite shapes, nets, volume and capacity	Similarity; trigonometry; optimisation
V · Data and decisions	**10 Data, tables and graphs**	Data types, frequency and two-way tables, charts, averages, trends, misleading graphs	Statistics; statistical literacy

Approaches to learning

The approaches to learning (ATL) skills are the methods a student uses to learn mathematics, not an extra subject. Each chapter names one skill focus and builds it through a specific activity, in the form: *In order for a student to …, students must …* The table shows how the ten chapters develop the five skill categories.

Skill category	Chapters	What the activity requires
Communication	1, 3, 8	Explaining a method to a named audience, choosing notation that a reader can follow.
Research	9, 10	Deciding which measurement or sample answers the question, and stating the limits of the data.
Thinking (critical)	2, 5, 6, 10	Testing a claim, checking a counterexample, distinguishing a pattern from a proof.
Thinking (transfer)	2, 4	Using one representation in a new context, such as a multiplier in a shop and in a science measurement.
Self-management	all	Recording working, keeping units, and returning to a question that has been left incomplete.
Collaboration	all investigations	Agreeing a method, dividing the recording, and reconciling two different answers.

APPROACHES TO LEARNING

One example of the structure in practice

Chapter 3, critical thinking: *In order for a student to test whether a classification is sound, students must construct a set of categories, apply them to awkward cases such as 1 and 91, and revise the categories when a case fits two of them or none.* The activity is the classification task, not a sentence about being careful.

Skills are assessed through the work they produce: the explanation a student writes, the table they record, the justification they give. A skill that leaves no trace in the work has not been developed, only mentioned.

WHERE EACH SKILL FOCUS IS DEVELOPED

Chapter	Skill focus	Activity that builds it
1	Critical thinking	Investigation 1 · counting the numbers digit cards can make
2	Transfer	Application 2 · comparing binary and decimal storage claims
3	Critical thinking	Investigation 3 · why squares have an odd number of factors
4	Transfer	Application 4 · scaling a canteen recipe and costing the packs
5	Critical thinking	Investigation 5 · whether two discounts add up
6	Critical thinking	Investigation 6 · generalising the matchstick chain
7	Thinking skills	Investigation 7 · the triangle inequality
8	Thinking skills	Investigation 8 · how much overlap two sets can have
9	Research skills	Application 9 · choosing a water tank from measured data
10	Critical thinking	Application 10 · designing and reporting a survey

Command terms and assessment

Command terms tell you what kind of answer is expected. They are not decoration: “state” requires a short answer with no working, while “explain” requires a reason. The descriptors below are teaching interpretations written for this course; the authoritative definitions are in the current MYP Mathematics guide.

Command term	What your answer must contain	Typical instruction
State / Write down	A short answer, no working required	State the value of the digit 4
Calculate	Working shown, answer with units	Calculate the area of the triangle
Describe	What happens, in words, without explaining why	Describe the pattern in the table
Explain	A reason or mechanism, linked to the mathematics	Explain why 1 is not prime
Justify / Verify	Evidence that supports a stated conclusion	Justify that the lines are parallel
Compare	Both similarities and differences, with reference to values	Compare the two averages
Show that	A logical chain that ends at the given result	Show that the angles sum to 180°
Investigate	A systematic process, a generalisation and a test	Investigate the number of matches
Evaluate / Discuss	A judgement supported by evidence, with limitations	Evaluate the two proposals

NOTE

How the four criteria are used in this book

A Knowing and understanding — fluency questions and worked examples. B Investigating patterns — the investigation in each chapter, which asks for a generalisation and a test. C Communicating — the Task C writing task. D Applying mathematics in real-life contexts — the application task, which always names the constraints and asks for a justified decision. Criteria are attached to activities that genuinely require them, not to ordinary exercises.

Levels are awarded on the evidence a student produces in the activity, using the success criteria printed beside each task. A task that has been attempted but not completed still provides evidence; a blank page provides none.

Notation, working and habits

Mathematical notation is a shared agreement. Using it precisely makes your reasoning checkable by someone else, including your teacher and your future self. The table records the conventions used throughout this book.

Symbol or convention	Meaning	Example
$\times$, $\cdot$	multiplication	$7 \times 4 = 28$
$\div$ or fraction bar	division	$12 \div 3 = {}^{12}\!/_{3} = 4$
$=$	is equal to; both sides have the same value	$3 + 4 = 7$
$\approx$	is approximately equal to	$\pi \approx 3.14$
$<$, $>$	is less than; is greater than	$-3 < 2$, $5 > -1$
$\leq$, $\geq$	is less than or equal to; is greater than or equal to	$x \leq 5$
$\neq$	is not equal to	$0.5 \neq 0.05$
x^2, x^3	power: repeated multiplication	$5^2 = 25$
$\sqrt{49}$	square root	$\sqrt{49} = 7$
%	per cent, out of 100	$25\% = 0.25 = {}^{1}\!/_{4}$
$\angle ABC$	the angle at vertex B	$\angle ABC = 90°$
$\triangle ABC$	the triangle with vertices A, B, C	
$\parallel$, $\perp$	is parallel to; is perpendicular to	$AB \parallel CD$
$\in$, $\notin$	is an element of; is not an element of	$3 \in \mathbb{N}$
$A \cup B$, $A \cap B$	union; intersection	$1,2 \cup 2,3 = 1,2,3$
π	the ratio of circumference to diameter	$\pi \approx 3.14159$
0.363636…	a recurring decimal: the block 36 repeats for ever	${}^{4}\!/_{11}$
cm, m, km, cm^2, m^3, L	units of length, area, volume and capacity	$1\ \text{m}^2 = 10{,}000\ \text{cm}^2$

METHOD

A dependable way to write up work

1. **Record the question.** Copy the important information, including units.
2. **Sketch or tabulate.** A small diagram or table prevents more errors than any other single habit.
3. **Estimate.** Write the size you expect.
4. **Calculate.** Show each step, one line at a time.
5. **State the answer with its unit.**
6. **Check.** Compare with your estimate, or substitute back into the original statement.

APPROACHES TO LEARNING

Academic honesty

Copying a solution without understanding it creates the appearance of progress and hides a gap. Use sample answers to compare methods, not to replace your attempt. In investigations, report your own data and reasoning and acknowledge every source you use.

PART I

THE GRAMMAR OF NUMBER

I

01 Number systems and place value

02 Powers, expanded form and order of operations

03 Factors, multiples and primes

PART I · THE GRAMMAR OF NUMBER

CHAPTER 01

Number systems and place value

Ten digits are enough to write any number we can name. Their power comes from position: the same symbol can mean seven, seventy or seven thousandths.

KEY CONCEPT	**Form**
RELATED	**Quantity**
RELATED	**Representation**
GLOBAL CONTEXT	**Orientation in space and time**

01
CHAPTER

INQUIRY QUESTION

How does a digit's position change its value, and how can we describe that structure so precisely that comparisons and calculations become reliable?

A museum label reads “about 12,000 objects”. A catalogue entry reads “11,846 objects”. Both describe the same collection. This chapter asks what each digit contributes, what zero is doing, and when “about” is the more honest answer.

IN THIS CHAPTER

1.1 Number systems and the number line
1.2 Place value and the role of zero
1.3 Expanded form and equivalent representations
1.4 Comparing, ordering and rounding
1.5 Estimation and sensible accuracy
Investigation 1 How many numbers can a set of digit cards make?
Application 1 The community noticeboard
Task C Explaining place value to a younger student
Review Consolidating Chapter 1
Review questions Practice across the whole chapter
Mixed 1 Fluency, reasoning and problem solving
Cumulative practice 1 Questions that mix this chapter with earlier ones

APPROXIMATELY 12 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

Place-value systems are technologies of representation. They make large quantities manageable, comparisons quick and written calculation possible. The history of zero is not an optional curiosity: it explains why our written system can distinguish 305 from 35 and 3.05 without inventing a new symbol for every value.

KEY CONCEPT · FORM

Form. A number's written form is not a single symbol but an organised system of positions. Each position carries a value ten times the position to its right.

RELATED CONCEPT · QUANTITY

Quantity. Place value lets a small set of digits represent any magnitude. Rounding records an appropriate level of precision rather than every last unit.

RELATED CONCEPT · REPRESENTATION

Representation. A number may appear as words, digits, an expanded sum, a position on a number line or a rounded claim. Choosing the form is part of solving the problem.

GLOBAL CONTEXT

Orientation in space and time

EXPLORATION

How different cultures have recorded quantity, and why the invention of a place-value system with a zero changed what numbers could do.

FACTUAL	CONCEPTUAL	DEBATABLE
What is the value of a digit in a given position? How does multiplying or dividing by 10 change a number's digits?	Why is zero essential in a place-value system, and how can one number have several equivalent representations?	When a number is reported as “about 2,000”, is the result more useful than the exact count?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Read, compare, round and rewrite integers and decimals; select sensible placements.
B · Investigating patterns	i, ii, iii	Investigate how counting possibilities grow with the number of digits; state and test a rule.
C · Communicating	i, ii, iii, iv	Move between words, digits, expanded form, number lines and rounded statements.
D · Applying mathematics	i, ii, iii, iv, v	Choose an appropriate accuracy for a public noticeboard and justify its effect on meaning.

ATL SKILL · CRITICAL THINKING

In order for a student to test the claim that adding a zero always multiplies a number by 10, students must try the claim on a whole number, a decimal and a number ending in zero, and then revise the claim so that it survives every case.

STATEMENT OF INQUIRY

The position of a digit in a base-ten system encodes its value, so a small set of symbols can represent any magnitude when place and zero are used consistently.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand how natural numbers, whole numbers and integers are related.
- Know that the value of a digit depends on its place in a base-ten number.
- Write whole numbers and decimals in expanded form and return to standard form.
- Compare and order integers and decimals using place value and a number line.
- Round to a stated place and choose a reasonable level of accuracy in context.
- Estimate a result and explain whether the estimate is likely to be too high or too low.

SUCCESS CRITERIA

- ✓ I can classify a number as natural, whole, integer or rational.
- ✓ I can state the value of any digit in a whole number or decimal and explain the role of zero.
- ✓ I can write a number in expanded form and reconstruct it accurately.
- ✓ I can order positive and negative numbers and decimals and justify the order.
- ✓ I can round to the nearest stated unit or place and avoid an over-precise answer.
- ✓ I can estimate, compare the estimate with an exact calculation, and explain the difference.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Write “four thousand and sixty” using digits.
2. Which is greater, 0.5 or 0.45? Explain in one sentence.
3. Calculate 8×7.
4. What is 100 more than 2,950?
5. Write $^{7}/_{10}$ as a decimal.
6. Put -2, 0 and 3 in order from least to greatest.

Answers are in the Chapter 1 answer section. If more than two are incorrect, review place value to thousands and simple fractions before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Counting, number bonds, place value to thousands and simple fractions of a whole.	Number families; decimal place value; integer ordering; expanded form; rounding and estimation.	Signed-number operations, powers and standard index form; scientific notation; absolute error.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Museum labels report collection size and years in different forms.	What does each digit in a displayed number actually contribute?	Base-ten place value, zero as placeholder, precision.	Count the possible numbers of different lengths.	Expanded form, number lines and justified rounding.	When does extra precision help, and when does it mislead?

A ENGAGE

A museum label, three ways of writing the same collection

The county museum has just finished counting its collection. The director needs one sentence for the new label, the cataloguer needs an exact record for the database, and the education officer needs a number a class can actually picture. The same collection produces three different numbers.

Label on the gallery wall	12,000 objects
Catalogue entry	11,846 objects
Education leaflet	nearly 12 thousand, of which about 4 hundred are on display

Figure 1.1 Three ways of describing one collection

Read the three records carefully. The wall label is short. The catalogue entry is exact. The leaflet adds a second idea: only part of the collection can be seen. All three are about the same museum, yet each answers a different question, and each has been written for a different reader.

Now look at the numbers themselves. **11,846** contains five digits. Change the position of one digit and the value of the collection changes by thousands. Remove the **8** and the number becomes 1,146: a different museum. Remove the **1** from the front and the number becomes 1,846. Nothing about the shape of the digits has changed; only their positions have.

The museum's education officer has a further problem. She wants a display board that reads **1,000 objects** beside **10,000 objects**, in the same size of digit.

Set side by side, the two labels look almost identical: one extra zero, one extra comma. A visitor glancing at the board may not notice the difference, yet the second collection is ten times the size of the first. Somewhere in this chapter is the reason that **1,000** and **10,000** differ by a factor of ten even though they differ by a single character on the page.

NOTICE AND WONDER

- Which of the three records would you put on the wall, and which would you put in the database? Why?
- The catalogue says 11,846. If the museum acquires 154 more objects, what would the wall label say?
- The leaflet says “about 4 hundred are on display”. Could the exact number be 384? Could it be 441?
- What does the zero in 11,846 contribute? What would happen to the number if that zero were removed?

> Keep this question open until the end of the chapter: when a number is reported as “about 12,000”, has information been lost, or has the museum simply told us what we need?

THE THREE NUMBERS BEHIND EVERY DIGIT

A written number carries three separate pieces of information, and good mathematical reading keeps them apart:

1. **The digit itself.** How many of that size of unit are there?
2. **The position of the digit.** Which size of unit is being counted — ones, tens, tenths?
3. **The precision of the report.** Has the number been rounded, and to what place?

Most early mistakes in this chapter come from confusing the first two. The digit 4 is worth four, forty, four hundred or four tenths depending entirely on where it stands. A useful habit is to read a number aloud in place-value language before calculating with it: 11,846 is "eleven thousand, eight hundred and forty-six", which is one ten-thousand, one thousand, eight hundreds, four tens and six ones.

The third piece of information is the one adults use most and explain least. When the museum writes "about 12,000", it is telling us that the exact count is not the point: the label is meant to convey scale. When a nurse records a temperature as 38.4 °C rather than 38.37 °C, the choice of precision is a decision about how much trust the instrument deserves. This chapter ends by asking you to make that decision deliberately.

B **EXPLORE**
How many different numbers can three cards make?

Work in pairs. You need three cards labelled **3**, **0** and **5**, and a table to record what you find. Do not read on until you have made and recorded at least eight numbers.

YOUR INVESTIGATION

Step 1. Arrange the three cards in a row to make a three-digit number. Record it. Rearrange them to make a different three-digit number. Continue until you believe you have found every possible three-digit number that uses all three cards exactly once.

Step 2. Now make two-digit numbers using any two of the cards. Record every possibility.

Step 3. Finally, make one-digit numbers. Record them.

Cards used	Number of different numbers	List them
1 card		
2 cards		
3 cards		

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

1. Which three-digit numbers did you find? Compare with another pair. Did either pair miss one?
2. Why can 305 be made but 035 cannot be a three-digit number? What is 035 actually worth?
3. How many three-digit numbers are possible? Explain how you know you have them all, without simply listing again.
4. How many two-digit numbers are possible? How does the presence of the zero change the count?
5. Predict how many three-digit numbers four cards would make, if the cards were 3, 0, 5 and 7. Then test your prediction with a systematic list.

MAKING A CONJECTURE

A **conjecture** is a statement you believe is true because of the cases you have tested, but which you have not yet proved. Write one conjecture that begins:

> "With n cards, the number of possible three-digit numbers is … because …"

Test your conjecture on a case you have not used yet. If it fails, revise it. A conjecture that survives a deliberate attempt to break it is worth more than one that has only been checked

against friendly examples.

WHAT YOU SHOULD HAVE NOTICED

- Zero cannot take the leading position in a number written in the usual way, because the leading position would then not be doing its job: 035 is simply 35, a two-digit number.
- The number of arrangements grows quickly, but not without pattern: with three different non-zero digits there are six three-digit numbers, and each of those six appears twice if you also allow a zero in the middle or at the end.
- The position of a digit, not the digit itself, decides its value. This is the idea the rest of the chapter makes precise.

INVESTIGATION

Explore further: a fair challenge

Choose three digits of your own, at least one of which is zero. Without listing, decide how many three-digit numbers they can make, then check by listing systematically. Write one sentence explaining how you decided, and one sentence explaining what you would need to change if leading zeros were allowed.



Three-digit arrangements made from the cards 3, 0 and 5. Zero cannot take the leading place in a three-digit number.

1.1 | Number systems and the number line

A · I

Mathematicians use different number sets for different purposes. Distinguishing them is not a vocabulary game: the classification tells you which operations are always possible and which values count as solutions.

DEFINITION

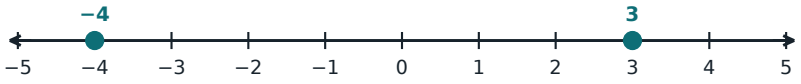
Definitions: the number families

The **natural numbers** are the counting numbers 1, 2, 3, 4, … The **whole numbers** are the natural numbers together with zero: 0, 1, 2, 3, … The **integers** are the whole numbers and their negatives: …, -3, -2, -1, 0, 1, 2, 3, … A **rational number** can be written as a fraction $\frac{a}{b}$, where a and b are integers and $b \neq 0$. Every integer is also rational, because it can be written with denominator 1: $7 = \frac{7}{1}$ and $-3 = \frac{-3}{1}$. Terminating decimals are rational too: $0.25 = \frac{25}{100}$. The families are nested — natural numbers sit inside the whole numbers, which sit inside the integers, which sit inside the rational numbers.

MATHEMATICAL LANGUAGE

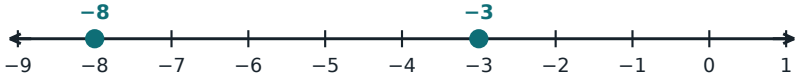
Vocabulary you will use

natural number · whole number · integer · rational number · negative · positive · number line · order · inequality · precision · estimate



On a horizontal number line, values increase from left to right. Negative numbers lie to the left of zero.

A number line is more than a display device. It gives a shared order to every integer, fraction and decimal. To compare two values, locate both: the one further to the right is greater. This stays true for negative numbers, where a larger digit means a smaller value.



−8 lies further left than −3, so −8 < −3 even though 8 is greater than 3.

KEY IDEA

Symbols describe position, not magnitude alone

"-5 < -2" is correct because -5 lies to the left of -2, even though 5 is greater than 2. The symbol always points towards the smaller value, and the smaller value is the one further left. Reversing the symbol reverses the claim: -5 > -2 is false.

READING A NEGATIVE NUMBER LINE

Negative numbers are the first place where a number's written size and its position disagree. The digits in -8 are larger than the digits in -3, yet -8 is the smaller number. Two habits prevent the error:

- Say "eight below zero" rather than "minus eight", so the direction is audible.
- Mark both numbers on a line before comparing. The further-left number is always the smaller one.

The same logic orders temperatures, floors of a building, bank balances and positions below sea level. In each case zero is a boundary rather than an absence: 0 °C is a temperature, floor 0 is a floor, and 0 metres above sea level is a height.

WORKED EXAMPLE 1.1.1

Classifying with evidence

GIVEN The numbers 0, 7, -3, 2.5 and $\frac{3}{4}$.

ASKED State whether each number is natural, whole, an integer or rational. A number may belong to more than one family.

METHOD Test the families in order from smallest to largest. A number that belongs to a larger family is written with evidence, not just a label.

1. 0 is not natural, because counting begins at 1. It is whole, an integer and rational: $0 = \frac{0}{1}$.
2. 7 is natural, whole, an integer and rational: $7 = \frac{7}{1}$.
3. -3 is an integer and rational ≤*ft*(-3 = $\frac{-3}{1}$*right*). It is not natural or whole because it is negative.
4. 2.5 is rational because $2.5 = \frac{5}{2}$. It is not an integer: the point lies between 2 and 3.
5. $\frac{3}{4}$ is rational because both 3 and 4 are integers and the denominator is not zero.

ANSWER 0: whole, integer, rational. 7: natural, whole, integer, rational. -3: integer, rational. 2.5 and $\frac{3}{4}$: rational.

CHECK Every number in the list belongs to at least one family, and the largest family, rational, contains all of them.

WORKED EXAMPLE 1.1.2

A context with negative values

GIVEN Overnight temperatures of -6 °C, -3 °C, -1 °C, 0 °C and 4 °C.

ASKED Order the temperatures from coldest to warmest.

METHOD Coldest means the least number, which is the furthest left on a number line.

1. Compare the negatives first: -6 < -3 < -1.
2. Zero is warmer than every negative value.
3. 4 is warmer than zero.

ANSWER -6 °C, -3 °C, -1 °C, 0 °C, 4 °C.

CHECK Read the list back against the line: each value lies to the right of the one before it.

WATCH OUT Reading the absolute size of the numerals gives “-8 is bigger than -3 because 8 is bigger than 3”. Mark both values on a number line: the further-left number is smaller, so -8 < -3.

WORKED EXAMPLE 1.1.3

Placing fractions and decimals on the line

GIVEN The values $\frac{1}{2}$, 0.45 and $\frac{3}{5}$.

ASKED Place them in order from least to greatest, and explain how the number line supports the order.

METHOD Convert to a common form. Decimals to hundredths make the comparison immediate.

1. $\frac{1}{2} = 0.5$ and $\frac{3}{5} = 0.6$.
2. Compare in hundredths: 0.45 < 0.50 < 0.60.
3. Check the positions on a number line between 0 and 1: 0.45 lies just left of the midpoint, 0.5 at the midpoint, 0.6 to its right.

ANSWER $0.45 < \frac{1}{2} < \frac{3}{5}$.

CHECK The differences are 0.05 and 0.10, so the second gap is twice the first. The line shows that too.

PRACTICE 1.1 · Number families and order

1	State whether each number is natural, whole, an integer or rational. Some answers will name more than one family: (a) 12 (b) 0 (c) $\frac{3}{4}$ (d) -4 (e) 1.2	FLUENCY
2	Write all the integers from -4 to 2 inclusive.	FLUENCY
3	Replace each box with <, > or =: (a) -7 □ -2 (b) 0 □ -1 (c) -5 □ 5	FLUENCY
4	Order these temperatures from coldest to warmest: 3 °C, -2 °C, -7 °C, 0 °C, 5 °C.	FLUENCY
5	Maya says, “Zero is a natural number because we can count zero items.” Explain whether Maya is using the mathematical definition of natural numbers correctly.	REASONING
6	Give an example to show that this statement is false: “If a number is written as a fraction, it must be less than 1.”	REASONING
7	A lift's buttons show floors -2, -1, 0, 1 and 3. A person enters at floor 1, travels to the lowest floor, then to floor 3. Describe the two movements on a number line and state the final floor relative to the starting floor.	PROBLEM SOLVING
8	Construct a number line from -1 to 5. Mark and label -0.5, $\frac{1}{2}$ and 2.5. Explain why 2.5 lies between 2 and 3 even though it is not an integer.	EXTENSION
9	A submarine is at -120 m, a diver at -18 m and a bird at 45 m. Write the three positions in order, then state the vertical distance between the submarine and the bird.	EXTENSION

1.2 | Place value and the role of zero

A · I, II

Our number system is called base ten because each position is worth ten times the position to its right. The digit 4 can stand for 4 ones, 4 tens, 4 hundreds or 4 tenths; the position supplies the value.

Thousands	Hundreds	Tens	Ones	Tenths	Hundredths	Thousandths
3	0	4	8	3	0	7

Place-value columns for 3,048.307. The decimal point marks the boundary between whole-number places and fractional places.

DEFINITION

Place value

A **place value** is the value of a digit determined by its position. In 3,048.307, the digit 4 is in the tens place, so its value is $4 \times 10 = 40$. The digit 7 is in the thousandths place, so its value is $7 \times 0.001 = 0.007$.

Moving one place to the left multiplies the value of a digit by 10. Moving one place to the right divides it by 10. This is why multiplying a number by 10 does not require inventing new digits: every digit shifts one place left. A zero appears at the end of a whole number because there are no units in that position, not because zero is being “added”.

KEY IDEA

Zero has two jobs

As a **number**, zero marks the position on the number line between the positive and negative integers. As a **digit**, zero holds an empty place so that the other digits keep their intended values. Without the zero in 3,048, the digits would read 348 and mean a completely different quantity. Without the zero in 3.05, the digit 5 would shift into the tenths place and produce 3.5 — ten times too large.



0.3 means three tenths

A tenths strip: 0.3 is three of ten equal parts, so 0.3 = 3/10.

SHIFTING THE DIGITS

Multiply by 10 and every digit moves one place left; divide by 10 and every digit moves one place right. The decimal point does not travel as a trick. It marks the boundary between whole-number places and fractional places, and that boundary stays where it is while the digits move across it.

Original number	Multiply by 10	Multiply by 100	Divide by 10
4.06	40.6	406	0.406
37.5	375	3,750	3.75
0.8	8	80	0.08

Multiplying and dividing by powers of ten

COMMON ERROR

Adding a zero is not the same as multiplying by ten

Adding a zero to the whole number 4 gives 40, which is ten times as large. Adding a zero after the decimal part of 4.5 gives 4.50, which is the same number. The rule depends on **where** the zero is inserted, not merely that one more zero appears. Insert a zero inside a number and the other digits change place: 4.05 is not 4.5.

WORKED EXAMPLE 1.2.1

Reading place value

GIVEN The number 6,502.94.

ASKED State the value of each underlined digit: (a) the 5, (b) the 4, (c) the 9.

METHOD Name the place first, then multiply the digit by the value of that place. Reading the position before the digit prevents most errors.

1. The 5 is in the hundreds place. Its value is $5 \times 100 = 500$.
2. The 4 is the second digit after the decimal point, in the hundredths place. Its value is $4 \times 0.01 = 0.04$.
3. The 9 is the first digit after the decimal point, in the tenths place. Its value is $9 \times 0.1 = 0.9$.

ANSWER (a) 500 (b) 0.04 (c) 0.9

CHECK The three values are very different even though the digits are all single digits. Position, not digit, decides value.

WORKED EXAMPLE 1.2.2

Shifting by powers of ten

GIVEN (a) 4.06 × 100 (b) 37.5 ÷ 10 (c) 0.8 × 1,000.

ASKED Calculate each value and explain the movement of the digits.

METHOD Each multiplication by 10 moves every digit one place left; each division by 10 moves every digit one place right. For 100, move two places.

1. 4.06 × 100 = 406: every digit moves two places left, so the tenths digit 0 becomes a unit and the hundredths digit 6 becomes a ten.
2. 37.5 ÷ 10 = 3.75: every digit moves one place right.
3. 0.8 × 1,000 = 800: eight tenths of a thousand is eight hundred.

ANSWER (a) 406 (b) 3.75 (c) 800

CHECK Estimate first. 4 × 100 = 400, and 406 is close. 37 ÷ 10 is about 3.7. Eight tenths of a thousand is 800, not 8,000.

WORKED EXAMPLE 1.2.3

Which is larger: 3.5 or 3.05?

GIVEN Two decimals that look similar when written quickly.

ASKED Decide which is larger, using place value rather than the number of digits.

METHOD Compare digit by digit from the left, in the same place. Never compare the total number of digits.

1. Both numbers have 3 units, so compare the tenths: 3.5 has 5 tenths, 3.05 has 0 tenths.
2. Since 5 > 0, the comparison is decided in the tenths place.
3. Write 3.05 as 3.050 if you need the same number of decimal places: 3.500 > 3.050.

ANSWER 3.5 > 3.05. The extra digit in 3.05 does not make it larger.

CHECK On a number line, 3.05 lies just past 3 and 3.5 lies halfway to 4.

WATCH OUT "3.05 has more digits, so it is bigger." Digits after the decimal point do not add size unless they are in a place that carries value.

PRACTICE 1.2 · Place value and the role of zero

1	State the value of the digit 4 in each number: (a) 4,271 (b) 9,403 (c) 0.4 (d) 12.04	FLUENCY
2	State the value of the digit 7 in each number: (a) 6,427 (b) 23.85 (c) 0.072 (d) 8,106.7	FLUENCY
3	Write the number represented by 8,000 + 10 + 6.	FLUENCY
4	Calculate: (a) 36.2 × 10 (b) 5.48 × 100 (c) 920 ÷ 10 (d) 7.3 ÷ 100	FLUENCY
5	In the number 2.06, what does the zero do? Write what the number would become if the zero were removed and explain the change in value.	REASONING
6	Which is larger, 3.5 or 3.05? Explain using place value rather than the number of digits.	REASONING
7	A shop prices an item at 12.50. The manager multiplies the price by 100 to work in cents. What is the new number, and what does it mean?	PROBLEM SOLVING
8	I multiply a number by 10 and then divide the result by 100. The final answer is 0.34. What was the original number?	EXTENSION

9 Amir writes 3,000 + 40 + 5 and says the number is 345. Explain what he has done wrong and write the correct number. EXTENSION

1.3 | Expanded form and equivalent representations

C · I, II

Expanded form takes a number apart so that every digit can be seen with the value it contributes. Written both ways, a number becomes something you can check, compare and reconstruct.

DEFINITION
Expanded form

Expanded form writes a number as the sum of the values of its digits. For whole numbers each term is a digit multiplied by a power of ten:

$$3{,}048 = 3 \times 1{,}000 + 0 \times 100 + 4 \times 10 + 8 \times 1$$

For decimals the same idea continues to the right of the decimal point:

$$40.607 = 4 \times 10 + 0 \times 1 + 6 \times 0.1 + 0 \times 0.01 + 7 \times 0.001$$

The zero terms may be left out, but writing them makes the structure visible: every place is accounted for, including the empty ones.

MATHEMATICAL LANGUAGE
Vocabulary you will use

place value · digit · expanded form · standard form · equivalent · tenths · hundredths · thousandths · round · word form

WHY EXPANDED FORM IS WORTH THE EFFORT

Expanded form is the written record of the place-value system. Three things become possible once a number is written this way.

- **Checking.** If a digit has been written in the wrong place, the expanded sum will not match the number you meant.
- **Reconstructing.** Given the expanded sum, you can rebuild the number without guessing where each digit belongs.
- **Comparing.** Two numbers in expanded form can be compared term by term, starting from the largest place.

In later chapters expanded form becomes the bridge to index notation: $3 \times 1{,}000$ is 3×10^3. The expanded sum you write now is the same idea that will later be compressed into powers.

KEY IDEA
Equivalent representations of the same quantity

0.7, 0.70 and $^{7}/_{10}$ all name the same number. So do $2^{1}/_{2}$, 2.5 and $^{5}/_{2}$. Writing zeros after the last decimal digit does not change the value, because those zeros contribute 0×0.01 and so on. This is exactly why expanded form can be used as a check: the zero terms add nothing.

COMMON ERROR

Zero terms are not optional in the reasoning

Leaving out a zero term when you write the number is fine; forgetting that the place exists is not. If 3,048 is expanded as $3 \times 1{,}000 + 4 \times 10 + 8$, the reader cannot tell whether the hundreds place is zero or missing. Writing 0×100 removes the doubt.

WORKED EXAMPLE 1.3.1

Writing a whole number in expanded form and in words

GIVEN The number 5,207.

ASKED (a) Write it in expanded form using all four places. (b) Write it in words. (c) Explain which digit is doing the least work.

METHOD Read the places from left to right: thousands, hundreds, tens, ones. Write each digit multiplied by its place value.

1. Thousands: $5 \times 1{,}000$. Hundreds: 2×100. Tens: 0×10. Ones: 7×1.
2. Join the terms: $5{,}207 = 5 \times 1{,}000 + 2 \times 100 + 0 \times 10 + 7 \times 1$.
3. In words: five thousand, two hundred and seven.

ANSWER (a) $5 \times 1{,}000 + 2 \times 100 + 0 \times 10 + 7 \times 1$ (b) five thousand, two hundred and seven (c) The zero contributes nothing to the total, but it holds the tens place so that the 2 keeps the value 200 rather than 20.

CHECK Add the terms: $5{,}000 + 200 + 0 + 7 = 5{,}207$, which matches the original number.

WATCH OUT Reading the digits without their places can produce “five thousand and twenty-seven”, which would be 5,027. Say each place aloud as you expand.

WORKED EXAMPLE 1.3.2

Rebuilding a number from an expanded sum

GIVEN $6 \times 100 + 0 \times 10 + 3 \times 1 + 9 \times 0.01 + 4 \times 0.001$.

ASKED Write the number in standard form and state the value of the 9.

METHOD Allocate each term to its column, then read the columns from left to right.

1. $6 \times 100 = 600$ goes in the hundreds column.
2. 0×10 means the tens column is empty, so write 0.
3. $3 \times 1 = 3$ goes in the ones column; the decimal point follows.
4. 9×0.01 goes in the hundredths column, so the tenths column needs a 0.
5. 4×0.001 goes in the thousandths column.

ANSWER 603.094. The 9 has value 0.09.

CHECK Read the number back: 603.094 has 6 hundreds, 0 tens, 3 ones, 0 tenths, 9 hundredths and 4 thousandths. Every term is accounted for.

WATCH OUT Writing 603.94 loses a place. The 9 was hundredths, not tenths, so a zero must be written between the decimal point and the 9.

WORKED EXAMPLE 1.3.3

The same quantity, three ways

GIVEN The quantity two and a half litres.

ASKED Write it as a mixed number, as a decimal and in expanded form; explain what changes and what does not.

METHOD Choose the form that makes the next operation easiest, then check that the value is unchanged.

1. As a mixed number: $2\frac{1}{2}$ litres.
2. As a decimal: 2.5 litres, because $\frac{1}{2} = 0.5$.
3. In expanded form: $2 \times 1 + 5 \times 0.1$ litres, which is $2 + 0.5$.

ANSWER $2\frac{1}{2}$ L = 2.5 L = $2 \times 1 + 5 \times 0.1$ L. The representation changes; the quantity does not.

CHECK Pour 2.5 litres into a 2.5-litre bottle: it fills exactly. The form of the number has no effect on the amount of water.

PRACTICE 1.3 · Expanded form and equivalent representations

1	Write in expanded form using all places: (a) 4,706 (b) 90,215 (c) 6.08	FLUENCY
2	Write each as a single number: (a) $8 \times 1{,}000 + 3 \times 100 + 0 \times 10 + 5 \times 1$ (b) $2 \times 10 + 7 \times 1 + 4 \times 0.1$ (c) $5 \times 0.01 + 3 \times 0.001$	FLUENCY
3	Write in words: (a) 3,060 (b) 12.5 (c) 0.09	FLUENCY
4	Write in figures: (a) seven thousand and forty (b) two hundred and six thousandths (c) ninety and nine tenths	FLUENCY
5	Layla writes 4.5 as $4 \times 1 + 5 \times 0.01$. Explain her error and write the correct expansion.	REASONING
6	Which number is larger: 0.70 or 0.7? Explain using expanded form.	REASONING
7	A till records 12.5 as 12.05. State how much money has been lost or gained on the record and explain the change in place value.	PROBLEM SOLVING
8	Write the smallest and the largest four-digit numbers that can be made from the digits 4, 0, 7 and 2 without repetition.	PROBLEM SOLVING
9	A number is written as $a \times 100 + b \times 10 + c \times 1$ where a, b and c are digits and $a \neq 0$. Explain why the number is always between 100 and 999.	EXTENSION
10	A calculator displays 3,050. A finger slips and the 5 becomes a 0. Describe the change in value, and state which place was affected.	EXTENSION

1.4 | Comparing, ordering and rounding

A · I, II, III

Comparing is a place-value skill, not a digit-counting skill. Rounding is a decision about precision: it replaces a number with the nearest value at a chosen place. Both depend on the same system of positions.

COMPARING DECIMALS

Compare from the left, one place at a time, and stop as soon as the digits differ. Never compare the number of digits after the decimal point: 0.8 has fewer digits than 0.75 but is

larger.

1. Write both numbers with the same number of decimal places, using trailing zeros.
2. Align the decimal points and compare the digits in each column from left to right.
3. The first column where the digits differ decides the comparison.

METHOD
A reliable comparison routine

Write 0.8 and 0.75 with the same number of places: 0.80 and 0.75. Compare the tenths: 8 tenths against 7 tenths. Since 8 > 7, we have 0.80 > 0.75, so 0.8 > 0.75. The comparison needed only one column.

ORDERING A MIXED LIST

When a list mixes positive and negative numbers, split it first. Every negative number is less than every positive number, and zero sits between them. Order the negatives by remembering that a larger digit after the minus sign means a smaller value.

ROUNDING

DEFINITION
Rounding to a place

To **round** a number to a given place, look at the digit immediately to the right of that place. If it is 5 or more, increase the digit in the place by 1; if it is 4 or less, leave it unchanged. Then replace every digit to the right with zeros (before the decimal point) or remove them (after the decimal point).



3,486 is closer to 3,500 than to 3,400, so it rounds to 3,500 when rounded to the nearest hundred.

Rounding is not “making the number bigger” or “making it smaller”. It is choosing the nearest value at the required precision. 3,486 rounded to the nearest hundred is 3,500 because 3,486 is 14 away from 3,500 and 86 away from 3,400. Rounded to the nearest thousand, the same number is 3,000.

KEY IDEA
Every rounded answer should be readable as a decision

When you round, say the place aloud: “to the nearest hundred”, “to two decimal places”. A rounded number without its place is a claim without evidence, because 3,500 could be the rounding of anything from 3,450 to 3,549.

COMMON ERROR
Two rounding mistakes to avoid

Mistake 1: rounding twice. Rounding 4.47 to one decimal place by first rounding to 4.5 and then to 5 gives the wrong answer; the correct answer is 4.5, because the digit after the tenths is 7. **Mistake 2: dropping the zeros.** 2,958 rounded to the nearest hundred is 3,000, not 30. The zeros record which places are now unknown.

WORKED EXAMPLE 1.4.1

Ordering a mixed list

GIVEN -2.5, 0.4, -2.05, 3, -0.5.

ASKED Order the numbers from least to greatest.

METHOD Separate the negatives, order them using place value, then attach zero and the positive numbers.

1. Negatives: -2.5, -2.05 and -0.5. Compare the sizes 2.5, 2.05 and 0.5: $2.5 > 2.05 > 0.5$.
2. For negatives the order reverses: $-2.5 < -2.05 < -0.5$.
3. The positive values are 0.4 and 3, with $0.4 < 3$.

ANSWER $-2.5 < -2.05 < -0.5 < 0.4 < 3$.

CHECK Place all five on a number line. Each value should lie to the right of the one before it.

WATCH OUT "-2.05 is bigger than -2.5 because 205 is bigger than 25." It is bigger in magnitude but further left on the line, so it is smaller: $-2.05 > -2.5$ is correct only as a magnitude statement, and wrong as a value statement.

WORKED EXAMPLE 1.4.2

Rounding a whole number to different places

GIVEN The number 3,846.

ASKED Round it to (a) the nearest ten (b) the nearest hundred (c) the nearest thousand.

METHOD Identify the place, then look at the single digit to its right.

1. Nearest ten: look at the ones digit 6, which is 5 or more, so 3,846 becomes 3,850.
2. Nearest hundred: look at the tens digit 4, which is less than 5, so 3,846 becomes 3,800.
3. Nearest thousand: look at the hundreds digit 8, which is 5 or more, so 3,846 becomes 4,000.

ANSWER (a) 3,850 (b) 3,800 (c) 4,000

CHECK Each answer lies within half a unit of the rounding place from 3,846: 4 from 3,850, 46 from 3,800 and 154 from 4,000.

WORKED EXAMPLE 1.4.3

Rounding decimals

GIVEN The number 0.4736.

ASKED Round it to (a) two decimal places (b) one decimal place (c) the nearest whole number.

METHOD Count decimal places from the point, then use the single digit immediately to the right of the last place you keep.

1. Two decimal places: keep 0.47, look at 3, which is less than 5, so the answer is 0.47.
2. One decimal place: keep 0.4, look at 7, which is 5 or more, so the answer is 0.5.
3. Nearest whole number: look at the tenths digit 4, which is less than 5, so the answer is 0.

ANSWER (a) 0.47 (b) 0.5 (c) 0

CHECK Each answer is the nearest value at that precision. 0.4736 is closer to 0.47 than 0.48, closer to 0.5 than 0.4, and closer to 0 than to 1.

WORKED EXAMPLE 1.4.4

Choosing a sensible accuracy

GIVEN A school records 1,247 students; a shop sells an item for 12.48; a nurse records a temperature of 37.6 °C.

ASKED State a sensible precision for each record and justify it.

METHOD Ask what decision the number will support. Extra digits are only useful if they change the decision.

1. Student numbers are counted in whole people, so 1,247 is exact; a newsletter may round it to "about 1,250" for readability.
2. Money is recorded to the nearest cent because that is the smallest unit in use: 12.48, not 12.481.
3. A clinical thermometer reads to one decimal place, so 37.6 °C records the instrument's precision; writing 37.63 would claim accuracy the instrument does not have.

ANSWER 1,247 (or 1,250 in a newsletter) · 12.48 · 37.6 °C. In each case the precision matches the unit that matters.

CHECK Would an extra digit change any decision? If not, the extra digit is decoration rather than information.

PRACTICE 1.4 · Comparing, ordering and rounding

1	Replace each box with <, > or =: (a) 0.6 □ 0.58 (b) 2.05 □ 2.5 (c) 3.4 □ 3.40 (d) -1.2 □ -1.02	FLUENCY
2	Round 7,463 to (a) the nearest ten (b) the nearest hundred (c) the nearest thousand.	FLUENCY
3	Round 5.278 to (a) two decimal places (b) one decimal place (c) the nearest whole number.	FLUENCY
4	Write these in order from least to greatest: 2.4, -3, 0.45, -0.9, 2.04.	FLUENCY
5	Tom says "0.305 is larger than 0.35 because it has more digits." Explain why he is wrong using place value.	REASONING
6	A number rounds to 6.8 when rounded to one decimal place. Write three possible values of the number.	REASONING
7	A runner's time is 12.46 s. The results board shows times to one decimal place. What will be displayed, and what information is lost?	PROBLEM SOLVING
8	A shop rounds every price up to the next whole unit. What does a customer pay in total for items priced 3.40, 2.65 and 1.20, and how much more than the exact total?	PROBLEM SOLVING
9	Find the smallest and largest whole numbers that round to 500 when rounded to the nearest hundred.	EXTENSION
10	A number rounds to 3.0 to one decimal place and to 3 to the nearest whole number. What can you say about the number, and what can you not say?	EXTENSION
11	Explain why rounding to the nearest hundred and then to the nearest thousand can give a different answer from rounding directly to the nearest thousand. Give an example.	EXTENSION
12	The population of a town is reported as 24,000 and later as 23,847. Which report is more useful for (a) a road sign (b) a funding application? Justify both choices.	PROBLEM SOLVING

1.5 | Estimation and sensible accuracy

D · I, II, III

An estimate is a calculation you can do in your head with numbers you can hold. Its purpose is not to replace the exact answer but to tell you whether the exact answer is plausible.

DEFINITION

Estimating by rounding

To **estimate** a calculation, round each number to a convenient place, carry out the simpler calculation, and state the rounding you used. An estimate should be labelled: “about 8,000, using numbers rounded to the nearest hundred”.

TWO HABITS THAT MAKE ESTIMATES USEFUL

- **Round in one direction on purpose.** If you round every number up, the estimate is certainly too high; if you round every number down, it is certainly too low. That knowledge is often more valuable than the estimate itself.
- **Compare, then decide.** An exact answer far from your estimate suggests a keying error, a misplaced decimal point or a wrong operation. An exact answer close to the estimate is evidence that the calculation is sound.

KEY IDEA

Front-end estimation

For a sum, add the leading digits first and then adjust: 4,876 + 3,142 is about 4,000 + 3,000 = 7,000, and the remaining parts add roughly 900 + 100 = 1,000, so the total is about 8,000. Front-end estimation keeps the size of the answer in view while you work.

WORKED EXAMPLE 1.5.1

Estimating a sum and judging the estimate

GIVEN 4,876 + 3,142.

ASKED Estimate the sum, then calculate it exactly and compare.

METHOD Round each number to the nearest hundred, add, then compare with the exact value.

1. 4,876 ≈ 4,900 and 3,142 ≈ 3,100.
2. Estimate: 4,900 + 3,100 = 8,000.
3. Exact: 4,876 + 3,142 = 8,018.

ANSWER The estimate is 8,000; the exact sum is 8,018, which is 18 more. The estimate is reasonable: it is within 1% of the exact value.

CHECK Both numbers were rounded in opposite directions — one up by 24, one down by 42 — so the estimate should be slightly low, and it is.

WORKED EXAMPLE 1.5.2

Estimating a product in a real context

GIVEN A class orders 38 workbooks at 21 units each.

ASKED Estimate the total cost and say whether the estimate is more likely to be too high or too low.

METHOD Round both factors to convenient numbers whose product is easy, then reason about the direction of the rounding.

1. $38 \approx 40$ (rounded up) and $21 \approx 20$ (rounded down).
2. Estimate: $40 \times 20 = 800$ units.
3. Exact: $38 \times 21 = 798$ units.

ANSWER About 800 units; the exact cost is 798, so the estimate is 2 units too high. Because one factor was rounded up by 2 and the other down by 1, the two errors nearly cancel.

CHECK $38 \times 21 = 38 \times 20 + 38 = 760 + 38 = 798$.

WORKED EXAMPLE 1.5.3

Choosing the accuracy the decision needs

GIVEN A community centre must buy paint for a wall of area 47.6 m². One litre covers 12 m² and paint is sold only in whole litres.

ASKED How many litres should be bought, and why is the rounded calculation not simply "round the answer"?

METHOD Calculate, then round in the direction that keeps the job possible.

1. $47.6 \div 12 \approx 3.97$ litres.
2. Rounding to the nearest whole litre gives 4 litres — but 4 litres cover 48 m², which is just enough.
3. Because paint cannot be bought in part-litres and running short would stop the job, round **up** to the next whole litre: 4 litres.

ANSWER 4 litres. Rounding up is a decision about risk, not about the nearest number.

CHECK $4 \times 12 = 48$ m², which is greater than 47.6 m². Three litres would cover only 36 m².

PRACTICE 1.5 · Estimation and sensible accuracy

1	Estimate by rounding to the nearest hundred: (a) 687 + 214 (b) 1,240 - 380	FLUENCY
2	Estimate by rounding to the nearest ten: (a) 38 × 21 (b) 612 ÷ 29	FLUENCY
3	Estimate: (a) 4,912 + 3,087 (b) 19 × 41 (c) 5,980 ÷ 31	FLUENCY
4	Without calculating, decide whether 49 × 21 is greater or less than 1,000. Explain your reasoning.	REASONING
5	A ticket costs 18.50. Estimate the cost of 24 tickets and state whether your estimate is above or below the exact cost.	PROBLEM SOLVING
6	A runner completes 12 laps of a 400 m track and says she ran "about 5 km". Is the claim reasonable? Show your estimate.	PROBLEM SOLVING
7	A shop assistant calculates 39 × 49 and writes 191. Use estimation to show the answer must be wrong, then give the correct value.	EXTENSION
8	A budget of 2,000 units is shared between 7 clubs. Estimate each share and explain whether the organiser can promise "about 300 each".	EXTENSION

9	Give an example where rounding both numbers up gives an estimate that is more than 10% too high, and explain why the error is larger than usual.	EXTENSION
10	A newspaper reports that “nearly 2,000 people attended”, while the organisers counted 1,614. Is the report acceptable? Justify your answer with an estimate of the error.	PROBLEM SOLVING

F INVESTIGATE AND APPLY

Two tasks: a criterion B investigation and a criterion D application

INVESTIGATION 1 | How many numbers can a set of digit cards make?

B · I, II, III

You have met this idea in the Explore activity. Now treat it as a proper investigation: record systematically, look for a pattern, state a rule and test it.

INVESTIGATION

Investigation brief

Question. Using each of the digit cards 1, 2, 3 and 4 exactly once, how many different numbers can be made? How does the answer change when a zero card is included? **What you need.** Four cards numbered 1, 2, 3 and 4; a second set that replaces one card with a 0; paper for systematic lists. **What to produce.** A table of results, a clear statement of any pattern, a general rule with a reason, and a test of the rule on a case you have not used.

PART 1 · RECORD SYSTEMATICALLY

Complete this table for sets of cards without a zero. Use a systematic list — for example, fix the first digit and vary the rest — rather than writing numbers at random.

Number of cards	One-digit numbers	Two-digit numbers	Three-digit numbers	Four-digit numbers
1		–	–	–
2			–	–
3				–
4				

Numbers made from distinct non-zero digit cards (copy and complete)

PART 2 · LOOK FOR STRUCTURE

1. How many two-digit numbers can be made from 1, 2 and 3? List them, then count them by fixing the first digit.
2. How many three-digit numbers can be made from 1, 2, 3 and 4? Explain your count without listing every number.
3. Describe how the number of arrangements changes as one more card is added. Write the change as a multiplication.
4. State a rule for the number of arrangements of n different non-zero cards, used all at once.

PART 3 · BRING IN THE ZERO

Now repeat the count with the cards 0, 1, 2 and 3. A number cannot begin with 0, so some arrangements are not numbers at all.

Cards	All arrangements	Arrangements beginning with 0	Valid numbers
0 and 1	2	1	1
0, 1 and 2	6	2	4
0, 1, 2 and 3			
0, 1, 2, 3 and 4			

Numbers made when one card is a zero

PART 4 · EXPLAIN, GENERALISE AND TEST

1. Explain why the number of arrangements that begin with zero is the number of arrangements of the remaining cards.
2. Write a rule for the number of valid numbers made from n cards, one of which is zero.
3. Test your rule on a set of five cards. Does it give the count you expect? Show your check.
4. A classmate claims: “Adding a zero card doubles the number of valid numbers.” Test the claim with two cases and decide whether it is true.
5. Explain in one paragraph why the zero card changes the count in the way it does.

CRITERION TASK

Criterion B task · What your report must contain

Investigating patterns (strands i–iii). A systematic record of results; a description of the pattern as a general rule; a verification of the rule against at least one further case; a statement of the limits of your rule, for example whether it requires all the cards to be used.

Level	What the work shows		
0	The report does not record results in a usable way.		
1–2	Some arrangements are listed, but the record is incomplete or unsystematic.		
3–4	The record is complete for the cases attempted	a pattern is described in words.	
5–6	A correct general rule is stated and supported by a reason	the rule is tested on a new case.	
7–8	The rule is justified	the effect of the zero is explained	the limits of the rule are stated and verified.

Formative rubric · Criterion B (classroom use, not an official IB descriptor)

APPLICATION 1 | The community noticeboard

D · I, II, III, IV

The Riverside Community Centre is printing a noticeboard for its annual open day. Every number on the board has to survive being read from three metres away — and every number has to remain honest.

THE BRIEF

Item	Exact figure
Visitors during the year	12,487
Volunteer hours	3,842
Open-day budget	950 units
Cost of printing per poster	2.35 units
Posters needed	24
Refreshment cost per visitor (planned)	1.20 units

Information the centre has collected

- The board has room for **six numbers only**.
- The director wants “numbers people can read at a glance”.
- The treasurer wants “no number on the board that could be challenged as inaccurate”.

TASK 1 · CHOOSE AND JUSTIFY

1. Choose six figures for the board and write each one in the precision you would print. For each, state the exact value and the rounding you used.
2. Which figures must stay exact? Explain why.
3. Which figures may be rounded, and what does the reader lose by the rounding?

TASK 2 · ESTIMATE THE OPEN-DAY COSTS

1. Estimate the total printing cost by rounding the price and the number of posters. State whether your estimate is above or below the exact total.
2. Calculate the exact printing cost and compare it with your estimate.
3. If 300 visitors attend, estimate the refreshment cost. Round in the direction that protects the budget, and justify the direction.
4. Is 950 units enough? Show the figures that support your answer.

TASK 3 · EVALUATE THE WORDING

The first draft of the board reads: **“Over 13,000 visitors this year — nearly 4,000 volunteer hours.”**

1. Are both statements accurate? Use the exact figures to check each one.
2. Rewrite the two statements so that they are both readable and honest.
3. A trustee says, “Rounding up makes the centre look better and nobody checks.” Write a two-sentence reply using the idea of precision from this chapter.

CRITERION TASK

Criterion D task · What your response must contain

Applying mathematics (strands i–iv). The relevant quantities identified; appropriate rounding chosen for each figure; calculations shown with units; the accuracy of each printed figure justified; a recommendation for the board supported by the numbers.

TASK C | Explaining place value to a younger student

C · I, II, III, IV

You have been asked to write one page for a Year 5 mathematics display. Your reader is ten years old and has just met decimals.

CRITERION TASK

The task

Write a one-page explanation that answers the question: **why does the position of a digit change its value?** Your page must contain:

1. One worked example using a whole number and one using a decimal, set out so that a reader can follow every step.
2. One diagram — a place-value table, a number line or an annotated number — that carries part of the explanation.
3. One correct sentence about what zero does.
4. One example of a common mistake, with the correction.
5. No more than 200 words of continuous text, not counting the diagram or the example.

Criterion C strand	Evidence in your page
Language and notation (i)	Place-value vocabulary used correctly: digit, place, value, tenths, hundredths.
Representation (ii)	A diagram is used and referred to in the text, not merely attached.
Reasoning (iii)	The explanation says why the value changes, not only what happens.
Organisation (iv)	A reader can move from the example to the rule without extra help.

Success criteria for this task

G REFLECT AND CONSOLIDATE

Summary, vocabulary, misconceptions, self-assessment and review

REVIEW | Consolidating Chapter 1

A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- Numbers belong to nested families: natural numbers inside whole numbers, inside integers, inside rational numbers.
- In base ten, each place is worth ten times the place on its right, so a digit's value depends on its position.
- Zero has two roles: a number on the line and a placeholder in a written number.
- Expanded form makes the place-value structure visible and allows a number to be checked and reconstructed.
- Comparing numbers is a left-to-right comparison of places; the number of digits is irrelevant.
- Rounding replaces a number by the nearest value at a chosen place; the choice of place is a decision about precision.

- An estimate is a check on plausibility, not a substitute for an exact answer.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Natural number	A counting number: 1, 2, 3, …
Whole number	Zero together with the natural numbers.
Integer	Any whole number, positive, negative or zero.
Rational number	A number that can be written as a fraction of two integers.
Place value	The value a digit has because of its position.
Expanded form	A number written as the sum of the values of its digits.
Standard form	A number written with digits in the usual way.
Round	Replace a number by the nearest value at a stated place.
Estimate	An approximate value used to judge plausibility.
Precision	The place to which a number has been given or rounded.

Vocabulary for Chapter 1

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence	
“A longer decimal is larger.”	Compare place by place.	$0.305 < 0.35$	
“Adding a zero always multiplies by 10.”	It depends where the zero is written.	$4.5 = 4.50$ but $4 \to 40$	
“Zero is a natural number.”	Counting begins at 1	zero is a whole number.	$0 \in W$, $0 \notin \mathbb{N}$
“-8 > -3 because 8 > 3.”	The further left, the smaller.	$-8 < -3$	
“Rounding makes an answer exact.”	Rounding records precision, not exactness.	3,500 could be 3,450 to 3,549	

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
classify a number as natural, whole, integer or rational.			
state the value of any digit, including after a decimal point.			
write a number in expanded form and rebuild it.			
order integers and decimals and justify the order.			

I can …	Not yet	Getting there	Confident
round to a stated place and choose a sensible accuracy.			
estimate a calculation and judge the estimate.			

How confident are you? Tick one column for each statement

REFLECTION

1. Which idea in this chapter changed the way you read a number? Give an example of the change.
2. Where did you make a mistake this chapter, and what was the first line at which your reasoning went wrong?
3. Write one sentence beginning “A digit's value depends on …” and one beginning “Zero matters because …”.
4. When would you prefer an exact number to a rounded one? Describe a situation from your own life.
5. Which question in this chapter would you like to revisit, and what would you do differently?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ1 · Applying the chapter

1	Write “nine thousand and seventy” in digits.	FLUENCY
2	State the value of the digit 6 in 4,632.85.	FLUENCY
3	Round 8,736 to the nearest hundred.	FLUENCY
4	Order from least to greatest: -3.4, 2.05, -0.8, 2.5, 0.	FLUENCY
5	Calculate 4.7×100.	FLUENCY
6	Write 0.36 in expanded form.	FLUENCY
7	Explain why 4,050 is not the same as 4,005, using place value.	REASONING
8	A number rounds to 700 to the nearest hundred and to 1,000 to the nearest thousand. What can you conclude?	REASONING
9	Two students round 2,951 to the nearest hundred. One writes 3,000 and one writes 2,900. Who is correct, and why might the other answer appear?	REASONING
10	A box holds 24 packs; each pack holds 18 cards. Estimate the number of cards, then calculate exactly.	PROBLEM SOLVING
11	A shop's takings are 1,247 units on Monday and 986 units on Tuesday. Estimate the two-day total and state whether the estimate is above or below the exact total.	PROBLEM SOLVING
12	The temperature at 06:00 is -4 °C. It rises 9 degrees by noon and falls 6 degrees by midnight. What is the midnight temperature?	PROBLEM SOLVING
13	Write a paragraph of no more than 60 words explaining to a younger student why zero is needed in 305.	COMMUNICATING

14	Write 6,084 in words and in expanded form.	COMMUNICATING
15	Find a number that is greater than 3.4 but less than 3.5, and explain how you know it lies between them.	EXTENSION
16	How many whole numbers round to 60 when rounded to the nearest ten?	EXTENSION
17	The digit 4 appears in the number 4,444 four times. State each of its four values and their sum.	EXTENSION
18	Explain why the statement “a number with more digits is always larger” is false for decimals but true for whole numbers.	EXTENSION
19	A museum's collection is 11,846 objects, of which 4,096 are on display. Estimate how many are in storage and state whether your estimate is above or below the exact answer.	PROBLEM SOLVING

MIXED 1 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX1 · End-of-chapter mixed set

1	Write the number that is 100 more than 9,950.	FLUENCY
2	Calculate: (a) 6.2 × 10 (b) 340 ÷ 100 (c) 0.07 × 1,000	FLUENCY
3	Round 12.478 to two decimal places and to one decimal place.	FLUENCY
4	Put in order: -2.5, 1.05, -2.05, 1.5, -0.25.	FLUENCY
5	Calculate 3,000 + 400 + 5 + 0.6 + 0.02.	FLUENCY
6	Sasha says “2.4 × 100 = 2.400”. Identify the error and give the correct answer.	REASONING
7	Is 0.7 of a litre more or less than 700 ml? Explain.	REASONING
8	A lift starts at floor 4, goes down 7 floors, then up 3. Where does it finish?	PROBLEM SOLVING
9	Three friends share 2,000 units equally. Estimate each share, then calculate exactly to the nearest unit and state the remainder.	PROBLEM SOLVING
10	A printer produces 68 pages a minute. Estimate how long it takes to print 2,000 pages.	PROBLEM SOLVING
11	Explain in two sentences why expanded form is useful when checking a written number.	COMMUNICATING
12	Describe one situation in which rounding to the nearest ten would be misleading.	COMMUNICATING
13	A number rounds to 5.6 to one decimal place. Write the smallest and largest possible values.	EXTENSION
14	Find two numbers whose rounded values to the nearest hundred are equal but whose difference is 90.	EXTENSION
15	Explain what happens to a number's value when a zero is inserted between the decimal point and the first non-zero digit. Use an example.	EXTENSION

16	A sign says “2,000 m to the summit”. The actual distance is 1,760 m. Estimate the percentage of overstatement and comment on whether the sign is acceptable.	**EXTENSION**

CUMULATIVE PRACTICE 1 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP1 · Cumulative practice

1	Write the value of the digit 7 in 27,405.	**FLUENCY**
2	Round 6.847 to two decimal places.	**FLUENCY**
3	Explain why 2.5 and 2.05 are not equal.	**REASONING**
4	Explain why rounding can change the answer to a subtraction.	**REASONING**
5	A shop rounds prices to the nearest unit. What is the greatest possible error for a price shown as 40?	**PROBLEM SOLVING**
6	Write 3,048 in words.	**PROBLEM SOLVING**
7	Explain how zero makes 305 different from 35.	**COMMUNICATING**
8	Find two numbers that round to 60 to the nearest ten.	**EXTENSION**

CHAPTER 02

Powers, expanded form and order of operations

A power is a compressed instruction: it records how many copies of a factor to multiply. A shared order of operations keeps that compression unambiguous.

KEY CONCEPT	**Form**
RELATED	**Simplification**
RELATED	**Quantity**
GLOBAL CONTEXT	**Scientific and technical innovation**

02
CHAPTER

INQUIRY QUESTION

When does a shorter notation make a calculation clearer, and what rules must everyone share before a compact expression can be read correctly?

A phone advertises 64 GB of storage. A computer reports the same memory as 59.6 GB. Both numbers are honest. This chapter explains why a power of ten and a power of two can describe the same machine, and why the order in which we apply operations changes the answer.

IN THIS CHAPTER

2.1 From repeated multiplication to powers
2.2 Powers of ten and expanded form
2.3 Squares, cubes and roots
2.4 Order of operations
2.5 Substitution and expressions with powers
Investigation 2 Does doubling always stay ahead?
Application 2 The school data store
Task C Explaining a calculator's answer
Review Consolidating Chapter 2
Review questions Practice across the whole chapter
Mixed 2 Fluency, reasoning and problem solving
Cumulative practice 2 Questions that mix this chapter with earlier ones

APPROXIMATELY 14 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

A computer stores information in binary, so its capacity grows by powers of two, while storage is often advertised in powers of ten. The difference is small at first and large at scale. The same compact notation that describes a memory chip also describes the distance to a star.

KEY CONCEPT · FORM

Form. A power has a precise form: a base and an exponent. Changing either part changes the number, and expanded form and index form are two ways of writing one quantity.

RELATED CONCEPT · SIMPLIFICATION

Simplification. Index notation shortens repeated multiplication, while an agreed order of operations makes a long expression readable in one consistent way.

RELATED CONCEPT · QUANTITY

Quantity. Powers of ten and powers of two describe and compare very large quantities such as storage capacity, distance and population.

GLOBAL CONTEXT

Scientific and technical innovation

EXPLORATION

How compact notation and agreed calculation rules allow people and machines to process quantities far beyond what can be written digit by digit.

FACTUAL

What does 2^{10} mean? Which operation is performed first in $6 + 2 \times 3$? What is the value of 10^4?

CONCEPTUAL

Why does an exponent of 1 leave the base unchanged? How can 2^{10} and 10^3 be compared fairly? Why is a shared order of operations necessary?

DEBATABLE

Should a calculator be trusted to decide the meaning of an expression, or should the person writing it make the meaning visible?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Evaluate powers, roots and expressions using the agreed order.
B · Investigating patterns	i, ii, iii	Investigate doubling from paper folding; generalise the thickness after n folds.
C · Communicating	i, ii, iii, iv	Translate between repeated multiplication, index form, expanded form and standard form.
D · Applying mathematics	i, ii, iii, iv	Interpret advertised and binary storage measures with justified assumptions.

ATL SKILL · TRANSFER

In order for a student to decide whether $(a+b)^2$ is always a^2+b^2, students must transfer the substitution method from one pair of values to a second pair, compare the two results and decide whether a single counterexample settles the claim.

STATEMENT OF INQUIRY

Compressing repeated multiplication into a power, and agreeing the order in which operations apply, allows complex calculations to be recorded compactly and reproduced reliably.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand a power as repeated multiplication and identify its base and exponent.
- Know the place values represented by powers of ten and use them in expanded form.
- Recognise square and cube numbers and find square roots of perfect squares.
- Apply the order of operations, including brackets and indices.
- Substitute values into expressions and formulas containing powers.
- Compare quantities written as different powers and justify the comparison.

SUCCESS CRITERIA

- ✓ I can write repeated multiplication using an exponent and expand a power correctly.
- ✓ I can evaluate powers of whole numbers, including squares and cubes.
- ✓ I can write a whole number in expanded form using powers of ten and return to standard form.
- ✓ I can find square roots of perfect squares and explain what the root means.
- ✓ I can evaluate expressions using brackets, indices, multiplication, division, addition and subtraction in the agreed order.
- ✓ I can substitute a value into an expression with a power and explain the meaning of the result.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Calculate 6×7 and 9×9.
2. Write $4{,}000 + 500 + 30 + 6$ as a single number.
3. Calculate $12 + 3 \times 4$.
4. Write $3 \times 3 \times 3$ as a product and evaluate it.
5. What is the area of a square with side 8 cm?
6. Insert brackets so that $2 + 4 \times 3$ has the value 18.

Answers are in the Chapter 2 answer section. If more than two are incorrect, review multiplication facts and the meaning of brackets before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Multiplication tables, place value, simple brackets and areas of squares and rectangles.	Index notation, powers of ten, squares, cubes and roots, order of operations, substitution.	Negative and fractional indices, standard form, algebraic expansion and sequences.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Storage, distance and population are reported in compact units.	How can repeated multiplication be recorded without losing meaning?	Base and exponent, powers of ten, inverse roots.	Model the thickness of folded paper as a power of two.	Move between expanded, index and standard forms.	When does compression clarify a quantity, and when does it hide it?

A ENGAGE
64 GB or 59.6 GB? Two honest numbers for one machine

A phone is advertised with 64 GB of storage. The same phone, connected to a computer, reports 59.6 GB. Neither number is a lie. Understanding why is the first step towards understanding powers.

Storage is counted in two different ways. Manufacturers count in powers of ten: 1 GB means 1,000 × 1,000 × 1,000 bytes. The computer's operating system counts in powers of two: 1 GiB means 1,024 × 1,024 × 1,024 bytes. The same physical memory, described on two scales.

Description	Advertised (powers of ten)	Reported by the computer (powers of two)
Bytes in one “GB”	$1{,}000^3 = 1{,}000{,}000{,}000$	$1{,}024^3 \approx 1{,}074{,}000{,}000$
64 units	64,000,000,000 bytes	about 59.6 GiB

Figure 2.1 The same storage, two scales

The discrepancy is about 7%. It comes entirely from the choice of base. A number written as a power compresses a multiplication into two small symbols, and the two symbols must be read precisely: the base tells you what is being multiplied, and the exponent tells you how many times.

The same compression appears everywhere. A square of side 8 cm has area $8^2 = 64$ cm². A cube of side 3 cm has volume $3^3 = 27$ cm³. A sheet of paper folded ten times would be $2^{10} = 1{,}024$ layers thick — about the height of a thick book, from a single sheet.

NOTICE AND WONDER

- Why does the exponent count factors rather than tell you what to multiply by?
- Which is larger: 2^{10} or 10^3? How could you decide without a calculator?
- A phone advertises 128 GB and reports 119.2 GB. Is the difference the same as for 64 GB? Why?
- What would 10^0 mean, if the pattern is $10^3 = 1{,}000$, $10^2 = 100$, $10^1 = 10$?

> Keep this question open: when does a shorter notation make a quantity clearer, and when does it hide the very difference that matters?

THE SECOND PUZZLE: WHICH OPERATION COMES FIRST?

Two students calculate $6 + 2 \times 3$. One writes 24, the other writes 12. Both have followed a rule; only one rule is shared. Without an agreed order, a written expression has more than one meaning, and a calculator's answer would depend on its design rather than on mathematics.

This chapter therefore does two things at once: it compresses repeated multiplication into a power, and it fixes the order in which operations are applied so that the compression can be read by everybody in the same way.

B EXPLORE
How thick is a folded sheet of paper?

Work in pairs with one sheet of paper and a ruler. Record every measurement in a table before drawing any conclusion.

YOUR INVESTIGATION

Step 1. Measure the thickness of one sheet. A single sheet is too thin for a ruler, so measure a stack of 100 sheets and divide by 100. Record the result in millimetres.

Step 2. Fold the sheet in half once. How many layers are there? Fold again. Count the layers after 1, 2, 3 and 4 folds. Write the counts as a sequence.

Step 3. Calculate the thickness after each fold: number of layers × thickness of one sheet.

Step 4. Continue the pattern in the table without folding — the paper becomes impossible to fold after about seven folds, but the arithmetic does not stop.

Folds	Layers	Layers as a power of 2	Thickness (mm)
0	1	2^0	
1	2	2^1	
2	4	2^2	
3			
4			
7			
10			
20			

Recording table (copy and complete; use your own sheet thickness)

QUESTIONS TO GUIDE YOUR OBSERVATION

1. How does the number of layers change at each fold? Write the change as a multiplication, then as a power.
2. How many layers after 10 folds? Show the calculation rather than the answer alone.
3. What is the thickness after 10 folds? After 20 folds? Compare the second answer with the first: by what factor has it grown?
4. After how many folds would the stack be taller than your table (about 75 cm)? Show your reasoning.
5. Does the paper's original size matter for the thickness? Does it matter for anything else?

MAKING A CONJECTURE

Write a conjecture in the form:

> "After n folds the number of layers is …, so the thickness is … . I know this because … ."

WHAT YOU SHOULD HAVE NOTICED

- Each fold doubles the number of layers, so the number of layers after n folds is 2^n — a product of n twos.
- Doubling grows slowly at first and then extremely fast: the first ten folds give about a thousand layers, the next ten give about a million.
- A power is a record of repeated multiplication, so it compresses a long product into two numbers: a base and an exponent.
- Comparing 2^{10} with 10^3 requires calculation, not intuition: 1,024 and 1,000 are close, but they are not equal.



Each fold doubles the number of layers, so the number of layers is a power of 2.

INVESTIGATION
Explore further

Repeat the table for a sheet of thickness 0.05 mm. How many folds would be needed for the stack to reach 1 metre? 1 kilometre? Record your method as well as your answers, and be ready to explain why the numbers grow faster than you expect.

2.1 | From repeated multiplication to powers

A · I, II

A power is a compressed instruction. It records which number is being multiplied and how many copies of it appear.

DEFINITION
Base, exponent and power

In the expression a^n, the number a is the **base** and n is the **exponent** (or index). The whole expression is called a **power**. It means $a^n = a \times a \times a \times \ldots \times a$, with n factors of a. $2^4 = 2 \times 2 \times 2 \times 2 = 16$; the base is 2, the exponent is 4, and 16 is the value.

$$2^4 = 2 \times 2 \times 2 \times 2$$

2 is the base; 4 is the exponent (the number of factors)

2^4 records four factors of 2. The base is 2; the exponent is 4.

MATHEMATICAL LANGUAGE
Vocabulary you will use

base · exponent · index · power · squared · cubed · expanded form · evaluate · factor

KEY IDEA
The exponent counts factors, not multiplications

Students often say "2^4 means multiply 2 by 4". It does not: it means multiply four copies of 2. The exponent is a counter. This is why $2^4 = 16$ but $2 \times 4 = 8$, and why $3^2 = 9$ but $3 \times 2 = 6$.

SPECIAL NAMES AND SPECIAL CASES

- a^2 is read "a squared", because it gives the area of a square of side a.
- a^3 is read "a cubed", because it gives the volume of a cube of edge a.
- $a^1 = a$: one factor leaves the number unchanged.
- $a^0 = 1$ for every non-zero a: dividing a^n by itself gives 1, and the exponents subtract to 0.

WRITING POWERS IN EXPANDED FORM

Rewriting a power as a product is called writing it in **expanded form**. It is the safest way to evaluate a power you are unsure of, and it protects you from the "multiply the base by the exponent" error.

Power	Read as	Expanded form	Value
5^2	five squared	5×5	25
2^5	two to the power five	$2 \times 2 \times 2 \times 2 \times 2$	32
10^4	ten to the power four	$10 \times 10 \times 10 \times 10$	10,000
3^3	three cubed	$3 \times 3 \times 3$	27
7^1	seven to the power one	7	7

Powers, expanded form and value

WORKED EXAMPLE 2.1.1

Evaluating powers

GIVEN (a) 4^3 (b) 2^6 (c) 10^5

ASKED Write each in expanded form and give its value.

METHOD Write the base as many times as the exponent says, then multiply step by step.

1. $4^3 = 4 \times 4 \times 4 = 16 \times 4 = 64$.
2. $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 32 \times 2 = 64$.
3. $10^5 = 10 \times 10 \times 10 \times 10 \times 10 = 100{,}000$.

ANSWER (a) 64 (b) 64 (c) 100,000

CHECK Notice that $4^3 = 64 = 2^6$. Different bases and exponents can give the same value; a power is a way of writing a number, not the number itself.

WORKED EXAMPLE 2.1.2

Comparing powers without a calculator

GIVEN 2^{10} and 10^3.

ASKED Decide which is larger, and by how much.

METHOD Evaluate both using known powers, then compare the values.

1. $2^{10} = 1{,}024$ (from the folding investigation, or by doubling five times from $2^5 = 32$).
2. $10^3 = 1{,}000$.
3. 1,024 - 1,000 = 24.

ANSWER 2^{10} is larger by 24. The two are close in value, although the numbers in the notation look very different.

CHECK Both are about one thousand. When two quantities are "about a thousand", the notation does not tell you which is larger; only the value does.

WORKED EXAMPLE 2.1.3

Reading a power in context

GIVEN A computer's memory is described as 2^6 GB in binary units (GiB).

ASKED How many binary units is that, and why is the number not 64 in the operating system's report?

METHOD Evaluate the power, then interpret the result in the context of the two counting systems.

1. $2^6 = 64$, so the memory is 64 GiB.
2. The manufacturer may advertise the same memory as 68.7 GB, because $64 \times 1{,}024^3 \div 1{,}000^3 \approx 68.7$.
3. Both numbers describe the same physical memory; each uses a different base.

ANSWER 64 GiB, which is about 68.7 GB when counted in powers of ten. The difference comes from the base, not from a fault.

CHECK A smaller number in GiB and a larger number in GB is exactly what the two bases predict.

PRACTICE 2.1 · From repeated multiplication to powers

1	Write each product as a power: (a) $3 \times 3 \times 3 \times 3$ (b) 7×7 (c) $10 \times 10 \times 10 \times 10 \times 10$	FLUENCY
2	Write each power in expanded form: (a) 5^3 (b) 2^5 (c) 8^2	FLUENCY
3	Evaluate: (a) 6^2 (b) 2^4 (c) 10^3 (d) 5^1	FLUENCY
4	Evaluate: (a) 3^4 (b) 7^3 (c) 2^7	FLUENCY
5	Ben says "3^5 means $3 \times 5 = 15$". Explain the error and give the correct value.	REASONING
6	Which is larger, 2^5 or 5^2? Explain without evaluating both completely.	REASONING
7	A cube has edge 5 cm. Write its volume as a power and calculate it.	PROBLEM SOLVING
8	Find two different powers with value 64.	EXTENSION
9	Without calculating, decide whether 2^{11} is greater or less than 2,000. Explain.	EXTENSION
10	Write 81 as a power in three different ways.	EXTENSION

2.2 | Powers of ten and expanded form

C · I, II

The place-value system is a system of powers of ten. Once that is visible, expanded form becomes a sum of powers, and multiplying or dividing by 10 becomes a change of exponent.

10^4	10^3	10^2	10^1	10^0
3	0	4	8	6

Every place is a power of ten: $10^0 = 1$, $10^1 = 10$, $10^2 = 100$, $10^3 = 1{,}000$.

DEFINITION

Powers of ten

$10^0 = 1$, $10^1 = 10$, $10^2 = 100$, $10^3 = 1{,}000$, $10^4 = 10{,}000$, and so on. Each power is ten times the previous one. In a base-ten number, the place of a digit records which power of ten multiplies it.

$$3{,}486 = 3 \times 10^3 + 4 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$$

FROM EXPANDED FORM TO INDEX FORM AND BACK

The expanded form you met in Chapter 1 and the index form are the same statement written at different lengths. Chapter 1 wrote 3,000 + 400 + 80 + 6; index form writes $3 \times 10^3 + 4 \times 10^2 + 8 \times 10^1 + 6 \times 10^0$.

Standard form	Expanded with values	Expanded with powers of ten	In words
2,507	2,000 + 500 + 0 + 7	$2 \times 10^3 + 5 \times 10^2 + 0 \times 10^1 + 7 \times 10^0$	two thousand, five hundred and seven
40,090	40,000 + 0 + 0 + 90	$4 \times 10^4 + 0 \times 10^3 + 0 \times 10^2 + 9 \times 10^1 + 0 \times 10^0$	forty thousand and ninety

The same number, four ways

KEY IDEA

Multiplying and dividing by powers of ten

Multiplying by 10^n adds n to the exponent of every place, which is why the digits move n places to the left. Dividing by 10^n subtracts n, so the digits move n places to the right. The rule you used in Chapter 1 is a consequence of index notation.

COMMON ERROR

Zero exponents and empty places

$10^0 = 1$, not 0. The pattern $10^3 = 1{,}000$, $10^2 = 100$, $10^1 = 10$ divides by 10 each time, so the next term must be 1. Writing $10^0 = 0$ would break the pattern and make expanded form collapse.

WORKED EXAMPLE 2.2.1

Writing a number using powers of ten

GIVEN The number 70,408.

ASKED Write it in expanded form using powers of ten.

METHOD Label each place with its power of ten, then multiply each digit by that power.

1. The digits occupy the places 10^4, 10^3, 10^2, 10^1, 10^0.
2. Allocate: 7×10^4, 0×10^3, 4×10^2, 0×10^1, 8×10^0.
3. Add the terms.

ANSWER $70{,}408 = 7 \times 10^4 + 0 \times 10^3 + 4 \times 10^2 + 0 \times 10^1 + 8 \times 10^0$.

CHECK 70,000 + 0 + 400 + 0 + 8 = 70,408.

WATCH OUT Writing 8×10^1 instead of 8×10^0 would make the number 70,480. Count the places from the right, starting at 10^0.

WORKED EXAMPLE 2.2.2

Reconstructing a number from index form

GIVEN $5 \times 10^3 + 2 \times 10^1 + 9 \times 10^0$.

ASKED Write the number in standard form and state the value of the missing place.

METHOD Each term fills one place; every place not named contains a zero.

1. $5 \times 10^3 = 5{,}000$ fills the thousands place.
2. $2 \times 10^1 = 20$ fills the tens place, so the hundreds place is empty (0).
3. $9 \times 10^0 = 9$ fills the ones place.

ANSWER 5,029. The hundreds place is empty, so its digit is 0.

CHECK The expanded sum gives $5{,}000 + 20 + 9 = 5{,}029$, which matches.

WORKED EXAMPLE 2.2.3

Multiplying by powers of ten

GIVEN (a) 3.6×10^2 (b) $47 \div 10^2$ (c) 0.05×10^3.

ASKED Calculate each value and explain the movement of the digits using exponents.

METHOD Multiplying by 10^n moves every digit n places left; dividing moves them n places right.

1. $3.6 \times 10^2 = 360$: two places left.
2. $47 \div 10^2 = 0.47$: two places right.
3. $0.05 \times 10^3 = 50$: three places left.

ANSWER (a) 360 (b) 0.47 (c) 50

CHECK Estimate first: 3.6×100 is a little under 400; 47 divided by 100 is about 0.5; 0.05 thousand is 50.

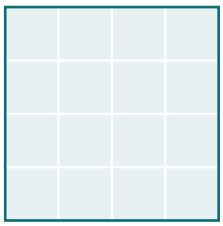
PRACTICE 2.2 · Powers of ten and expanded form

1	Write using powers of ten: (a) 3,500 (b) 61 (c) 8,004	FLUENCY
2	Evaluate: (a) 10^0 (b) 10^1 (c) 10^6	FLUENCY
3	Calculate: (a) 2.5×10^3 (b) $600 \div 10^3$ (c) 0.4×10^2	FLUENCY
4	Write in standard form: $4 \times 10^4 + 7 \times 10^2 + 3 \times 10^0$.	FLUENCY
5	Explain why $10^0 = 1$ using the pattern of powers of ten.	REASONING
6	A student writes $6{,}000 = 6^3$. Explain the error and write the correct index form.	REASONING
7	A number is $9 \times 10^4 + 9 \times 10^0$. Write it and state how many digits it has.	PROBLEM SOLVING
8	A warehouse holds 4×10^3 boxes, each with 10^2 items. How many items are stored? Give the answer as a power of ten and as a number.	PROBLEM SOLVING
9	Write 1,000,000 as a power of ten, then as a power of 100, then as a power of 1,000.	EXTENSION
10	A number has six digits and is written as $a \times 10^5 + \ldots$. What are the possible values of a, and why?	EXTENSION

2.3 | Squares, cubes and roots

A · I, II

A square number is a number of unit squares that can be arranged in a square. Its root is the side length. That picture makes “square root” a description rather than a symbol to memorise.



4 × 4 = 16
$4^2 = 16$

a square number

A square number can be arranged as a square of dots: 4 × 4 = 16.

DEFINITION
Square and cube numbers

A **square number** is the result of multiplying a whole number by itself: 1, 4, 9, 16, 25, … The **square root** of a square number is the side of that square: √49 = 7 because 7 × 7 = 49. A **cube number** is the result of multiplying a whole number by itself twice: 1, 8, 27, 64, … The **cube root** reverses that: *sqrt*[3]64 = 4.

n	1	2	3	4	5	6	7	8	9	10
n^2	1	4	9	16	25	36	49	64	81	100
n^3	1	8	27	64	125	216	343	512	729	1,000

The first square and cube numbers

KEY IDEA
A root answers “what side would make this area?”

√36 = 6 because a square of side 6 has area 36. The symbol asks a question about a shape, and the answer is a length. This is why roots are useful in measurement, and why √36 is not “half of 36”.

COMMON ERROR
Three root errors

Claim: “√16 = 8 because 16 ÷ 2 = 8.” A root is not a half. 4 × 4 = 16, so √16 = 4. **Claim:** “√2 is exactly 1.4.” In fact $1.4^2 = 1.96$, so 1.4 is only an approximation. √2 ≈ 1.414. **“Only squares have roots.”** Every positive number has a square root, but only square numbers have whole-number roots.

WORKED EXAMPLE 2.3.1

Finding roots of square numbers

GIVEN (a) $\sqrt{81}$ (b) $\sqrt{144}$ (c) $\sqrt{1}$

ASKED Evaluate each root and explain what it means geometrically.

METHOD Ask which number multiplied by itself gives the value under the root.

1. $9 \times 9 = 81$, so $\sqrt{81} = 9$: a square of side 9 has area 81.
2. $12 \times 12 = 144$, so $\sqrt{144} = 12$.
3. $1 \times 1 = 1$, so $\sqrt{1} = 1$: the unit square has side 1.

ANSWER (a) 9 (b) 12 (c) 1

CHECK Square each answer: $9^2 = 81$, $12^2 = 144$, $1^2 = 1$.

WORKED EXAMPLE 2.3.2

Working between squares and roots

GIVEN A square tile has area 225 cm².

ASKED Find its side length, then the perimeter of the tile.

METHOD Take the square root for the side, then use the perimeter formula for a square.

1. Side = $\sqrt{225} = 15$ cm, because $15 \times 15 = 225$.
2. Perimeter = $4 \times 15 = 60$ cm.

ANSWER Side 15 cm; perimeter 60 cm.

CHECK Area = $15^2 = 225$ cm², and four equal sides of 15 cm give 60 cm.

WORKED EXAMPLE 2.3.3

Estimating a root that is not a whole number

GIVEN $\sqrt{50}$.

ASKED Estimate the root to one decimal place and explain why it is not a whole number.

METHOD Find the two square numbers either side, then refine.

1. $7^2 = 49$ and $8^2 = 64$, so $\sqrt{50}$ lies between 7 and 8, very close to 7.
2. Try 7.1: $7.1^2 = 50.41$, which is slightly too large.
3. Try 7.07: $7.07^2 = 49.98$, slightly too small, so $\sqrt{50} \approx 7.07$.

ANSWER $\sqrt{50} \approx 7.07$. It is not a whole number because 50 is not a square number.

CHECK $7^2 = 49 < 50 < 64 = 8^2$, so the root must lie between 7 and 8.

PRACTICE 2.3 · Squares, cubes and roots

1	Evaluate: (a) 7^2 (b) 9^2 (c) 12^2 (d) 5^3	FLUENCY
2	Evaluate: (a) $\sqrt{64}$ (b) $\sqrt{100}$ (c) $\sqrt{169}$	FLUENCY
3	Which of these are square numbers? 36, 40, 49, 56, 64, 70	FLUENCY
4	Explain why 30 is not a square number.	REASONING
5	A student says $\sqrt{25} + \sqrt{9} = \sqrt{34}$. Is this correct? Explain.	REASONING
6	A square garden has area 196 m². Find its side and its perimeter.	PROBLEM SOLVING
7	A cube has volume 216 cm³. Find its edge.	PROBLEM SOLVING

8	Between which two whole numbers does √75 lie? Estimate it to one decimal place.	EXTENSION
9	Explain why there is no whole number whose square is greater than 100 but whose cube is less than 1,000.	EXTENSION
10	The first five square numbers are 1, 4, 9, 16, 25. Describe how the gaps between them change, and explain the pattern.	EXTENSION

2.4 | Order of operations

A · I, II, III

An expression is read by everyone in the same order, so that one written calculation has one value. The order is a shared agreement, not a mathematical discovery about nature.



The agreed order of operations. Within one step, work from left to right.

DEFINITION

The agreed order

1. **Brackets** — work inside them first.
2. **Indices** — powers and roots.
3. **Multiplication and division**, from left to right.
4. **Addition and subtraction**, from left to right.

• An expression in brackets is evaluated as if it were a single number.

KEY IDEA

Why the order matters

$6 + 2 \times 3$ has one value, 12, because multiplication is performed before addition: $6 + 6 = 12$. If addition came first the value would be 24. The order is chosen so that expressions such as $2 + 3x$ and $4 + 2 \times 5$ behave consistently as algebra develops, and so that expanding brackets later gives the answers everyone expects.

BRACKETS CHANGE THE STORY

Brackets are the only way to override the order. $6 + 2 \times 3 = 12$, but $(6 + 2) \times 3 = 24$. When you are unsure whether an expression needs brackets, ask which operation you intend to happen first, then write the brackets that enforce it.

METHOD

A dependable routine for evaluating an expression

1. Copy the expression and mark the operations you can see.
2. Underline the brackets and evaluate them, replacing each with its value.
3. Evaluate any indices.
4. Work through multiplication and division left to right, rewriting the expression at each step.
5. Finish with addition and subtraction, left to right.
6. Check the size of the answer against a rough estimate.

WORKED EXAMPLE 2.4.1

Evaluating an expression with two operations

GIVEN $20 - 3 \times 4$.

ASKED Evaluate, and explain which operation was performed first.

METHOD Multiplication before subtraction.

1. $3 \times 4 = 12$.
2. Replace: $20 - 12$.
3. $20 - 12 = 8$.

ANSWER 8

CHECK If subtraction came first the answer would be $17 \times 4 = 68$, which is far from the estimate: 3×4 is about 12, and $20 - 12$ is about 8.

WORKED EXAMPLE 2.4.2

An expression with brackets and an index

GIVEN $(4 + 5)^2 - 3 \times 6$.

ASKED Evaluate, showing each step.

METHOD Brackets, then indices, then multiplication, then subtraction.

1. Brackets: $4 + 5 = 9$, so the expression becomes $9^2 - 3 \times 6$.
2. Index: $9^2 = 81$, giving $81 - 3 \times 6$.
3. Multiplication: $3 \times 6 = 18$, giving $81 - 18$.
4. Subtraction: $81 - 18 = 63$.

ANSWER 63

CHECK Estimate: 9^2 is about 80, and 3×6 is about 20, so the answer should be about 60. It is.

WATCH OUT Calculating $4 + 5^2$ would give 29, a completely different expression. The brackets belong to the whole sum.

WORKED EXAMPLE 2.4.3

Inserting brackets to obtain a stated value

GIVEN The digits and operations 2 + 4 × 3.

ASKED Insert brackets so that the expression has the value 18, then so that it has the value 14.

METHOD Try each possible bracket position and evaluate.

1. Without brackets: 2 + 12 = 14.
2. With brackets around the sum: (2 + 4) × 3 = 6 × 3 = 18.
3. There is no bracket position that gives a third value, because there are only two operations.

ANSWER (2 + 4) × 3 = 18; the unbracketed expression gives 14.

CHECK Evaluate both expressions by the agreed order, not by reading left to right.

PRACTICE 2.4 · Order of operations

1	Evaluate: (a) 5 + 2 × 3 (b) 12 - 2 × 5 (c) 8 ÷ 2 + 3	FLUENCY
2	Evaluate: (a) (5 + 2) × 3 (b) 12 ÷ (2 + 4) (c) 20 - (3 + 4)	FLUENCY
3	Evaluate: (a) $2^3 + 4$ (b) 3×2^2 (c) $(2 + 3)^2$	FLUENCY
4	Evaluate: 30 - 2 × (3 + 4).	FLUENCY
5	Two students evaluate 18 - 6 ÷ 2. One gets 6, the other gets 15. Which is correct, and what mistake produced the other answer?	REASONING
6	Add one pair of brackets to make 3 + 4 × 2 equal 14.	REASONING
7	A taxi charges 3 units to start plus 2 units per kilometre. Write an expression for a 7 km journey and evaluate it.	PROBLEM SOLVING
8	Evaluate $100 - (4 + 6)^2 \div 4$.	PROBLEM SOLVING
9	Insert brackets in 24 ÷ 6 + 2 × 3 to obtain two different values.	EXTENSION
10	Explain why 2 + 3 × 4 and (2 + 3) × 4 differ, using the order of operations in your answer.	EXTENSION
11	A calculator gives 24 for 6 + 2 × 3. What has the calculator done, and what does this tell you about relying on a calculator?	EXTENSION
12	The school's print charge is 5 units per page plus a 12-unit setup fee. A teacher's bill is 112 units. How many pages were printed? Show the equation and its solution.	PROBLEM SOLVING

2.5 | Substitution and expressions with powers

C · III, IV

Substitution replaces each letter with a number. When the expression contains powers, the order of operations decides what the power applies to, so brackets matter.

DEFINITION
Substitution

To **substitute** is to replace every occurrence of a variable with a given value, then evaluate using the agreed order. In $3x^2$, the power applies only to x, so $x = 4$ gives $3 \times 16 = 48$. In $(3x)^2$, the power applies to the whole product, so $x = 4$ gives $12^2 = 144$.

KEY IDEA
Read the exponent's reach

An exponent applies only to the symbol immediately before it, unless brackets extend its reach. $2x^2$ means $2 \times (x^2)$; $(2x)^2$ means $(2x) \times (2x)$. The two expressions differ by a factor of 4 for every non-zero x, and the difference is the most common source of error in this section.

Expression	Value when $x = 3$	Why
$2x^2$	$2 \times 9 = 18$	the power applies to x only
$(2x)^2$	$6^2 = 36$	the power applies to $2x$
$2x^2 + 1$	$18 + 1 = 19$	addition last
$(2x + 1)^2$	$7^2 = 49$	the power applies to the whole bracket

The same substitution, different answers

WORKED EXAMPLE 2.5.1
Substituting a positive value

GIVEN $4x^2 + 3$ with $x = 5$.

ASKED Evaluate the expression.

METHOD Substitute, evaluate the index, multiply, then add.

1. Substitute: $4 \times 5^2 + 3$.
2. Index: $5^2 = 25$, giving $4 \times 25 + 3$.
3. Multiply: $4 \times 25 = 100$, giving $100 + 3$.
4. Add: 103.

ANSWER 103

CHECK $x^2 = 25$ is about 25; four of them is about 100; add 3 gives about 103.

WORKED EXAMPLE 2.5.2
Substituting a negative value

GIVEN $x^2 - 2x$ with $x = -3$.

ASKED Evaluate the expression.

METHOD Substitute inside brackets to keep the sign visible, then apply the index before multiplying.

1. Substitute: $(-3)^2 - 2 \times (-3)$.
2. Index: $(-3)^2 = 9$, because a negative times a negative is positive.
3. Multiply: $2 \times (-3) = -6$, giving $9 - (-6)$.
4. Subtract a negative: $9 + 6 = 15$.

ANSWER 15

CHECK The square contributes 9 and the “subtract twice a negative” adds 6, so the total must exceed 9.

WATCH OUT Writing -3^2 gives -9. The brackets are what make the square positive.

WORKED EXAMPLE 2.5.3

Using a formula with a power

GIVEN The area of a circle is $A = \pi r^2$. A circular table has radius 0.8 m. Use $\pi \approx 3.14$.

ASKED Find the area, with its unit.

METHOD Substitute the radius, square it, then multiply by π.

1. $A = 3.14 \times 0.8^2$.
2. $0.8^2 = 0.64$.
3. $3.14 \times 0.64 = 2.0096$.

ANSWER $A \approx 2.01$ m² (to two decimal places).

CHECK A square of side 1.6 m would have area 2.56 m²; a circle inside it must have a smaller area, and 2.01 m² is smaller.

PRACTICE 2.5 · Substitution and expressions with powers

1	Evaluate $3x^2$ when $x = 4$.	FLUENCY
2	Evaluate $5x^2 - 2$ when $x = 3$.	FLUENCY
3	Evaluate $a^2 + b^2$ when $a = 6$ and $b = 8$.	FLUENCY
4	Explain why $2x^2$ and $(2x)^2$ give different answers when $x = 3$.	REASONING
5	Evaluate $x^2 - 2x + 1$ when $x = 4$ and when $x = -4$.	PROBLEM SOLVING
6	The volume of a cube is $V = s^3$. Find V when $s = 2.5$ cm.	PROBLEM SOLVING
7	Write a sentence explaining to a classmate why brackets change the value of a substituted expression.	COMMUNICATING
8	Find a value of x for which $x^2 = 2x$.	EXTENSION
9	Evaluate $(a + b)^2$ and $a^2 + b^2$ when $a = 2$ and $b = 3$, then explain why the results differ.	EXTENSION
10	A formula is $P = 2(l + w)$. Explain why $P = 2l + 2w$ gives the same result, using the idea of expanding brackets.	EXTENSION

F **INVESTIGATE AND APPLY**
A criterion B investigation and a criterion D application

INVESTIGATION 2 | Does doubling always stay ahead?

B · I, II, III

Powers of two grow quickly, but powers of ten start larger. Investigate where they meet, and how the gap behaves.

INVESTIGATION

Investigation brief

Question. Compare 2^n with 10^n and with n^2 for $n = 1, 2, 3, \ldots$. Where does each pair cross? What rule describes the crossing point? **What you need.** A table with columns for n, 2^n, 10^n and n^2; a calculator for the larger values. **What to produce.** A completed table, a description of each pattern, a conjecture about the crossing points, and a verification.

PART 1 · RECORD

n	2^n	10^n	n^2	Which is largest?
1	2	10	1	
2	4	100	4	
3	8	1,000	9	
4			16	
5		100,000	25	
6		1,000,000	36	
10			100	

Copy and complete

PART 2 · DESCRIBE

1. How does 2^n change when n increases by 1? Give two examples from the table.
2. How does n^2 change when n increases by 1? Give two examples.
3. Which grows faster at first, 2^n or n^2? Where does the order change?
4. Compare 2^n and 10^n. Which is larger for every n, and by what factor?

PART 3 · CONJECTURE, TEST AND JUSTIFY

1. Write a conjecture describing when 2^n overtakes n^2, and test it for $n = 5$ and $n = 6$.
2. Explain why 10^n is always larger than 2^n for $n \geq 1$. Your explanation should use the bases, not a list of examples.
3. A classmate says "2^n always doubles but n^2 only adds a little, so 2^n must win for every n". Test the claim at $n = 3$ and decide whether it is true as stated. Revise it if necessary.

CRITERION TASK

Criterion B task · Requirements

A complete, systematic table; patterns described in words with evidence; a general statement that predicts rather than lists; at least one test of the statement on a case not used to form it; an explanation of any limitation.

APPLICATION 2

The school data store

D · I, II, III, IV

The school's server is being replaced. The business manager has three quotes, and each describes capacity in different units. Your task is to compare them and recommend one.

THE BRIEF

Quote	Advertised capacity	Reported capacity (binary)	Annual cost
A	8 TB (powers of ten)	7.28 TiB	1,200 units
B	6 TiB (powers of two)	6 TiB	1,050 units
C	2^{43} bytes	2^{43} bytes	900 units

Three quotes for the same server room

- The school stores about 5.5 TB of video each year.
- The manager wants the cheapest option that will hold three years of video.
- The technician wants “no advertised figure that overstates the real space by more than 10%”.

TASK 1 · CONVERT AND COMPARE

1. Write each capacity in bytes using powers of ten. Show the power you used.
2. Convert quote C from bytes to TB, using 1 TB = 10^{12} bytes.
3. Which quote offers the largest real capacity? Which offers the smallest?

TASK 2 · CHECK THE CLAIMS

1. Quote A advertises 8 TB but reports 7.28 TiB. Calculate the percentage overstatement and decide whether it passes the technician's test.
2. Is the difference caused by a fault, a rounding, or a choice of base? Explain in two sentences.
3. Rewrite quote A's advertisement so that it is accurate and still persuasive.

TASK 3 · RECOMMEND

1. How much storage does three years of video require? Round sensibly and state your rounding.
2. Which quote is the cheapest that meets the requirement? Justify with figures.
3. Write a recommendation of no more than 80 words to the business manager, including one risk you have identified.

CRITERION TASK
Criterion D task · Requirements

Relevant quantities identified; conversions correct and shown; a decision justified with calculated evidence; rounding and accuracy explained; a recommendation that acknowledges one limitation.

TASK C | Explaining a calculator's answer

C · I, II, III, IV

CRITERION TASK
The task

A classmate enters 6 + 2 × 3 into a calculator and gets 24. Write a short explanation (no more than 150 words) for the class noticeboard that:

1. Shows the correct value and the order of operations used.
2. Explains why the calculator produced a different value, without blaming the calculator.

3. States how the expression should be written so that no reader can misread it.
4. Includes one example in which brackets are necessary and one in which they are optional.

Criterion C strand	Evidence	
Notation (i)	Correct symbols for multiplication and brackets	no ambiguous expressions.
Representation (ii)	At least one worked layout showing the steps.	
Reasoning (iii)	The explanation says why the order exists, not only what it is.	
Organisation (iv)	A reader can apply the rule from your explanation alone.	

Success criteria

G REFLECT AND CONSOLIDATE
Summary, vocabulary, misconceptions and review

REVIEW | Consolidating Chapter 2
A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- A power records repeated multiplication: the base is the number multiplied and the exponent counts the factors.
- Powers of ten are the place values of the base-ten system, so expanded form can be written in index notation.
- Square and cube numbers have geometric meanings; roots reverse them.
- The order of operations is an agreement that makes a written expression unambiguous: brackets, indices, multiplication and division, addition and subtraction.
- Brackets change what an expression means, and therefore what a substitution produces.
- Notation choices carry meaning: $2x^2$ and $(2x)^2$ are different quantities.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Base	The number that is multiplied repeatedly in a power.
Exponent (index)	The number of factors of the base in a power.
Power	A number written as a base with an exponent.
Square number	The result of multiplying a whole number by itself.
Square root	The side of a square with a given area.
Cube number	The result of multiplying a whole number by itself twice.
Expanded form	A number or expression written as a sum of products.
Substitution	Replacing a variable with a number and evaluating.
Order of operations	The agreed sequence for evaluating an expression.

Vocabulary for Chapter 2

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence
"3^4 means 3×4."	The exponent counts factors.	$3^4 = 81$, not 12
"$10^0 = 0$."	Any non-zero number to the power 0 is 1.	$10^2 \div 10^2 = 1$
"$\sqrt{16} = 8$."	A root is not a half.	$4^2 = 16$
"Work left to right."	Brackets, indices, then $\times$ $\div$, then $+$ $-$.	$6 + 2 \times 3 = 12$
"$2x^2 = (2x)^2$."	The exponent reaches only the symbol before it.	$x = 3$: 18 against 36

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
write repeated multiplication as a power and expand a power.			
evaluate powers, squares, cubes and roots.			
write a whole number in index form using powers of ten.			
apply the order of operations, including brackets and indices.			
substitute a value into an expression with a power.			
explain why two expressions that look similar have different values.			

How confident are you?

REFLECTION

1. Which notation in this chapter saved you the most writing? Give an example.
2. Describe a mistake you made with the order of operations and the rule that now prevents it.
3. Write one sentence beginning "An exponent tells you …" and one beginning "Brackets matter because …".
4. Why might a manufacturer prefer one base and an operating system another? Is either dishonest?
5. Which investigation result surprised you, and what would you test next?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ2 · Applying the chapter

1 Evaluate: (a) 4^2 (b) 2^5 (c) 10^4 FLUENCY

2 Write $7 \times 7 \times 7$ as a power. FLUENCY

3	Evaluate $\sqrt{121}$.	FLUENCY
4	Evaluate $16 - 2 \times 3$.	FLUENCY
5	Evaluate $(3 + 4)^2$.	FLUENCY
6	Calculate 3.4×10^2.	FLUENCY
7	Explain why 5^2 and 2^5 are not equal.	REASONING
8	Is 2^6 greater or less than 6^2? Show the values.	REASONING
9	A student writes $8{,}000 = 8^3$. What has been confused? Give the correct form.	REASONING
10	Evaluate $2x^2 + 3$ when $x = 4$.	PROBLEM SOLVING
11	Evaluate $x^2 - 3x$ when $x = 5$.	PROBLEM SOLVING
12	A square room has area 169 m². Find the length of a skirting board needed for all four walls.	PROBLEM SOLVING
13	A cube-shaped tank has edge 3 m. Find its volume in m³ and in litres.	PROBLEM SOLVING
14	Explain in two sentences why the order of operations must be agreed rather than discovered.	COMMUNICATING
15	Insert brackets to make $2 + 3 \times 4^2$ equal 400.	EXTENSION
16	Find two powers of 2 whose sum is 96.	EXTENSION
17	Explain why $(-2)^2 = 4$ but $-2^2 = -4$.	EXTENSION
18	The thickness of a sheet is 0.1 mm. How many folds would make the stack taller than 1 m? Show your reasoning with powers of 2.	EXTENSION
19	Write a 40-word explanation of why $10^0 = 1$ for a classmate who thinks it should be 0.	COMMUNICATING
20	A shop advertises a 32 GB card. The computer reports 29.8 GB. Estimate the percentage difference and explain its cause.	PROBLEM SOLVING

MIXED 2 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX2 · End-of-chapter mixed set

1	Evaluate: (a) 3^3 (b) 6^2 (c) 2^8	FLUENCY
2	Evaluate: (a) $45 - 5 \times 6$ (b) $(45 - 5) \times 6$	FLUENCY
3	Write 60,304 using powers of ten.	FLUENCY
4	Calculate: (a) 0.7×10^3 (b) $250 \div 10^2$	FLUENCY
5	Evaluate $\sqrt{196}$.	FLUENCY
6	What is the largest two-digit square number? Explain how you decided.	REASONING
7	Explain why the value of 2^n doubles when n increases by 1, but the value of n^2 does not.	REASONING
8	A page holds about 10^3 words. How many pages hold 10^6 words?	PROBLEM SOLVING

9	Evaluate $3(x + 2)^2$ when $x = 3$.	PROBLEM SOLVING
10	A square tile has area 400 cm². How many tiles are needed to cover a rectangle 4 m by 2 m?	PROBLEM SOLVING
11	Explain, in no more than 50 words, why 10^0 must be 1 for expanded form to work.	COMMUNICATING
12	Write the expression for “five more than three times the square of a number” and evaluate it when the number is 4.	COMMUNICATING
13	Find n such that $2^n = 512$.	EXTENSION
14	A sheet folded 12 times has 4,096 layers. Write this as a power and explain what the exponent counts.	EXTENSION
15	Which is larger, $2^{10} \times 2^3$ or 2^{12}? Explain using the meaning of the exponents.	EXTENSION
16	Explain why $\sqrt{9} \times \sqrt{4} = \sqrt{36}$, and test whether the same rule works for $\sqrt{9} + \sqrt{4}$.	EXTENSION

CUMULATIVE PRACTICE 2 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP2 · Cumulative practice

1	Evaluate 3^4.	FLUENCY
2	Evaluate $\sqrt{144}$.	FLUENCY
3	Evaluate $20 - 3 \times 4$.	FLUENCY
4	Explain why $2^3 \times 2^2 = 2^5$.	REASONING
5	A square has area 81 cm². Find its side length.	PROBLEM SOLVING
6	Write 5,000 + 300 + 4 in standard form.	PROBLEM SOLVING
7	Explain the order in which operations are carried out in $4 + 2(3 + 1)^2$.	COMMUNICATING
8	Find the value of n if $2^n = 64$.	EXTENSION

CHAPTER 03

Factors, multiples and primes

Every whole number is built from smaller whole numbers. Learning to see that structure turns division, fair sharing and repeating schedules into questions with exact answers.

KEY CONCEPT	**Relationships**
RELATED	**System**
RELATED	**Representation**
GLOBAL CONTEXT	**Fairness and development**

03
CHAPTER

INQUIRY QUESTION

When is a number best understood by what it is made of, and when does the way we group things decide whether a division is fair?

A school has 84 Year 7 students, 60 Year 8 students and 48 Year 9 students. It wants identical mixed teams with no student left out. The number of teams is not a matter of opinion: it must divide all three totals. Arithmetic decides what is possible before anyone negotiates.

IN THIS CHAPTER

3.1 Factors, multiples and factor pairs
3.2 Divisibility tests
3.3 Prime and composite numbers
3.4 Prime factorisation
3.5 Highest common factor and lowest common multiple
Investigation 3 The odd number of factors
Application 3 The fairness relay
Task C Writing the divisibility guide
Review Consolidating Chapter 3
Review questions Practice across the whole chapter
Mixed 3 Fluency, reasoning and problem solving
Cumulative practice 3 Questions that mix this chapter with earlier ones

APPROXIMATELY 13 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

A school cannot make identical mixed teams from 84, 60 and 48 students unless the number of teams divides all three. Two bells ring together only at common multiples. Arithmetic decides what is possible before anyone negotiates.

KEY CONCEPT · RELATIONSHIPS

Relationships. Factors and multiples describe how whole numbers are connected: one number divides another exactly, or two numbers share a common structure.

RELATED CONCEPT · SYSTEM

System. The primes are building blocks: every whole number greater than 1 is assembled from them in exactly one way, apart from order.

RELATED CONCEPT · REPRESENTATION

Representation. Factor trees, index notation and Venn diagrams display the same factorisation in different forms.

GLOBAL CONTEXT

Fairness and development

EXPLORATION

How number structure shapes fair division, shared timetables and the allocation of resources.

FACTUAL	CONCEPTUAL	DEBATABLE
What is a factor of 36? What is a multiple of 7? Which numbers below 100 are prime?	Why is 1 neither prime nor composite? Why is every prime factorisation unique? How are HCF and LCM related?	Is a number with many factors more useful than a prime number, or does usefulness depend entirely on the problem?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Find factors and multiples, apply divisibility tests, identify primes, calculate HCF and LCM.
B · Investigating patterns	i, ii, iii	Investigate how many factors a number has; generalise why squares have an odd number of factors.
C · Communicating	i, ii, iii, iv	Represent factorisations with trees, index notation and diagrams, using correct vocabulary.
D · Applying mathematics	i, ii, iii, iv	Solve grouping and scheduling problems, justify assumptions and evaluate the fairness of a plan.

ATL SKILL · CRITICAL THINKING

In order for a student to judge the claim that every number with a 3 in it is divisible by 3, students must test it on numbers that support it and on numbers that refute it, then replace the claim with a test that looks at the digit sum.

STATEMENT OF INQUIRY

Whole numbers are built from factors in a structured way, and that structure makes it possible to divide quantities fairly, schedule repeating events and communicate precisely.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand the relationship between factors and multiples, and list factor pairs systematically.
- Know the divisibility tests for 2, 3, 4, 5, 6, 8, 9 and 10 and explain how they work.
- Classify numbers as prime, composite or neither, and test whether a number is prime.
- Write any composite number as a product of primes using a factor tree or division.
- Find the highest common factor and lowest common multiple of two or three numbers.
- Choose and justify HCF or LCM in sharing and scheduling contexts.

SUCCESS CRITERIA

- ✓ I can list all factors of a number without missing any, using factor pairs.
- ✓ I can explain the difference between a factor and a multiple with examples.
- ✓ I can decide whether a number is divisible by 2, 3, 4, 5, 6, 8, 9 or 10 and give evidence.
- ✓ I can identify prime numbers and explain why 1 is excluded.
- ✓ I can write a prime factorisation in index form and reconstruct the number from it.
- ✓ I can calculate HCF and LCM and explain which one a context requires.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Calculate 7×8 and 9×6.
2. Divide 96 by 8.
3. List the first five multiples of 4.
4. Write 100 as a product of two equal factors.
5. Which is larger, 3^3 or 4^2?
6. Is 56 in the 7 times table? Explain how you know.

Answers are in the Chapter 3 answer section. If more than two are incorrect, review multiplication and division facts before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Multiplication tables, division facts and simple fractions of amounts.	Factor pairs, divisibility, primes, prime factorisation, HCF and LCM.	Laws of indices, algebraic factorisation, number theory and cryptography.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Groups, timetables and shared resources must divide exactly.	What smaller numbers build a given whole number?	Factors, multiples, primes and common structure.	Count the factors of the numbers 1 to 30 and explain the pattern.	Represent factorisations as trees, index products and diagrams.	When does the largest common group produce the fairest result?

A ENGAGE

Three year groups, identical teams

A school has 84 students in Year 7, 60 in Year 8 and 48 in Year 9. It wants identical mixed teams — every team with the same number of students, no student left out, and no team made from a single year group. How many teams can there be?

The answer is not a matter of preference. A team size that divides 84 but not 60 would leave some students out. A team size that divides all three produces a fair arrangement. Before any negotiation about who plays with whom, arithmetic has already decided which team sizes are possible.

Team size	Teams from 84	Teams from 60	Teams from 48	Possible?
4	21	15	12	yes
6	14	10	8	yes
8	10.5	7.5	6	no
10	8.4	6	4.8	no

Figure 3.1 Three possible plans

Two numbers in that table are not whole numbers, and those rows are impossible. The same idea appears in timetables: two bells ring every 12 minutes and every 18 minutes. They ring together only at times that are common multiples of 12 and 18 — 36 minutes, 72 minutes, and so on.

This chapter is about the structure that decides such questions: which smaller numbers build a given number, which numbers are divisible by which, and how the answers can be organised so that nothing is missed.

NOTICE AND WONDER

- Why does the team size have to divide each year group exactly?
- Which team size gives the greatest number of teams? Which gives the largest teams?
- 12 and 18 both divide 36. Is 36 the first time the two bells ring together?
- What is special about a number whose only divisors are 1 and itself?

> Keep this question open: when is a number best understood by what it is made of, and when does the way we group things decide whether a division is fair?

B EXPLORE

How many rectangles can a number make?

Work in pairs with squared paper. You need to find every rectangle that can be made from a given number of unit squares, with no gaps and no overlaps.

YOUR INVESTIGATION

Step 1. Take 12 unit squares. Arrange all of them into a rectangle. Record the dimensions. Rearrange to make a different rectangle. Continue until you are sure you have them all.

Step 2. Repeat for 16 squares, then for 17 squares, then for 18 squares.

Step 3. A rectangle with dimensions a by b uses $a \times b$ squares, so each rectangle records a **factor pair** of the number. Record every pair you found.

Number of squares	Rectangles found	Factor pairs	Number of factor pairs
12			
16			
17			
18			
24			

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

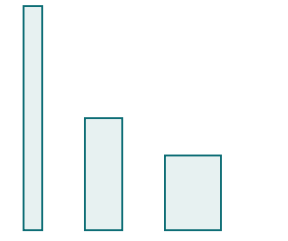
1. Which numbers produced only one rectangle? What is special about those numbers?
2. Which number produced a square? What does that tell you about its factors?
3. How can you be sure you have found every factor pair, without guessing?
4. Why does it stop being useful to try dimensions longer than the side of the square?
5. Predict the number of factor pairs for 36, then check by listing systematically.

MAKING A CONJECTURE

> “A number has an odd number of factors when … because … .”

WHAT YOU SHOULD HAVE NOTICED

- Every rectangle records two factors that multiply to the number, so factor pairs come in twos.
- When the number is a square, one pair has equal factors, so the count of factors is odd rather than even.
- 17 produced only the 1 × 17 rectangle: its only factors are 1 and 17.
- Testing dimensions up to the square root is enough, because any larger factor would already have appeared in a pair with a smaller one.



rectangles for 12: 3 factor pairs

The factor pairs of 12 as rectangles: 1 × 12, 2 × 6 and 3 × 4.

INVESTIGATION

Explore further

Investigate the numbers from 1 to 30 and record how many factors each has. Sort them into groups: one factor pair, two factor pairs, three, and so on. What do the numbers in each group have in common?

3.1 | Factors, multiples and factor pairs

A · I, II

Factors and multiples describe the same relationship from two directions. Reading the relationship in both directions prevents most confusion.

DEFINITION
Factors and multiples

A **factor** of a whole number is a whole number that divides it exactly, with no remainder. A **multiple** of a number is the result of multiplying it by a whole number. For 24: the factors are 1, 2, 3, 4, 6, 8, 12, 24. The multiples of 24 begin 24, 48, 72, 96, … The statement “6 is a factor of 24” and “24 is a multiple of 6” say exactly the same thing.

KEY IDEA
The relationship, not the word

Students who memorise “factors are small, multiples are big” eventually meet a counterexample: 24 is a factor of 24, and 24 is a multiple of 24. A reliable test is the division: if $a \div b$ has no remainder, then b is a factor of a and a is a multiple of b.



the factors of 24 are the positions where the divisions are exact

Testing each number up to 24 to see whether it divides 24 exactly.

METHOD
Finding every factor without missing one

1. Start with 1 and the number itself: this is always a factor pair.
2. Test 2, 3, 4, … in order, writing each pair as you find it.
3. Stop when you reach a number whose square is larger than the number — beyond that point, every factor has already appeared in a pair.
4. Check your list: each factor pairs with exactly one other, and the pairs multiply to the number.

COMMON ERROR
Missing factors by stopping too early

Listing the factors of 36 as 1, 2, 3, 4, 6, 9, 12, 18, 36 and then forgetting 36 itself is a common slip. Writing factor pairs — (1, 36), (2, 18), (3, 12), (4, 9), (6, 6) — makes the complete list visible and shows why 36 has nine factors, an odd number.

WORKED EXAMPLE 3.1.1

Listing factors systematically

GIVEN The number 36.

ASKED List all its factors, and state how many there are.

METHOD Use factor pairs, testing upwards and stopping at the square root.

1. (1, 36); (2, 18); (3, 12); (4, 9); (6, 6).
2. Beyond 6 the pairs repeat in reverse, so the list is complete.
3. Collect the factors: 1, 2, 3, 4, 6, 9, 12, 18, 36.

ANSWER Nine factors.

CHECK The square pair (6, 6) appears once, which is why the total is odd.

WORKED EXAMPLE 3.1.2

Using the factor–multiple relationship

GIVEN The numbers 7 and 56.

ASKED State whether 7 is a factor of 56 and whether 56 is a multiple of 7, with evidence.

METHOD Divide and interpret the remainder.

1. $56 \div 7 = 8$ with no remainder.
2. Therefore 7 is a factor of 56, and 56 is a multiple of 7.
3. The multiplier 8 is also a factor: $7 \times 8 = 56$.

ANSWER Yes to both; 7 and 8 are a factor pair of 56.

CHECK $7 \times 8 = 56$ confirms the pair.

WORKED EXAMPLE 3.1.3

A context with factors

GIVEN A baker has 48 rolls and wants to arrange them in equal rows on trays, with more than one row and more than one roll per row.

ASKED List the possible arrangements and state which uses the most trays.

METHOD Each arrangement is a factor pair; exclude 1×48 and 48×1.

1. Factor pairs of 48: (1, 48), (2, 24), (3, 16), (4, 12), (6, 8).
2. Excluding the single-row and single-column arrangements leaves four pairs: 2×24, 3×16, 4×12, 6×8.
3. Each pair can be read in two ways, so there are eight arrangements in total.

ANSWER Eight arrangements; the arrangement with the most trays is 48 rows of 1, which is excluded, so the most useful is 24 rows of 2 (24 trays).

CHECK $24 \times 2 = 48$; all 48 rolls are used.

WATCH OUT Reading a factor pair as one arrangement rather than two halves the answer. A 2×24 tray is not the same layout as a 24×2 tray.

PRACTICE 3.1 · Factors, multiples and factor pairs

1	List all the factors of (a) 18 (b) 40 (c) 25	FLUENCY
2	Write the first five multiples of (a) 6 (b) 9 (c) 11	FLUENCY
3	Write all the factor pairs of 30.	FLUENCY
4	Is 8 a factor of 96? Give evidence.	FLUENCY

5	Explain the difference between a factor and a multiple using 15 and 45.	REASONING
6	A number has exactly two factor pairs. What kind of number is it? Give two examples.	REASONING
7	A gardener has 60 bulbs and wants equal rows of more than 3 and fewer than 20 bulbs. List the possible row lengths.	PROBLEM SOLVING
8	Two lights flash every 6 seconds and every 8 seconds. After how many seconds do they next flash together?	PROBLEM SOLVING
9	Find the smallest number with exactly six factors.	EXTENSION
10	Explain why a square number always has an odd number of factors.	EXTENSION

3.2 | Divisibility tests

A · I, III

A divisibility test is a shortcut that avoids long division. Each test works because of the structure of our base-ten number system, and each can be explained rather than merely memorised.

DEFINITION
Divisibility tests

A whole number is divisible by:

- **2** if its last digit is even (0, 2, 4, 6 or 8).
- **3** if the sum of its digits is divisible by 3.
- **4** if its last two digits form a number divisible by 4.
- **5** if its last digit is 0 or 5.
- **6** if it is divisible by both 2 and 3.
- **8** if its last three digits form a number divisible by 8.
- **9** if the sum of its digits is divisible by 9.
- **10** if its last digit is 0.

KEY IDEA
Why the digit-sum test for 3 works

10 = 9 + 1, so every power of ten is one more than a multiple of 9, and therefore one more than a multiple of 3. Expanding 472 gives 4 × 100 + 7 × 10 + 2 = 4(99+1) + 7(9+1) + 2. The multiples of 9 contribute nothing to the remainder, so the remainder depends only on 4 + 7 + 2 = 13, which leaves remainder 1. That is why the digit sum decides.

Number	÷ 2	÷ 3	÷ 4	÷ 5	÷ 9	÷ 10
1,236	yes (6)	yes (1+2+3+6 = 12)	yes (36)	no	no (12)	no
2,340	yes (0)	yes (9)	yes (40)	yes (0)	yes (9)	yes (0)
5,115	no (5)	yes (12)	no (15)	yes (5)	no (12)	no

The same number tested six ways

COMMON ERROR

Two divisibility traps

Trap 1: "divisible by 4" is not "even". 26 is even but not divisible by 4, because 26 ÷ 4 leaves 2. **Trap 2: a number can pass two tests but fail their product.** Passing the tests for 2 and 3 does guarantee divisibility by 6, because 2 and 3 share no factor other than 1. Passing the tests for 4 and 6 does **not** guarantee divisibility by 24: 12 passes both but is not divisible by 24.

WORKED EXAMPLE 3.2.1

Applying several tests

GIVEN The number 3,528.

ASKED Decide whether it is divisible by 2, 3, 4, 5, 6, 9 and 10, giving a reason each time.

METHOD Apply each test to the digits, not by dividing.

1. Last digit 8 is even, so divisible by 2.
2. Digit sum 3+5+2+8=18, divisible by 3, so divisible by 3 and by 6.
3. Last two digits 28 divisible by 4, so divisible by 4.
4. Last digit is not 0 or 5, so not divisible by 5 or 10.
5. Digit sum 18 is divisible by 9, so divisible by 9.

ANSWER Divisible by 2, 3, 4, 6 and 9; not by 5 or 10.

CHECK 3,528 ÷ 8 is not required, but 3,528 = 8 × 441, so it is also divisible by 8 — a reminder to test beyond the list when a question asks.

WORKED EXAMPLE 3.2.2

Using divisibility to find a missing digit

GIVEN The three-digit number 4□2 is divisible by 3.

ASKED Find the possible digits for the box.

METHOD Use the digit-sum test and solve an inequality.

1. Digit sum: 4 + □ + 2 = 6 + □.
2. This must be divisible by 3, and the digit is between 0 and 9.
3. 6 + □ can be 6, 9, 12 or 15, giving □ = 0, 3, 6 or 9.

ANSWER 0, 3, 6 or 9.

CHECK 402, 432, 462 and 492 all have digit sums divisible by 3; each divides by 3 exactly.

WORKED EXAMPLE 3.2.3

Explaining a test

GIVEN The claim "a number is divisible by 5 when its last digit is 0 or 5".

ASKED Explain why the test works for every whole number.

METHOD Use place value: every place above the ones is a multiple of 10, and 10 is a multiple of 5.

1. Any whole number can be written as $10k + d$, where d is the last digit.
2. $10k$ is always divisible by 5.
3. So the whole number is divisible by 5 exactly when d is divisible by 5, that is, when d is 0 or 5.

ANSWER The test works because all higher places contribute multiples of 10, which are already divisible by 5.

CHECK Test on 3,715: 3,710 is divisible by 5 and so is 5, so the total is.

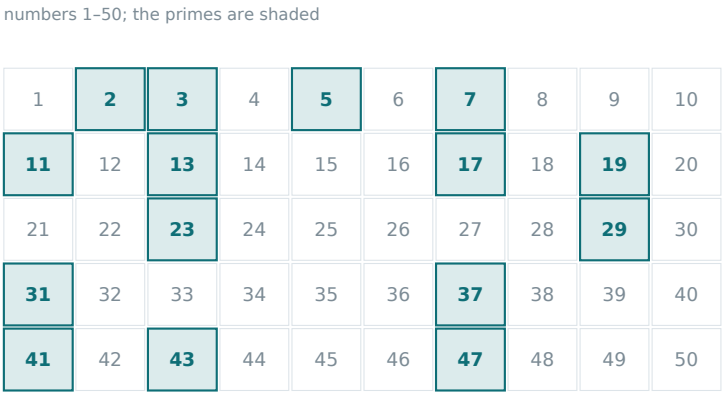
PRACTICE 3.2 · Divisibility tests

1	Which of 84, 95, 120, 132 are divisible by 3?	FLUENCY
2	Which of 216, 315, 420 are divisible by 9?	FLUENCY
3	Is 1,236 divisible by 4? By 8?	FLUENCY
4	Find the smallest digit to replace the box so that 5□ 4 is divisible by 3.	FLUENCY
5	Explain why the digit-sum test works for 9 but not for 4.	REASONING
6	A student says “48 is divisible by 2 and by 4, so it must be divisible by 8.” Is this reasoning sound?	REASONING
7	A number is divisible by 5 and by 9. What is the smallest two-digit number that satisfies both?	PROBLEM SOLVING
8	A year group of 372 students is split into equal groups of 6. How many groups? Use divisibility to justify that the split is exact.	PROBLEM SOLVING
9	Find a three-digit number divisible by 2, 3, 4 and 5 but not by 9.	EXTENSION
10	Explain why a number divisible by 6 must be even, and why a number divisible by 4 need not be divisible by 8.	EXTENSION

3.3 | Prime and composite numbers

A · I, II

Primes are the numbers that cannot be built by multiplying smaller whole numbers. They are the atoms of arithmetic, and every other number is assembled from them.



A sieve of Eratosthenes for the numbers 1 to 50: each prime is shaded.

DEFINITION

Prime and composite

A **prime number** is a whole number greater than 1 with exactly two factors: 1 and itself. A **composite number** is a whole number greater than 1 with more than two factors. The number 1 is neither prime nor composite, because it has only one factor. The primes begin 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, …

KEY IDEA
Why 1 is excluded

If 1 were counted as prime, a number could be written as a product of primes in many ways — $6 = 2 \times 3 = 1 \times 2 \times 3 = 1 \times 1 \times 2 \times 3$ — and the unique building blocks would lose their uniqueness. Excluding 1 keeps the fundamental theorem of arithmetic true: every composite number has exactly one prime factorisation, apart from the order of the factors.

METHOD
Testing whether a number is prime

1. Check the small primes in order: 2, 3, 5, 7, 11, 13, ...
2. Stop when the square of the prime you are testing is larger than the number.
3. If no prime divides the number exactly, the number is prime.

Why can you stop at the square root? If a number n has a factor larger than $\sqrt{n}$, the paired factor must be smaller than $\sqrt{n}$, and you would already have found it.

Prime tested	Square of the prime	Does it divide 97?
2	4	no (last digit is 7)
3	9	no (digit sum 16)
5	25	no (last digit is not 0 or 5)
7	49	no ($97 = 7 \times 13 + 6$)
11	121	stop: $121 > 97$, so no larger prime can divide 97

Testing 97 for primality

COMMON ERROR
The commonest primality error

“91 is prime because it looks prime.” $91 = 7 \times 13$. Numbers that look irregular can still be composite, so a test is always needed. Conversely, 2 is prime although it is even: evenness is not the same as compositeness.

WORKED EXAMPLE 3.3.1
Classifying numbers

GIVEN The numbers 1, 21, 29, 51 and 2.

ASKED Classify each as prime, composite or neither, with evidence.

METHOD Test for factors rather than judging by appearance.

1. 1 has one factor, so it is neither prime nor composite.
2. $21 = 3 \times 7$, composite.
3. 29 is not divisible by 2, 3 or 5; $7^2 = 49 > 29$, so 29 is prime.
4. $51 = 3 \times 17$, composite.
5. 2 has factors 1 and 2 only, so it is prime.

ANSWER neither, composite, prime, composite, prime.

CHECK Each composite has a named factor pair; each prime survived testing up to its square root.

WORKED EXAMPLE 3.3.2

Testing a larger number

GIVEN 143.

ASKED Decide whether 143 is prime.

METHOD Test primes up to $\sqrt{143} \approx 12$.

1. Not divisible by 2, 3 (digit sum 8) or 5.
2. $143 \div 7 = 20$ remainder 3, so not divisible by 7.
3. $143 \div 11 = 13$ exactly, so 143 is composite.

ANSWER $143 = 11 \times 13$, composite.

CHECK $11 \times 13 = 143$.

WORKED EXAMPLE 3.3.3

Reasoning about primes

GIVEN The statement “every prime number greater than 2 is odd”.

ASKED Explain why the statement is true, and why it does not say that every odd number is prime.

METHOD Use the definition of prime and a counterexample for the converse.

1. An even number greater than 2 has 2 as a factor, as well as 1 and itself, so it has at least three factors.
2. Therefore no even number greater than 2 can be prime.
3. The converse fails: 9 is odd but composite, because $9 = 3 \times 3$.

ANSWER The statement is true because even numbers above 2 are composite; oddness is necessary but not sufficient for primality.

CHECK 2 is the only even prime; 9, 15, 21 and 25 are odd composites.

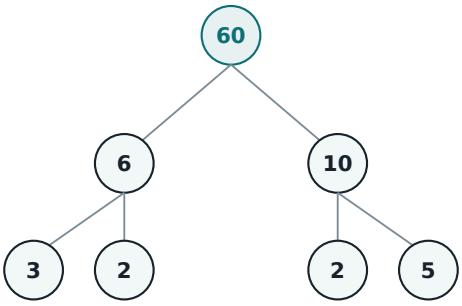
PRACTICE 3.3 · Prime and composite numbers

1	List the prime numbers between 20 and 40.	FLUENCY
2	Classify each as prime or composite: 33, 37, 39, 41.	FLUENCY
3	Write 10 as the sum of two primes.	FLUENCY
4	Explain why 1 is not a prime number.	REASONING
5	Is 121 prime? Justify your answer.	REASONING
6	Two students disagree about 87. One says “it is prime because it is odd”; the other says “it is composite because 8 + 7 = 15”. Who is right, and what is the correct reason?	REASONING
7	A number between 40 and 50 is the product of two primes. Find all possibilities.	PROBLEM SOLVING
8	Explain why 2 is the only even prime.	PROBLEM SOLVING
9	Find the smallest number greater than 100 that is prime.	EXTENSION
10	Is the sum of two primes always even? Test with examples and explain.	EXTENSION

3.4 | Prime factorisation

C · I, II

Every composite number can be broken into primes, and the result is unique. That uniqueness is what makes primes useful: they are an identity card for a number.



A factor tree for 60. The primes at the ends of the branches give $60 = 2 \times 2 \times 3 \times 5$.

DEFINITION
Prime factorisation

A **prime factorisation** writes a number as a product of prime factors. It is usual to write the primes in order and to use index notation for repeated factors:

$$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$$

METHOD
Two reliable methods

Factor tree. Split the number into any factor pair, then split each composite branch until every branch ends in a prime. Collect the primes at the ends. **Division by primes.** Divide repeatedly by the smallest prime that divides exactly, writing the quotient each time, until the quotient is 1. The primes used are the factors.

Both methods give the same prime factorisation. If they appear to disagree, one of the branches has been left composite or a factor has been lost.

KEY IDEA
Reconstructing a number from its primes

The prime factorisation is a complete recipe. $2^2 \times 3 \times 5$ can be rebuilt as $4 \times 3 \times 5 = 60$, and it also tells you every factor of 60: any factor is built from a selection of the primes, such as $2 \times 5 = 10$.

COMMON ERROR
Leaving a composite branch

A tree that stops at $60 = 6 \times 10$ is not finished, because 6 and 10 are composite. Continue until every branch ends in a prime: $6 = 2 \times 3$ and $10 = 2 \times 5$, giving $2 \times 3 \times 2 \times 5 = 2^2 \times 3 \times 5$.

WORKED EXAMPLE 3.4.1

Prime factorisation by factor tree

GIVEN The number 84.

ASKED Write its prime factorisation in index form.

METHOD Build a tree and continue until every branch is prime.

1. $84 = 4 \times 21$.
2. $4 = 2 \times 2$ and $21 = 3 \times 7$.
3. Collect the primes: 2, 2, 3, 7.

ANSWER $84 = 2^2 \times 3 \times 7$.

CHECK $4 \times 3 \times 7 = 84$.

WORKED EXAMPLE 3.4.2

Prime factorisation by division

GIVEN The number 360.

ASKED Write its prime factorisation and use it to list three factors other than 1 and 360.

METHOD Divide repeatedly by the smallest prime that divides exactly.

1. $360 \div 2 = 180$; $\div 2 = 90$; $\div 2 = 45$ (three twos).
2. $45 \div 3 = 15$; $\div 3 = 5$ (two threes).
3. 5 is prime, so stop.

ANSWER $360 = 2^3 \times 3^2 \times 5$. Factors include $2^3 \times 3 = 24$, $3^2 \times 5 = 45$ and $2 \times 5 = 10$.

CHECK $8 \times 9 \times 5 = 360$.

WORKED EXAMPLE 3.4.3

Using prime factors to find a perfect square

GIVEN The number 72.

ASKED Find the smallest whole number by which 72 must be multiplied to make a square number.

METHOD Write the prime factorisation and look for unpaired primes.

1. $72 = 2^3 \times 3^2$.
2. In a square number every prime must appear an even number of times; here 2 appears three times.
3. Multiply by 2 to complete the pair: $72 \times 2 = 144 = 2^4 \times 3^2 = 12^2$.

ANSWER Multiply by 2 to obtain 144.

CHECK $\sqrt{144} = 12$ exactly.

PRACTICE 3.4 · Prime factorisation

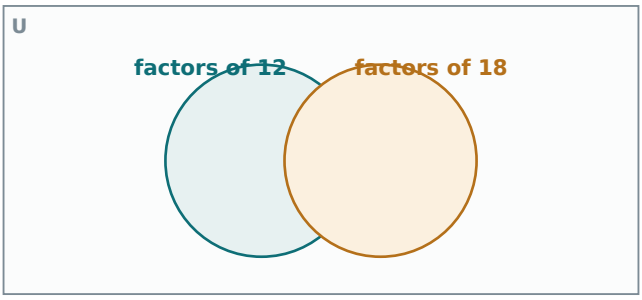
1	Write the prime factorisation of (a) 24 (b) 45 (c) 100	FLUENCY
2	Write 126 as a product of primes.	FLUENCY
3	Reconstruct the number from $2^3 \times 5^2$.	FLUENCY
4	Explain why $60 = 6 \times 10$ is not a prime factorisation.	REASONING
5	Two students factorise 90 as $2 \times 3^2 \times 5$ and $2 \times 3 \times 3 \times 5$. Are both correct?	REASONING
6	Use a prime factorisation to decide whether 252 is divisible by 7.	PROBLEM SOLVING

7	Find the smallest number that must be multiplied by 108 to make a square number.	PROBLEM SOLVING
8	A number is $2^2 \times 3 \times 5^2$. List four of its factors and explain how you chose them.	EXTENSION
9	Explain why the prime factorisation of a number is unique.	EXTENSION
10	Find two numbers whose prime factorisations use only 2 and 3 and whose product is 324.	EXTENSION

3.5 | Highest common factor and lowest common multiple

D · I, III, IV

The HCF answers “what is the largest group we can make from both numbers?”, and the LCM answers “when do two repeating events coincide?”. Both can be found from prime factorisations.



The factors of 12 and 18. The overlap contains the common factors 1, 2, 3 and 6, so the highest common factor is 6.

DEFINITION

HCF and LCM

The **highest common factor (HCF)** of two or more numbers is the largest number that divides all of them exactly. The **lowest common multiple (LCM)** is the smallest number that all of them divide exactly. For 12 and 18: the common factors are 1, 2, 3, 6, so HCF = 6. The multiples of 12 are 12, 24, 36, … and of 18 are 18, 36, …, so LCM = 36.

METHOD

Finding HCF and LCM from prime factors

1. Write each number as a product of primes in index form.
2. For the **HCF**, take the primes that appear in both, each to the **lower** power.
3. For the **LCM**, take every prime that appears in either, each to the **higher** power.
4. Multiply the chosen primes.

KEY IDEA

A check that always works

For any two positive whole numbers, HCF × LCM = product of the numbers. For 12 and 18: 6 × 36 = 216 = 12 × 18. The check catches most arithmetic slips, and it explains why the two ideas are partners rather than opposites.

Numbers	Prime factorisations	HCF	LCM	HCF × LCM	
12 and 18	$2^2 \times 3$	2×3^2	6	36	$6 \times 36 = 216$
24 and 36	$2^3 \times 3$	$2^2 \times 3^2$	12	72	$12 \times 72 = 864$
8 and 9	2^3	3^2	1	72	$1 \times 72 = 72$

Two numbers, both quantities

The third row shows two numbers with no common factor other than 1. Such numbers are called **coprime**, and their LCM is simply their product.

WORKED EXAMPLE 3.5.1

Finding both values from prime factors

GIVEN The numbers 24 and 36.

ASKED Find the HCF and the LCM.

METHOD Compare the prime factorisations, then apply the lower-power and higher-power rules.

1. $24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$.
2. HCF: common primes are 2 and 3, to the lower powers: $2^2 \times 3 = 12$.
3. LCM: all primes to the higher powers: $2^3 \times 3^2 = 8 \times 9 = 72$.

ANSWER HCF = 12, LCM = 72.

CHECK $12 \times 72 = 864 = 24 \times 36$.

WORKED EXAMPLE 3.5.2

Choosing the right quantity

GIVEN (a) A shop has 84 pens and 60 pencils to pack into identical packs with no items left over.
(b) Two buses leave at 6:00, one every 12 minutes and one every 18 minutes.

ASKED Decide whether each problem needs the HCF or the LCM, then solve it.

METHOD Ask whether the answer is a size of group (HCF) or a moment in time (LCM).

1. Packs: the number of items per pack must divide both totals, so this is an HCF problem. $84 = 2^2 \times 3 \times 7$ and $60 = 2^2 \times 3 \times 5$, so HCF = 12.
2. Buses: the next coincidence is a common multiple of 12 and 18. $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$, so LCM $= 2^2 \times 3^2 = 36$ minutes.

ANSWER (a) 12 items per pack (7 packs of pens, 5 of pencils) (b) 6:36.

CHECK $84 \div 12 = 7$ and $60 \div 12 = 5$; $36 \div 12 = 3$ and $36 \div 18 = 2$.

WORKED EXAMPLE 3.5.3

Three numbers at once

GIVEN The numbers 8, 12 and 18.

ASKED Find the HCF and the LCM of all three.

METHOD Extend the same rules to three factorisations.

1. $8 = 2^3$, $12 = 2^2 \times 3$, $18 = 2 \times 3^2$.
2. HCF: the only prime in all three is 2, to the lowest power 1: HCF = 2.
3. LCM: every prime that appears, to the highest power: $2^3 \times 3^2 = 72$.

ANSWER HCF = 2, LCM = 72.

CHECK 72 is divisible by 8, 12 and 18; no smaller number is: 36 is not divisible by 8.

WATCH OUT The rule HCF × LCM = product of the numbers applies to two numbers, not three. Use it only as a check for pairs.

PRACTICE 3.5 · Highest common factor and lowest common multiple

1	Find the HCF of (a) 12 and 20 (b) 18 and 27 (c) 16 and 24	FLUENCY
2	Find the LCM of (a) 4 and 6 (b) 6 and 9 (c) 8 and 12	FLUENCY
3	Find the HCF and LCM of 10 and 15.	FLUENCY
4	Explain why the LCM of two coprime numbers is their product.	REASONING
5	A student finds the HCF of 24 and 36 as 72. What has been found instead?	REASONING
6	48 girls and 36 boys are split into equal teams, each team containing both. What is the largest possible team size?	PROBLEM SOLVING
7	Two traffic lights change every 40 and 60 seconds. They change together at 8:00. When do they next change together?	PROBLEM SOLVING
8	A classroom is 84 cm by 60 cm on a plan; square tiles must fit exactly along both sides. Find the largest tile size.	PROBLEM SOLVING
9	Find two numbers whose HCF is 6 and whose LCM is 36.	EXTENSION
10	Three bells ring every 6, 8 and 12 minutes. They ring together at noon. When is the next time all three ring together?	EXTENSION

F INVESTIGATE AND APPLY

A criterion B investigation and a criterion D application

INVESTIGATION 3 | The odd number of factors

B · I, II, III

In the Explore activity you noticed that some numbers make a square. Investigate the connection between square numbers and the number of factors they have.

INVESTIGATION

Investigation brief

Question. Which numbers have an odd number of factors, and why? **What you need.** The numbers 1 to 36 arranged in a table; space to record the factor pairs. **What to produce.** A table of factor counts, a conjecture, an explanation and a test on a number larger than 36.

PART 1 · RECORD

Number	Factors	How many	Odd or even?
1	1	1	odd
2	1, 2	2	even
3			
4			
5			
6			
8			
9			
12			
16			
20			
25			
36			

Copy and complete the factor counts

PART 2 · DESCRIBE AND CONJECTURE

1. List every number in your table with an odd number of factors. What do they have in common?
2. Explain why factor pairs usually come in twos.
3. What happens to the factor pair when a number is a square? Use 16 as your example.
4. Write a conjecture beginning: “A number has an odd number of factors exactly when …”.

PART 3 · TEST AND JUSTIFY

1. Test your conjecture on 49 and on 64.
2. Test it on a number greater than 100, such as 100 or 121.
3. Explain why a non-square cannot have an unpaired factor. Your explanation should refer to the multiplication, not to a list.
4. A classmate says “every odd number has an odd number of factors”. Find a counterexample and explain what the correct statement is.

CRITERION TASK

Criterion B task · Requirements

A complete table; a pattern described in words; a general statement that predicts new cases; a verification on at least two cases not used to form the conjecture; a justification that explains the pairing.

APPLICATION 3 | The fairness relay

D · I, II, III, IV

The sports committee must organise a relay for three year groups of 84, 60 and 48 students. Teams must be identical, every student must run, and the committee wants the largest possible team size — but there are constraints on time and space.

THE BRIEF

Constraint	Value
Students in each year group	84, 60, 48
Track lanes available	8
Time available after the changeover	45 minutes
Minutes per race	12
Maximum team size that still fits a lane	12

Constraints from the committee

- Teams must be identical across all three year groups.
- Every student must race exactly once.
- The committee wants the greatest number of teams possible.

TASK 1 · FIND THE POSSIBLE TEAM SIZES

1. List all common factors of 84, 60 and 48.
2. State the HCF and explain what it represents in the context.
3. Which of the common factors satisfy the lane limit of 12?

TASK 2 · PLAN THE RELAY

1. Choose a team size and calculate the number of teams in each year group.
2. With 8 lanes and 12 minutes per race, how many races can be held in 45 minutes? Show your reasoning.
3. Does your plan fit the time available? If not, revise the team size and recalculate.

TASK 3 · EVALUATE THE FAIRNESS

1. Two proposals are made: the largest possible team size, and the smallest possible team size above 4. Compare them on time, fairness and organisation.
2. Which proposal would you recommend? Justify with figures.
3. Write two sentences explaining why the HCF is the right mathematical tool for this problem rather than the LCM.

CRITERION TASK

Criterion D task · Requirements

The relevant numbers identified; the HCF used correctly and explained in context; calculations shown with units; a decision justified against the constraints; limitations of the plan acknowledged.

TASK C | Writing the divisibility guide

C · I, II, III, IV

CRITERION TASK

The task

Write a one-page guide, for students in the year below, called **"How to tell whether a number divides exactly"**. Your guide must include:

1. Tests for 2, 3, 4, 5, 6, 9 and 10, each with one worked example.
2. A short explanation of why the test for 3 works, written without algebra.
3. A warning about one test that is easy to misuse, with a corrected example.
4. A table of four numbers tested against all seven divisors, with the results.

Criterion C strand	Evidence
Notation (i)	Correct use of "divisible by" and "factor".
Representation (ii)	A table organises the tests so they can be compared.
Reasoning (iii)	The explanation of the digit-sum test says why, not just what.
Organisation (iv)	A reader can use the guide without asking a question.

Success criteria

G REFLECT AND CONSOLIDATE

Summary, vocabulary, misconceptions and review

REVIEW | Consolidating Chapter 3

A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- Factors divide exactly; multiples are produced by multiplying. Each statement implies the other.
- Factor pairs make a complete factor list possible without guessing.
- Divisibility tests are shortcuts based on place value, and each can be explained.
- Primes have exactly two factors; 1 is neither prime nor composite.
- Every composite number has exactly one prime factorisation, apart from order.
- The HCF is the largest common divisor; the LCM is the smallest common multiple. For two numbers, HCF × LCM = product.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Factor	A whole number that divides another exactly.
Multiple	The result of multiplying a number by a whole number.
Factor pair	Two factors that multiply to give a number.
Divisible	Divides exactly, with no remainder.
Prime	A whole number greater than 1 with exactly two factors.

Term	Meaning in this course
Composite	A whole number greater than 1 with more than two factors.
Prime factorisation	A number written as a product of primes.
HCF	The largest number dividing two or more numbers exactly.
LCM	The smallest number that two or more numbers divide exactly.
Coprime	Having no common factor other than 1.

Vocabulary for Chapter 3

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence
“1 is prime.”	A prime has exactly two factors.	1 has one factor
“91 is prime.”	Test primes up to the square root.	$91 = 7 \times 13$
“Factors are always smaller than the number.”	The number is a factor of itself.	$24 \div 24 = 1$
“HCF and LCM are the same thing.”	They answer different questions.	12 and 18: 6 and 36
“Two tests guarantee the product.”	Only when the divisors are coprime.	$12 \div 4$ and $\div 6$ but not $\div 24$

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
list all factors of a number using factor pairs.			
explain the difference between a factor and a multiple.			
apply the divisibility tests and say why they work.			
identify primes and explain why 1 is excluded.			
write a prime factorisation in index form.			
calculate HCF and LCM and choose the right one in context.			

How confident are you?

REFLECTION

1. Which divisibility test do you find easiest to explain, and why?
2. Describe a moment when prime factorisation made a hard question easy.
3. Write one sentence beginning “A prime number is …” and one beginning “The HCF matters because …”.
4. In the relay problem, which constraint mattered most? Would a different committee have valued a different number?
5. Which of your factor lists this chapter contained a mistake, and how did you find it?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ3 · Applying the chapter

1	List the factors of 45.	FLUENCY
2	Write the first four multiples of 12.	FLUENCY
3	Is 132 divisible by 3? By 4?	FLUENCY
4	Write 150 as a product of primes.	FLUENCY
5	Find the HCF of 36 and 54.	FLUENCY
6	Find the LCM of 8 and 10.	FLUENCY
7	Explain why 2 is prime but 4 is not.	REASONING
8	A number is divisible by 9. Must it be divisible by 3? Must a number divisible by 3 be divisible by 9?	REASONING
9	Explain why the LCM of 6 and 15 is 30 rather than 90.	REASONING
10	A rectangle is 72 cm by 48 cm. Square tiles of equal size fit exactly. Find the largest possible tile side.	PROBLEM SOLVING
11	Two machines beep every 45 and 60 seconds. They beep together now. When next?	PROBLEM SOLVING
12	Use prime factors to decide whether 441 is a square number.	PROBLEM SOLVING
13	Explain in two sentences why every composite number can be written as a product of primes.	COMMUNICATING
14	Write 84 as a product of primes and use it to list all its factors.	COMMUNICATING
15	Find the smallest number with exactly eight factors.	EXTENSION
16	Two numbers have HCF 4 and LCM 48. Find both numbers.	EXTENSION
17	Explain why the product of two consecutive numbers is always even.	EXTENSION
18	120 students and 72 teachers are split into identical groups containing both. What is the largest number of groups?	PROBLEM SOLVING
19	Explain why a divisibility test for 7 is harder to state than the test for 5.	COMMUNICATING

MIXED 3 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX3 · End-of-chapter mixed set

1	Write the factors of 60 that are also multiples of 5.	FLUENCY
2	Find the HCF of 16, 24 and 40.	FLUENCY
3	Find the LCM of 5, 6 and 10.	FLUENCY
4	Is 2,345 divisible by 5? By 9?	FLUENCY

5	Write 96 in index form using primes.	FLUENCY
6	Explain why 1 is a factor of every number but not a prime.	REASONING
7	Is the statement “every number with an even digit sum is divisible by 3” true? Test it.	REASONING
8	A chocolate bar is 24 cm long and 18 cm wide. Squares of equal size cover it exactly. Find the largest possible square side.	PROBLEM SOLVING
9	Buses leave every 15 and 25 minutes. They leave together at 7:00. When next?	PROBLEM SOLVING
10	A number between 60 and 70 has exactly four factors. Find it.	PROBLEM SOLVING
11	Explain how a factor tree shows that a prime factorisation is complete.	COMMUNICATING
12	Write two sentences explaining the difference between HCF and LCM to a classmate.	COMMUNICATING
13	Find two numbers whose product is 180 and whose HCF is 6.	EXTENSION
14	Explain why adding two odd numbers always gives an even number, using factors of 2.	EXTENSION
15	A number has prime factorisation $2^a \times 3^b$. How many factors does it have? Test your rule.	EXTENSION
16	A school hall floor is 5.4 m by 4.2 m. Square tiles of whole-centimetre side cover it exactly. Find the largest tile side and the number of tiles.	PROBLEM SOLVING

CUMULATIVE PRACTICE 3 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP3 · Cumulative practice

1	List the factors of 36.	FLUENCY
2	Find the HCF of 18 and 30.	FLUENCY
3	Find the LCM of 9 and 12.	FLUENCY
4	Explain why 51 is not prime.	REASONING
5	Two buses leave every 8 and 12 minutes. When do they next leave together if they leave together now?	PROBLEM SOLVING
6	A rectangle is 84 cm by 60 cm. Find the largest square tile that fits exactly.	PROBLEM SOLVING
7	Explain how a factor tree shows the prime factorisation is complete.	COMMUNICATING
8	Find the smallest number with exactly four factors.	EXTENSION

PART II

PARTS AND PROPORTIONS

II

04 Fractions and decimals

05 Percentages and applications

PART II · PARTS AND PROPORTIONS

CHAPTER 04

Fractions and decimals

A fraction and a decimal can name the same point on a number line. Learning to move between forms makes it easier to compare quantities, calculate and judge how precise an answer really is.

KEY CONCEPT	**Relationships**
RELATED	**Equivalence**
RELATED	**Representation**
GLOBAL CONTEXT	**Scientific and technical innovation**

04
CHAPTER

INQUIRY QUESTION

When two numbers look different but describe the same amount, how do we choose the form that is easiest to compare, calculate with or trust?

A 3/8 inch drill bit and a 0.375 inch drill bit are the same size. A laboratory that records one third of a litre as 0.33 litres has changed the quantity slightly. In measurement, the form of a number carries information about precision.

IN THIS CHAPTER

4.1 What a fraction means
4.2 Equivalent fractions and simplifying
4.3 Comparing and ordering fractions
4.4 Fractions and decimals
4.5 Calculating with decimals
4.6 Adding and subtracting fractions
4.7 Multiplying and dividing fractions

Investigation 4 Which product is larger?
Application 4 Scaling the canteen recipe
Task C Explaining the rule for dividing
Review Consolidating Chapter 4
Review questions Practice across the whole chapter
Mixed 4 Fluency, reasoning and problem solving
Cumulative practice 4 Questions that mix this chapter with earlier ones

APPROXIMATELY 16 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

A fraction and a decimal are two records of one point on the number line. Choosing between them is not a matter of taste: the choice decides what can be seen, what cancels neatly and how much precision the reader is being offered.

KEY CONCEPT · RELATIONSHIPS

Relationships. A fraction records a relationship between part and whole: the denominator sets the part size, the numerator counts parts.

RELATED CONCEPT · EQUIVALENCE

Equivalence. Different fractions and decimals can name the same number, so a calculation can be rewritten in a convenient form.

RELATED CONCEPT · REPRESENTATION

Representation. A number line, an area diagram, a fraction and a decimal make different features visible; the form should serve the task.

GLOBAL CONTEXT

Scientific and technical innovation

EXPLORATION

Precision and measurement: how the way a quantity is recorded affects comparison, calculation and trust in a result.

FACTUAL

What is an equivalent fraction? What does each digit after a decimal point represent? How can a fraction be written as a decimal?

CONCEPTUAL

Why do different fractions name the same number? How does the form of a number influence the calculation you choose?

DEBATABLE

Is a fraction always a more exact record of a quantity than a decimal?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Generate equivalent fractions, order fractions, calculate with fractions and decimals, round decimals.
B · Investigating patterns	i, ii	Investigate which fractions produce terminating decimals and generalise from the prime factors of denominators.
C · Communicating	i, ii, iii	Explain equivalence and method choice, use number lines and diagrams, record results with suitable notation.
D · Applying mathematics	i, ii, iii, iv	Scale recipes, compare measurements, justify the choice of fraction or decimal and evaluate the effect of rounding.

ATL SKILL · TRANSFER

In order for a student to decide whether a recipe scaled for 4 or for 15 should be recorded in fractions or decimals, students must transfer the same scaling method to an awkward quantity such as $^{4}/_{9}$ of a cup and compare which record a cook and a supplier could act on.

STATEMENT OF INQUIRY

The same quantity can be represented in equivalent forms, and choosing an appropriate form improves the precision of comparison, calculation and communication.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand equivalent fractions and write a fraction in simplest form.
- Compare and order fractions using common denominators, benchmarks or decimals.
- Add and subtract fractions, including mixed numbers.
- Multiply and divide fractions and interpret both operations.
- Know decimal place value to thousandths, and order, round and calculate with decimals.
- Convert between fractions and decimals and explain which decimals terminate.

SUCCESS CRITERIA

- ✓ I can generate equivalent fractions and simplify using a common factor.
- ✓ I can compare and order fractions with unlike denominators and justify my method.
- ✓ I can add and subtract fractions and mixed numbers, regrouping when needed.
- ✓ I can multiply and divide fractions, using the reciprocal correctly.
- ✓ I can read, order, round and calculate with decimals to thousandths.
- ✓ I can convert between fractions and decimals and explain which decimals terminate.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Shade $\frac{3}{4}$ of a rectangle divided into 8 equal parts.
2. Write 0.5 and 0.25 as fractions in simplest form.
3. Find $\frac{1}{2}$ of 20.
4. Which is larger, $\frac{3}{4}$ or $\frac{3}{8}$?
5. Calculate $\frac{1}{4} + \frac{1}{4}$.
6. Write “seven tenths” as a decimal and as a fraction.

Answers are in the Chapter 4 answer section. If more than two are incorrect, review simple equivalent fractions and decimal tenths before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Halves, quarters and tenths; simple fractions of shapes and amounts; decimal notation to hundredths.	Equivalence, ordering, the four operations with fractions, decimal place value and fraction–decimal links.	Percentages, ratio and proportion, rational expressions and algebraic fractions.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Recipes, measurements and shared quantities rarely arrive as whole numbers.	How can different-looking numbers describe exactly the same amount?	Equivalence, representation and precision.	Which fractions produce terminating decimals, and how can denominators predict it?	Number lines, fraction diagrams and decimal notation.	When is an exact fraction more useful than a rounded decimal?

A ENGAGE
Half a pizza, or two quarters?

Two friends order the same size pizza. Ana eats $\frac{2}{4}$ of hers and Ben eats $\frac{1}{2}$ of his. Ben says he ate more because “one half sounds bigger than two quarters”. Ana disagrees.

Both ate the same amount. The fractions look different because they are cut differently, but the quantity is identical. This is the central idea of the chapter: a fraction is a **number**, not a pair of numbers, and it can be written in many equivalent ways.

The same confusion appears in other clothes. A phone battery at 0.5 is at $\frac{1}{2}$; a sale of 50% is also a half. Decimals, fractions and percentages are three notations for one quantity. Choosing between them is a matter of convenience, not of mathematics.

Fraction	Decimal	Percentage	Where it appears
$\frac{1}{4}$	0.25	25%	A quarter of an hour, 15 minutes
$\frac{3}{10}$	0.3	30%	Three tenths of a litre, 300 ml
$\frac{2}{5}$	0.4	40%	Two fifths of a class of 30, 12 students
$\frac{5}{8}$	0.625	62.5%	Five eighths of a metre, 62.5 cm

Figure 4.1 The same quantity, four notations

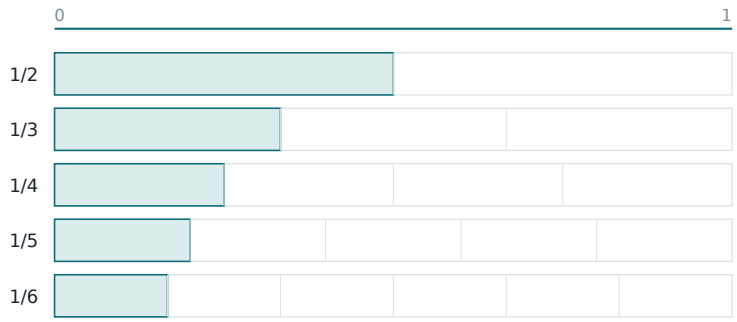
NOTICE AND WONDER

- Why does $\frac{2}{4}$ equal $\frac{1}{2}$ even though the numbers are different?
- Which notation would you choose to compare two quantities quickly? Which to calculate exactly?
- Does a fraction get larger when its denominator gets larger? Explain with an example.
- What would a fraction larger than 1 mean in the pizza situation?

Keep this question open: why can one number have so many names, and how do we choose the name that helps?

B EXPLORE
The fraction wall

Work in pairs. You need a strip of paper the same length as the fraction wall below, a ruler and a pencil.



A fraction wall. Aligned bars show which fractions are equal and which are larger: each half-line matches a line on the bar below it.

YOUR INVESTIGATION

Step 1. Fold your strip in half, then in quarters, then in eighths. Shade one half.

Step 2. Without unfolding differently, find how many eighths cover the same length as one half. Record the result.

Step 3. Repeat for one third against sixths and ninths, and for one quarter against twelfths.

Step 4. Record every equality you discover in a table like the one below.

One fraction	Equal fractions found	What was done to the numerator and denominator?
$\frac{1}{2}$		
$\frac{1}{3}$		
$\frac{1}{4}$		
$\frac{2}{3}$		

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

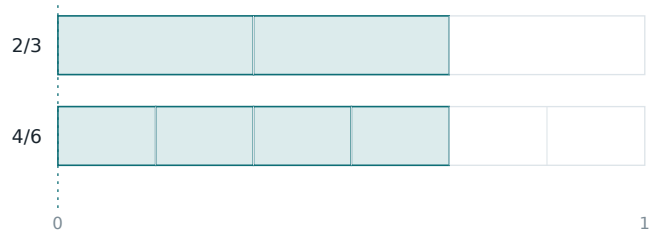
1. In $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$, what happens to the top number each time? To the bottom number?
2. Is $\frac{1}{2}$ equal to $\frac{1}{3}$? Use the wall to explain, not the numbers alone.
3. Which is larger, $\frac{3}{4}$ or $\frac{5}{8}$? How does the wall show this?
4. Find two fractions on the wall that are equal to $\frac{2}{3}$.
5. Predict an equivalent fraction for $\frac{3}{5}$ that is not on the wall, then test it by reasoning.

MAKING A CONJECTURE

> “Multiplying the numerator and denominator by the same number … because … .”

WHAT YOU SHOULD HAVE NOTICED

- Multiplying top and bottom by the same number keeps the fraction equal: $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$.
- The pieces get smaller as the denominator grows, but there are more of them, and the total length is unchanged.
- A larger denominator alone does not mean a larger fraction: $\frac{1}{8}$ is smaller than $\frac{1}{3}$.



Two thirds and four sixths cover exactly the same length of the bar, so they are equivalent fractions.

INVESTIGATION
Explore further

Cut a square of paper into quarters, then shade $\frac{2}{3}$ of the square as accurately as you can without measuring. Compare your method with a partner's. Can you shade $\frac{2}{3}$ using a grid of twelfths? What does the grid tell you about $\frac{2}{3}$ and $\frac{8}{12}$?

4.1 | What a fraction means

A · I, II

A fraction can be read as a part of a whole, as a division, as an operator on a quantity, or as a position on a number line. Reading it in more than one way prevents most errors.

DEFINITION

Fraction, numerator, denominator

In the fraction $\frac{a}{b}$ ($b \neq 0$), a is the **numerator** and b is the **denominator**. The denominator says how many equal parts the whole is divided into; the numerator says how many of those parts are taken. A **proper fraction** has numerator smaller than denominator, so its value is less than 1. An **improper fraction** has numerator greater than or equal to the denominator, so its value is at least 1. A **mixed number** is a whole number and a proper fraction written together, such as $2\frac{3}{5}$.

KEY IDEA

Four readings of $\frac{3}{4}$

- **Part of a whole:** three of four equal parts.
- **Division:** $3 \div 4$.
- **Operator:** three quarters of a quantity — $\frac{3}{4}$ of 20 is 15.
- **Number:** a point on the number line, three quarters of the way from 0 to 1.

The fourth reading matters most later: $\frac{3}{4}$ is a single number that can be added, multiplied and placed on a line, not a sum of two numbers.

METHOD

Converting between mixed numbers and improper fractions

1. Mixed to improper: multiply the whole number by the denominator, add the numerator, and keep the denominator. $2\frac{3}{5} = \frac{2 \times 5 + 3}{5} = \frac{13}{5}$.
2. Improper to mixed: divide the numerator by the denominator; the quotient is the whole number, the remainder goes over the denominator. $\frac{17}{4} = 4\frac{1}{4}$.
3. Check by estimating: $\frac{13}{5}$ is between 2 and 3, which matches $2\frac{3}{5}$.

COMMON ERROR

Adding the whole numbers to the denominator

A common error is to write $2\frac{3}{5} = \frac{5}{5}$, adding 2 and 3. The whole number 2 means two whole units, each of which is $\frac{5}{5}$, so the total is $\frac{10}{5} + \frac{3}{5} = \frac{13}{5}$.

WORKED EXAMPLE 4.1.1

Reading a fraction in context

GIVEN A 600 ml bottle is $\frac{3}{5}$ full.

ASKED How much liquid is in the bottle? Explain using two readings of the fraction.

METHOD Use the fraction as an operator: divide by the denominator, multiply by the numerator.

1. Divide 600 ml into 5 equal parts: 600 ÷ 5 = 120 ml per part.
2. Take 3 parts: 3 × 120 = 360 ml.
3. Reading it as a multiplication: $\frac{3}{5} \times 600 = 360$.

ANSWER 360 ml.

CHECK $\frac{3}{5}$ is more than half, and 360 ml is more than 300 ml.

WORKED EXAMPLE 4.1.2

Mixed numbers and improper fractions

GIVEN The mixed number $3\frac{2}{7}$ and the improper fraction $\frac{23}{6}$.

ASKED Write each in the other form.

METHOD Apply the two conversion rules and check by estimation.

1. $3\frac{2}{7} = \frac{3 \times 7 + 2}{7} = \frac{23}{7}$.
2. 23 ÷ 6 = 3 remainder 5, so $\frac{23}{6} = 3\frac{5}{6}$.

ANSWER $\frac{23}{7}$ and $3\frac{5}{6}$.

CHECK $\frac{23}{7}$ is between 3 and 4; $3\frac{5}{6}$ is between 3 and 4.

WORKED EXAMPLE 4.1.3

A fraction of a quantity with a remainder

GIVEN A class of 32 students; $\frac{5}{8}$ walk to school.

ASKED How many walk, and how many do not?

METHOD Find one eighth first, then scale.

1. One eighth of 32 is 32 ÷ 8 = 4.
2. Five eighths is 5 × 4 = 20 students.
3. The remainder is 32 - 20 = 12, which is $\frac{3}{8}$ of the class.

ANSWER 20 walk; 12 do not.

CHECK $\frac{3}{8} \times 32 = 12$.

WATCH OUT The denominator tells you the number of parts, not the size of the class. Always divide the quantity by the denominator first.

PRACTICE 4.1 · What a fraction means

1	Write $\frac{4}{9}$ in words and state the numerator and the denominator.	FLUENCY
2	Convert to improper fractions: (a) $2\frac{1}{3}$ (b) $5\frac{3}{4}$ (c) $1\frac{7}{8}$	FLUENCY
3	Convert to mixed numbers: (a) $\frac{17}{5}$ (b) $\frac{29}{9}$ (c) $\frac{11}{11}$	FLUENCY
4	Find $\frac{2}{3}$ of 45.	FLUENCY
5	Explain why $\frac{5}{4}$ is greater than 1.	REASONING
6	A student says $\frac{1}{8} > \frac{1}{3}$ “because 8 is bigger than 3”. Explain the error.	REASONING

7	A tank holds 240 litres and is $\frac{5}{6}$ full. How many litres are needed to fill it?	PROBLEM SOLVING
8	Sam spends $\frac{3}{7}$ of a 56-page book. How many pages remain?	PROBLEM SOLVING
9	Find a fraction of an hour equal to 50 minutes.	EXTENSION
10	Explain why $\frac{4}{4}$, $\frac{7}{7}$ and $\frac{12}{12}$ are all equal to 1.	EXTENSION

4.2 | Equivalent fractions and simplifying

A · II, III

Equivalent fractions are different names for the same number. Simplifying chooses the simplest name; expanding chooses a convenient one. Both are done by multiplying or dividing top and bottom by the same number.

DEFINITION
Equivalent fractions and simplest form

Fractions are **equivalent** when they represent the same value, such as $\frac{2}{3}$ and $\frac{4}{6}$. Multiplying or dividing the numerator and denominator by the same non-zero whole number produces an equivalent fraction. A fraction is in **simplest form** when the numerator and denominator have no common factor other than 1.

KEY IDEA
Why the value does not change

$\frac{2}{3} = \frac{2 \times 2}{3 \times 2} = \frac{4}{6}$. Multiplying both parts by 2 cuts every third into two sixths: the number of pieces doubles and each piece halves, so the total is unchanged. The same reasoning explains division: $\frac{6}{9} = \frac{6 \div 3}{9 \div 3} = \frac{2}{3}$ groups the pieces in threes.

METHOD
Simplifying in one step

1. Find the highest common factor of the numerator and denominator.
2. Divide both by it.
3. Check that the result is in simplest form by testing for a common factor again.

Using the HCF does the job in one step. Dividing repeatedly by small factors also works and can be safer when the HCF is hard to see.

Fraction	HCF of numerator and denominator	Simplest form	Check
$\frac{18}{24}$	6	$\frac{3}{4}$	$3 \times 6 = 18$, $4 \times 6 = 24$
$\frac{45}{60}$	15	$\frac{3}{4}$	both divide by 15
$\frac{28}{42}$	14	$\frac{2}{3}$	$28 = 14 \times 2$, $42 = 14 \times 3$
$\frac{51}{68}$	17	$\frac{3}{4}$	$51 = 17 \times 3$, $68 = 17 \times 4$

Simplifying with the HCF

COMMON ERROR

Dividing only one part

$\frac{8}{12}$ is not $\frac{2}{12}$: dividing the numerator alone changes the value. Both parts must be divided by the same number. A useful check is to compare the decimal values before and after; they must match.

WORKED EXAMPLE 4.2.1

Simplifying

GIVEN The fraction $\frac{56}{72}$.

ASKED Write it in simplest form.

METHOD Find the HCF of 56 and 72 and divide both parts.

1. $56 = 2^3 \times 7$ and $72 = 2^3 \times 3^2$, so the HCF is $2^3 = 8$.
2. $\frac{56 \div 8}{72 \div 8} = \frac{7}{9}$.

ANSWER $\frac{7}{9}$.

CHECK 7 and 9 share no factor other than 1; $56 \div 72 = 0.777\ldots = 7 \div 9$.

WORKED EXAMPLE 4.2.2

Finding a missing part

GIVEN $\frac{3}{7} = \frac{\square}{42}$.

ASKED Find the missing numerator.

METHOD Ask what the denominator was multiplied by, then do the same to the numerator.

1. $7 \times 6 = 42$, so the multiplier is 6.
2. $3 \times 6 = 18$.

ANSWER 18, so $\frac{3}{7} = \frac{18}{42}$.

CHECK $18 \div 42 = 0.42857$ and $3 \div 7 = 0.42857$.

WORKED EXAMPLE 4.2.3

Simplifying in context

GIVEN A survey found that 48 out of 120 students walk to school.

ASKED Write the proportion as a fraction in simplest form.

METHOD Write the fraction, then simplify with the HCF.

1. Fraction: $\frac{48}{120}$.
2. HCF of 48 and 120 is 24.
3. $\frac{48 \div 24}{120 \div 24} = \frac{2}{5}$.

ANSWER $\frac{2}{5}$ of the students.

CHECK $\frac{2}{5} \times 120 = 48$.

PRACTICE 4.2 · Equivalent fractions and simplifying

1	Simplify: (a) $\frac{6}{9}$ (b) $\frac{10}{25}$ (c) $\frac{21}{28}$	FLUENCY
2	Fill the missing numbers: (a) $\frac{1}{2} = \frac{\square}{10}$ (b) $\frac{2}{5} = \frac{8}{\square}$	FLUENCY
3	Write two fractions equivalent to $\frac{3}{5}$.	FLUENCY
4	Simplify $\frac{36}{48}$.	FLUENCY

5	Explain why dividing the top and bottom by the same number keeps the fraction equal.	REASONING
6	Is $\frac{4}{6}$ in simplest form? Justify your answer.	REASONING
7	A recipe uses 15 g of sugar in 40 g of mixture. Write this as a fraction in simplest form.	PROBLEM SOLVING
8	Write $\frac{48}{60}$ in simplest form and as a percentage.	PROBLEM SOLVING
9	Find the fraction equal to $\frac{5}{8}$ whose numerator is 40.	EXTENSION
10	Two fractions are equivalent and the first is $\frac{7}{9}$. The second has denominator 63. Find its numerator and explain.	EXTENSION

4.3 | Comparing and ordering fractions

C · II, III

Comparing fractions is comparing numbers, and there is more than one reliable method. Choosing the method to suit the numbers is part of the skill.

METHOD

Four methods for comparing fractions

1. **Common denominator:** rewrite both fractions with the LCM of the denominators, then compare numerators.
2. **Common numerator:** if the numerators are equal, the fraction with the smaller denominator is larger.
3. **Benchmark:** compare each fraction with a familiar value such as $\frac{1}{2}$ or 1.
4. **Decimal:** divide to compare as decimals when the numbers are awkward.

KEY IDEA

Choosing a method

Common denominators give an exact answer and show the size of the difference. Decimals are quick but can require rounding, which risks a wrong conclusion when two fractions are very close — $\frac{7}{16}$ and $\frac{11}{25}$ differ by only 0.0025.

Fraction	Decimal	Comparison with $\frac{1}{2}$
$\frac{5}{8}$	0.625	above
$\frac{2}{3}$	0.667	above
$\frac{7}{10}$	0.7	above
$\frac{3}{4}$	0.75	above

Ordering four fractions

All four are above $\frac{1}{2}$, so the benchmark alone cannot order them. With a common denominator of 120: $\frac{75}{120}$, $\frac{80}{120}$, $\frac{84}{120}$, $\frac{90}{120}$, giving $\frac{5}{8} < \frac{2}{3} < \frac{7}{10} < \frac{3}{4}$.

COMMON ERROR

Comparing numerators only

$\frac{3}{8}$ and $\frac{2}{5}$: the numerators 3 and 2 do not decide which is larger, because the denominators differ. With a common denominator of 40, $\frac{15}{40}$ and $\frac{16}{40}$, so $\frac{2}{5}$ is larger — the opposite of what the numerators suggest.

WORKED EXAMPLE 4.3.1

Comparing two fractions

GIVEN $\frac{5}{8}$ and $\frac{7}{12}$.

ASKED State which is larger and by how much.

METHOD Use the LCM of the denominators.

1. LCM of 8 and 12 is 24.
2. $\frac{5}{8} = \frac{15}{24}$ and $\frac{7}{12} = \frac{14}{24}$.
3. The difference is $\frac{1}{24}$.

ANSWER $\frac{5}{8}$ is larger by $\frac{1}{24}$.

CHECK As decimals, $0.625 > 0.583$.

WORKED EXAMPLE 4.3.2

Ordering three fractions

GIVEN $\frac{3}{5}$, $\frac{5}{9}$, $\frac{7}{12}$.

ASKED Write them in ascending order.

METHOD Convert to decimals because the denominators are unrelated.

1. $\frac{3}{5} = 0.6$.
2. $\frac{5}{9} = 0.555\ldots$
3. $\frac{7}{12} = 0.5833\ldots$
4. Ordering: $0.555 < 0.583 < 0.6$.

ANSWER $\frac{5}{9} < \frac{7}{12} < \frac{3}{5}$.

CHECK With a common denominator of 180: $\frac{100}{180} < \frac{105}{180} < \frac{108}{180}$.

WORKED EXAMPLE 4.3.3

Comparing with a mixed number

GIVEN $2\frac{2}{3}$ and $\frac{17}{6}$.

ASKED Decide which is larger.

METHOD Write both as improper fractions over a common denominator.

1. $2\frac{2}{3} = \frac{8}{3} = \frac{16}{6}$.
2. Compare $\frac{16}{6}$ with $\frac{17}{6}$.

ANSWER $\frac{17}{6}$ is larger by $\frac{1}{6}$.

CHECK $2\frac{2}{3} \approx 2.667$ and $\frac{17}{6} \approx 2.833$.

WATCH OUT Comparing only the fractional parts (both thirds or sixths) without the whole numbers gives the wrong order.

PRACTICE 4.3 · Comparing and ordering fractions

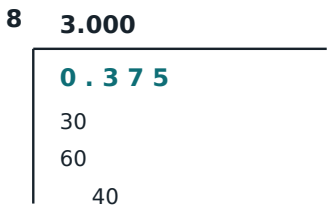
1 Insert < or >: (a) $\frac{2}{5} \square \frac{3}{5}$ (b) $\frac{1}{4} \square \frac{1}{3}$ (c) $\frac{3}{4} \square \frac{2}{3}$ **FLUENCY**

2	Write in ascending order: $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$, $\frac{1}{8}$.	FLUENCY
3	Write $\frac{7}{10}$ and $\frac{2}{3}$ with denominator 30.	FLUENCY
4	Which is larger: $\frac{5}{6}$ or $\frac{6}{7}$?	FLUENCY
5	Explain how to compare two fractions with the same numerator.	REASONING
6	A student compares $\frac{7}{8}$ and $\frac{8}{9}$ by saying "8 > 7 so the second is larger". Is the conclusion right? Is the reasoning sound?	REASONING
7	Three students read $\frac{2}{5}$, $\frac{3}{8}$ and $\frac{1}{3}$ of the same book. Order the amounts read.	PROBLEM SOLVING
8	Which is closer to 1: $\frac{11}{12}$ or $\frac{13}{14}$?	PROBLEM SOLVING
9	Find a fraction between $\frac{2}{5}$ and $\frac{1}{2}$.	EXTENSION
10	Explain why there is always another fraction between any two different fractions.	EXTENSION

4.4 | Fractions and decimals

A · II · C · II

Every fraction can be written as a decimal by division, and every terminating decimal can be written as a fraction over a power of ten. Some fractions produce decimals that never end but repeat.



Converting 3/8 to a decimal by division: the remainder is carried and the division continues past the decimal point, giving 0.375.

DEFINITION

Terminating and recurring decimals

A **terminating decimal** has a finite number of digits after the point, such as 0.375. A **recurring decimal** repeats a block of digits for ever, such as 0.333... where the 3 never stops. This book writes the repeating block twice and ends with an ellipsis: 0.363636... for the block 36. A fraction in simplest form gives a terminating decimal exactly when its denominator has no prime factors other than 2 and 5.

37 hundredths shaded = 0.37

A hundred square with 37 hundredths shaded. The same quantity is written 0.37, 37/100 or 37%.

METHOD

Converting in both directions

Fraction to decimal: divide the numerator by the denominator, continuing past the point until the remainder is 0 or a cycle appears. **Terminating decimal to fraction:** write over a power of ten and simplify. $0.45 = {}^{45}\!/_{100} = {}^{9}\!/_{20}$. **Recurring decimal to fraction:** use the repeating block. If $x = 0.333\ldots$, then $10x = 3.333\ldots$, so $9x = 3$ and $x = {}^{1}\!/_{3}$.

KEY IDEA

Why 2s and 5s matter

${}^{3}\!/_{8} = frac32^{3}$. Multiplying by $frac5^{3}5^{3}$ gives ${}^{375}\!/_{1000} = 0.375$ — a terminating decimal, because 8 divides a power of ten. A denominator with a factor of 3 or 7 never divides a power of ten, so the division must recur.

Fraction	Decimal	Terminating?
${}^{3}\!/_{8}$	0.375	yes, denominator 2^3
${}^{7}\!/_{20}$	0.35	yes, denominator $2^2 \times 5$
${}^{5}\!/_{6}$	0.8333...	no, denominator has a factor 3
${}^{2}\!/_{7}$	0.285714285714...	no, denominator has a factor 7

Fractions, decimals and the reason

COMMON ERROR

Rounding too early

${}^{2}\!/_{3} \approx 0.67$ is fine for reporting, but using 0.67 in a later calculation gives slight errors that accumulate. Keep the fraction, or keep more digits than you need, until the final step.

WORKED EXAMPLE 4.4.1

Fraction to terminating decimal

GIVEN $\frac{7}{20}$.

ASKED Write it as a decimal and explain why it terminates.

METHOD Scale to a denominator of 100, or divide.

1. $\frac{7}{20} = \frac{7 \times 5}{20 \times 5} = \frac{35}{100}$.
2. $\frac{35}{100} = 0.35$.
3. The denominator $20 = 2^2 \times 5$ has only 2s and 5s, so the decimal terminates.

ANSWER 0.35.

CHECK $7 \div 20 = 0.35$.

WORKED EXAMPLE 4.4.2

Fraction to recurring decimal

GIVEN $\frac{5}{6}$.

ASKED Write it as a decimal, using dot notation for any repeating part.

METHOD Divide and watch for the repeating remainder.

1. $5 \div 6 = 0.8$ remainder 2.
2. The remainder 2 recurs, giving 0.8333…
3. The digit 3 repeats, so the decimal is 0.8333….

ANSWER 0.8333….

CHECK Multiplying by 6 gives 4.999…, which equals 5.

WORKED EXAMPLE 4.4.3

Decimal to fraction

GIVEN 2.6 and 0.3636….

ASKED Write each as a fraction in simplest form.

METHOD Use a power of ten for terminating decimals; use the repeating block for recurring ones.

1. $2.6 = \frac{26}{10} = \frac{13}{5} = 2\frac{3}{5}$.
2. For $x = 0.3636\ldots$: $100x = 36.3636\ldots$, so $99x = 36$ and $x = \frac{36}{99} = \frac{4}{11}$.

ANSWER $\frac{13}{5}$ and $\frac{4}{11}$.

CHECK $4 \div 11 = 0.363636\ldots$

WATCH OUT 0.3636… is not $\frac{36}{100}$; the block repeats for ever, so 99, not 100, is the correct denominator in the subtraction.

PRACTICE 4.4 · Fractions and decimals

1	Write as decimals: (a) $\frac{1}{4}$ (b) $\frac{3}{5}$ (c) $\frac{9}{20}$	FLUENCY
2	Write as fractions in simplest form: (a) 0.7 (b) 0.25 (c) 0.6	FLUENCY
3	Which of $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{5}$ give terminating decimals?	FLUENCY
4	Write $\frac{5}{9}$ as a recurring decimal.	FLUENCY
5	Explain why $\frac{1}{3}$ does not terminate but $\frac{1}{4}$ does.	REASONING
6	A student writes 0.3 for $\frac{1}{3}$. What is the error and why does it matter?	REASONING

7	Order 0.65, $\frac{5}{8}$, $\frac{3}{5}$ from smallest to largest.	PROBLEM SOLVING
8	A 1 kg bag is 0.375 full. Write the mass as a fraction in simplest form.	PROBLEM SOLVING
9	Write 0.66... as a fraction in simplest form.	EXTENSION
10	Explain why every terminating decimal can be written as a fraction.	EXTENSION

4.5 | Calculating with decimals

A · I, II · C · I

Decimal calculations follow the same rules as whole-number calculations, with one extra job: keeping track of where the decimal point belongs. Lining up place values, or counting decimal places, or converting to fractions, all do that job.

DEFINITION
Aligning place values

To add or subtract decimals, write the numbers one under the other with the **decimal points in a line**, then fill empty places with zeros. Adding zeros after the last digit does not change a number: 4.5 = 4.50 = 4.500.

METHOD
Four decimal calculations

Add or subtract: line up the points, pad with zeros, then calculate as for whole numbers. 12.4 + 3.75 becomes 12.40 + 3.75 = 16.15. **Multiply by a whole number:** multiply as if there were no point, then place the point so the answer has the same number of decimal places as the number you started with. 0.6 × 7 = 4.2. **Multiply two decimals:** multiply as whole numbers and count the total decimal places in both factors. 0.4 × 0.3 = 0.12, because 4 tenths of 3 tenths is 12 hundredths. **Divide:** make the divisor a whole number by multiplying both numbers by the same power of ten, then divide. 2.4 ÷ 0.6 = 24 ÷ 6 = 4.

KEY IDEA
Multiplying and dividing by 10, 100, 1000

Multiplying by 10 moves every digit one place to the left, so the decimal point appears to move one place to the right: 3.47 × 10 = 34.7. Dividing by 10 moves it one place to the left: 3.47 ÷ 10 = 0.347. The digits never change; only their place values do. Trailing zeros may be dropped, but zeros between digits may not.

Calculation	Quick estimate	Why the estimate is safe
8.9 + 12.35	9 + 12 = 21	the answer must be slightly under 21
0.7 × 0.8	about 1 × 1 = 1 but under	two numbers under 1 give a product under 1
4.5 ÷ 0.9	about 4.5 ÷ 1 = 4.5	dividing by less than 1 makes the answer larger
0.36 ÷ 4	about 0.4 ÷ 4 = 0.1	the answer is near one tenth

Which operation, which estimate

COMMON ERROR

Two decimal-point errors

Misaligned points. Writing 12.4 + 3.75 as 12.4 over 3.75 right-aligned gives 15.65 — wrong, because the 5 of 3.75 then sits in the tenths column. **Counting places wrongly.** 0.2 × 0.3 is not 0.6: two decimal places in total means 0.06.

WORKED EXAMPLE 4.5.1

Adding and subtracting decimals

GIVEN 7.4 - 2.86.

ASKED Calculate, showing the alignment.

METHOD Line up the decimal points and pad with a zero.

1. Write 7.40 above 2.86 with the points in line.
2. 7.40 - 2.86 = 4.54.

ANSWER 4.54.

CHECK 4.54 + 2.86 = 7.40.

WORKED EXAMPLE 4.5.2

Multiplying two decimals

GIVEN 1.6 × 0.25.

ASKED Calculate without a calculator and check by estimation.

METHOD Multiply as whole numbers, then place the point.

1. 16 × 25 = 400.
2. There are 1 + 2 = 3 decimal places in the factors.
3. So 400 with 3 decimal places is 0.400 = 0.4.

ANSWER 0.4.

CHECK A quarter of 1.6 is 0.4, and 0.25 is a quarter.

WORKED EXAMPLE 4.5.3

Dividing by a decimal

GIVEN 3.6 ÷ 0.04.

ASKED Calculate and explain why the answer is larger than 3.6.

METHOD Multiply divisor and dividend by 100 to clear the decimal.

1. 3.6 ÷ 0.04 = 360 ÷ 4.
2. 360 ÷ 4 = 90.
3. Dividing by 0.04 asks how many hundredths fit into 3.6; there are 90 of them.

ANSWER 90.

CHECK 90 × 0.04 = 3.6.

WATCH OUT The answer is larger than the number you started with. Dividing by a number less than 1 always enlarges, and that is a fast check on any decimal division.

PRACTICE 4.5 · Calculating with decimals

1 Calculate: (a) 3.7 + 1.45 (b) 9.2 - 4.65 (c) 0.8 + 12.36 FLUENCY

2 Calculate: (a) 0.7 × 6 (b) 0.4 × 0.5 (c) 0.25 × 8 FLUENCY

3	Calculate: (a) 4.8 ÷ 6 (b) 1.2 ÷ 0.3 (c) 6 ÷ 0.5	FLUENCY
4	Multiply by 10, 100 and 1000: start with 2.405.	FLUENCY
5	A student writes 5.6 + 0.42 = 0.98. What did they do, and what is the correct answer?	REASONING
6	Explain why 0.2 × 0.3 = 0.06 and not 0.6.	REASONING
7	A bottle holds 1.25 litres and another holds 0.85 litres. How much do they hold together, and how much more is in the larger bottle?	PROBLEM SOLVING
8	Petrol costs 1.45 per litre. Find the cost of 12.5 litres.	PROBLEM SOLVING
9	Show that 0.36 ÷ 0.4 and 3.6 ÷ 4 give the same answer, and explain why.	EXTENSION
10	Without calculating, decide whether 0.9 × 0.9 is larger or smaller than 0.9, and justify.	EXTENSION

4.6 | Adding and subtracting fractions

A · I, II

Fractions can only be added when the parts are the same size. Finding a common denominator makes the parts comparable; it does not change the quantities.

DEFINITION

Adding fractions

To add or subtract fractions, write them with a common denominator, then add or subtract the numerators and keep the denominator:

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$$

KEY IDEA

Why not add the denominators?

$\frac{1}{2} + \frac{1}{3}$ is not $\frac{2}{5}$: a half plus a third is more than a half, while $\frac{2}{5}$ is less than a half. With sixths, $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$, which is close to 1 and passes the estimation check.

METHOD

Adding and subtracting mixed numbers

1. Convert to improper fractions, or handle whole numbers and fractions separately.
2. Use the LCM of the denominators.
3. Add or subtract; simplify the result; convert back to a mixed number if needed.
4. If subtracting produces a negative fraction, borrow one whole unit from the whole number.

Stage	Work
Convert	$\frac{13}{4} - \frac{11}{6}$
Common denominator	LCM of 4 and 6 is 12: $\frac{39}{12} - \frac{22}{12}$
Subtract	$\frac{17}{12}$
Simplify	$1\frac{5}{12}$

Worked steps for $3\frac{1}{4} - 1\frac{5}{6}$

COMMON ERROR

Adding numerators and denominators

"$\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$" is the most common fraction error. The check is estimation: the answer must lie between the two fractions. $\frac{2}{5} = 0.4$ lies below both, so it cannot be right.

WORKED EXAMPLE 4.6.1

Adding fractions with unlike denominators

GIVEN $\frac{2}{3} + \frac{1}{4}$.

ASKED Find the sum in simplest form.

METHOD Use the LCM of the denominators.

1. LCM of 3 and 4 is 12.
2. $\frac{2}{3} = \frac{8}{12}$ and $\frac{1}{4} = \frac{3}{12}$.
3. $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$.

ANSWER $\frac{11}{12}$, already in simplest form.

CHECK $0.667 + 0.25 = 0.917$, close to $\frac{11}{12} = 0.917$. The answer is just below 1, as expected.

WORKED EXAMPLE 4.6.2

Subtracting mixed numbers with borrowing

GIVEN $4\frac{1}{5} - 2\frac{2}{3}$.

ASKED Find the difference.

METHOD Convert to improper fractions to avoid negative parts.

1. $4\frac{1}{5} = \frac{21}{5}$ and $2\frac{2}{3} = \frac{8}{3}$.
2. LCM of 5 and 3 is 15: $\frac{63}{15} - \frac{40}{15} = \frac{23}{15}$.
3. $\frac{23}{15} = 1\frac{8}{15}$.

ANSWER $1\frac{8}{15}$.

CHECK $4.2 - 2.667 = 1.533$, and $\frac{23}{15} = 1.533$.

WATCH OUT Subtracting the whole numbers and then the fractions gives $2 - \frac{7}{15}$, which invites the sign error $-\frac{2}{15}$ appearing out of nowhere. Improper fractions avoid it.

WORKED EXAMPLE 4.6.3

A context with three fractions

GIVEN A student spends $\frac{1}{4}$ of an evening on mathematics, $\frac{3}{8}$ on a project and $\frac{1}{6}$ reading.

ASKED What fraction of the evening is left?

METHOD Add the three fractions, then subtract from 1.

1. LCM of 4, 8 and 6 is 24.
2. $\frac{6}{24} + \frac{9}{24} + \frac{4}{24} = \frac{19}{24}$.
3. $1 - \frac{19}{24} = \frac{5}{24}$.

ANSWER $\frac{5}{24}$ of the evening.

CHECK $0.25 + 0.375 + 0.167 = 0.792$ and $\frac{19}{24} = 0.792$.

PRACTICE 4.6 · Adding and subtracting fractions

1 Add: (a) $\frac{1}{5} + \frac{2}{5}$ (b) $\frac{3}{8} + \frac{1}{8}$ (c) $\frac{5}{9} + \frac{2}{9}$ **FLUENCY**

2	Add: (a) $\frac{1}{2} + \frac{1}{4}$ (b) $\frac{1}{3} + \frac{1}{6}$	FLUENCY
3	Subtract: (a) $\frac{5}{6} - \frac{1}{3}$ (b) $\frac{7}{10} - \frac{2}{5}$	FLUENCY
4	Calculate $2\frac{1}{2} + 1\frac{1}{3}$.	FLUENCY
5	Explain why $\frac{1}{2} + \frac{1}{3} \neq \frac{2}{5}$.	REASONING
6	A student adds numerators and denominators and gets $\frac{5}{6}$ for $\frac{1}{2} + \frac{1}{3}$. Is the answer right? Is the method?	REASONING
7	A recipe needs $\frac{2}{3}$ cup of flour and $\frac{3}{4}$ cup of sugar. What is the combined amount?	PROBLEM SOLVING
8	A tank is $\frac{5}{8}$ full. After using $\frac{1}{4}$ of a tank, how full is it?	PROBLEM SOLVING
9	Calculate $\frac{5}{6} + \frac{7}{8} - \frac{1}{2}$ in simplest form.	EXTENSION
10	Explain how to subtract $1\frac{5}{6}$ from $3\frac{1}{4}$ without getting a negative fraction.	EXTENSION

4.7 | Multiplying and dividing fractions

C · I, III

Multiplication of fractions answers "a part of a part", and division answers "how many of these fit inside that". Both are easier than addition, because no common denominator is needed.

DEFINITION

Multiplying and dividing fractions

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

To divide by a fraction, multiply by its **reciprocal** — the fraction turned upside down.

KEY IDEA

Two pictures that explain the rules

Multiplication: $\frac{2}{3} \times \frac{3}{4}$ means two thirds of three quarters. Take a square, cut it into quarters and shade three, then cut the square into thirds and take two of them. The overlap is 6 of 12 small squares, or $\frac{1}{2}$. **Division:** $\frac{3}{4} \div \frac{1}{8}$ asks how many eighths fit in three quarters. Each quarter contains two eighths, so there are 6.

Expression	Meaning	Answer
$\frac{2}{3} \times \frac{3}{4}$	two thirds of three quarters	$\frac{1}{2}$
$\frac{3}{4} \div \frac{1}{8}$	how many eighths in three quarters	6
$\frac{3}{4} \div \frac{2}{5}$	how many two fifths fit in three quarters	$\frac{15}{8} = 1\frac{7}{8}$

Multiplication and division compared

COMMON ERROR

Inverting the wrong fraction

When dividing, the **second** fraction is inverted: $\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2}$. Inverting the first fraction gives $\frac{4}{3} \times \frac{2}{5} = \frac{8}{15}$, which is less than 1 — an impossible answer to "how many $\frac{2}{5}$ fit in $\frac{3}{4}$".

WORKED EXAMPLE 4.7.1

Multiplying fractions

GIVEN $\frac{5}{6} \times \frac{3}{10}$.

ASKED Multiply and simplify.

METHOD Multiply numerators and denominators, then simplify; or cancel common factors first.

1. Cancel: $\frac{5}{6} \times \frac{3}{10} = \frac{1 \times 1}{2 \times 2}$ after dividing 5 and 10 by 5, and 3 and 6 by 3.
2. Result: $\frac{1}{4}$.

ANSWER $\frac{1}{4}$.

CHECK $0.833 \times 0.3 = 0.25$.

WORKED EXAMPLE 4.7.2

Dividing by a fraction

GIVEN $2\frac{1}{2} \div 1\frac{1}{4}$.

ASKED Find the quotient and interpret the answer.

METHOD Convert to improper fractions, then multiply by the reciprocal.

1. $2\frac{1}{2} = \frac{5}{2}$ and $1\frac{1}{4} = \frac{5}{4}$.
2. $\frac{5}{2} \times \frac{4}{5} = \frac{20}{10} = 2$.

ANSWER 2, meaning $1\frac{1}{4}$ fits exactly twice into $2\frac{1}{2}$.

CHECK $1\frac{1}{4} \times 2 = 2\frac{1}{2}$.

WORKED EXAMPLE 4.7.3

A fraction of a fraction in context

GIVEN A school has 480 students. Of these, $\frac{3}{8}$ are in the first year, and $\frac{2}{5}$ of the first year walk to school.

ASKED How many first-year students walk to school?

METHOD Multiply the fractions to find the overall part, then apply it to the total.

1. $\frac{3}{8} \times 480 = 180$ first-year students.
2. $\frac{2}{5} \times 180 = 72$.
3. As a single calculation: $\frac{3}{8} \times \frac{2}{5} = \frac{3}{20}$, and $\frac{3}{20} \times 480 = 72$.

ANSWER 72 students.

CHECK $\frac{3}{20}$ of 480 is $480 \div 20 \times 3 = 72$.

PRACTICE 4.7 · Multiplying and dividing fractions

1	Multiply: (a) $\frac{2}{3} \times \frac{3}{5}$ (b) $\frac{1}{2} \times \frac{4}{9}$	FLUENCY
2	Divide: (a) $\frac{1}{2} \div \frac{1}{4}$ (b) $\frac{3}{4} \div \frac{3}{8}$	FLUENCY
3	Find $\frac{2}{3}$ of $\frac{9}{10}$.	FLUENCY
4	Evaluate $1\frac{1}{2} \times \frac{2}{3}$.	FLUENCY

5	Explain why dividing by $\frac{1}{4}$ is the same as multiplying by 4.	REASONING
6	A student says “multiplying always makes a number bigger”. Provide a counterexample using fractions.	REASONING
7	A tank holds $2\frac{1}{3}$ litres. Bottles hold $\frac{1}{6}$ litre. How many bottles can be filled?	PROBLEM SOLVING
8	Three fifths of a class of 35 are girls; two thirds of the girls play a sport. How many girls play a sport?	PROBLEM SOLVING
9	Calculate $\frac{3}{4} \div \frac{9}{16}$ and explain the result without inverting.	EXTENSION
10	Find a fraction that gives $\frac{5}{6}$ when multiplied by $\frac{2}{3}$.	EXTENSION

F INVESTIGATE AND APPLY

A criterion B investigation and a criterion D application

INVESTIGATION 4 | Which product is larger?

B · I, II, III

$\frac{3}{4}$ of $\frac{2}{3}$ and $\frac{2}{3}$ of $\frac{3}{4}$ look like different calculations. Investigate whether they are.

INVESTIGATION
Investigation brief

Question. Does the order of the fractions in a “fraction of a fraction” calculation change the result? **What you need.** Squared paper, two colours, a ruler. **What to produce.** Two area diagrams, a table of results for four pairs of fractions, a general statement and a justification.

PART 1 · DRAW

1. Draw a square 12 cm by 12 cm. Shade $\frac{2}{3}$ of it with horizontal strips.
2. Shade $\frac{3}{4}$ of the same square with vertical strips in a second colour.
3. Count the squares covered by both colours. Express that count as a fraction of the whole square.
4. Repeat with the roles reversed. What do you notice?

PART 2 · RECORD AND CONJECTURE

First fraction	Second fraction	Product	Decimal value
$\frac{2}{3}$	$\frac{3}{4}$		
$\frac{3}{5}$	$\frac{1}{2}$		
$\frac{5}{8}$	$\frac{4}{5}$		
$\frac{1}{4}$	$\frac{7}{9}$		

Copy and complete

PART 3 · JUSTIFY AND TEST

1. Write your conjecture about the order of the fractions.
2. Explain, using the area diagram, why the order cannot change the number of small squares in the overlap.
3. Test the conjecture with one pair of mixed numbers.

4. Would the same be true for division? Test with $\frac{3}{4} \div \frac{1}{2}$ and $\frac{1}{2} \div \frac{3}{4}$.

CRITERION TASK
Criterion B task · Requirements

Accurate diagrams; a table with correct values; a conjecture stated generally; a justification that refers to the structure of the calculation; a test that could have refuted the conjecture.

APPLICATION 4 | Scaling the canteen recipe

D · I, II, III, IV

A canteen recipe serves 24 students. The school now has 90 students at lunch and must scale the recipe without buying wastefully.

THE BRIEF

Ingredient	Amount	Pack size sold	Price per pack
Rice	$1\frac{1}{2}$ kg	2 kg	3.20
Beans	$\frac{3}{4}$ kg	1 kg	2.10
Oil	$\frac{1}{3}$ litre	1 litre	4.50
Tomato	$\frac{5}{8}$ kg	$\frac{1}{2}$ kg	1.40

The recipe for 24 servings

- The recipe must be scaled to 90 servings.
- Packs are sold only in the sizes shown, so some ingredients will be left over.
- The budget for these four ingredients is 60 units of currency.

TASK 1 · SCALE THE RECIPE

1. Find the scale factor from 24 to 90 servings as a fraction in simplest form.
2. Calculate the scaled amount of each ingredient.
3. Convert each amount to a mixed number or decimal where that makes it easier to read.

TASK 2 · COST THE SHOPPING

1. Decide how many packs of each ingredient must be bought to cover the scaled amount.
2. Calculate the total cost.
3. Compare the total with the budget and state whether it fits.

TASK 3 · EVALUATE AND RECOMMEND

1. Which ingredient produces the most waste, as a fraction of what is bought?
2. Suggest one change to the plan that reduces waste without changing the menu.
3. Write three sentences justifying your recommendation, using your calculations as evidence.

CRITERION TASK
Criterion D task · Requirements

The scale factor identified and applied correctly; all four amounts calculated; pack counts justified with reference to the quantities; cost compared with the budget; a recommendation supported by evidence and a limitation stated.

TASK C | Explaining the rule for dividing

C · I, II, III, IV

CRITERION TASK

The task

Write a two-page explanation, for a student who believes that “dividing always makes numbers smaller”, showing why dividing by a fraction can make a number larger.

1. A numerical example with a diagram: $\frac{3}{4} \div \frac{1}{8} = 6$.
2. A statement of the rule using the word reciprocal.
3. An explanation of why the rule works, using the idea of “how many fit”.
4. A check that the same rule gives sensible answers when the divisor is a whole number, such as $6 \div 2$.

Criterion C strand	Evidence
Notation (i)	Correct fraction notation, including mixed numbers.
Representation (ii)	A diagram that shows the division as fitting parts into a whole.
Reasoning (iii)	The explanation shows why inverting works, not only how.
Organisation (iv)	The argument moves from example to rule to justification.

Success criteria

G REFLECT AND CONSOLIDATE

Summary, vocabulary, misconceptions and review

REVIEW | Consolidating Chapter 4

A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- A fraction is one number with many names; equivalent fractions come from multiplying or dividing both parts by the same number.
- Simplest form uses the HCF; expanding is useful for comparison and for addition.
- Fractions can be compared by common denominators, common numerators, benchmarks or decimals.
- Fractions convert to terminating decimals when the denominator has only factors 2 and 5, and recur otherwise.
- Addition and subtraction need a common denominator; multiplication and division do not.
- Dividing by a fraction is multiplying by its reciprocal.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Numerator	The top number of a fraction: how many parts are taken.

Term	Meaning in this course
Denominator	The bottom number: how many equal parts the whole is divided into.
Proper fraction	A fraction less than 1, with numerator smaller than denominator.
Improper fraction	A fraction at least 1, with numerator not smaller than denominator.
Mixed number	A whole number and a proper fraction written together.
Equivalent fractions	Fractions with the same value.
Simplest form	No common factor other than 1 in numerator and denominator.
Terminating decimal	A decimal with a finite number of digits.
Recurring decimal	A decimal whose digits repeat for ever.
Reciprocal	The fraction turned upside down.

Vocabulary for Chapter 4

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence
“A bigger denominator means a bigger fraction.”	The parts are smaller.	$\frac{1}{8} < \frac{1}{3}$
“Adding means adding tops and bottoms.”	Parts must be the same size first.	$\frac{1}{2}+\frac{1}{3}=\frac{5}{6}$
“Multiplying always enlarges.”	Multiplying by a fraction below 1 shrinks.	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
“0.3 equals $\frac{1}{3}$.”	$0.3 = \frac{3}{10}$.	$\frac{1}{3} = 0.333\ldots$
“Division makes smaller.”	Dividing by a fraction below 1 enlarges.	$\frac{3}{4} \div \frac{1}{8} = 6$

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
convert between mixed numbers and improper fractions.			
simplify a fraction using the HCF.			
compare and order fractions using a method I can justify.			
convert between fractions and decimals, including recurring decimals.			
add and subtract fractions and mixed numbers.			
multiply and divide fractions and explain why the rules work.			

How confident are you?

REFLECTION

1. Which comparison method do you now prefer, and for which kinds of numbers?
2. Describe a mistake you made with fractions this chapter and what you changed.
3. Write one sentence explaining why $^{2}/_{4}$ and $^{1}/_{2}$ are the same number.
4. In the canteen task, which was harder: scaling the amounts or buying the packs? Why?
5. Which notation — fraction, decimal or percentage — do you find clearest, and does it depend on the context?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ4 · Applying the chapter

1	Simplify $^{24}/_{36}$.	FLUENCY
2	Convert $3\,^{4}/_{5}$ to an improper fraction.	FLUENCY
3	Write $^{9}/_{20}$ as a decimal.	FLUENCY
4	Find $^{3}/_{4}$ of 60.	FLUENCY
5	Calculate $^{2}/_{5} + {}^{1}/_{2}$.	FLUENCY
6	Calculate $^{3}/_{4} \times {}^{8}/_{9}$.	FLUENCY
7	Explain why $^{6}/_{8}$ and $^{9}/_{12}$ are equivalent.	REASONING
8	Which is larger, $^{4}/_{7}$ or $^{5}/_{9}$? Justify.	REASONING
9	Explain the difference between 0.333… and 0.3.	REASONING
10	A 3 litre bottle is $^{2}/_{5}$ full. How much is in it?	PROBLEM SOLVING
11	A worker completes $^{5}/_{12}$ of a job on Monday and $^{1}/_{3}$ on Tuesday. What fraction remains?	PROBLEM SOLVING
12	A ribbon $2\,^{1}/_{4}$ m long is cut into pieces $^{3}/_{8}$ m long. How many pieces?	PROBLEM SOLVING
13	Explain how the area model explains $^{2}/_{3} \times {}^{3}/_{4} = {}^{1}/_{2}$.	COMMUNICATING
14	Explain why the product of two proper fractions is always smaller than either fraction.	COMMUNICATING
15	Find two fractions between $^{1}/_{3}$ and $^{1}/_{2}$.	EXTENSION
16	Write 0.2727… as a fraction in simplest form.	EXTENSION
17	Explain why $^{3}/_{4} \div {}^{3}/_{4} = 1$ using the “how many fit” meaning.	EXTENSION
18	A recipe for 8 people uses $^{2}/_{3}$ cup of oil. How much for 20 people?	PROBLEM SOLVING
19	Two fractions have a product of $^{1}/_{2}$ and one of them is $^{3}/_{4}$. Find the other and explain your method.	COMMUNICATING
20	Explain why $^{1}/_{2} + {}^{1}/_{3}$ must be greater than $^{1}/_{2}$ and less than 1, without calculating the exact answer.	COMMUNICATING

MIXED 4 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX4 · End-of-chapter mixed set

1	Order $\frac{2}{3}$, $\frac{3}{5}$, $\frac{7}{10}$ from smallest to largest.	FLUENCY
2	Calculate $\frac{7}{8} - \frac{1}{4}$.	FLUENCY
3	Calculate $\frac{5}{6} \div \frac{10}{9}$.	FLUENCY
4	Write 0.85 as a fraction in simplest form.	FLUENCY
5	Find $\frac{2}{7}$ of 42.	FLUENCY
6	Is $\frac{4}{5}$ closer to $\frac{3}{4}$ or to 1? Explain.	REASONING
7	A student writes $\frac{2}{3} \times \frac{3}{4} = \frac{6}{7}$. What has gone wrong and what check would catch it?	REASONING
8	A 2.4 kg bag is $\frac{7}{8}$ full. What mass of contents is in it?	PROBLEM SOLVING
9	Three quarters of a class of 36 chose football; of those, $\frac{2}{9}$ chose goalkeeper. How many chose goalkeeper?	PROBLEM SOLVING
10	A tap fills $\frac{2}{9}$ of a tank per hour. How long to fill $\frac{2}{3}$ of the tank?	PROBLEM SOLVING
11	Explain why dividing by $\frac{1}{2}$ doubles a number.	COMMUNICATING
12	Show that $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$ is less than 1, and say by how much.	COMMUNICATING
13	Find a fraction greater than $\frac{5}{8}$ and less than $\frac{2}{3}$.	EXTENSION
14	A fraction equals $\frac{3}{7}$ and its denominator exceeds its numerator by 24. Find the fraction.	EXTENSION
15	Explain why $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} = \frac{1}{5}$ without multiplying all the numerators.	EXTENSION
16	A recipe uses $\frac{3}{4}$ kg of flour for 12 cakes. How much for 30 cakes, and how many 2 kg bags are needed?	PROBLEM SOLVING

CUMULATIVE PRACTICE 4 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP4 · Cumulative practice

1	Order $\frac{2}{3}$, $\frac{3}{5}$ and $\frac{7}{10}$.	FLUENCY
2	Calculate $\frac{5}{8} + \frac{1}{4}$.	FLUENCY
3	Calculate $\frac{2}{3} \div \frac{4}{9}$.	FLUENCY
4	Explain why $\frac{3}{4} \times \frac{4}{3} = 1$.	REASONING
5	A 2 kg bag is $\frac{3}{8}$ full. Find the mass inside.	PROBLEM SOLVING
6	Three quarters of 48 students walk; one third of those cycle part of the way. How many cycle part of the way?	PROBLEM SOLVING
7	Explain why $\frac{1}{12}$ is a recurring decimal.	COMMUNICATING

8 Explain why dividing by $\frac{1}{5}$ multiplies a number by 5. **EXTENSION**

PART II · PARTS AND PROPORTIONS

CHAPTER 05

Percentages and applications

A percentage compares a quantity with 100. Once a price, a budget or a change is written on that scale, very different quantities can be judged side by side — provided we ask what the percentage is of.

KEY CONCEPT	**Change**
RELATED	**Quantity**
RELATED	**Proportion**
GLOBAL CONTEXT	**Fairness and development**

05
CHAPTER

INQUIRY QUESTION

When a shop advertises “20% off”, what exactly is being reduced, and how can a buyer tell whether the offer is genuinely good?

A 30% discount on a 20-unit item saves 6 units; the same 30% on a 200-unit item saves 60. The percentage is identical, but the effect on a household is not. Reading the base is part of reading the offer honestly.

IN THIS CHAPTER

5.1 What a percentage is
5.2 Finding a percentage of a quantity
5.3 One quantity as a percentage of another
5.4 Percentage increase and decrease
5.5 Discounts, taxes and budgets
Investigation 5 Do two discounts add up?
Application 5 Choosing a phone plan
Task C Interrogating an advertisement
Review Consolidating Chapter 5
Review questions Practice across the whole chapter
Mixed 5 Fluency, reasoning and problem solving
Cumulative practice 5 Questions that mix this chapter with earlier ones

APPROXIMATELY 14 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

The word itself is the definition: per centum, by the hundred. A percentage is a fraction with a denominator of 100. That single idea connects a discount, a tax rate, a survey result and a test score — and it is the idea that makes it possible to compare quantities of very different sizes.

KEY CONCEPT · CHANGE

Change. A percentage describes a change or a part relative to a base; the same absolute change is a different percentage when the base changes.

RELATED CONCEPT · QUANTITY

Quantity. A percentage is not an amount until it is applied to something. 20% of 50 and 20% of 80 are the same proportion but not the same money.

RELATED CONCEPT · PROPORTION

Proportion. Percentages compare a part with a whole scaled to 100, linking fractions, decimals and ratio.

GLOBAL CONTEXT

Fairness and development

EXPLORATION

Financial decisions and access: how percentage claims in prices, taxes and budgets shape what people can afford, and whether an offer is fair.

FACTUAL

What does per cent mean? How do you convert between percentages, fractions and decimals? How do you find a percentage of a quantity?

CONCEPTUAL

How can the same percentage produce different amounts, and why do successive changes not add?

DEBATABLE

Is a larger percentage discount always the better deal? Should a budget be judged by its percentages or by its amounts?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Convert between forms, calculate a percentage of a quantity, increase and decrease amounts, calculate discounts and taxes.
B · Investigating patterns	i, ii	Investigate successive percentage changes, record multipliers and generalise an equivalent single change.
C · Communicating	i, ii, iii	Explain the choice of base and method, represent percentages with grids and bars, justify a financial claim.
D · Applying mathematics	i, ii, iii, iv	Price a stall, compare offers, build a budget, evaluate rounding and recommend a course of action.

ATL SKILL · CRITICAL THINKING

In order for a student to compare “20% off, then an extra 10% at the till” with “30% off”, students must decide on a method before calculating, test it on two different prices and explain why the two claims are not the same.

STATEMENT OF INQUIRY

Percentages express change and comparison on a common scale of 100, and interpreting the base and the multiplier determines whether a financial claim is understood or misleading.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand per cent as a proportion of 100 and convert between percentages, fractions and decimals.
- Find a percentage of a quantity using benchmarks and multipliers.
- Increase and decrease quantities by a percentage and explain the multiplier.
- Use percentages in discounts, sale prices, taxes and budgets.
- Compare offers and justify a financial decision with evidence.
- Investigate what happens when two percentage changes are applied one after the other.

SUCCESS CRITERIA

- ✓ I can convert between percentages, fractions and decimals.
- ✓ I can find a percentage of a quantity and choose an efficient method.
- ✓ I can increase or decrease a quantity by a percentage using a multiplier.
- ✓ I can calculate a discount, a sale price and a price including tax.
- ✓ I can build and interpret a simple budget using percentages.
- ✓ I can compare two offers and explain which is better and why.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Write $^{3}/_{4}$ as a decimal.
2. Find $^{1}/_{10}$ of 80.
3. Calculate 0.2×150.
4. Write $^{45}/_{100}$ in simplest form.
5. Increase 50 by 10.
6. Which is larger, 0.6 or 0.06?

Answers are in the Chapter 5 answer section. If more than two are incorrect, review fraction–decimal conversion and finding a fraction of an amount before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Fractions and decimals; finding a fraction of an amount; simple money calculations and comparisons.	Percent as per hundred, conversions, percentage of a quantity, increase and decrease, discounts, tax and budgets.	Ratio and proportion, repeated and compound change, interest, and percentages in statistics.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Offers, taxes and household budgets are written as percentages of something.	What is the percentage of, and does the claim tell the whole story?	Change, quantity and proportion.	Do two successive discounts add up to one larger discount?	Hundred grids, bar models, multipliers and clear financial statements.	When is a percentage the honest way to describe a change?

A ENGAGE
Seventy per cent off — off what?

A shop window carries three signs: “30% off”, “a further 20% off the reduced price”, and “save 50%”. A jacket marked at 200 units of currency is in the window. A customer reads all three signs and expects to pay 0.

That cannot be right, and the arithmetic explains why. Each percentage is calculated on a different amount: the first on the marked price, the second on what remains. Percentages are comparisons, and a percentage is meaningless until the amount it compares with is identified.

Advertisers know this. “50% more free” sounds like double, but if the original pack is 400 g and the new pack is 600 g, the increase is 50% of 400 g — a half as much again, not double. The wording is not false; it is a comparison with an unstated base.

Statement	Base of the comparison	Effect on a 200-unit price
30% off	the marked price	pay 140
20% off, then a further 20%	the price after the first reduction	pay 128
“Half price”	the marked price	pay 100

Figure 5.1 The same discount, described three ways

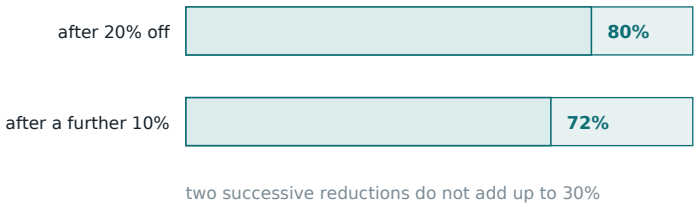
NOTICE AND WONDER

- Why do the first two statements give different answers if both mention 20%?
- What base is being used when a news report says “crime rose by 200%”?
- Which is more useful to a customer: the percentage saved or the final price?
- Can a percentage be greater than 100? What would that mean?

Keep this question open: when does a percentage clarify a comparison, and when does it hide the base that makes it meaningful?

B EXPLORE
Do two discounts add up?

Work in groups of three. You need a calculator, a sheet of marked prices and three coloured pens.



two successive reductions do not add up to 30%

A price reduced by 20% and then by a further 10% ends at 72% of the original price, not at 70%.

YOUR INVESTIGATION

Step 1. Choose five marked prices between 80 and 400 units. Record them.

Step 2. For each price, calculate the result of a 20% discount followed by a further 10% discount on the reduced price.

Step 3. For each price, calculate the result of a single 30% discount.

Step 4. Compare the two results and record the difference.

Marked price	After 20% off	After a further 10%	Single 30% off	Difference
100				
150				
200				
250				
400				

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

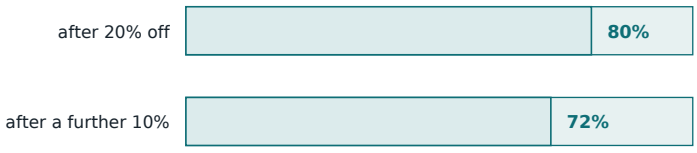
1. Is the difference between the two methods the same for every marked price, or does it grow?
2. What percentage of the marked price does the double discount actually remove?
3. Why is the second discount smaller in money terms than the first, even though both are described as percentages?
4. Would the order of the two discounts matter: 10% then 20%?
5. How would you explain the result to a customer in one sentence?

MAKING A CONJECTURE

> "Two successive discounts of a% and b% are equivalent to a single discount of … because … ."

WHAT YOU SHOULD HAVE NOTICED

- The double discount leaves 72% of the price, so it is equivalent to a single 28% discount, not 30%.
- The second discount applies to a smaller amount, so it removes less money.
- The order does not matter: $0.8 \times 0.9 = 0.9 \times 0.8 = 0.72$.
- Multiplying the multipliers is the reliable method: $0.8 \times 0.9 = 0.72$.



two successive reductions do not add up to 30%

A price reduced by 20% and then by a further 10% ends at 72% of the original price, not at 70%.

INVESTIGATION

Explore further

Investigate what single percentage increase is equivalent to a 10% increase followed by a 10% increase. Then test whether "a 10% rise and a 10% fall" returns a price to where it started.

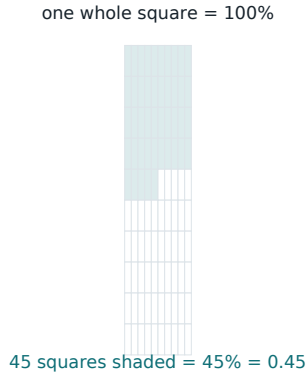
5.1 | What a percentage is

A · I, II

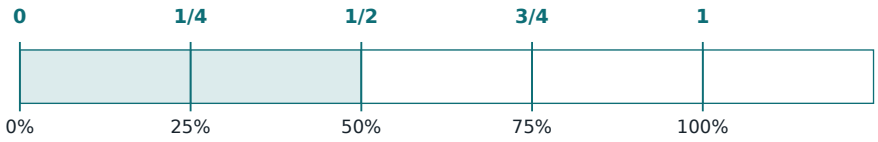
A percentage is a fraction with denominator 100. That single definition explains every conversion, every comparison and every calculation in the chapter.

DEFINITION
Percentage

A **percentage** is a fraction whose denominator is 100. The symbol % means “out of 100”, so $45\% = {}^{45}\!/_{100} = 0.45$. To convert a fraction or decimal to a percentage, multiply by 100. To convert a percentage to a fraction or decimal, divide by 100.



45 of 100 squares shaded. A percentage is a fraction with denominator 100, so this is 45%, or 0.45, or 45/100.



percentages, fractions and decimals name the same positions

The same positions on a number line written as percentages, fractions and decimals.

KEY IDEA
Benchmarks worth knowing by heart

- $1\% = 0.01$; $10\% = 0.1$; $50\% = {}^{1}\!/_{2}$; $25\% = {}^{1}\!/_{4}$; $75\% = {}^{3}\!/_{4}$.
- $20\% = {}^{1}\!/_{5}$; $12.5\% = {}^{1}\!/_{8}$; $33{}^{1}\!/_{3}\% = {}^{1}\!/_{3}$; $66{}^{2}\!/_{3}\% = {}^{2}\!/_{3}$.
- 100% is the whole; 200% is double; 150% is one and a half times.

METHOD
Converting between the three forms

1. Percentage to decimal: divide by 100 (move the point two places left). $37\% = 0.37$.
2. Decimal to percentage: multiply by 100 (move the point two places right). $0.6 = 60\%$.
3. Fraction to percentage: multiply by 100, or scale to a denominator of 100. ${}^{3}\!/_{8} = 37.5\%$.
4. Percentage to fraction: write over 100 and simplify. $35\% = {}^{35}\!/_{100} = {}^{7}\!/_{20}$.

COMMON ERROR

Moving the wrong way

Multiplying by 100 when converting a percentage to a decimal turns 45% into 4500%, an error of two factors of ten. A useful check: a percentage below 100% must become a decimal below 1.

WORKED EXAMPLE 5.1.1

Converting in both directions

GIVEN $\frac{7}{20}$, 0.08 and 62.5%.

ASKED Write each as a percentage, and each percentage as a fraction in simplest form.

METHOD Use the definition of percentage as a fraction over 100.

1. $\frac{7}{20} = \frac{35}{100} = 35\%$.
2. $0.08 \times 100 = 8\%$.
3. $62.5\% = \frac{62.5}{100} = \frac{625}{1000} = \frac{5}{8}$.

ANSWER 35%, 8%, $62.5\% = \frac{5}{8}$.

CHECK $5 \div 8 = 0.625$, which is 62.5%.

WORKED EXAMPLE 5.1.2

Reading a percentage in context

GIVEN A school reports that 92% of its 250 students use the library.

ASKED State the number of students who use the library and the number who do not, then write each as a fraction.

METHOD Apply the percentage to the total, using the complement for the remainder.

1. 92% of 250 = $0.92 \times 250 = 230$.
2. The remainder is 250 - 230 = 20, which is 8%.
3. As fractions: $\frac{230}{250} = \frac{23}{25}$ use it; $\frac{20}{250} = \frac{2}{25}$ do not.

ANSWER 230 use the library; 20 do not.

CHECK $\frac{23}{25} + \frac{2}{25} = 1$, and 230 + 20 = 250.

WORKED EXAMPLE 5.1.3

A percentage above 100

GIVEN A club had 40 members last year and 60 this year.

ASKED Express this year's membership as a percentage of last year's, and describe the change.

METHOD Divide the new value by the base and multiply by 100.

1. $\frac{60}{40} = 1.5$.
2. $1.5 \times 100 = 150\%$.

ANSWER The membership is 150% of last year's, an increase of 50%.

CHECK A 50% increase on 40 is $40 \times 1.5 = 60$.

PRACTICE 5.1 · What a percentage is

1	Write as percentages: (a) 0.35 (b) 0.7 (c) 1.25	FLUENCY
2	Write as decimals: (a) 6% (b) 45% (c) 120%	FLUENCY

3	Write as fractions in simplest form: (a) 20% (b) 75% (c) 8%	FLUENCY
4	Write $^{3}/_{5}$ as a percentage.	FLUENCY
5	Explain why 0.5% is not the same as 50%.	REASONING
6	A student says “a percentage larger than 100 is impossible”. Give a counterexample and explain.	REASONING
7	A jacket costs 80 units. A sign says “pay 120% of the marked price”. What is the new price and what does the sign mean?	PROBLEM SOLVING
8	45% of a class of 40 walk to school. How many is that?	PROBLEM SOLVING
9	A quantity increases from 25 to 40. By what percentage did it increase?	EXTENSION
10	Write $33\,^{1}/_{3}$% as a fraction and explain why it is not exactly $^{1}/_{3}$ in decimal form.	EXTENSION

5.2 | Finding a percentage of a quantity

A · I · C · I, III

Finding a percentage of a quantity is a multiplication. Three methods give the same answer; the multiplier method is fastest and the most useful later.

METHOD

Three methods, one answer: 15% of 240

1. **Fraction method:** $^{15}/_{100} \times 240 = {}^{3600}/_{100} = 36$.
2. **Ten-percent method:** 10% of 240 is 24; 5% is half of that, 12; so 15% is 24 + 12 = 36.
3. **Multiplier method:** 15% = 0.15, and 0.15 × 240 = 36.

KEY IDEA

The multiplier is the idea worth keeping

A percentage of an amount is the amount multiplied by the decimal form of the percentage. This is why 15% of 240 and 240 × 0.15 are the same calculation, and why calculators can handle any percentage in one step.

Percentage	Multiplier	Calculation	Result
1%	0.01	240 × 0.01	2.4
5%	0.05	240 × 0.05	12
10%	0.10	240 × 0.10	24
15%	0.15	240 × 0.15	36
25%	0.25	240 × 0.25	60

Percentages of 240 by multiplier

COMMON ERROR

Dividing by the percentage

“15% of 240” is not 240 ÷ 15. Dividing by 15 finds how many groups of 15 fit into 240, which is a different question. Estimation catches it: 10% of 240 is about 24, so 15% must be a little over 24, not 16.

WORKED EXAMPLE 5.2.1

Percentage of a quantity

GIVEN 12.5% of 96.

ASKED Calculate the value using a multiplier.

METHOD Convert the percentage to a decimal and multiply.

1. 12.5% = 0.125.
2. 0.125 × 96 = 12.

ANSWER 12.

CHECK 12.5% = $^{1}/_{8}$, and 96 ÷ 8 = 12.

WORKED EXAMPLE 5.2.2

Choosing a method

GIVEN A restaurant bill of 84 units; a service charge of 17.5% is added.

ASKED Find the total bill.

METHOD Find the charge, then add it; or use a single multiplier of 1.175.

1. Charge: 0.175 × 84 = 14.7.
2. Total: 84 + 14.7 = 98.7.
3. Single step: 84 × 1.175 = 98.7.

ANSWER 98.7 units.

CHECK 17.5% is a little under a fifth, and a fifth of 84 is 16.8, so a charge near 15 is reasonable.

WORKED EXAMPLE 5.2.3

Percentage of a percentage problem

GIVEN In a school of 600 students, 30% are in the first year. Of the first year, 45% take the bus.

ASKED How many first-year students take the bus?

METHOD Apply the percentages one at a time, or combine the multipliers.

1. First year: 0.3 × 600 = 180.
2. Bus: 0.45 × 180 = 81.
3. Combined: 0.3 × 0.45 = 0.135, and 0.135 × 600 = 81.

ANSWER 81 students.

CHECK 13.5% of 600 is 600 × 0.1 + 600 × 0.035 = 60 + 21 = 81.

WATCH OUT Adding the percentages gives 75%, which is far too large. A percentage of a percentage multiplies.

PRACTICE 5.2 · Finding a percentage of a quantity

1	Find 10% of (a) 80 (b) 250 (c) 64	FLUENCY
2	Find 25% of (a) 60 (b) 44	FLUENCY
3	Find 15% of 40.	FLUENCY
4	Find 12.5% of 200.	FLUENCY
5	Explain why finding 5% is the same as finding 10% and halving.	REASONING

6	A student calculates 20% of 50 as 50 ÷ 20 = 2.5. What has been calculated, and what is the correct answer?	REASONING
7	A phone costs 320 units and a case costs 18.5% of that. What is the total?	PROBLEM SOLVING
8	In a survey of 480 people, 35% chose option A and 22.5% chose option B. How many chose neither?	PROBLEM SOLVING
9	Find 7.5% of 240 without a calculator, showing your method.	EXTENSION
10	A jacket is reduced by 30% to 63 units. What was the original price?	EXTENSION

5.3 | One quantity as a percentage of another

C · II, III

"What percentage of the class are girls?" is a different question from "how many girls are in the class?". The first is a comparison, and the second is a quantity.

DEFINITION

Expressing a ratio as a percentage

To express a as a percentage of b: divide a by b, then multiply by 100.

$$\text{percentage} = {}^{a}\!/_{b} \times 100\%$$

The amount b is the **base** of the comparison. Changing the base changes the percentage, even when the quantities stay the same.

KEY IDEA

The base always matters

20 out of 25 is ${}^{20}\!/_{25} = 80\%$, but 20 out of 50 is ${}^{20}\!/_{50} = 40\%$. The numerator is the same; the base has doubled, so the percentage has halved. Every percentage statement should be read as "this, compared with that".

Part	Base	Fraction	Percentage
18	24	${}^{3}\!/_{4}$	75%
18	36	${}^{1}\!/_{2}$	50%
18	45	${}^{2}\!/_{5}$	40%
18	90	${}^{1}\!/_{5}$	20%

The same part, different bases

COMMON ERROR

Dividing the larger by the smaller

To express 12 as a percentage of 30, the base is 30, so ${}^{12}\!/_{30} = 40\%$. Writing ${}^{30}\!/_{12} = 250\%$ answers a different question — what percentage 30 is of 12 — and the wording is what decides. Look for "as a percentage of".

WORKED EXAMPLE 5.3.1

Expressing a part as a percentage

GIVEN A football team wins 17 of its 25 matches.

ASKED Express the wins as a percentage of the matches played.

METHOD Divide part by base and multiply by 100.

1. $^{17}/_{25} = 0.68$.
2. $0.68 \times 100 = 68\%$.

ANSWER 68%.

CHECK $^{17}/_{25} = {}^{68}/_{100}$ by multiplying both parts by 4.

WORKED EXAMPLE 5.3.2

Comparing two percentages

GIVEN Class A has 18 girls out of 30 students; Class B has 14 girls out of 20.

ASKED Which class has the higher proportion of girls?

METHOD Express both as percentages of their own totals.

1. Class A: $^{18}/_{30} = 60\%$.
2. Class B: $^{14}/_{20} = 70\%$.
3. The counts (18 and 14) cannot be compared directly because the class sizes differ.

ANSWER Class B, with 70% against 60%.

CHECK In classes of equal size, the higher count would give the higher percentage.

WORKED EXAMPLE 5.3.3

A percentage change of a percentage

GIVEN A shop's sales this month are 40% higher than last month; last month they were 20% higher than the month before.

ASKED Describe the sales this month compared with two months ago.

METHOD Combine the multipliers in order.

1. Two months ago = 100 units.
2. Last month = $1.2 \times 100 = 120$.
3. This month = $1.4 \times 120 = 168$.

ANSWER Sales are 168% of the level two months ago, an increase of 68%.

CHECK Adding the percentages gives 60%, but the true figure is 68% because the second increase is calculated on a larger base.

WATCH OUT Percentages of different bases cannot be added. Convert to amounts or to multipliers first.

PRACTICE 5.3 · One quantity as a percentage of another

1	Express 7 out of 20 as a percentage.	FLUENCY
2	Express 9 out of 12 as a percentage.	FLUENCY
3	Express 45 out of 60 as a percentage.	FLUENCY
4	A test has 40 marks and a student scores 34. What percentage is that?	FLUENCY
5	Explain why 20 out of 30 is not the same percentage as 30 out of 20.	REASONING

6	Two students say “half the class”. One means 15 of 30, the other 20 of 40. Are they describing the same proportion?	REASONING
7	A shop sells 240 of 300 items in stock. What percentage remains?	PROBLEM SOLVING
8	In a survey, 84 of 120 students walk and 27 of 60 cycle. Which group is the larger proportion?	PROBLEM SOLVING
9	A student scores 68% on a test of 50 marks. How many marks did they score?	EXTENSION
10	A number is increased from 45 to 54. By what percentage did it increase?	EXTENSION

5.4 | Percentage increase and decrease

A · II · D · II

A percentage change is always calculated on the original amount. Using multipliers makes increases and decreases a single multiplication, and combining them becomes simple.

DEFINITION
Percentage increase and decrease

An increase of p% multiplies the original amount by $1 + {}^{p}/_{100}$. A decrease of p% multiplies by $1 - {}^{p}/_{100}$. An increase of 15% is a multiplier of 1.15; a decrease of 15% is a multiplier of 0.85.

METHOD
Working with percentage change

1. Identify the original amount — the base.
2. Convert the change to a multiplier.
3. Multiply once. Do not add the change separately unless you need to show it.
4. To find the percentage change between two amounts, divide the change by the **original** amount and multiply by 100.

Change	Multiplier	Example on 200
+5%	1.05	210
+20%	1.20	240
−10%	0.90	180
−25%	0.75	150
+100%	2.00	400
−100%	0	0

Multipliers for common changes

COMMON ERROR
Calculating the change on the new amount

A 120-unit item increases by 10%. The increase is 12 units, not 12 on 132. When a question reverses direction — “a price rises 10% and then falls 10%” — the fall is calculated on the raised price, so the result is not the original.

WORKED EXAMPLE 5.4.1

A single increase

GIVEN A rent of 850 units rises by 6%.

ASKED Find the new rent.

METHOD Multiply by 1 + 0.06.

1. Multiplier = 1.06.
2. 850 × 1.06 = 901.

ANSWER 901 units.

CHECK 6% of 850 is 51, and 850 + 51 = 901.

WORKED EXAMPLE 5.4.2

A decrease in a shop

GIVEN A bicycle priced at 280 units is reduced by 15% in a sale, then reduced by a further 20% at the end of the sale.

ASKED Find the final price and the overall percentage reduction.

METHOD Multiply the multipliers in sequence.

1. 280 × 0.85 = 238.
2. 238 × 0.80 = 190.4.
3. Combined multiplier: 0.85 × 0.8 = 0.68, so the price is 68% of the original.

ANSWER 190.4 units; an overall reduction of 32%.

CHECK 280 × 0.68 = 190.4. Adding the percentages would give 35%, which is too much.

WORKED EXAMPLE 5.4.3

Finding the percentage change

GIVEN A town's population fell from 12,500 to 11,750.

ASKED Find the percentage decrease.

METHOD Divide the change by the original, then multiply by 100.

1. Change: 12,500 - 11,750 = 750.
2. *frac*75012,500 = 0.06.
3. 0.06 × 100 = 6%.

ANSWER A decrease of 6%.

CHECK 12,500 × 0.94 = 11,750.

PRACTICE 5.4 · Percentage increase and decrease

1	Increase 60 by 10%.	FLUENCY
2	Decrease 80 by 25%.	FLUENCY
3	Increase 250 by 12%.	FLUENCY
4	A price falls from 90 to 72. Find the percentage decrease.	FLUENCY
5	Explain why a 20% increase followed by a 20% decrease does not return to the original amount.	REASONING
6	A student finds a 10% increase on 40 as 40 + 10 = 50. What has gone wrong?	REASONING

7	A laptop costs 450 units; a 12% discount is applied and then 8% tax on the reduced price. Find the final price.	PROBLEM SOLVING
8	A salary rises from 1,800 to 2,070. Find the percentage increase.	PROBLEM SOLVING
9	A price is increased by 25% and then reduced by 20%. Find the overall change.	EXTENSION
10	Two successive increases of 10% are equivalent to what single increase?	EXTENSION

5.5 | Discounts, taxes and budgets

D · I, III, IV

Real problems rarely announce which quantity is the base. Deciding the base, and keeping track of it through several changes, is the mathematical work.

KEY IDEA

Three questions to ask of any percentage problem

1. **What is the base?** The percentage is always of something; find the amount it compares with.
2. **Is the percentage of the base, or is the base itself unknown?** If the base is unknown, divide by the multiplier rather than multiplying.
3. **How many changes are there, and on which amount is each calculated?** Multiply the multipliers in order.

METHOD

Finding the original amount

If a price after a 20% increase is 96 units, then $96 = 1.2 \times \textit{original}$, so the original is $96 \div 1.2 = 80$. Dividing undoes the multiplication. Estimating the answer before dividing — the original must be less than 96 — prevents most errors.

Situation	Equation	Original
After +15% the value is 92	$1.15x = 92$	80
After −25% the value is 45	$0.75x = 45$	60
After +8% the value is 270	$1.08x = 270$	250
After two changes the value is 68	$1.7x = 68$	40

Working backwards through a change

COMMON ERROR

Dividing by the percentage change

“After a 20% increase the price is 96, so the original was $96 \times 0.8 = 76.8$” is wrong twice over: it applies the change again and applies it in the same direction. Working backwards divides by the multiplier.

WORKED EXAMPLE 5.5.1

Finding the original price

GIVEN In a sale, a coat is reduced by 30% and sells for 84 units.

ASKED Find the original price.

METHOD Write the relationship and divide by the multiplier.

1. Sale price = 0.7 × original.
2. Original = 84 ÷ 0.7 = 120.

ANSWER 120 units.

CHECK 30% of 120 is 36, and 120 - 36 = 84.

WORKED EXAMPLE 5.5.2

Comparing two offers

GIVEN Offer 1: 25% off a price of 160 units. Offer 2: 20% off a price of 150 units, then a further 10% off the reduced price.

ASKED Which offer is cheaper, and by how much?

METHOD Calculate both final prices with a single multiplier chain.

1. Offer 1: 160 × 0.75 = 120.
2. Offer 2: 150 × 0.8 × 0.9 = 108.
3. Difference: 120 - 108 = 12.

ANSWER Offer 2 is cheaper by 12 units.

CHECK As percentages of the marked prices, offer 1 removes 25% and offer 2 removes 28%.

WORKED EXAMPLE 5.5.3

Interpreting a percentage claim

GIVEN A news report says “the number of visitors rose 200% to 900”.

ASKED Find the number of visitors before the rise and explain what the percentage means.

METHOD Interpret 200% as a multiplier of 3.

1. A rise of 200% means the new value is 100% + 200% = 300% of the old.
2. $3x = 900$, so $x = 300$.

ANSWER 300 visitors before; the number tripled.

CHECK 900 ÷ 3 = 300, and the increase is 600, which is 200% of 300.

WATCH OUT “Rose 200%” and “rose to 200%” differ by a factor: the first triples, the second doubles.

DEFINITION

Discount, tax and budget

A **discount** is an amount taken off a marked price, usually quoted as a percentage of it. A **tax** — VAT or sales tax in many countries — is an amount added to a price, also a percentage. A **budget** is a plan for spending a fixed total, and percentages let each item be compared with that total.

KEY IDEA

Adding and removing tax

Adding tax: multiply by 1 + *rate*. At 18% VAT, a price of 250 becomes 250 × 1.18 = 295. **Removing tax:** divide by 1 + *rate*, never multiply by 1 - *rate*. If a till receipt of 295 includes 18% VAT, the price before tax is 295 ÷ 1.18 = 250, and the tax itself is 45. Removing tax and taking a discount are different operations: a discount of 18% and a tax of 18% do not cancel, because they are calculated on different amounts.

Case	Working	Result
Discount of 20% on 90	90 × 0.8	72
Tax of 18% added to 72	72 × 1.18	84.96
Tax of 18% included in 84.96	84.96 ÷ 1.18	72

One price, three ways of quoting it

WORKED EXAMPLE 5.5.4

Reading a receipt with tax

GIVEN A receipt shows a total of 236 units, which includes 18% VAT.

ASKED Find the price before tax and the amount of VAT.

METHOD Divide by the tax multiplier.

1. 1.18 × *price* = 236.
2. Price = 236 ÷ 1.18 = 200.
3. VAT = 236 - 200 = 36, which is 18% of 200.

ANSWER 200 units before tax, 36 units of VAT.

CHECK 200 × 1.18 = 236.

WATCH OUT Taking 18% of 236 gives 42.48, which is too much — the 18% must be taken of the pre-tax price, not of the total.

WORKED EXAMPLE 5.5.5

A budget with percentages

GIVEN A family budget of 2,400 units a month: 35% housing, 20% food, 15% transport, 10% school costs, and the rest saved.

ASKED Find each amount and the percentage saved.

METHOD Convert each percentage to an amount, then check the total.

1. Housing = 0.35 × 2400 = 840; food = 480; transport = 360; school = 240.
2. Spent = 840 + 480 + 360 + 240 = 1920.
3. Saved = 2400 - 1920 = 480, which is 480 ÷ 2400 = 20%.

ANSWER 840, 480, 360 and 240 units; 480 units saved, 20% of the budget.

CHECK 35 + 20 + 15 + 10 + 20 = 100%.

PRACTICE 5.5 · Discounts, taxes and budgets

1	After a 10% increase a value is 55. Find the original.	FLUENCY
2	After a 25% decrease a value is 60. Find the original.	FLUENCY
3	A price of 240 is reduced by 15%. Find the sale price.	FLUENCY
4	A population of 800 grows by 5%. Find the new population.	FLUENCY

5	Explain why “rose 200%” is not the same as “rose to 200%”.	REASONING
6	A shop raises prices 10% and later cuts them 10%. Is the final price the same as the original? Explain.	REASONING
7	A bicycle is discounted 20% and now costs 176 units. What was the original price, and how much was saved?	PROBLEM SOLVING
8	A student's mark improves from 45 to 54; a second student's from 60 to 72. Who improved by the larger percentage?	PROBLEM SOLVING
9	A price rises 10%, then 10% again. What single percentage increase is equivalent?	EXTENSION
10	A shop's sales fall 20% in a bad month, then rise 20% the next month. Is the shop back to its starting level?	EXTENSION
11	A television costs 480 units before tax. VAT is 18%. Find the total paid, and the VAT.	PROBLEM SOLVING
12	A hotel bill of 590 units includes 18% VAT. Find the price before tax.	PROBLEM SOLVING
13	A monthly budget of 1,800 units allows 25% for rent and 30% for food. How much is left for everything else?	PROBLEM SOLVING
14	A shop advertises "18% off, VAT included". Explain why the saving is less than 18% of the final price.	EXTENSION

F INVESTIGATE AND APPLY

A criterion B investigation and a criterion D application

INVESTIGATION 5 | Do two discounts add up?

B · I, II, III

Extend the Explore activity into a general investigation of successive percentage changes, including rises and falls mixed together.

INVESTIGATION
Investigation brief

Question. When two percentage changes are applied one after the other, what single change is equivalent to them? **What you need.** A calculator, a table of starting values, and the discount diagram. **What to produce.** A table of combined results, a rule in words, a formula, and a test on a case where one change is a rise.

PART 1 · COLLECT RESULTS

First change	Second change	Value after both	Single equivalent change
−20%	−10%		
+10%	+10%		
+20%	−20%		
−25%	+25%		
+50%	−50%		

Copy and complete (start value 500)

PART 2 · DESCRIBE AND GENERALISE

1. What single multiplier combines a change of -a% with a change of -b%?
2. Write the single equivalent change for +a% followed by -b%.
3. Explain why the order of the two changes does not affect the final value.
4. A shopkeeper claims “20% off, then 20% off, is 40% off”. Calculate the true equivalent and explain the difference.

PART 3 · TEST AND EVALUATE

1. Test your rule on a change of +30% followed by +30%.
2. Find a pair of changes that returns a value exactly to its starting point, other than no change at all.
3. Explain what happens to a value after a 100% decrease.
4. Write two sentences a shopkeeper could use to describe successive discounts honestly.

CRITERION TASK

Criterion B task · Requirements

Complete table with correct values; a general rule stated in words; a formula using multipliers; verification on at least two new cases; a justification that explains why multipliers multiply.

APPLICATION 5 | Choosing a phone plan

D · I, II, III, IV

Three companies offer plans with different prices, different data allowances and different percentage discounts. The cheapest sticker price is not always the cheapest plan.

THE BRIEF

Company	Monthly price	Data included	Offer	Extra data cost per GB
Alpha	24	8 GB	20% off for the first year	3.50
Beta	30	12 GB	15% off, plus 2 GB free	3.00
Gamma	20	5 GB	no discount, but 50% extra data free	4.00

The three offers (monthly)

- The student uses about 11 GB each month.
- The plan will be kept for 12 months.
- Discounts apply only to the monthly price, not to extra data.

TASK 1 · CALCULATE THE EFFECTIVE PRICE

1. Calculate the discounted monthly price for Alpha and for Beta.
2. Calculate how much extra data each plan needs for 11 GB per month, and its cost.
3. Find the total monthly cost of each plan for this student.

TASK 2 · COMPARE OVER A YEAR

1. Calculate the total cost of each plan for 12 months.
2. Rank the plans from cheapest to most expensive.
3. State how much the cheapest saves compared with the most expensive.

TASK 3 · EVALUATE THE OFFERS

1. Describe each offer as a percentage saving on the monthly price, using a common basis where possible.
2. Which offer looks best in advertising, and which is best in practice? Explain the difference.
3. Recommend a plan and justify it in three sentences, acknowledging one assumption you have made.

CRITERION TASK
Criterion D task · Requirements

Discounts applied correctly; extra data costed with units; totals for 12 months; a ranking supported by figures; a recommendation with an assumption and a limitation stated.

TASK C | Interrogating an advertisement

C · I, III, IV

CRITERION TASK
The task

A supermarket advertises: **“Now 50% more free! Was 400 g, now 600 g.”** Write an analysis for a consumer magazine explaining what the claim does and does not say.

1. State the base of the 50% comparison and calculate the increase.
2. Explain why “50% more free” is not the same as “double the amount”.
3. Calculate the price per 100 g before and after if the old pack cost 1.60 and the new pack costs 2.10, and state whether the customer gains.
4. Rewrite the advertisement in a form that states the comparison honestly.

Criterion C strand	Evidence
Notation (i)	Correct percentage notation and units throughout.
Reasoning (iii)	The analysis separates the claim from the comparison it uses.
Organisation (iv)	The four parts follow a logical order for a reader.

Success criteria

G REFLECT AND CONSOLIDATE

Summary, vocabulary, misconceptions and review

REVIEW

Consolidating Chapter 5

A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- A percentage is a fraction over 100; it converts to decimals and fractions by dividing or multiplying by 100.
- A percentage of a quantity is a multiplication by the decimal form.
- A percentage compares a part with a base; changing the base changes the percentage.
- An increase of p% multiplies by $1 + {}^{p}/_{100}$; a decrease multiplies by $1 - {}^{p}/_{100}$.
- Successive percentage changes combine by multiplying multipliers, never by adding percentages.
- Working backwards from a changed amount divides by the multiplier.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Percentage	A fraction with denominator 100, written with %.
Base	The amount a percentage compares with.
Multiplier	The decimal form of a percentage change, such as 1.15.
Percentage increase	A rise expressed as a percentage of the original amount.
Percentage decrease	A fall expressed as a percentage of the original amount.
Complement	The part that remains, such as 8% when 92% is taken.
Reverse percentage	Finding the original amount from the changed amount.

Vocabulary for Chapter 5

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence
“Percentages can be added.”	They apply to different bases.	$0.8 \times 0.9 = 0.72$
“A percentage of a quantity means divide by it.”	Multiply by the decimal form.	15% of 240 is 36
“+20% then −20% returns to the start.”	The base changes.	$1.2 \times 0.8 = 0.96$
“50% more means double.”	It means half as much again.	400 g → 600 g
“After a 20% rise, divide by 20%.”	Divide by 1.2.	$96 \div 1.2 = 80$

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
convert between percentages, fractions and decimals.			
find a percentage of a quantity using a multiplier.			
express one quantity as a percentage of another.			

I can …	Not yet	Getting there	Confident
increase and decrease amounts by a percentage.			
combine successive percentage changes correctly.			
find an original amount after a percentage change.			

How confident are you?

REFLECTION

1. Which percentage benchmark do you now use most often in mental arithmetic?
2. Describe a percentage claim you have seen outside school and whether the base was clear.
3. Write one sentence explaining why percentages cannot simply be added.
4. In the phone plan task, did the percentage discount or the data allowance matter more? Why?
5. Which error — using the wrong base or forgetting to convert to a decimal — did you make more often?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ5 · Applying the chapter

1	Write 0.72 as a percentage.	FLUENCY
2	Write 35% as a fraction in simplest form.	FLUENCY
3	Find 20% of 85.	FLUENCY
4	Increase 140 by 10%.	FLUENCY
5	Decrease 90 by 30%.	FLUENCY
6	Express 16 out of 40 as a percentage.	FLUENCY
7	Explain why 25% of 80 is the same as 80% of 25.	REASONING
8	Is a 10% discount followed by a 10% increase the same as no change?	REASONING
9	Explain what “a 150% increase” means for a quantity of 40.	REASONING
10	A shirt costs 60 units after a 25% reduction. Find the original price.	PROBLEM SOLVING
11	A school has 720 students; 45% are boys. How many girls?	PROBLEM SOLVING
12	A price rises 5% to 84 units. Find the original price.	PROBLEM SOLVING
13	Explain why multiplying by 1.15 finds a 15% increase in one step.	COMMUNICATING
14	Explain why successive percentage changes can be combined by multiplying multipliers.	COMMUNICATING
15	Find two successive changes equivalent to a single 44% increase.	EXTENSION
16	A price is reduced 20%, then increased 20%. Find the overall change.	EXTENSION

17	A student scores 45%, then 54%. Describe the change in the score as a percentage of the first score.	EXTENSION
18	A shop buys an item for 40 and sells it for 52. Find the profit as a percentage of the cost price.	PROBLEM SOLVING
19	A discount of 30% is followed by a discount of 10%. Explain why the total discount is 37%, not 40%.	COMMUNICATING
20	A country's emissions fall 5% each year for two years. Explain why the total fall is not 10%.	COMMUNICATING

MIXED 5 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX5 · End-of-chapter mixed set

1	Find 12% of 150.	FLUENCY
2	Write $\frac{5}{8}$ as a percentage.	FLUENCY
3	Increase 250 by 24%.	FLUENCY
4	Decrease 75 by 40%.	FLUENCY
5	Express 42 out of 56 as a percentage.	FLUENCY
6	A price is 25% more than another. Is the second price 25% less than the first?	REASONING
7	Explain why “30% off the sale price” is not the same as “30% off the original price”.	REASONING
8	A class of 32 has 12 boys. What percentage are girls?	PROBLEM SOLVING
9	A phone is reduced 15% to 340 units. Find the original price.	PROBLEM SOLVING
10	A town of 4,500 grows by 6%, then by 4%. Find the population.	PROBLEM SOLVING
11	Explain how to find 15% of an amount using 10% and 5%.	COMMUNICATING
12	Two shops offer 20% off and 15% off respectively on prices of 100 and 110. Which is cheaper?	COMMUNICATING
13	Find the single percentage change equivalent to +50% then −50%.	EXTENSION
14	A value increases by 20% and then decreases by 20%. Write the result as a percentage of the original.	EXTENSION
15	A price after a 12% rise is 224 units. Find the original.	EXTENSION
16	A shop marks a price up 25% and then offers 20% off. What is the final price as a percentage of the original cost to the shop?	PROBLEM SOLVING

CUMULATIVE PRACTICE 5 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP5 · Cumulative practice

1	Write 0.08 as a percentage.	FLUENCY
2	Find 35% of 60.	FLUENCY
3	Decrease 240 by 15%.	FLUENCY
4	Explain why 0.5% is much smaller than 50%.	REASONING
5	A price of 96 follows a 20% increase. Find the original price.	PROBLEM SOLVING
6	A shop offers 30% off, then a further 10% off. Find the overall discount.	PROBLEM SOLVING
7	Explain why percentages cannot simply be added when they apply in turn.	COMMUNICATING
8	Two increases of 10% equal what single increase?	EXTENSION

PART III

PATTERNS AND SYMBOLS

III

06 Patterns, expressions and equations

PART III · PATTERNS AND SYMBOLS

CHAPTER 06

Patterns, expressions and equations

Algebra begins by noticing what stays the same while something else changes. A pattern can be shown in shapes, written in a table, described in words and compressed into a rule — and an equation lets us work backwards from a result to the unknown.

KEY CONCEPT	**Relationships**
RELATED	**Patterns**
RELATED	**Equivalence**
GLOBAL CONTEXT	**Scientific and technical innovation**

06
CHAPTER

INQUIRY QUESTION

A tiled border grows by the same number of tiles at every step. How can you predict the hundredth stage without drawing it, and how can you find which stage uses exactly 100 tiles?

A square pond of side n metres needs a one-slab border counted by $4n + 4$. The rule lets a builder order slabs without drawing the plan; an equation reverses the count when the delivery contains only 96 slabs.

IN THIS CHAPTER

6.1 Patterns and rules
6.2 Expressions and substitution
6.3 Simplifying expressions
6.4 Solving linear equations
6.5 Forming and using equations
Investigation 6 The matchstick chain
Application 6 The school event budget
Task C Explaining the balancing method
Review Consolidating Chapter 6
Review questions Practice across the whole chapter
Mixed 6 Fluency, reasoning and problem solving
Cumulative practice 6 Questions that mix this chapter with earlier ones

APPROXIMATELY 16 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

Algebra begins with a question that arithmetic cannot answer comfortably: what stays the same while something else changes? Once that is written as a rule, a table of twenty cases becomes one line, and a question about the hundredth case becomes a calculation.

KEY CONCEPT · RELATIONSHIPS

Relationships. A pattern relates a position to a value. An equation states that two expressions name the same number, so a solution can be checked by substitution.

RELATED CONCEPT · PATTERNS

Patterns. Regular change can be described, continued and predicted. The step and the starting value together determine the rule.

RELATED CONCEPT · EQUIVALENCE

Equivalence. Expressions may look different and still be equal for every value of the variable. Collecting like terms and expanding rewrite an expression without changing its value.

GLOBAL CONTEXT

Scientific and technical innovation

EXPLORATION

How systems and patterns work: using rules to model growth, cost and design in construction.

FACTUAL	CONCEPTUAL	DEBATABLE
What is a variable, a term and a coefficient? How do you substitute a value? What separates an expression from an equation?	How can different-looking expressions always be equal, and what does it mean to solve an equation?	Is a rule guessed from several terms trustworthy? When is a formula more useful than a table of values?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Continue and generalise sequences, write and simplify expressions, substitute values, solve one- and two-step equations.
B · Investigating patterns	i, ii	Investigate a matchstick chain, record counts in a table, justify the general rule and test it against a claim.
C · Communicating	i, ii, iii	Move between diagram, table, words and symbols; explain each step in a solution; justify that two expressions are equivalent.
D · Applying mathematics	i, ii, iii, iv	Model a courtyard border and a service charge, solve design equations, evaluate a budget and recommend a size.

ATL SKILL · CRITICAL THINKING

In order for a student to explain the sequence 4, 7, 10, 13 in two different ways, students must compare “add 3 each time” with $3n + 1$, use both descriptions to find the 40th term and state what each description can and cannot predict.

STATEMENT OF INQUIRY

Patterns follow algebraic rules, and equivalent expressions and equations let us predict later stages and recover unknown values.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand how a sequence is built from a position and a rule, and distinguish term-to-term from position-to-term descriptions.
- Write, interpret and substitute into algebraic expressions.
- Collect like terms and expand a single bracket.
- Form and solve one- and two-step equations, and check the solution.
- Use a formula that links quantities in a context.
- Investigate a growing matchstick pattern and justify a general rule.

SUCCESS CRITERIA

- ✓ I can continue a sequence and explain the rule in words.
- ✓ I can find a position-to-term rule and use it for a distant term.
- ✓ I can write and evaluate expressions, including with negative values.
- ✓ I can simplify expressions by collecting like terms and expanding brackets.
- ✓ I can solve linear equations and check my answer by substitution.
- ✓ I can form an equation from a context and interpret the solution.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Calculate 6×7.
2. Calculate $20 - 3 \times 4$.
3. Write $5 + 5 + 5 + 5$ as a multiplication.
4. Find the next two terms of 2, 5, 8, 11, …
5. Calculate $3 \times 4 + 2$.
6. Solve $6 + \square = 15$.

Answers are in the Chapter 6 answer section. If more than two are incorrect, review multiplication facts, order of operations and simple missing-number sentences before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Number patterns, missing-number sentences, multiplication facts and the order of operations.	Sequences and rules, expressions, substitution, simplifying, expanding, equations and formulas.	Linear graphs, simultaneous equations, inequalities, quadratic expressions and algebraic proof.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Borders, chains and service charges grow by a regular rule.	How can one rule describe every stage, and how can we reverse it?	Relationships, pattern and equivalence.	How many matches make a chain of n squares?	Diagrams, tables, words, expressions and balanced equations.	When does a pattern count as explained rather than merely continued?

A ENGAGE
The matchstick chain that grows by three

A chain of squares is built from matchsticks. The first stage uses 4 matches, the second 7, the third 10. A student claims the fortieth stage will use 120 matches.

It will not. The number grows by 3 at each stage, and the rule that produces the chain is $3n + 1$, not $3n$: the first stage uses 4 matches because of the extra match at the start. At stage 40 the count is $3 \times 40 + 1 = 121$.

Algebra exists to make that reasoning possible without drawing all forty stages. A **pattern** is described by a **rule**; the rule is written as an **expression**; an unknown in the rule is found by solving an **equation**. Those three moves — describe, express, solve — are what this chapter teaches.

Stage n	Squares drawn	Matches used	Rule $3n + 1$
1	1	4	4
2	2	7	7
3	3	10	10
10	10	31	31

Figure 6.1 Three stages of the chain

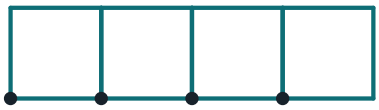
NOTICE AND WONDER

- Why does the count go up by 3 each time rather than 4?
- Which is easier to extend to stage 100: the drawing or the rule?
- What would a stage 0 mean, and would the rule still give a sensible answer?
- If a chain used 100 matches, how many squares would it have?

> Keep this question open: when does a pattern become a rule, and what makes a rule trustworthy beyond the cases we have checked?

B EXPLORE
The matchstick chain

Work in pairs with matchsticks (or with a pencil and ruler) and a large sheet of paper.



stage 4

The first four stages of the matchstick chain. Each new square shares one match, so the number of matches grows by 3 at every stage.

YOUR INVESTIGATION

Step 1. Build stages 1 to 5 of the chain. Record the number of matches used at each stage in a table.

Step 2. Record the **difference** between successive counts. What do you notice?

Step 3. Predict the number of matches for stage 6, then build it and check.

Step 4. Find a rule that gives the number of matches for any stage n.

Stage n	1	2	3	4	5	6
Matches						
Difference from previous	—					

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

1. What does the constant difference of 3 tell you about the rule?
2. Why is the rule not simply $3n$? What does the extra 1 represent in the drawing?
3. Use your rule to predict stage 20, then check it by continuing the table.
4. A chain uses 61 matches. Write an equation and solve it to find the stage.
5. Design a different chain — for example, triangles that share a side — and find its rule.

MAKING A CONJECTURE

> "The number of matches at stage n is … because each new square adds … matches, and … ."

WHAT YOU SHOULD HAVE NOTICED

- Each new square adds 3 matches, because one match is shared.
- A constant difference of 3 means the rule contains $3n$.
- The first stage uses one more than 3×1, so the rule is $3n + 1$.
- Checking the rule against every recorded stage is a test, not a proof; the justification comes from the structure of the drawing.

INVESTIGATION
Explore further

A different chain uses hexagons that share one side. Build the first four stages, record the counts and find a rule. Then compare your rule with a partner's and explain any difference in the constant.

6.1 | Patterns and rules

B · I, II

A sequence of numbers is a pattern; a rule describes it. Two kinds of description matter: how to get from one term to the next, and how to get to any term directly.

DEFINITION
Sequences and rules

A **sequence** is a list of numbers in order, called **terms**. A **term-to-term rule** says how to get from one term to the next. A **position-to-term rule** gives the term from its position n. For 5, 8, 11, 14, …: the term-to-term rule is "add 3"; the position-to-term rule is $3n + 2$.

KEY IDEA

Why the difference points to the rule

If successive terms differ by a constant d, the rule contains dn. Subtracting dn from each term then shows what must be added to reach the first term. For 5, 8, 11, 14, … the difference is 3, and $5 - 3 \times 1 = 2$, so the rule is $3n + 2$.

Position n	1	2	3	4	5
Term	5	8	11	14	17
$3n$	3	6	9	12	15
Term - $3n$	2	2	2	2	2

Two descriptions of the same sequence

METHOD

Finding a position-to-term rule

1. Write the terms in a table with their positions.
2. Find the constant difference d between successive terms.
3. Subtract dn from each term; the constant remainder is the number added to the rule.
4. Check the rule on at least two positions, including one you did not use.

COMMON ERROR

Assuming the first term is the multiplier

In the sequence 4, 7, 10, 13, … the first term is 4, but the rule is not $4n$. The constant difference is 3, so the rule is $3n + 1$. The first term is used at the end of the process, not the beginning.

WORKED EXAMPLE 6.1.1

Finding a rule from terms

GIVEN The sequence 7, 12, 17, 22, …

ASKED Find the position-to-term rule and the 20th term.

METHOD Use the constant difference and adjust for the first term.

1. The difference is 5, so the rule contains $5n$.
2. $7 - 5 \times 1 = 2$, so the rule is $5n + 2$.
3. 20th term: $5 \times 20 + 2 = 102$.

ANSWER Rule $5n + 2$; 20th term 102.

CHECK The 4th term by rule is 22, which matches the sequence.

WORKED EXAMPLE 6.1.2

Using a rule to find a position

GIVEN A sequence has rule $4n + 3$. One of its terms is 51.

ASKED Find the position of that term.

METHOD Form an equation from the rule and solve it.

1. $4n + 3 = 51$.
2. $4n = 48$.
3. $n = 12$.

ANSWER The 12th term.

CHECK $4 \times 12 + 3 = 51$.

WORKED EXAMPLE 6.1.3

A sequence that decreases

GIVEN The sequence 30, 26, 22, 18, …

ASKED Find the rule, and state whether the sequence ever reaches 0.

METHOD Use the constant difference, which is negative.

1. The difference is -4, so the rule contains $-4n$.
2. $30 - (-4 \times 1) = 34$, so the rule is $34 - 4n$.
3. The 8th term is $34 - 32 = 2$ and the 9th is -2, so the sequence passes from 2 to -2 and never equals 0.

ANSWER Rule $34 - 4n$; the sequence skips 0 because 34 is not a multiple of 4.

CHECK The 1st term is $34 - 4 = 30$, and the 4th is $34 - 16 = 18$.

PRACTICE 6.1 · Patterns and rules

1	Write the next two terms: (a) 3, 6, 9, 12, … (b) 20, 17, 14, …	FLUENCY
2	Find the term-to-term rule: 2, 9, 16, 23, …	FLUENCY
3	Find the position-to-term rule for 5, 9, 13, 17, …	FLUENCY
4	Find the 10th term of the sequence with rule $6n - 1$.	FLUENCY
5	Explain why a constant difference of 4 means the rule contains $4n$.	REASONING
6	A student writes the rule for 6, 10, 14, 18 as $6n$. Test the rule and explain the error.	REASONING
7	A pattern of tiles uses 5 tiles in stage 1, 8 in stage 2, 11 in stage 3. How many in stage 12?	PROBLEM SOLVING
8	A sequence has rule $5n - 2$. Which term equals 43?	PROBLEM SOLVING
9	Find the rule for 100, 91, 82, 73, …	EXTENSION
10	Two sequences have rules $3n + 4$ and $4n + 1$. Which is larger at the 1st term, and which at the 10th?	EXTENSION

6.2 | Expressions and substitution

A · I, II

An expression is a mathematical phrase built from numbers, letters and operations. It has no equals sign: it describes a calculation, and it can be evaluated once the letters are known.

DEFINITION

Expression, term, coefficient, substitution

An **expression** combines numbers and letters with operations, such as $3x + 5$ or $2(a - 4)$. A **term** is a part separated by + or − signs. The **coefficient** is the number multiplying a letter: in $7y$ the coefficient is 7. **Substituting** means replacing a letter with a number and calculating the result.

KEY IDEA

Notation that prevents errors

- $3x$ means $3 \times x$; the multiplication sign is not written.
- x^2 means $x \times x$, not $2x$. If $x = 5$, $x^2 = 25$ but $2x = 10$.
- $\frac{x}{4}$ means $x \div 4$.
- Brackets show that the whole bracket is multiplied: $2(x + 3)$ means $2 \times (x + 3)$.

METHOD

Substituting into an expression

1. Rewrite the expression with the number in brackets where the letter was.
2. Follow the order of operations: brackets, indices, multiplication and division, then addition and subtraction.
3. Include the units if the expression represents a quantity.

Value of x	Value of y	Working	Result
2	1	$3 \times 4 - 2$	10
3	5	$3 \times 9 - 10$	17
0	4	$0 - 8$	-8
-1	2	$3 \times 1 - 4$	-1

Substituting into $3x^2 - 2y$

COMMON ERROR

Squaring the coefficient

In $3x^2$ with $x = 4$, the square applies only to x: $3 \times 16 = 48$. Writing $(3 \times 4)^2 = 144$ squares the 3 as well, which the notation does not say.

WORKED EXAMPLE 6.2.1

Substituting two values

GIVEN The expression $5a - 2b + 7$.

ASKED Evaluate it when $a = 4$ and $b = 3$, and when $a = 0$ and $b = 6$.

METHOD Substitute in brackets and follow the order of operations.

1. $5(4) - 2(3) + 7 = 20 - 6 + 7 = 21$.
2. $5(0) - 2(6) + 7 = 0 - 12 + 7 = -5$.

ANSWER 21 and -5.

CHECK The first is positive because $5a$ dominates; the second is negative because $-2b$ dominates.

WORKED EXAMPLE 6.2.2

Substituting a negative number

GIVEN The expression $x^2 - 3x + 1$.

ASKED Evaluate it when $x = -2$.

METHOD Put the negative number in brackets so the signs are visible.

1. $(-2)^2 - 3(-2) + 1$.
2. $4 + 6 + 1$.

ANSWER 11.

CHECK Both the square and the product with -3 are positive here, so the result exceeds the value at $x = 2$, which is $4 - 6 + 1 = -1$.

WATCH OUT -2^2 without brackets means $-(2^2) = -4$. The brackets change the meaning, and they change the answer.

WORKED EXAMPLE 6.2.3

Using an expression in context

GIVEN A taxi charges 3 units to start and 2 units for each kilometre. The cost for k kilometres is $2k + 3$.

ASKED Find the cost of an 8 km journey, and the distance travelled for 21 units.

METHOD Substitute to find the cost; solve an equation to find the distance.

1. $2(8) + 3 = 19$ units.
2. $2k + 3 = 21$, so $2k = 18$ and $k = 9$ km.

ANSWER 19 units for 8 km; 9 km for 21 units.

CHECK $2 \times 9 + 3 = 21$.

PRACTICE 6.2 · Expressions and substitution

1	Evaluate $4x + 3$ when $x = 6$.	FLUENCY
2	Evaluate $2a - b$ when $a = 5$ and $b = 7$.	FLUENCY
3	Evaluate x^2 when (a) $x = 7$ (b) $x = 12$	FLUENCY
4	Evaluate $3y^2$ when $y = 4$.	FLUENCY
5	Explain the difference between $2x$ and x^2 when $x = 6$.	REASONING
6	A student evaluates $2x + 5$ at $x = 3$ as 16. What has gone wrong?	REASONING
7	Evaluate $5 - 2n$ for $n = -3$.	PROBLEM SOLVING

8	A phone plan costs $0.1m + 5$ for m minutes. Find the cost for 120 minutes.	PROBLEM SOLVING
9	Evaluate $\frac{a+b}{2}$ for $a = 7$ and $b = 12$.	EXTENSION
10	Evaluate $2x^2 - y^2$ for $x = 3$ and $y = 5$.	EXTENSION

6.3 | Simplifying expressions

A · II · C · I

Simplifying collects like terms and removes brackets. It does not change the value of the expression: it changes how the same value is written.

DEFINITION

Like terms and simplifying

Like terms contain exactly the same letters raised to the same powers, such as $4a$ and $-7a$. They can be combined by adding or subtracting their coefficients.

$$4a + 3b - a + 2b = 3a + 5b$$

Terms with different letters, such as $3a$ and $5b$, cannot be combined into one term.

KEY IDEA

Why collecting works

$4a - a$ means four lots of a minus one lot of a, which leaves three lots: $3a$. This is the distributive idea in disguise, and it is why only like terms combine: $3a + 5b$ cannot collapse, because a and b count different things.

METHOD

Simplifying an expression

1. Identify the like terms, including their signs.
2. Collect them, keeping the sign in front of each term with it.
3. Expand any brackets first, then collect.
4. Check by substituting a value into the original and the simplified form; the results must agree.

Expression	Expanded	Why
$3(x + 4)$	$3x + 12$	every term in the bracket is multiplied by 3
$5(2a - 1)$	$10a - 5$	the 5 multiplies both terms
$2(x + 3) + 4(x - 1)$	$6x + 2$	expand, then collect like terms
$-3(y + 2)$	$-3y - 6$	a negative multiplier changes both signs

Expanding brackets

COMMON ERROR

Sign errors when collecting

In $7x - 3 - 2x + 8$, the correct collection is $5x + 5$. Treating the minus as belonging to the 3 but not the 2 gives wrong answers. Circling each sign with its term prevents the error.

WORKED EXAMPLE 6.3.1

Collecting like terms

GIVEN $8a + 5 - 3a + 2b - 7$.

ASKED Simplify.

METHOD Group like terms and combine coefficients.

1. a terms: $8a - 3a = 5a$.
2. b terms: $2b$.
3. Numbers: $5 - 7 = -2$.

ANSWER $5a + 2b - 2$.

CHECK At $a = 1$, $b = 1$: original $8 + 5 - 3 + 2 - 7 = 5$; simplified $5 + 2 - 2 = 5$.

WORKED EXAMPLE 6.3.2

Expanding and simplifying

GIVEN $4(2x + 3) - 2(x - 5)$.

ASKED Expand and simplify.

METHOD Multiply every term in each bracket, then collect.

1. $4(2x + 3) = 8x + 12$.
2. $-2(x - 5) = -2x + 10$.
3. $8x + 12 - 2x + 10 = 6x + 22$.

ANSWER $6x + 22$.

CHECK At $x = 2$: original $4(7) - 2(-3) = 28 + 6 = 34$; simplified $12 + 22 = 34$.

WATCH OUT The second bracket is multiplied by -2, so $-2 \times -5 = +10$. Losing that sign is the commonest error in this topic.

WORKED EXAMPLE 6.3.3

Forming and simplifying from a context

GIVEN A rectangle has width w cm and length 3 cm more than twice the width.

ASKED Write and simplify an expression for the perimeter.

METHOD Write each side, then add.

1. Width w; length $2w + 3$.
2. Perimeter $= 2w + 2(2w + 3)$.
3. Expand: $2w + 4w + 6 = 6w + 6$.

ANSWER $6w + 6$ cm.

CHECK If $w = 4$, the rectangle is 4 by 11, perimeter 30; the expression gives 30.

PRACTICE 6.3 · Simplifying expressions

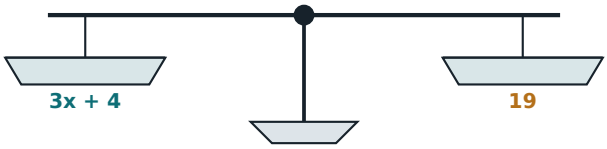
1	Simplify: (a) $3x + 5x$ (b) $9y - 4y$ (c) $2a + 3a - a$	FLUENCY
2	Simplify $4p + 3q + 2p - q$.	FLUENCY

3	Expand: (a) $3(x + 2)$ (b) $5(2y - 1)$	FLUENCY
4	Simplify $2(x + 4) + 3(x + 1)$.	FLUENCY
5	Explain why $3a + 4b$ cannot be simplified further.	REASONING
6	A student simplifies $5x - 2 + 3x$ as $8x - 2$ and another as $6x$. Which is right?	REASONING
7	Simplify $3(2a - 1) - 2(a - 4)$.	PROBLEM SOLVING
8	The perimeter of a square with side $s + 2$ is written as $4(s + 2)$. Expand it.	PROBLEM SOLVING
9	Simplify $5 - 2(3x - 4) + x$.	EXTENSION
10	Show that $2(3x + 1)$ and $3(2x + 1)$ are not equal, and find their difference.	EXTENSION

6.4 | Solving linear equations

A · III · C · III

An equation states that two expressions are equal. Solving it means finding the value of the letter that makes the statement true, and inverse operations are the tool.



both sides must stay equal

An equation is a balance. Whatever is done to one side must be done to the other, so the two sides stay equal.

DEFINITION

Equation and solution

An **equation** contains an equals sign. A **solution** is a value of the letter that makes the equation true. Equations with the letter appearing only to the first power are **linear**. $3x + 4 = 19$ has the solution $x = 5$, because $3 \times 5 + 4 = 19$.

KEY IDEA

The balance idea

An equation is a balance. Whatever is done to one side must be done to the other, or the balance is destroyed. Solving means undoing the operations in reverse order: if the expression multiplied by 3 and then added 4, undo the adding first, then the multiplying.

$3x + 4 = 19$	
$3x = 15$	subtract 4 from both sides
$x = 5$	divide both sides by 3

The same equation solved in two inverse operations, with the reason for each step recorded beside it.

METHOD

Solving a linear equation

1. Simplify each side if necessary.
2. Undo addition or subtraction first, then multiplication or division.
3. Keep the equation balanced by doing the same operation to both sides.
4. Check by substituting the solution into the original equation.

Equation	First step	Second step	Solution
$2x + 7 = 15$	subtract 7	divide by 2	$x = 4$
$5(x - 2) = 20$	divide by 5	add 2	$x = 6$
$\frac{x}{4} + 3 = 7$	subtract 3	multiply by 4	$x = 16$
$9 - 2x = 1$	subtract 9	divide by -2	$x = 4$

Two-step and bracket equations

COMMON ERROR

Working on one side only

Subtracting 4 from the left of $3x + 4 = 19$ without subtracting from the right changes the equation into a false statement. A useful habit is to write the operation beside the equation, as in the figure, so the balance is visible.

WORKED EXAMPLE 6.4.1

A two-step equation

GIVEN $4x - 7 = 21$.

ASKED Solve for x.

METHOD Undo the subtraction, then the multiplication.

1. Add 7 to both sides: $4x = 28$.
2. Divide both sides by 4: $x = 7$.

ANSWER $x = 7$.

CHECK $4 \times 7 - 7 = 21$.

WORKED EXAMPLE 6.4.2

An equation with brackets

GIVEN $3(x + 4) = 27$.

ASKED Solve for x, using two different first steps.

METHOD Either divide by 3 first, or expand the bracket.

1. Dividing first: $x + 4 = 9$, so $x = 5$.
2. Expanding first: $3x + 12 = 27$, so $3x = 15$ and $x = 5$.

ANSWER $x = 5$.

CHECK $3(5 + 4) = 27$.

WATCH OUT Both routes are valid; choosing one and finishing it is better than starting both.

WORKED EXAMPLE 6.4.3

An equation with the letter on both sides

GIVEN $5x - 3 = 2x + 12$.

ASKED Solve for x.

METHOD Collect the letter terms on one side and the numbers on the other.

1. Subtract $2x$ from both sides: $3x - 3 = 12$.
2. Add 3 to both sides: $3x = 15$.
3. Divide by 3: $x = 5$.

ANSWER $x = 5$.

CHECK Left $= 25 - 3 = 22$; right $= 10 + 12 = 22$.

PRACTICE 6.4 · Solving linear equations

1	Solve (a) $x + 7 = 12$ (b) $x - 4 = 9$ (c) $3x = 21$	FLUENCY
2	Solve (a) $2x + 3 = 11$ (b) $5x - 4 = 16$	FLUENCY
3	Solve $\frac{x}{3} = 8$.	FLUENCY
4	Solve $4(x - 1) = 16$.	FLUENCY
5	Explain why subtracting 5 from one side of $x + 5 = 12$ is allowed but subtracting 5 from only one term of $2x + 5$ is not.	REASONING
6	A student solves $3x + 2 = 14$ by dividing by 3 first, getting $x + 2 = \frac{14}{3}$. What has gone wrong?	REASONING
7	Solve $7 - 2x = 1$.	PROBLEM SOLVING
8	Solve $2(x + 5) = 3x$.	PROBLEM SOLVING
9	Solve $5x + 2 = 3x + 10$.	EXTENSION
10	Solve $\frac{x}{2} + \frac{x}{3} = 10$.	EXTENSION

6.5 | Forming and using equations

D · I, II

The difficult part of an applied problem is rarely the solving. It is translating the words into an equation, which requires naming the unknown and stating the relationship it satisfies.

METHOD

Translating words into equations

1. Name the unknown: “let x be the number of …”.
2. Write the relationship as an equation, checking that both sides measure the same kind of quantity.
3. Solve, then answer the question that was asked, with units.
4. Check that the answer is sensible in the context, not merely arithmetically correct.

Statement	Equation	
A number increased by 8 is 21.	$x + 8 = 21$	
Three times a number, less 5, is 16.	$3x - 5 = 16$	
Two consecutive numbers add to 35.	$x + (x + 1) = 35$	
A taxi costs 3 plus 2 per km	a journey costs 21.	$2k + 3 = 21$
The perimeter of a square is 36.	$4s = 36$	

Words and their equations

COMMON ERROR

Answering the wrong question

"A number is tripled and then 4 is added, giving 25" leads to $3x + 4 = 25$ and $x = 7$. If the question then asks for the value after tripling, the answer is 21, not 7. Always return to the question at the end.

WORKED EXAMPLE 6.5.1

A number problem

GIVEN Three consecutive whole numbers have a sum of 72.

ASKED Find the numbers.

METHOD Name the middle number x so the others are $x - 1$ and $x + 1$.

1. $(x - 1) + x + (x + 1) = 72$.
2. $3x = 72$.
3. $x = 24$, so the numbers are 23, 24 and 25.

ANSWER 23, 24, 25.

CHECK $23 + 24 + 25 = 72$.

WORKED EXAMPLE 6.5.2

A geometry problem

GIVEN A rectangle is twice as long as it is wide, and its perimeter is 54 cm.

ASKED Find the dimensions.

METHOD Name the width w, write the perimeter, and solve.

1. Width w, length $2w$.
2. Perimeter: $2w + 2(2w) = 6w$.
3. $6w = 54$, so $w = 9$ and the length is 18.

ANSWER 9 cm by 18 cm.

CHECK $2(9) + 2(18) = 54$.

WORKED EXAMPLE 6.5.3

A comparison problem

GIVEN A school hires a hall for a fixed 120 units plus 3 units per student. A second hall charges 6 units per student with no fixed charge.

ASKED For how many students do the two halls cost the same?

METHOD Write both costs as expressions and set them equal.

1. First hall: $3n + 120$; second: $6n$.
2. $3n + 120 = 6n$.
3. $120 = 3n$, so $n = 40$.

ANSWER 40 students, when each costs 240 units.

CHECK First hall $3(40) + 120 = 240$; second $6(40) = 240$.

WATCH OUT Forming the equation as $6n + 120 = 3n$ inverts the two halls and gives a negative answer, which is a signal that the expressions have been swapped.

PRACTICE 6.5 · Forming and using equations

1	A number plus 12 is 30. Write and solve an equation.	FLUENCY
2	Four times a number is 52.	FLUENCY
3	The perimeter of a square is 48 cm. Find the side.	FLUENCY
4	Two consecutive numbers sum to 41. Find them.	FLUENCY
5	Explain why the equation for "5 less than three times a number is 16" is $3x - 5 = 16$ rather than $5 - 3x = 16$.	REASONING
6	A student's answer to a length problem is -4 cm. What does that tell you?	REASONING
7	A phone costs 20 units plus 0.05 per minute. After how many minutes does the bill reach 32 units?	PROBLEM SOLVING
8	A rectangle has perimeter 40 cm and length 4 cm more than its width. Find its dimensions.	PROBLEM SOLVING
9	The sum of three consecutive even numbers is 84. Find them.	EXTENSION
10	A father is three times as old as his son; in 12 years he will be twice as old. Find their present ages.	EXTENSION

F INVESTIGATE AND APPLY

A criterion B investigation and a criterion D application

INVESTIGATION 6 | The matchstick chain

B · I, II, III

Return to the chain from the Explore activity and turn the conjecture into a general result with a justification.

INVESTIGATION
Investigation brief

Question. How does the number of matches in a chain of squares depend on the number of squares, and can the rule be justified rather than merely observed? **What you need.** Matchsticks or a drawing grid, a table, and the four stages you built. **What to produce.** A table to stage 8, a general rule, a direct justification from the structure of the drawing, and a test of a prediction.

PART 1 · EXTEND THE EVIDENCE

Stage n	1	2	3	4	5	6	7	8
Matches	4	7						

Copy and complete

PART 2 · GENERALISE

1. State the term-to-term rule in words.
2. State the position-to-term rule as an expression in n.
3. Use the rule to predict stage 25, then check by extending the table twice more.
4. Use an equation to find the stage that uses 100 matches. Is your answer a whole number, and what does that tell you?

PART 3 · JUSTIFY

1. The first square uses 4 matches. How many does each additional square use, and why?
2. Write a sentence that derives $3n + 1$ from the drawing rather than from the numbers.
3. A classmate says “the rule is $4n$ because each square uses 4 matches”. Explain what this rule counts.
4. Test the rule at stage 0 and interpret the answer in the drawing.

CRITERION TASK
Criterion B task · Requirements

Accurate table; a rule stated as an expression; predictions tested at two new stages; a justification that derives the rule from the structure; an evaluation of a rival rule.

APPLICATION 6 | The school event budget

D · I, II, III, IV, V

A student council is planning a concert. Costs are partly fixed and partly per ticket, and the ticket price must cover both.

THE BRIEF

Item	Cost
Hall hire	420 units
Sound equipment	180 units
Performer	300 units
Programme per ticket	1.20 units

Item	Cost
Ticket price under consideration	6 units

Costs and prices

- Fixed costs must be paid whatever the attendance.
- Each ticket sold adds the programme cost and brings in the ticket price.
- The council wants to know the break-even attendance and the profit at three possible attendances.

TASK 1 · BUILD THE EXPRESSIONS

1. Write an expression for the total cost when n tickets are sold.
2. Write an expression for the income when n tickets are sold.
3. Write an equation that represents breaking even.

TASK 2 · SOLVE AND INTERPRET

1. Solve the break-even equation. Round up if necessary and explain why.
2. Calculate the profit at 200, 300 and 400 tickets.
3. State the profit as a percentage of income at 300 tickets.

TASK 3 · EVALUATE THE PLAN

1. The council considers raising the ticket price to 8 units. Recalculate the break-even attendance.
2. Discuss one advantage and one risk of the higher price.
3. Recommend a price with two pieces of evidence from your calculations.

CRITERION TASK

Criterion D task · Requirements

Both expressions correct and interpreted; break-even solved and rounded with justification; profits calculated; a percentage used accurately; a recommendation with evidence and a stated assumption about attendance.

TASK C | Explaining the balancing method

C · I, III, IV

CRITERION TASK

The task

Write an explanation, for a student who has missed the lesson, of why an equation can be solved by doing the same operation to both sides.

1. Use a balance diagram to explain the idea.
2. Solve $4x + 5 = 29$ step by step, writing the reason beside each step.
3. Explain why subtracting 5 from only one side gives a wrong answer.
4. Show a check that proves your solution is correct.

Criterion C strand	Evidence
Notation (i)	Equations written with correct symbols and aligned steps.

Criterion C strand	Evidence
Reasoning (iii)	Each step has a reason, not just a result.
Organisation (iv)	The explanation can be followed by someone who was absent.

Success criteria

G REFLECT AND CONSOLIDATE
Summary, vocabulary, misconceptions and review

REVIEW | Consolidating Chapter 6
A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- A sequence can be described by a term-to-term rule or by a position-to-term rule.
- A constant difference d means the rule contains dn; the remainder fixes the constant.
- Expressions are evaluated by substitution, following the order of operations.
- Like terms can be collected; brackets are removed by multiplying every term inside.
- Linear equations are solved by inverse operations applied to both sides.
- Word problems become equations by naming the unknown and stating the relationship.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Sequence	A list of numbers in order, called terms.
Term	One number or element of a sequence.
Position-to-term rule	A rule giving the term from its position n.
Expression	Numbers and letters combined with operations, without an equals sign.
Coefficient	The number multiplying a letter.
Substitution	Replacing letters with numbers.
Like terms	Terms with identical letters and powers.
Equation	A statement that two expressions are equal.
Solution	A value that makes an equation true.
Linear	The letter appears only to the first power.

Vocabulary for Chapter 6

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence
"x^2 means $2x$."	It means $x \times x$.	$6^2 = 36$, $2 \times 6 = 12$
"The rule is the first term times n."	Find the difference first.	4, 7, 10 gives $3n + 1$
"$3a + 4b = 7ab$."	Unlike terms do not combine.	a and b count different quantities
"Subtract from one side."	Both sides must change.	$3x + 4 = 19$
"$-2(x - 5) = -2x - 10$."	Both signs change.	$-2 \times -5 = +10$

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
find the rule for a sequence and the term at any position.			
substitute values, including negatives, into an expression.			
simplify expressions and expand brackets.			
solve linear equations, including those with brackets.			
form an equation from a word problem and interpret the answer.			

How confident are you?

REFLECTION

1. Which is more useful to you: a term-to-term rule or a position-to-term rule? Why?
2. Describe a mistake you made with signs this chapter and how you now avoid it.
3. Write one sentence explaining what an equation is.
4. In the concert task, which unknown was hardest to name? Why?
5. Where else in mathematics have you seen a rule built from a pattern?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ6 · Applying the chapter

1	Write the next term: 6, 11, 16, …	FLUENCY
2	Find the rule for 9, 13, 17, 21, …	FLUENCY
3	Evaluate $7x - 4$ when $x = 5$.	FLUENCY
4	Simplify $6a + 2b - 2a + 3b$.	FLUENCY
5	Solve $5x + 3 = 28$.	FLUENCY
6	Solve $2(x + 3) = 18$.	FLUENCY
7	Explain why $4(x + 2)$ is not $4x + 2$.	REASONING
8	A sequence has rule $5n - 3$. Is 47 a term in it?	REASONING
9	Explain the difference between an expression and an equation.	REASONING
10	A taxi charges 5 plus 1.5 per km. Find the cost of 12 km.	PROBLEM SOLVING
11	Solve $7x - 2 = 4x + 13$.	PROBLEM SOLVING
12	Three consecutive numbers have a sum of 63. Find them.	PROBLEM SOLVING
13	Explain why collecting like terms does not change the value of an expression.	COMMUNICATING

14	Explain how to check a solution without solving the equation again.	COMMUNICATING
15	Solve $3(2x - 1) = 4x + 9$.	EXTENSION
16	Solve $\frac{x}{3} + 4 = \frac{x}{2}$.	EXTENSION
17	Find the rule for a sequence whose 3rd term is 11 and whose difference is 4.	EXTENSION
18	A rectangle has perimeter 48 cm and length twice its width. Find the area.	PROBLEM SOLVING
19	Explain why an equation with the letter on both sides can still be solved by balancing.	COMMUNICATING
20	A student says “the 100th term is always 100 times the first term”. Refute or support this with two sequences.	COMMUNICATING

MIXED 6 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX6 · End-of-chapter mixed set

1	Find the 12th term of $7n - 2$.	FLUENCY
2	Evaluate $4a^2$ when $a = 3$.	FLUENCY
3	Simplify $8 - 3(x - 2)$.	FLUENCY
4	Solve $\frac{x}{5} = 9$.	FLUENCY
5	Solve $6x - 5 = 3x + 10$.	FLUENCY
6	Is $2(3x + 4)$ equal to $6x + 8$? Justify.	REASONING
7	Explain why $x = -2$ is not a solution of $3x + 4 = 10$.	REASONING
8	A pattern uses 7 tiles in stage 1 and 11 in stage 2. How many in stage 15?	PROBLEM SOLVING
9	After how many months does a 30-unit monthly fee equal a 90-unit annual payment?	PROBLEM SOLVING
10	Solve $4(x - 3) = 2(x + 5)$.	PROBLEM SOLVING
11	Explain how to decide whether an expression can be simplified.	COMMUNICATING
12	Explain why the answer to a word problem should be checked against the context, not only the equation.	COMMUNICATING
13	Find two consecutive odd numbers whose sum is 64.	EXTENSION
14	Solve $\frac{2x}{3} - 1 = 5$.	EXTENSION
15	A number is multiplied by 4, then 7 is subtracted; the result is the same as multiplying by 3 and adding 5. Find the number.	EXTENSION
16	A rectangle's length is 5 cm more than its width and its perimeter is 46 cm. Find its area.	PROBLEM SOLVING

CUMULATIVE PRACTICE 6 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP6 · Cumulative practice

1	Find the rule for 8, 13, 18, 23, ...	FLUENCY
2	Simplify $7x + 4 - 3x + 6$.	FLUENCY
3	Solve $3x + 8 = 26$.	FLUENCY
4	Explain why an expression has no solution.	REASONING
5	A pattern uses 6 tiles in stage 1 and 10 in stage 2. How many in stage 12?	PROBLEM SOLVING
6	Solve $5(x - 3) = 2x + 6$.	PROBLEM SOLVING
7	Explain how to check that $x = 4$ solves $3x - 5 = 7$.	COMMUNICATING
8	Explain why $4n + 2$ and $4(n + 2)$ are different.	EXTENSION

PART IV

SHAPE, STRUCTURE AND MEASURE

IV

07 Lines, angles, triangles and constructions

08 Sets and classification

09 Perimeter, area, surface area and volume

PART IV · SHAPE, STRUCTURE AND MEASURE

CHAPTER 07

Lines, angles, triangles and constructions

A single straight line contains 180°, and a full turn contains 360°. From those two facts, and one statement about parallel lines, an entire system of shape reasoning follows. Constructions turn that reasoning into a repeatable physical procedure.

KEY CONCEPT	**Relationships**
RELATED	**Space**
RELATED	**Measurement**
GLOBAL CONTEXT	**Orientation in space and time**

07
CHAPTER

INQUIRY QUESTION

A workshop needs a triangular roof truss with equal slopes and a horizontal collar beam. Which angles must the carpenter set, how can the shape be checked without measuring every angle, and how can a full-size template be reproduced exactly?

A roof truss is a triangle of timbers. Equal base angles make the slopes symmetrical; a parallel collar creates equal angles; a perpendicular from the apex locates the centre. None of these facts needs measuring if the geometry is right.

IN THIS CHAPTER

7.1 Points, lines, angles and notation
7.2 Angle relationships
7.3 Parallel lines and transversals
7.4 Triangles
7.5 Polygons and their angles
7.6 Constructions with compasses

Investigation 7 The triangle inequality

Application 7 Laying out the school garden
Task C Writing up a construction
Review Consolidating Chapter 7
Review questions Practice across the whole chapter
Mixed 7 Fluency, reasoning and problem solving
Cumulative practice 7 Questions that mix this chapter with earlier ones

APPROXIMATELY 16 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

One fact generates a system. If a straight line holds 180°, then angles at a point hold 360°, a triangle holds 180°, and every polygon angle sum follows. Geometry is the study of what must be true, not what happens to look true in one drawing.

KEY CONCEPT · RELATIONSHIPS

Relationships. Angles are related by position: on a line, at a point, between parallel lines, in triangles and in polygons.

RELATED CONCEPT · SPACE

Space. Lines, angles and shapes describe how parts of a structure fit together. A construction locates a point using distance, not guesswork.

RELATED CONCEPT · MEASUREMENT

Measurement. Degrees measure turn. A construction enforces an exact relationship by repeating a radius; a measuring instrument only approximates.

GLOBAL CONTEXT

Orientation in space and time

EXPLORATION

How spatial relationships and precise construction allow structures to be designed, checked and reproduced.

FACTUAL

What is an angle, and how is it measured? Which angle facts hold on a line, at a point and between parallel lines?

CONCEPTUAL

Why is the angle sum of a triangle fixed even when the triangle changes? How does a construction guarantee a result that measurement can only approximate?

DEBATABLE

When is an exact construction more trustworthy than a careful measurement? Should a design follow the standard construction or adapt to the materials available?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Apply angle facts on lines, at points, between parallel lines, in triangles and in polygons; classify triangles; describe constructions accurately.
B · Investigating patterns	i, ii	Test sets of three lengths, record whether a triangle is possible, conjecture and justify the triangle inequality.
C · Communicating	i, ii, iii	Name angles and lines correctly, label diagrams, and write reasoned angle chains and construction steps.
D · Applying mathematics	i, ii, iii, iv	Design and check a roof truss, use scale and construction constraints, and evaluate a template.

ATL SKILL · THINKING SKILLS

In order for a student to explain why two measured angle sums of 179° and 181° do not refute the triangle angle sum, students must distinguish a measurement from a deduction and name one procedure that would reduce the variation.

STATEMENT OF INQUIRY

Spatial relationships obey angle and shape rules, and precise constructions let those rules be checked and reproduced in real structures.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand how angles are named and measured, and why angle facts on a line and at a point must be true.
- Calculate angles between parallel lines, in triangles and in polygons, giving a reason at each step.
- Classify triangles by sides and angles and use the exterior angle relationship.
- Construct a perpendicular bisector, an angle bisector and a triangle from three side lengths.
- Know the triangle inequality and use it to decide whether a construction is possible.
- Investigate which triples of lengths form a triangle and justify the general condition.

SUCCESS CRITERIA

- ✓ I can name and classify an angle using correct notation.
- ✓ I can find missing angles on a straight line, at a point and between parallel lines, with reasons.
- ✓ I can use the triangle and polygon angle sums to solve problems.
- ✓ I can explain why a construction produces the required point or line.
- ✓ I can decide whether three lengths can form a triangle.
- ✓ I can produce a clear scale drawing and evaluate its accuracy.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Solve $x + 40 = 180$.
2. Calculate 180 - 65.
3. Solve $3x = 180$.
4. Name a triangle with two equal sides.
5. How many degrees are in a quarter turn?
6. Calculate $2 \times (180 - 70)$.

Answers are in the Chapter 7 answer section. If more than two are incorrect, review angle measurement, solving one-step equations and the names of common triangles before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Recognising and measuring angles with a protractor; naming triangles and quadrilaterals; simple angle sums.	Angle facts with reasons; parallel lines; triangle and polygon sums; ruler-and-compass constructions; triangle inequality.	Congruence and similarity, Pythagoras' theorem, circle theorems, trigonometric ratios and geometric proof.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Roofs, ramps and frames are built from straight lines and triangles.	Which angles are forced once one angle is known, and when can a shape be built at all?	Relationships, space and measurement.	Which triples of lengths can make a triangle?	Labelled diagrams, angle chains with reasons, and construction instructions.	When does a drawing prove a relationship, and when does it only suggest one?

A ENGAGE
Three lines, one impossible triangle

A student cuts three strips of card: 4 cm, 5 cm and 12 cm, and tries to make a triangle. The strips will not close: however the two shorter strips are arranged, they cannot reach across the longest one.

The reason is arithmetic. The two shorter sides together (4 + 5 = 9 cm) are shorter than the longest side (12 cm), so the triangle cannot exist. Geometry is not only drawing: it makes statements that either can or cannot be true, and the check is often a calculation.

The same is true of angles. The three angles of any triangle add to exactly 180°, whatever the triangle looks like. That is not a coincidence of drawing but a consequence of how straight lines behave — and this chapter builds the argument.

Sides given	Sum of two shorter	Longest side	Possible?
4, 5, 12	9	12	no
4, 5, 6	9	6	yes
7, 7, 13	14	13	yes
3, 8, 11	11	11	no, the strips lie flat

Figure 7.1 Three attempted triangles

NOTICE AND WONDER

- What has to be true of three lengths before they can form a triangle?
- Is there a largest possible third side once two sides are fixed?
- Why do the angles of a triangle always total 180°?
- What would a triangle with angles 100°, 50° and 40° look like, and why can it not exist?

Keep this question open: how can a property of a shape be certain for every case, when no one can check every case?

B EXPLORE
The triangle inequality

Work in pairs. You need strips of card or straws cut to different lengths, a ruler and a protractor.

YOUR INVESTIGATION

Step 1. Cut strips of 3 cm, 4 cm, 5 cm, 6 cm, 8 cm, 10 cm and 12 cm.

Step 2. Choose three strips and try to form a triangle. Record whether you succeed.

Step 3. Repeat with at least eight different combinations, including cases where one strip is much longer than the other two.

Step 4. For each combination, record the sum of the two shorter strips and compare it with the longest strip.

Lengths	Sum of two shorter	Longest	Triangle formed?
3, 4, 5			
3, 4, 8			
5, 6, 10			

Lengths	Sum of two shorter	Longest	Triangle formed?
5, 5, 10			
6, 8, 12			
4, 7, 10			

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

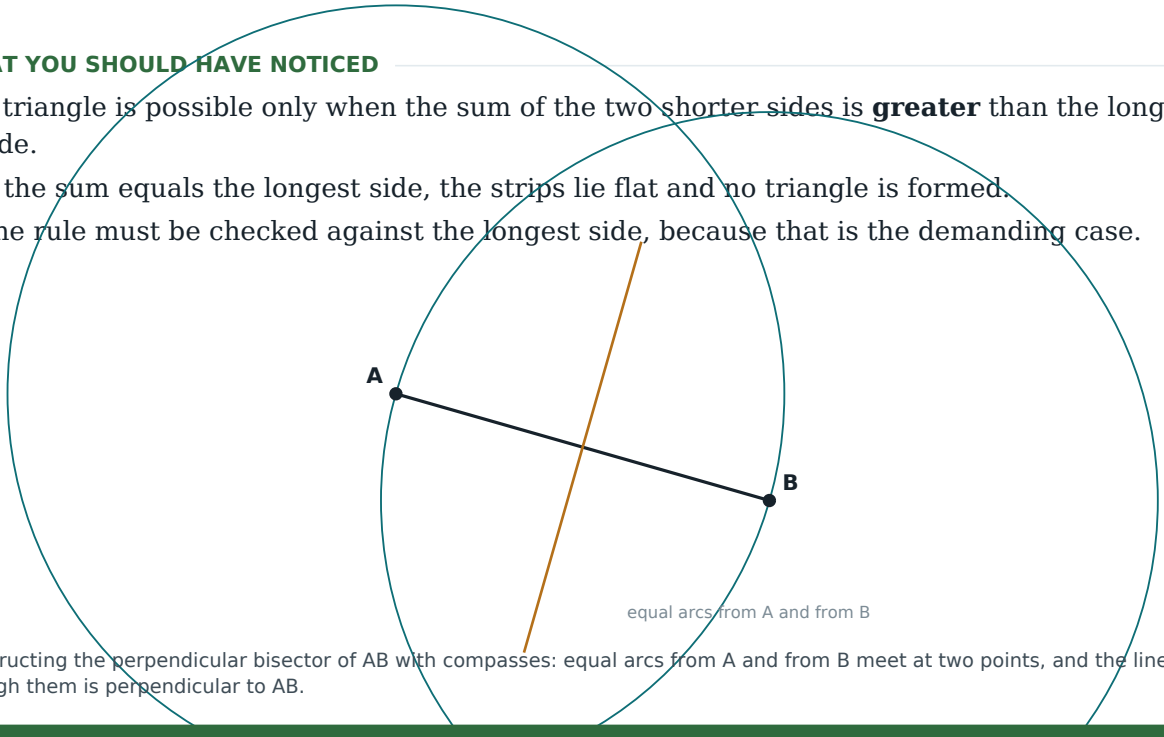
1. In the cases where a triangle was formed, how did the sum of the two shorter lengths compare with the longest?
2. What happens when the sum equals the longest length exactly? Build it and describe the result.
3. If two sides are 6 cm and 9 cm, what is the smallest and largest whole-number third side possible?
4. Does the rule depend on which side you call the longest? Explain.
5. A triangle has sides 5 cm, 5 cm and 5 cm. Does the rule hold? What kind of triangle is it?

MAKING A CONJECTURE

> “Three lengths form a triangle exactly when … because … .”

WHAT YOU SHOULD HAVE NOTICED

- A triangle is possible only when the sum of the two shorter sides is **greater** than the longest side.
- If the sum equals the longest side, the strips lie flat and no triangle is formed.
- The rule must be checked against the longest side, because that is the demanding case.



Constructing the perpendicular bisector of AB with compasses: equal arcs from A and from B meet at two points, and the line through them is perpendicular to AB.

INVESTIGATION

Explore further

Using a ruler and compasses, try to construct a triangle with sides 5 cm, 6 cm and 13 cm. Describe exactly where the construction fails, and explain why the failure was predictable before you began.

7.1 | Points, lines, angles and notation

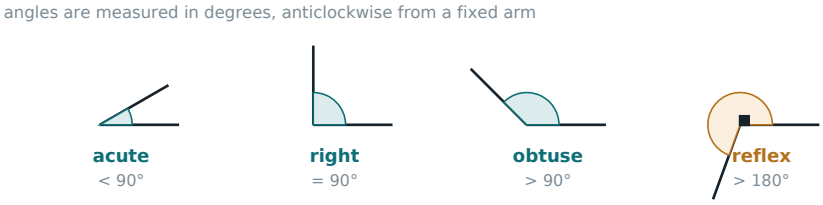
A · I, II

Geometry has its own vocabulary and notation, and precision in naming is part of the mathematics. A wrong label can make a correct calculation unreadable.

DEFINITION

Naming and measuring

A **point** marks a position and is named with a capital letter. A **line** extends without end in both directions; a **line segment** has two endpoints; a **ray** starts at a point and extends in one direction. An **angle** is formed where two rays meet at a **vertex**. It is measured in **degrees** (°) and named with three letters, with the vertex in the middle: angle ABC has its vertex at B.



Angles are classified by size: acute, right, obtuse and reflex.

Type	Size	Example in a clock face
Acute	between 0° and 90°	2 o'clock (60°)
Right	exactly 90°	3 o'clock
Obtuse	between 90° and 180°	5 o'clock (150°)
Straight	exactly 180°	6 o'clock
Reflex	between 180° and 360°	8 o'clock (240°)

Angle types

KEY IDEA

Notation that is worth learning exactly

- AB means the length of the segment from A to B; the line through them is written $overleftrightarrow{AB}$.
- $\angle ABC$ names the angle at B between BA and BC.
- Two lines meeting at right angles are **perpendicular**; lines that never meet however far they extend are **parallel**.
- Angles on the same arc are marked with matching arcs; equal angles are shown with identical arc marks.

METHOD

Measuring and drawing an angle

1. Place the centre of the protractor on the vertex, with the baseline along one arm.
2. Read the scale that starts at 0° on that arm — there are two scales, and using the wrong one gives an answer that is wrong by the supplement.
3. To draw an angle, draw the first arm, place the centre at the vertex, mark the required number of degrees, then join.
4. Check the result is sensible: an acute angle must look smaller than a right angle.

COMMON ERROR

Reading the wrong scale

A protractor has two scales, inner and outer. Reading the outer scale when the arm lies on the inner zero gives $180° - \theta$ instead of θ. If the measured angle is 150° when the drawing is clearly acute, the wrong scale has been read.

WORKED EXAMPLE 7.1.1

Naming and classifying

GIVEN Three rays from a point O: OA at 0°, OB at 62°, OC at 145°.

ASKED Name and classify angle AOB, angle BOC and angle AOC.

METHOD Subtract the readings and classify by size.

1. $\angle AOB = 62° - 0° = 62°$, acute.
2. $\angle BOC = 145° - 62° = 83°$, acute.
3. $\angle AOC = 145° - 0° = 145°$, obtuse.

ANSWER 62° acute, 83° acute, 145° obtuse.

CHECK $\angle AOB + \angle BOC = \angle AOC$, as the rays lie in order.

WORKED EXAMPLE 7.1.2

Using a right angle

GIVEN Two straight lines crossing, with one angle marked 37°.

ASKED Find the three other angles at the crossing.

METHOD Use the straight-line property and the equality of vertically opposite angles.

1. The angle next to 37° on a straight line is $180° - 37° = 143°$.
2. Vertically opposite angles are equal, so the other two are 37° and 143°.

ANSWER 37°, 143°, 143°.

CHECK $37 + 143 + 37 + 143 = 360°$, a full turn.

WORKED EXAMPLE 7.1.3

A reasoning step with notation

GIVEN A, B and C lie on a straight line in that order, with $\angle ABD = 118°$.

ASKED Find $\angle DBC$, giving a reason.

METHOD Angles on a straight line sum to 180°.

1. $\angle ABD + \angle DBC = 180°$ (angles on a straight line).
2. $\angle DBC = 180° - 118° = 62°$.

ANSWER 62°.

CHECK $118 + 62 = 180$.

WATCH OUT Naming the reason is part of the answer. A figure without a reason earns fewer marks than a figure with one.

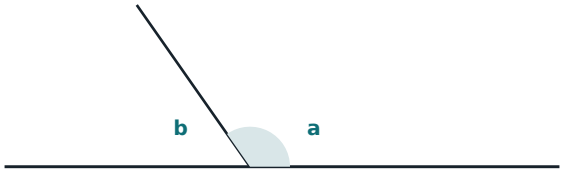
PRACTICE 7.1 · Points, lines, angles and notation

1	Classify each angle: 45°, 90°, 120°, 200°.	FLUENCY
2	Two angles on a straight line are 68° and x. Find x.	FLUENCY
3	Four angles meet at a point. Three are 80°, 95° and 110°. Find the fourth.	FLUENCY
4	Write the angle at vertex B between BA and BC using three letters.	FLUENCY
5	Explain why two angles on a straight line cannot both be obtuse.	REASONING
6	A student measures an acute angle as 130°. What has probably gone wrong?	REASONING
7	The angles at a point are $2x$, $3x$ and $4x$. Find x and each angle.	PROBLEM SOLVING
8	A clock shows 4 o'clock. Find the angle between the hands.	PROBLEM SOLVING
9	Two straight lines cross. One angle is $3y$ and the adjacent is $2y + 30$. Find y.	EXTENSION
10	Explain why vertically opposite angles are equal, using a straight line twice.	EXTENSION

7.2 | Angle relationships

A · II · C · III

Three relationships do most of the work in angle problems: angles on a straight line, angles at a point, and vertically opposite angles. Each is a consequence of a full turn being 360°.



Two angles on a straight line. However the line is divided, the two angles add to 180°.

DEFINITION
Three angle facts

- **Angles on a straight line** add to 180°.
- **Angles at a point** add to 360°.
- **Vertically opposite angles** — the angles opposite each other where two lines cross — are equal.

KEY IDEA
Why all three follow from one idea

A full turn is 360°. Two angles that together make a straight line make half a turn, so they add to 180°. Angles around a point make a whole turn, so they add to 360°. Vertically opposite angles are equal because each is 180° minus the same adjacent angle.

METHOD
Solving an angle problem with reasons

1. Mark on the diagram what you know and what you want.
2. Choose the relationship that connects them: straight line, point, or vertically opposite.
3. Write an equation; solve it.
4. State the reason with the answer, in words or in brackets.
5. Check that the angles are plausible: no angle in a triangle or on a line can be negative.

Diagram feature	Relationship	Equation
Two angles making a straight line	sum to 180°	$a + b = 180$
Angles all the way round a point	sum to 360°	$a + b + c + d = 360$
Two crossing lines	vertically opposite equal	$a = c$
Three angles on a line	sum to 180°	$a + b + c = 180$

Which relationship to use

COMMON ERROR
Assuming a diagram is to scale

A diagram may show an angle that looks like 90° but is labelled 85°. Never measure a diagram to find an answer, and never assume two lines are parallel because they look parallel. Use the given information and the named relationships only.

WORKED EXAMPLE 7.2.1

Angles on a straight line

GIVEN Three angles on a straight line are x, $2x$ and $3x + 30$.

ASKED Find x and the three angles.

METHOD Use the straight-line sum and solve the equation.

1. $x + 2x + 3x + 30 = 180$.
2. $6x = 150$.
3. $x = 25$, so the angles are 25°, 50° and 105°.

ANSWER $x = 25$; angles 25°, 50°, 105°.

CHECK $25 + 50 + 105 = 180$.

WORKED EXAMPLE 7.2.2

Vertically opposite angles

GIVEN Two straight lines cross. One angle is $4a - 10$ and the vertically opposite angle is $2a + 30$.

ASKED Find a and the size of both angles.

METHOD Set the vertically opposite angles equal.

1. $4a - 10 = 2a + 30$.
2. $2a = 40$, so $a = 20$.
3. Each angle is $4(20) - 10 = 70°$.

ANSWER $a = 20$; both angles 70°.

CHECK $2(20) + 30 = 70$; the adjacent angles are 110° each, and $70 + 110 + 70 + 110 = 360$.

WORKED EXAMPLE 7.2.3

Angles at a point

GIVEN Five angles meet at a point. Four are 40°, 65°, 90° and 105°.

ASKED Find the fifth, with a reason.

METHOD Use the point sum.

1. $40 + 65 + 90 + 105 = 300$.
2. $360 - 300 = 60$.

ANSWER 60°, because angles at a point sum to 360°.

CHECK $300 + 60 = 360$.

WATCH OUT A common slip is to use 180° at a point. Check whether the angles surround the point completely or lie on one side of a line.

PRACTICE 7.2 · Angle relationships

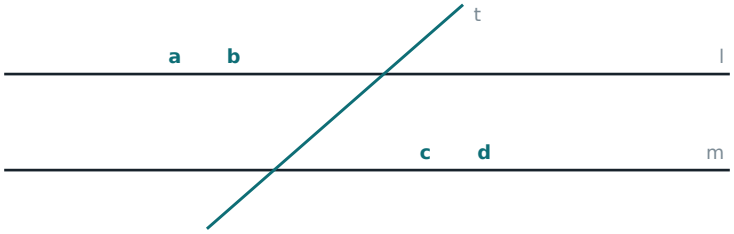
1	Two angles on a straight line: one is 47°. Find the other.	FLUENCY
2	Angles at a point: 120°, 130° and x. Find x.	FLUENCY
3	Two lines cross; one angle is 55°. Find the other three.	FLUENCY
4	Three angles on a line are 90°, x and 35°. Find x.	FLUENCY
5	Explain why vertically opposite angles are equal.	REASONING
6	A student finds an angle of 200° on a straight line. Is that possible?	REASONING

7	The angles at a point are x, $2x$, $3x$ and $4x$. Find each.	PROBLEM SOLVING
8	Two crossing lines form an angle of $3x + 10$ and its adjacent angle $2x$. Find x.	PROBLEM SOLVING
9	Four angles at a point are in the ratio $1 : 2 : 3 : 4$. Find the largest.	EXTENSION
10	Explain why the angles at a point cannot all be acute.	EXTENSION

7.3 | Parallel lines and transversals

A · II, III

When a line crosses two parallel lines, eight angles appear, and they fall into only two sizes. That is a strong result, and it is the basis of all reasoning about parallel lines.



A transversal crossing two parallel lines makes eight angles. Corresponding angles are equal, and alternate angles are equal.

DEFINITION

Corresponding and alternate angles

A **transversal** is a line that crosses two or more other lines. When it crosses **parallel** lines:

- **Corresponding angles** — in the same position at each crossing — are equal.
- **Alternate angles** — on opposite sides of the transversal, between the parallel lines — are equal.
- **Co-interior angles** — on the same side of the transversal, between the parallel lines — add to 180°.

KEY IDEA

Why only two sizes appear

Corresponding angles are equal because a translation along the transversal maps one crossing onto the other. Once corresponding angles are equal, alternate angles follow by vertically opposite angles, and co-interior angles follow from a straight line. One fact generates the other two.

METHOD

Finding angles between parallel lines

1. Identify the parallel lines and mark them with matching arrows.
2. Identify the transversal.
3. Find a corresponding or alternate partner for the angle you know.
4. Continue step by step, naming the relationship each time, until you reach the required angle.

Relationship	Position	What is true
Corresponding	same corner at each crossing	equal
Alternate	opposite sides, inside	equal
Co-interior	same side, inside	sum to 180°

The three relationships in one diagram

COMMON ERROR

Using alternate angles that are not alternate

Z-shaped pairs of angles are alternate and equal. C-shaped pairs inside the parallel lines are co-interior and supplementary, not equal. The letter shape is a useful check: Z means equal, C means add to 180°.

WORKED EXAMPLE 7.3.1

One angle, then the rest

GIVEN A transversal crosses two parallel lines, and one angle is 68°.

ASKED Find the alternate angle, the corresponding angle and the co-interior angle, with reasons.

METHOD Apply each relationship in turn.

1. Alternate angle: 68°, alternate angles are equal.
2. Corresponding angle: 68°, corresponding angles are equal.
3. Co-interior angle: 180° - 68° = 112°, co-interior angles sum to 180°.

ANSWER 68°, 68°, 112°.

CHECK All eight angles are 68° or 112°, and the sizes alternate around each crossing.

WORKED EXAMPLE 7.3.2

An algebraic parallel-line problem

GIVEN Two parallel lines cut by a transversal. A corresponding pair is $3x + 15$ and $5x - 25$.

ASKED Find x and the angle.

METHOD Set corresponding angles equal.

1. $3x + 15 = 5x - 25$.
2. $40 = 2x$, so $x = 20$.
3. Angle: $3(20) + 15 = 75°$.

ANSWER $x = 20$; the angles are 75°.

CHECK $5(20) - 25 = 75$.

WORKED EXAMPLE 7.3.3

Deciding whether lines are parallel

GIVEN A transversal crosses two lines. A pair of alternate angles measures 74° and 76°.

ASKED Decide whether the lines are parallel.

METHOD Use the converse of the alternate-angle property.

1. If the lines were parallel, the alternate angles would be equal.
2. $74° \neq 76°$, so the lines are not parallel.

ANSWER Not parallel; the alternate angles differ by 2°.

CHECK Two lines meeting at a very small angle look nearly parallel, which is why the calculation matters.

WATCH OUT "They look parallel" is not a reason. The converse properties are the test.

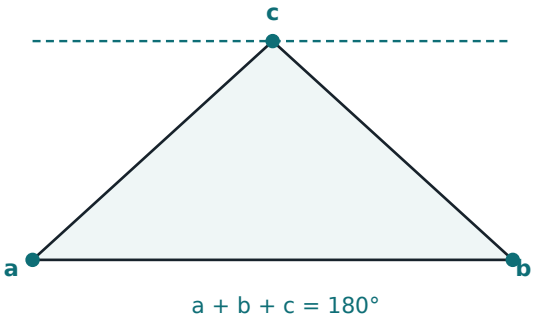
PRACTICE 7.3 · Parallel lines and transversals

1	Two parallel lines are cut by a transversal; one angle is 110°. Find the co-interior angle.	FLUENCY
2	One alternate angle is 42°. Find its partner.	FLUENCY
3	One corresponding angle is 128°. Find the vertically opposite angle to its partner.	FLUENCY
4	Two parallel lines; a transversal makes an angle of 65°. Find the four angles at the second crossing.	FLUENCY
5	Explain the difference between alternate and co-interior angles.	REASONING
6	A student says two lines are parallel because the corresponding angles "look equal". What should they do instead?	REASONING
7	Corresponding angles are $4x$ and $2x + 40$. Find x.	PROBLEM SOLVING
8	Co-interior angles are $3x$ and $2x + 30$. Find x.	PROBLEM SOLVING
9	Alternate angles are $5y - 10$ and $3y + 20$. Are the lines parallel? Find y if they are.	EXTENSION
10	Prove that co-interior angles sum to 180°, using corresponding and straight-line angles.	EXTENSION

7.4 | Triangles

A · II, III · B · III

A triangle is the simplest closed shape and the most constrained: three angles summing to 180° and three sides obeying the triangle inequality. Both facts generate a great deal of reasoning.



The three angles of any triangle add to 180°. The dashed line through the apex, parallel to the base, shows why: the three angles fit on a straight line.

DEFINITION
Triangles by sides and by angles

By sides: **equilateral** (three equal sides), **isosceles** (two equal sides), **scalene** (no equal sides). By angles: **acute**, **right-angled**, **obtuse**. The angles of any triangle add to 180°. In an isosceles triangle the angles opposite the equal sides are equal.

KEY IDEA
Why the angle sum is 180°

Draw a line through the apex parallel to the base. The two base angles then match alternate angles on that line, and the apex angle sits between them. Together the three angles make a straight line, so they sum to 180°. The proof uses only the parallel-line property, which is why that property was established first.

Triangle	Angles	Reason
Equilateral	60°, 60°, 60°	three equal angles summing to 180°
Right-angled	one angle 90°, other two sum to 90°	angle sum
Isosceles	two equal base angles	equal sides give equal angles
Obtuse	one angle above 90°	definition

Angle facts for special triangles

METHOD
Finding unknown angles in a triangle

1. Write the angle sum equation: the three angles total 180°.
2. Use any extra information: equal sides give equal angles, a right angle is 90°.
3. Solve for the unknown.
4. Check the answer against the type of triangle: an equilateral triangle must give 60° three times.

COMMON ERROR
Using 360° for a triangle

The 360° figure applies to angles at a point, not inside a triangle. A quick check: a triangle fits inside a straight angle, so its total must be less than 180° plus the exterior angle — and exactly 180°.

WORKED EXAMPLE 7.4.1

Two angles given

GIVEN A triangle has angles 48° and 67°.

ASKED Find the third angle and classify the triangle.

METHOD Subtract from 180°.

1. 48 + 67 = 115.
2. 180 - 115 = 65.

ANSWER 65°; all angles acute, so the triangle is acute-angled.

CHECK 48 + 67 + 65 = 180.

WORKED EXAMPLE 7.4.2

An isosceles triangle

GIVEN An isosceles triangle has a vertex angle of 40°.

ASKED Find the base angles, with reasons.

METHOD Use the angle sum and the equality of base angles.

1. The base angles are equal, so each is x with $2x + 40 = 180$.
2. $2x = 140$, so $x = 70$.

ANSWER Each base angle is 70°, because the angles opposite the equal sides are equal.

CHECK 40 + 70 + 70 = 180.

WORKED EXAMPLE 7.4.3

A triangle with algebra

GIVEN The angles of a triangle are x, $2x + 10$ and $3x - 10$.

ASKED Find each angle and classify the triangle.

METHOD Form the angle-sum equation.

1. $x + 2x + 10 + 3x - 10 = 180$.
2. $6x = 180$, so $x = 30$.
3. Angles: 30°, 70°, 80°.

ANSWER 30°, 70°, 80°; acute-angled.

CHECK 30 + 70 + 80 = 180.

WATCH OUT A negative or zero angle is a signal that the equation is wrong, not that the triangle is unusual.

PRACTICE 7.4 · Triangles

1	Two angles of a triangle are 55° and 65°. Find the third.	FLUENCY
2	A right-angled triangle has one acute angle of 32°. Find the other.	FLUENCY
3	An equilateral triangle: find each angle.	FLUENCY
4	An isosceles triangle has base angles of 50°. Find the vertex angle.	FLUENCY
5	Explain why a triangle cannot have two right angles.	REASONING
6	A triangle has angles 100°, 45° and 35°. What type is it?	REASONING
7	The angles of a triangle are in the ratio 2 : 3 : 4. Find them.	PROBLEM SOLVING

8	An isosceles triangle has a vertex angle of 110°. Find the base angles.	PROBLEM SOLVING
9	The angles of a triangle are $2x$, $3x$ and $4x + 45$. Find x and the largest angle.	EXTENSION
10	Explain why the largest angle of a triangle must be at least 60°.	EXTENSION

7.5 | Polygons and their angles

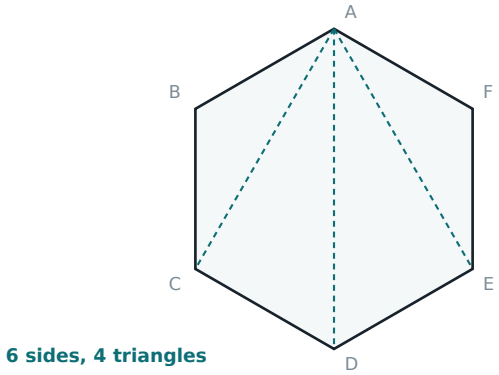
A · II, III · C · II

A polygon is a closed flat shape made of straight sides that do not cross. The angles of any polygon can be found from the angles of triangles, and that is why the triangle result in 7.4 is worth so much.

DEFINITION

Polygons, regular and irregular

A **polygon** has straight sides and no curves. It is **regular** when every side is the same length and every interior angle is equal; otherwise it is **irregular**. A polygon is **convex** when every interior angle is less than 180°, so no corner points inwards.



A hexagon split into triangles from one vertex. The diagonals divide it into four triangles, so its interior angles total 4 × 180° = 720°.

KEY IDEA

The interior angle sum

Drawing all the diagonals from one vertex splits an n-sided polygon into n - 2 triangles. Each triangle holds 180°, so the interior angles of the polygon total

$$(n - 2) \times 180°$$

For a quadrilateral: 2 × 180° = 360°. For a pentagon: 540°. For a hexagon: 720°. For a decagon: 1440°.

METHOD

Working with regular polygons

One interior angle: divide the total by n. A regular pentagon has 540° ÷ 5 = 108°. **Exterior angle:** each exterior angle is 360° ÷ n, because the exterior angles of any convex polygon add to a full turn. A regular pentagon has 360° ÷ 5 = 72°. **Check:** interior + exterior = 180° at every vertex, since they sit on a straight line.

Polygon	Sides n	Interior sum	Each interior	Each exterior
Triangle	3	180°	60°	120°
Quadrilateral	4	360°	90°	90°
Pentagon	5	540°	108°	72°
Hexagon	6	720°	120°	60°
Octagon	8	1080°	135°	45°
Decagon	10	1440°	144°	36°

Angle facts for regular polygons

COMMON ERROR

Assuming every quadrilateral is a rectangle

A rectangle has four right angles, but a quadrilateral need not. In a kite or a trapezium the four angles still total 360°, yet none of them may be 90°. The sum is fixed; the individual angles are not.

KEY IDEA

Why the exterior angles total 360°

Walk once around the outside of any convex polygon. At each vertex you turn through the exterior angle, and after the last vertex you are facing the way you started — one full turn, 360°. This is why 360 ÷ n gives the exterior angle of a regular polygon and why the number of sides can be recovered from one exterior angle.

WORKED EXAMPLE 7.5.1

Interior angles of a hexagon

GIVEN A regular hexagon.

ASKED Find each interior angle and each exterior angle.

METHOD Use the interior sum, then divide.

1. $n = 6$, so the interior sum is $(6 - 2) \times 180° = 720°$.
2. Each interior angle is $720° \div 6 = 120°$.
3. Each exterior angle is $180° - 120° = 60°$, which also equals $360° \div 6$.

ANSWER 120° interior, 60° exterior.

CHECK Six exterior angles of 60° make 360°, a full turn.

WORKED EXAMPLE 7.5.2

Working backwards to the number of sides

GIVEN A regular polygon whose interior angles are each 156°.

ASKED How many sides does it have?

METHOD Find the exterior angle first; it is smaller and divides 360 exactly.

1. Exterior angle = 180° - 156° = 24°.
2. $n = 360° \div 24° = 15$.
3. Check: (15 - 2) × 180° = 2340°, and 2340° ÷ 15 = 156°.

ANSWER 15 sides.

WATCH OUT Dividing 360 by the interior angle would give 2.3, which cannot be a number of sides. The exterior angle is the one that shares the full turn.

WORKED EXAMPLE 7.5.3

An irregular polygon

GIVEN A pentagon with angles 100°, 115°, 95°, 130° and x.

ASKED Find x.

METHOD Interior sum for five sides is 540°.

1. 100 + 115 + 95 + 130 = 440.
2. $x = 540 - 440 = 100$.

ANSWER $x = 100°$.

CHECK 440 + 100 = 540, and every angle is below 180°, so the pentagon is convex.

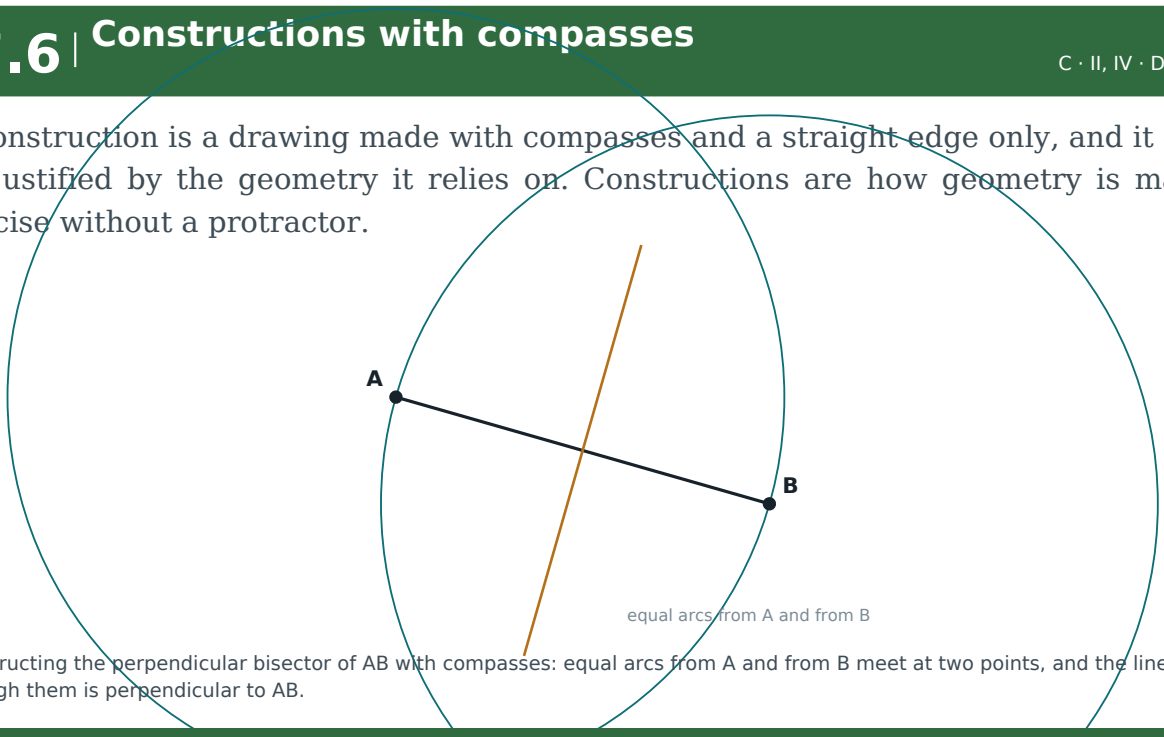
PRACTICE 7.5 · Polygons and their angles

1	Find the interior angle sum of: (a) a pentagon (b) an octagon (c) a dodecagon (12 sides)	FLUENCY
2	Find each interior angle of a regular: (a) hexagon (b) octagon (c) nonagon (9 sides)	FLUENCY
3	Find each exterior angle of a regular: (a) pentagon (b) decagon (c) 20-sided polygon	FLUENCY
4	A quadrilateral has angles 85°, 95° and 110°. Find the fourth.	FLUENCY
5	Explain why the interior angles of a quadrilateral total 360°.	REASONING
6	A student says a regular polygon can have interior angles of 150°. How many sides would it have, and how do you know?	REASONING
7	A regular polygon has an exterior angle of 24°. How many sides has it, and what is each interior angle?	PROBLEM SOLVING
8	A pentagon has four angles of 100°. Find the fifth angle and explain why the pentagon is not regular.	PROBLEM SOLVING
9	Two regular polygons have exterior angles of 30° and 45°. Find the difference between their numbers of sides.	EXTENSION
10	Explain why a convex polygon cannot have an interior angle of 200°.	EXTENSION

7.6 | Constructions with compasses

C · II, IV · D · II

A construction is a drawing made with compasses and a straight edge only, and it can be justified by the geometry it relies on. Constructions are how geometry is made precise without a protractor.



Constructing the perpendicular bisector of AB with compasses: equal arcs from A and from B meet at two points, and the line through them is perpendicular to AB.

DEFINITION

Construction

A **construction** is an accurate drawing made using only a straight edge and compasses. **Arcs** mark equal distances, and the intersections of arcs locate points exactly. Two constructions appear in this chapter: the **perpendicular bisector** of a segment, and the **bisector of an angle**.

METHOD

Perpendicular bisector of *AB*

1. Open the compasses to more than half of AB.
2. With the point at A, draw arcs above and below the segment.
3. With the same opening and the point at B, draw arcs to cut the first pair.
4. Draw the line through the two intersections. It is perpendicular to AB and passes through its midpoint.

METHOD

Bisector of an angle

1. With the point at the vertex, draw an arc cutting both arms.
2. With the point at each intersection, draw arcs of equal radius that cross inside the angle.
3. Draw the line from the vertex through the crossing point. It divides the angle into two equal parts.

KEY IDEA

Why constructions can be trusted

Every arc is drawn with a fixed compass opening, so every point on it is the same distance from the centre. The perpendicular bisector works because its two points are each equidistant from A and B, and the set of such points is exactly that line. Nothing is estimated; the accuracy comes from equal distances.

Construction	What is drawn	Why it works
Perpendicular bisector	arcs of equal radius from both ends	points equidistant from the endpoints
Angle bisector	arcs at the vertex, then equal arcs inside	equal distances from both arms
Equilateral triangle	two arcs of radius equal to the base	three equal sides
Triangle from three sides	arcs at each end of the base	the third vertex is fixed by two distances

Construction and its justification

COMMON ERROR

Rubbing out the construction lines

The arcs are the evidence that the construction is valid. Removing them leaves a drawing that cannot be checked. In an examination, construction arcs are usually required for full marks, and they should be drawn lightly but visibly.

WORKED EXAMPLE 7.6.1

Constructing a perpendicular bisector

GIVEN A segment AB of length 8 cm.

ASKED Construct its perpendicular bisector and state the distance from A to the midpoint.

METHOD Follow the standard construction.

1. Set the compasses to more than 4 cm, say 6 cm.
2. Draw arcs from A and from B, above and below.
3. Join the two intersections.
4. The bisector crosses AB at 4 cm from A.

ANSWER The bisector meets AB at its midpoint, 4 cm from A, at 90°.

CHECK Measure both halves and both angles; each pair should be equal.

WORKED EXAMPLE 7.6.2

Constructing a triangle from three sides

GIVEN Sides 7 cm, 5 cm and 4 cm.

ASKED Construct the triangle and decide whether a different triangle with the same sides is possible.

METHOD Draw the longest side as base, then locate the third vertex with two arcs.

1. Check the triangle inequality: $5 + 4 = 9 > 7$, so a triangle exists.
2. Draw $AB = 7$ cm.
3. With centre A and radius 5 cm draw an arc; with centre B and radius 4 cm draw an arc.
4. Join the intersection to A and B.

ANSWER The triangle is fixed: the two arcs meet at one point above AB and its mirror image below, giving the same triangle.

CHECK Measuring the angles of both versions gives the same three angles.

WATCH OUT If the two shorter sides barely exceed the longest, the arcs meet at a very shallow angle and small errors in compass opening produce a visibly wrong triangle.

WORKED EXAMPLE 7.6.3

Using a construction to solve a problem

GIVEN A school wants a flagpole equidistant from two gateposts A and B, and also on the path that runs between them.

ASKED Describe where the flagpole should stand.

METHOD Translate the conditions into a geometric locus.

1. Equidistant from A and B means on the perpendicular bisector of AB.
2. On the path means on the segment AB.
3. The only point on both is the midpoint of AB.

ANSWER At the midpoint of AB, found by constructing the perpendicular bisector.

CHECK The midpoint is 4 cm from each end on a drawing of an 8 cm path.

PRACTICE 7.6 · Constructions with compasses

1	What compass opening is needed to construct the perpendicular bisector of a 10 cm segment?	FLUENCY
2	How many arcs are drawn in the angle-bisector construction before the bisector itself?	FLUENCY
3	Construct an equilateral triangle of side 5 cm. Which two arcs locate the apex?	FLUENCY
4	State the size of the angles formed by a perpendicular bisector.	FLUENCY
5	Explain why the arcs must not be rubbed out.	REASONING
6	Why must the compass opening exceed half the segment?	REASONING
7	Construct a triangle with sides 6 cm, 6 cm and 4 cm. What type is it?	PROBLEM SOLVING
8	A point must be equidistant from two towns 30 km apart. Describe its possible positions.	PROBLEM SOLVING
9	Explain how to construct an angle of 45° using only compasses and a straight edge.	EXTENSION

10 Explain why the three angle bisectors of a triangle meet at a single point. **EXTENSION**

F INVESTIGATE AND APPLY

A criterion B investigation and a criterion D application

INVESTIGATION 7 | The triangle inequality

B · I, II, III

Turn the Explore activity into a general statement with a justification, and test its limits.

INVESTIGATION

Investigation brief

Question. Exactly which triples of lengths can form a triangle, and why does the rule hold for every case? **What you need.** Strips or compasses, a table of lengths, and your recording sheet. **What to produce.** A general statement, a justification from the geometry, a test on a limiting case, and a prediction for a new triple.

PART 1 · COLLECT AND DESCRIBE

Lengths	Two shorter sides	Their sum	Longest	Triangle?
2, 3, 4				
2, 3, 5				
2, 3, 6				
6, 8, 10				
5, 12, 13				
7, 9, 15				

Copy and complete

PART 2 · GENERALISE

1. Write the rule as a statement about the sum of the two shorter sides.
2. Write the rule as three inequalities, one for each side.
3. Two sides are 8 cm and 11 cm. Between which values must the third side lie? Give the answer as an inequality.
4. Predict, without drawing, whether 9 cm, 14 cm and 24 cm form a triangle, then check.

PART 3 · JUSTIFY AND EXTEND

1. Explain, using the shortest path between two points, why the rule must be true.
2. Describe the shape formed when the sum of the two shorter sides equals the longest exactly.
3. What happens to the area of a triangle as the third side approaches the limiting value?
4. Is there an equivalent rule for four lengths forming a quadrilateral? Test it with strips.

CRITERION TASK
Criterion B task · Requirements

Complete and correct table; the rule stated both in words and as inequalities; a justification from the shortest-path idea; a limiting case described; a prediction tested.

APPLICATION 7 | Laying out the school garden

D · I, II, III, IV

A garden is to be laid out in a triangular plot. The design must satisfy the client's conditions and be constructible accurately on the ground.

THE BRIEF

Constraint	Value
Two sides of the plot	9 m and 12 m
Area must fit inside a corner bounded by two paths	the angle between the sides must be at least 40° and at most 60°
Fence panels	2 m each, no cutting
Path along the longest side	1.5 m wide

The plot

- The third side must allow the angle condition to be met.
- Fence panels must span the perimeter exactly, so the perimeter must be a multiple of 2 m where possible.
- The design must be drawn to scale and justified with angle calculations.

TASK 1 · FIX THE TRIANGLE

1. Using the triangle inequality, state the possible range for the third side.
2. Choose a third side that gives an angle of about 50° between the 9 m and 12 m sides, and draw the triangle to scale.
3. Measure the third side on your drawing and compare it with the length you chose.

TASK 2 · CHECK THE CONSTRAINTS

1. Calculate the perimeter and decide how many fence panels are needed, and whether any adjustment is necessary.
2. Check that the angle lies within the range allowed.
3. State one reason why an exact fit may be impossible and how you would resolve it.

TASK 3 · JUSTIFY THE DESIGN

1. Explain the role of the triangle inequality in your design, in two sentences.
2. Explain the role of the construction in making the design buildable.
3. Present your recommendation with a scale drawing, a table of measurements and one stated assumption.

CRITERION TASK
Criterion D task · Requirements

Correct use of the triangle inequality; an accurate scale drawing with arcs shown; angle checked against the constraint; perimeter and panels calculated; a decision justified with reference to the constraints.

TASK C | Writing up a construction

C · I, II, IV

CRITERION TASK
The task

Write a set of instructions, with a diagram for each stage, showing how to construct the perpendicular bisector of a segment and how to use it to find the midpoint of a line on a playing field.

1. Numbered instructions that a classmate could follow without help.
2. A diagram for each stage, with the compass arcs shown.
3. A short paragraph explaining why the construction gives exactly the midpoint.
4. A note on the accuracy of the method compared with measuring with a ruler.

Criterion C strand	Evidence
Notation (i)	Correct geometric notation, including labels and arcs.
Representation (ii)	Diagrams that show the construction state at each stage.
Organisation (iv)	Instructions in a workable order, with no step assumed.

Success criteria

G REFLECT AND CONSOLIDATE

Summary, vocabulary, misconceptions and review

REVIEW | Consolidating Chapter 7

A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- Angles are measured in degrees and classified as acute, right, obtuse, straight or reflex.
- Angles on a straight line sum to 180°; angles at a point sum to 360°; vertically opposite angles are equal.
- When a transversal crosses parallel lines, corresponding and alternate angles are equal and co-interior angles sum to 180°.
- The angles of a triangle sum to 180°, and an isosceles triangle has two equal base angles.
- Three lengths form a triangle exactly when the sum of the two shorter exceeds the longest.
- Constructions with compasses produce exact figures, justified by equal distances.

ESSENTIAL VOCABULARY

Term | Meaning in this course	
Acute | An angle between 0° and 90°.	
Obtuse | An angle between 90° and 180°.	
Reflex | An angle between 180° and 360°.	
Vertically opposite | Angles opposite each other where two lines cross	equal.
Transversal | A line crossing two or more lines.	
Corresponding angles | Angles in the same position at each crossing of parallel lines	equal.
Alternate angles | Angles on opposite sides of a transversal between parallel lines	equal.
Co-interior angles | Angles on the same side of a transversal between parallel lines	sum to 180°.
Isosceles | A triangle with two equal sides and two equal base angles.	
Construction | An exact drawing made with compasses and a straight edge.	

Vocabulary for Chapter 7

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence	
“The angles of a triangle add to 360°.”	They add to 180°.	60 + 60 + 60 = 180	
“Alternate angles add to 180°.”	They are equal	co-interior angles add.	Z against C
“A diagram proves the angle.”	Only given information and reasons do.	Labelled 85° may look 90°	
“3, 4, 7 form a triangle.”	The sum equals the longest, so flat.	3 + 4 = 7	
“Constructions are approximate.”	They are exact if the arcs are kept.	Equal radii fix the points	

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
name, measure and classify angles accurately.			
use straight-line, point and vertically opposite angle facts with reasons.			
find angles between parallel lines and justify each step.			
use the angle sum of a triangle and properties of isosceles triangles.			
test whether three lengths can form a triangle.			

I can …	Not yet	Getting there	Confident
carry out compass constructions and explain why they work.			

How confident are you?

REFLECTION

1. Which angle fact do you find easiest to explain to someone else?
2. Describe a diagram that misled you this chapter and what you did about it.
3. Write one sentence explaining why the triangle inequality must be true.
4. In the garden task, which constraint was hardest to satisfy? Why?
5. Where in the world outside school have you seen a construction used for accuracy?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ7 · Applying the chapter

1	Find the complement of 37°.	FLUENCY
2	Two angles on a straight line: one is 128°. Find the other.	FLUENCY
3	Two angles of a triangle are 71° and 64°. Find the third.	FLUENCY
4	An isosceles triangle has base angles of 65°. Find the vertex angle.	FLUENCY
5	Do 5 cm, 6 cm and 11 cm form a triangle?	FLUENCY
6	One angle between parallel lines is 73°. State the alternate angle.	FLUENCY
7	Explain why a triangle cannot have angles 90°, 90° and 0°.	REASONING
8	A quadrilateral is split into two triangles. What is the angle sum of a quadrilateral?	REASONING
9	Explain the difference between a drawing and a construction.	REASONING
10	The angles of a triangle are in the ratio 1 : 2 : 6. Find the largest.	PROBLEM SOLVING
11	An isosceles triangle has a vertex angle of 30°. Find the base angles.	PROBLEM SOLVING
12	Two parallel lines are cut by a transversal. Co-interior angles are $2x$ and $3x + 20$. Find x.	PROBLEM SOLVING
13	Explain how to find the exterior angle of a triangle.	COMMUNICATING
14	Explain why the perpendicular bisector passes through the midpoint.	COMMUNICATING
15	The angles of a triangle are $x + 10$, $2x$ and $2x - 30$. Find each.	EXTENSION
16	Can a triangle have sides 8 cm, 8 cm and 16 cm? Explain.	EXTENSION
17	A triangle has two equal sides and a vertex angle of 100°. Sketch it and find the base angles.	EXTENSION
18	A flagpole must be equidistant from two corners of a rectangular field. Describe its possible positions.	PROBLEM SOLVING

MIXED 7 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX7 · End-of-chapter mixed set

1	Find the supplement of 68°.	FLUENCY
2	Classify 205°.	FLUENCY
3	Two angles at a point are 210° and x. Find x.	FLUENCY
4	Two angles of a triangle are 39° and 58°. Find the third.	FLUENCY
5	Are 7 cm, 8 cm and 9 cm possible sides of a triangle?	FLUENCY
6	Explain why the sum of two angles of a triangle must be less than 180°.	REASONING
7	A student claims a triangle has angles 100°, 60° and 30°. What is wrong?	REASONING
8	A straight line is divided into three angles x, $2x$ and $3x + 30$. Find the largest angle.	PROBLEM SOLVING
9	A ladder makes an angle of 65° with the ground. What angle does it make with the wall?	PROBLEM SOLVING
10	An isosceles triangle has an exterior angle of 110° at the vertex. Find the base angles.	PROBLEM SOLVING
11	Explain why the angles of an equilateral triangle are each 60°.	COMMUNICATING
12	Explain how to decide whether a triangle is right-angled from its angles.	COMMUNICATING
13	Two parallel lines are cut by two different transversals. Can you conclude anything about the angles formed?	EXTENSION
14	A triangle has angles $3x$, $4x$ and $5x$. Find the type of triangle.	EXTENSION
15	Construct, in writing, the steps needed to draw an angle of 60° without a protractor.	EXTENSION
16	A triangular plot has sides 13 m, 14 m and 15 m. Is it a valid triangle, and how many 2 m fence panels cover the perimeter?	PROBLEM SOLVING

CUMULATIVE PRACTICE 7 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP7 · Cumulative practice

1	Two angles of a triangle are 47° and 63°. Find the third.	FLUENCY
2	Two angles on a straight line: one is 96°. Find the other.	FLUENCY
3	Explain why alternate angles are equal between parallel lines.	REASONING
4	An isosceles triangle has a vertex angle of 44°. Find a base angle.	PROBLEM SOLVING
5	Do 6 cm, 7 cm and 14 cm form a triangle?	PROBLEM SOLVING
6	Explain why construction arcs are left visible.	COMMUNICATING
7	The angles of a triangle are x, $x + 20$ and $2x$. Find the largest.	EXTENSION

8 Explain why the perpendicular bisector is a line of symmetry. **EXTENSION**

CHAPTER 08

Sets and classification

A set is a collection described by a rule. Once the rule is precise, belonging can be tested, overlap can be counted and a decision can be defended. Sets give classification a language and a diagram.

KEY CONCEPT	**Form**
RELATED	**Representation**
RELATED	**Systems**
GLOBAL CONTEXT	**Fairness and development**

08
CHAPTER

INQUIRY QUESTION

A community centre offers two activity programmes. Some members use only one, some use both and some use neither. How can the centre record this fairly, count each group without double-counting, and decide what a combined membership should include?

A centre must know how many members use each programme, how many use both and how many use neither. Counting the overlap twice would overstate demand; ignoring it would understate it. Both errors cost money and both are avoidable.

IN THIS CHAPTER

8.1 Sets and set notation
8.2 Subsets, union, intersection and complement
8.3 Venn diagrams and survey problems
8.4 Classification and fair decisions
8.5 Number sets and the real line
Investigation 8 How much overlap is possible?
Application 8 The library survey
Task C Explaining a misleading headline
Review Consolidating Chapter 8
Review questions Practice across the whole chapter
Mixed 8 Fluency, reasoning and problem solving
Cumulative practice 8 Questions that mix this chapter with earlier ones

APPROXIMATELY 14 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

Classification is a mathematical act. The moment a centre writes “member of the music programme”, it has defined a set, and every later count depends on how carefully that definition was written. Sets make the definition visible and the counting checkable.

KEY CONCEPT · FORM

Form. A set is defined by the property its elements share. The form of the definition determines which objects belong and which do not.

RELATED CONCEPT · REPRESENTATION

Representation. Roster notation, set-builder notation and Venn diagrams represent the same collection in different ways. Each makes a different feature visible.

RELATED CONCEPT · SYSTEMS

Systems. Number sets form a nested system. Each larger set contains the smaller ones and adds new possibilities.

GLOBAL CONTEXT

Fairness and development

EXPLORATION

How clear classification and accurate counting support fair access to shared resources.

FACTUAL

What is a set, and how are its elements and subsets written? What do union, intersection and complement mean?

CONCEPTUAL

How can one collection be represented by a list, a rule and a diagram? Why does the overlap need special treatment when groups are combined?

DEBATABLE

When does a classification system help a decision, and when does it exclude people unfairly? Should a shared service be priced by use or by membership?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Use set notation, identify subsets, classify numbers and calculate counts from Venn diagrams.
B · Investigating patterns	i, ii	Investigate how overlap constrains the possible counts in a survey, and generalise the addition rule.
C · Communicating	i, ii, iii	Represent sets by roster, rule and diagram; label a universal set; justify a count in words.
D · Applying mathematics	i, ii, iii, iv	Audit a membership scheme, interpret survey data transparently, and evaluate a classification guide.

ATL SKILL · THINKING SKILLS

In order for a student to explain how two students can count 42 and 28 uses from the same 30 classmates, students must distinguish a count of uses from a count of people and name the information missing from the total.

STATEMENT OF INQUIRY

Precise classification reveals relationships between groups, and accurate set reasoning supports fair decisions about shared resources.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand sets as collections defined by a property, and use element and subset notation correctly.
- Represent a set by roster, set-builder notation and a Venn diagram.
- Find unions, intersections, complements and differences of sets.
- Know how the number sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are related.
- Use the addition rule to count overlapping groups without double-counting.
- Investigate how the overlap between two groups is constrained by the total surveyed.

SUCCESS CRITERIA

- ✓ I can list the elements of a set and state its cardinality.
- ✓ I can decide whether one set is a subset of another.
- ✓ I can read and draw a two-set Venn diagram with a universal set.
- ✓ I can calculate $n(A \cup B)$, $n(A \cap B)$ and $n(A')$.
- ✓ I can classify a number into the correct number sets.
- ✓ I can solve a two-group survey problem and explain each count.

NOTE

Prerequisite check · 10 minutes, no calculator

1. List the first five even numbers greater than zero.
2. Write 0.75 as a fraction in its simplest form.
3. Calculate 18 + 14 - 6.
4. Which is greater, -4 or -9?
5. Write two multiples of 3 that are less than 10.
6. Calculate 10 - 3 - 2.

Answers are in the Chapter 8 answer section. If more than two are incorrect, review factors and multiples, fraction equivalence and subtraction with compensation before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Sorting objects by one or two properties; using tables and simple lists; recognising odd, even, prime and square numbers.	Set notation and subsets; Venn diagrams; union, intersection and complement; number sets; two-group survey reasoning.	Three-set Venn diagrams, set algebra, logic and truth tables, probability using sets, and functions defined on number sets.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
A community centre records who uses which activity programme.	How can overlapping groups be counted once each, and what does “neither” mean?	Form, representation and systems.	How much overlap is possible when two totals and the overall total are known?	Roster lists, set-builder rules, labelled Venn diagrams and count statements.	When does a neat classification clarify a decision, and when does it hide an exception?

A ENGAGE
Who is in the group?

A club advertises two activities: chess and football. Of 40 members, 22 play chess and 25 play football. A member reads the notice and says “that is 47 members, but the club only has 40 — the numbers must be wrong.”

The numbers are not wrong. Some members do both activities, and those members are counted in both figures. The overlap is the missing information: 22 + 25 - 40 = 7 members play both. This is the arithmetic of sets, and it explains a great many surveys, registers and reports.

Sets are also how mathematics classifies. A set is defined by a rule that decides membership without ambiguity, which is exactly what a fair classification needs: no member left out, no member counted twice, and the same decision every time.

Group	Count
Members who play chess	22
Members who play football	25
Members who play both	7
Members who play at least one	22 + 25 - 7 = 40
Members who play neither	40 - 40 = 0

Figure 8.1 The club, counted two ways

NOTICE AND WONDER

- Why was 7 subtracted rather than added?
- Could the overlap be 30? Could it be 0? What limits the overlap?
- What would it mean if the total of the two activities were less than 40?
- How would the answer change if some members played neither activity?

Keep this question open: when does adding two counts mislead, and what has to be known about the overlap before a total can be trusted?

B EXPLORE
How much overlap is possible?

Work in pairs. You need a class list, two coloured pencils and a large sheet of paper.

YOUR INVESTIGATION

Step 1. Agree on two questions with yes-or-no answers that every member of the class can answer, such as “Do you walk to school?” and “Do you cycle to school?”.

Step 2. Collect the answers and record four counts: yes to both, yes to the first only, yes to the second only, no to both.

Step 3. Display the counts in a Venn diagram with a rectangle for everyone in the class.

Step 4. Now invent a hypothetical class of 30 in which 20 answer yes to A and 18 answer yes to B. Find the smallest and largest possible overlap.

Number who say yes to A	Number who say yes to B	Smallest overlap	Largest overlap	Number who say no to both
20	18			

Number who say yes to A	Number who say yes to B	Smallest overlap	Largest overlap	Number who say no to both
30	25			
12	12			
25	5			

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

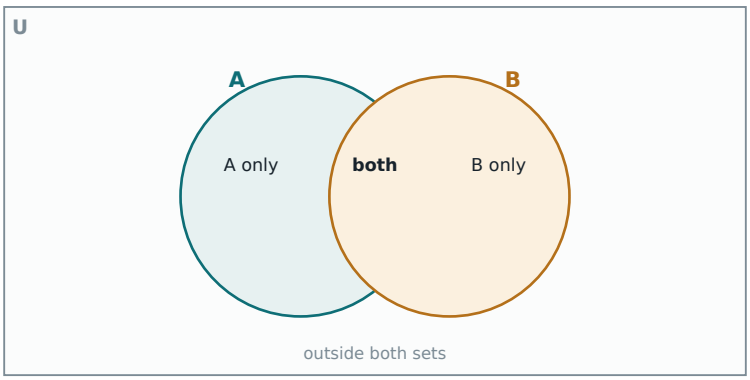
1. What is the largest possible number of people who could answer yes to both questions?
2. What is the smallest possible number? Which total makes the smallest overlap 0?
3. In your class data, how many answered no to both? Where does that number sit in the diagram?
4. If 20 out of 30 say yes to A and 18 say yes to B, what is the smallest number who could say no to both?
5. Can the two overlaps — “both yes” and “both no” — both be large? Explain.

MAKING A CONJECTURE

> “When two sets are chosen from a group of n, the overlap is at least … and at most … because … .”

WHAT YOU SHOULD HAVE NOTICED

- The largest possible overlap is the smaller of the two set sizes.
- The smallest possible overlap is 0 when the two sizes together do not exceed the group; otherwise it is the excess.
- The number outside both sets is the group size minus the union.
- Counts must be checked against the total: the four regions must add to the group size.



The four regions of a two-set Venn diagram: in A only, in both, in B only, and outside both but inside the universal set.

INVESTIGATION

Explore further

Design a survey with three overlapping questions and see how many regions the diagram needs. For three sets there are eight regions. Can you find a set of counts in which one region must be empty? Explain why.

8.1 | Sets and set notation

A · I, II

A set is a collection defined precisely enough that membership is never in doubt. The notation exists to make that precision visible.

DEFINITION
Sets and their notation

A **set** is a well-defined collection of objects, called its **elements**. Sets are named with capital letters and listed in braces: $A = 2, 4, 6, 8$.

- $x \in A$ means x is an element of A; $x \notin A$ means it is not.
- The **universal set** U contains everything under discussion.
- The **empty set** or $\varnothing$ has no elements.
- The **cardinality** $n(A)$ is the number of elements in A.

KEY IDEA
Two ways to describe the same set

Listing (roster method) names each element: $B = 3, 6, 9, 12$. **Rule** (set-builder method) states the condition: $B = x : x \text{ is a multiple of } 3 \text{ and } 0 < x < 15$. The listing is easier to read; the rule is easier to extend and to test. Both describe exactly the same set, and neither leaves membership in doubt — which is the whole point.

Set	Rule	Listed
Even numbers below 10	$x : x$ *even*, $0 < x < 10$	2, 4, 6, 8
Factors of 12	$x : x$ *divides* 12	1, 2, 3, 4, 6, 12
Prime numbers below 20	$x : x$ *prime*, $x < 20$	2, 3, 5, 7, 11, 13, 17, 19
Whole numbers between 4 and 5	$x : 4 < x < 5$, x *whole*	$\varnothing$

Sets described by rules

METHOD
Working with sets

1. Decide the universal set first: “multiples of 3” means nothing until the range is known.
2. List elements without repetition; order does not matter in a set.
3. Use a rule when the set is large or infinite, and a list when it is small.
4. Check the cardinality by counting the listed elements.

COMMON ERROR
A rule that does not decide

“The set of tall students” is not a set in the mathematical sense, because “tall” is not defined precisely enough to decide every case. “Students taller than 1.70 m” is a set. Precision, not subject matter, is what makes a collection a set.

WORKED EXAMPLE 8.1.1

Listing and describing

GIVEN The set $A = x : x$ *is a factor* of 18 with $U = 1, 2, 3, \ldots, 20$.

ASKED List the elements of A and state $n(A)$.

METHOD Test each whole number up to 18.

1. Factors: 1, 2, 3, 6, 9, 18.
2. Count: six elements.

ANSWER $A = 1, 2, 3, 6, 9, 18$ and $n(A) = 6$.

CHECK Each listed number divides 18 exactly; no other number below 20 does.

WORKED EXAMPLE 8.1.2

Membership and the empty set

GIVEN $B = x : x$ *is an even prime number* and $C = x : x$ *is a multiple* of 4, $10 < x < 12$.

ASKED Describe each set.

METHOD Test the definition against the possible values.

1. The only even prime is 2, so $B = 2$.
2. There is no multiple of 4 strictly between 10 and 12, so $C = \varnothing$.

ANSWER B has one element; C is empty.

CHECK $n(B) = 1$ and $n(C) = 0$.

WORKED EXAMPLE 8.1.3

Sets in a school context

GIVEN A year group of 120 students; U is the year group. Let F = students who study French and S = students who study Spanish.

ASKED Describe the meaning of $n(F) = 64$, $n(S) = 51$ and $n(F \cap S) = 20$.

METHOD Translate each symbol into words.

1. $n(F) = 64$: 64 students study French.
2. $n(S) = 51$: 51 study Spanish.
3. $n(F \cap S) = 20$: 20 study both.

ANSWER 64 French, 51 Spanish, 20 both.

CHECK 64 + 51 - 20 = 95 study at least one, so 25 study neither.

PRACTICE 8.1 · Sets and set notation

1	List the elements of $x : x$ *is an odd number*, $0 < x < 12$.	FLUENCY
2	State $n(a, b, c, d)$.	FLUENCY
3	Write the set of factors of 16.	FLUENCY
4	Is $7 \in 2, 3, 5, 7, 11$?	FLUENCY
5	Explain why “the set of interesting numbers” is not a set in mathematics.	REASONING
6	Explain the difference between $\varnothing$ and 0.	REASONING
7	Write a rule that describes 4, 8, 12, 16, 20.	PROBLEM SOLVING
8	$U = 1, 2, \ldots, 12$ and $A = x : x$ *is prime*. List A and $n(A)$.	PROBLEM SOLVING

9	A set has elements that are multiples of both 3 and 5 below 40. List it.	EXTENSION
10	Explain why the order of elements in a set does not matter.	EXTENSION

8.2 | Subsets, union, intersection and complement

A · II · C · I

Four operations describe how sets relate: being inside, combining, overlapping and excluding. Each has a symbol, and each has a picture in a Venn diagram.

DEFINITION
Set operations

- **Subset:** $A \subseteq B$ means every element of A is in B.
- **Union:** $A \cup B$ contains everything in A or in B or in both.
- **Intersection:** $A \cap B$ contains only what is in both.
- **Complement:** A' contains everything in U that is not in A.

KEY IDEA
The rule that links them

For any two sets, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. The subtraction corrects for double counting: elements in both sets appear twice in $n(A) + n(B)$ and should appear once in the union. This single formula answers most survey questions.

Operation	Result	Cardinality
$A \cap B$	3, 4	2
$A \cup B$	1, 2, 3, 4, 5, 6	6
A' with $U = 1, \ldots, 8$	5, 6, 7, 8	4
$A \cap B'$	1, 2	2

Operations on $A = 1, 2, 3, 4$ and $B = 3, 4, 5, 6$

COMMON ERROR
Adding without correcting

$n(A) + n(B)$ counts the overlap twice. A survey in which 18 like tea, 14 like coffee and 6 like both gives $18 + 14 - 6 = 26$ who like at least one, not 32. If the group has 30 members, 4 like neither — a figure that is easy to forget and easy to check.

WORKED EXAMPLE 8.2.1

Finding union and intersection

GIVEN $A = 2, 4, 6, 8, 10$ and $B = 4, 8, 12$.

ASKED Find $A \cap B$, $A \cup B$ and their cardinalities.

METHOD Compare the lists element by element.

1. Common elements: 4 and 8, so $A \cap B = 4, 8$ and $n(A \cap B) = 2$.
2. All elements: 2, 4, 6, 8, 10, 12, so $n(A \cup B) = 6$.

ANSWER 4, 8 and 2, 4, 6, 8, 10, 12.

CHECK 5 + 3 - 2 = 6, which matches the union count.

WORKED EXAMPLE 8.2.2

A survey with neither group

GIVEN In a class of 32, 20 study biology and 15 study chemistry; 8 study both.

ASKED Find how many study at least one, and how many study neither.

METHOD Use the union formula, then subtract from the total.

1. At least one: 20 + 15 - 8 = 27.
2. Neither: 32 - 27 = 5.

ANSWER 27 study at least one; 5 study neither.

CHECK The four regions are 12 + 8 + 7 + 5 = 32.

WORKED EXAMPLE 8.2.3

Using the complement

GIVEN $U = 1, 2, \ldots, 20$, A = multiples of 3 and B = multiples of 5.

ASKED Find $A \cap B$, $A \cup B$ and $n((A \cup B)')$.

METHOD List both sets, then apply the operations.

1. $A = 3, 6, 9, 12, 15, 18$ and $B = 5, 10, 15, 20$.
2. $A \cap B = 15$.
3. $A \cup B = 3, 5, 6, 9, 10, 12, 15, 18, 20$, with 9 elements.
4. Complement: 20 - 9 = 11.

ANSWER 15; nine elements; 11 elements outside.

CHECK 6 + 4 - 1 = 9.

WATCH OUT A number that is in both sets must appear once in the union. Listing it twice inflates the count.

PRACTICE 8.2 · Subsets, union, intersection and complement

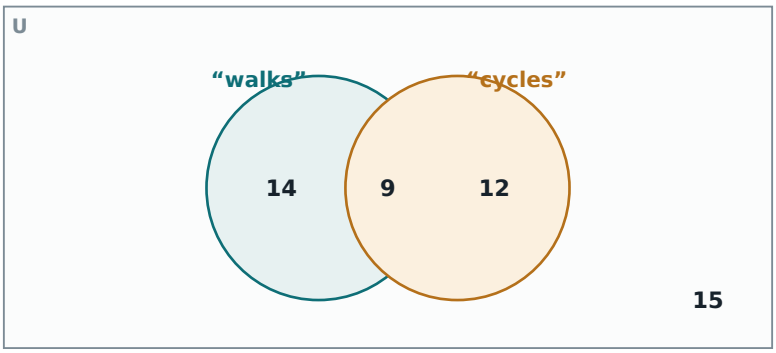
1	$A = 1, 3, 5$, $B = 3, 5, 7$. Find $A \cap B$ and $A \cup B$.	FLUENCY
2	$U = 1, \ldots, 10$, $A = 2, 4, 6, 8, 10$. Find A'.	FLUENCY
3	Is $1, 2 \subseteq 1, 2, 3$?	FLUENCY
4	$n(A) = 12$, $n(B) = 9$, $n(A \cap B) = 4$. Find $n(A \cup B)$.	FLUENCY
5	Explain why $n(A \cup B)$ is not $n(A) + n(B)$ in general.	REASONING
6	A student says $A \cap B$ is always smaller than A. Find a case where they are equal.	REASONING

7	In a group of 40, 25 like football and 22 like cricket; 10 like both. How many like neither?	PROBLEM SOLVING
8	A has 18 elements, B has 14, and $A \cup B$ has 26. Find $n(A \cap B)$.	PROBLEM SOLVING
9	$U = 1, \ldots, 30$. A = multiples of 4, B = multiples of 6. Find $n(A \cap B)$ and $n(A \cup B)$.	EXTENSION
10	Explain why $n(A \cup B) \leq n(U)$ must always hold.	EXTENSION

8.3 | Venn diagrams and survey problems

D · I, II, IV

A two-set Venn diagram has four regions, and every survey question about two groups can be solved by filling them in the right order.



A survey of 50 students. 14 walk only, 9 both walk and cycle, 12 cycle only and 15 do neither, so 23 walk and 21 cycle.

METHOD

Solving a survey problem

1. Draw the rectangle for U and two overlapping circles.
2. Put the number for “both” in the overlap first.
3. Subtract to fill “only A” and “only B”.
4. Add the three regions and subtract from the total to find “neither”.
5. Check that all four numbers add to the total.

KEY IDEA

Why the overlap comes first

The overlap is the only region that belongs to both sets, so every other region depends on it. Filling it first means the other numbers are found by subtraction, with no simultaneous unknowns. Starting anywhere else produces answers that do not add up.

Information	Region	Count
50 students in total	outside the circles	
23 walk	A only + both	
21 cycle	B only + both	
9 both	overlap	9
A only	23 - 9	14

Information	Region	Count
B only	21 - 9	12
Neither	50 - (14 + 9 + 12)	15

A worked survey

COMMON ERROR

Forgetting the rectangle

The number who belong to neither set has no circle to live in, and it is easy to leave it out. Always draw the rectangle, and always check that the regions add to the total.

WORKED EXAMPLE 8.3.1

Filling a Venn diagram

GIVEN In a class of 36, 19 play an instrument and 17 play a sport; 6 do both.

ASKED Draw a Venn diagram and state how many do neither.

METHOD Overlap first, then subtract, then check.

1. Both: 6.
2. Instrument only: 19 - 6 = 13.
3. Sport only: 17 - 6 = 11.
4. Neither: 36 - (13 + 6 + 11) = 6.

ANSWER 6 students do neither.

CHECK 13 + 6 + 11 + 6 = 36.

WORKED EXAMPLE 8.3.2

Working backwards from the diagram

GIVEN A diagram shows 12 in A only, 5 in both, 9 in B only and 4 outside both.

ASKED Find $n(A)$, $n(B)$, $n(A \cup B)$ and the total.

METHOD Read each region and combine.

1. $n(A) = 12 + 5 = 17$.
2. $n(B) = 9 + 5 = 14$.
3. $n(A \cup B) = 12 + 5 + 9 = 26$.
4. Total: 26 + 4 = 30.

ANSWER 17, 14, 26, 30.

CHECK 17 + 14 - 5 = 26.

WORKED EXAMPLE 8.3.3

A problem with the group size missing

GIVEN 40 people were asked about two newspapers. 24 read paper A, 18 read paper B and 7 read both.

ASKED Find how many read exactly one paper, and how many read neither.

METHOD Fill the regions; the total is given, so "neither" follows.

1. A only: 24 - 7 = 17.
2. B only: 18 - 7 = 11.
3. Exactly one: 17 + 11 = 28.
4. Neither: 40 - (28 + 7) = 5.

ANSWER 28 read exactly one; 5 read neither.

CHECK 17 + 7 + 11 + 5 = 40.

WATCH OUT "Exactly one" excludes the overlap while "at least one" includes it. Reading the question carefully is part of the mathematics.

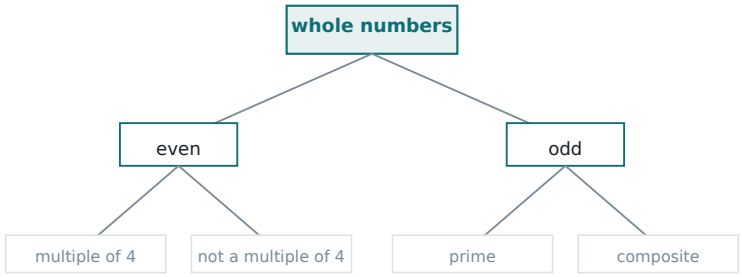
PRACTICE 8.3 · Venn diagrams and survey problems

1	30 students; 18 like maths, 14 like science, 8 like both. How many like neither?	FLUENCY
2	A diagram has 10 in A only, 4 in both and 6 in B only. Find $n(A \cup B)$.	FLUENCY
3	25 people; 15 read a magazine, 12 read a newspaper, 6 read both. How many read exactly one?	FLUENCY
4	In a class of 28, 20 play football. How many do not?	FLUENCY
5	Explain why the overlap must be filled in first.	REASONING
6	A diagram gives A only 15, both 8, B only 9, outside 3. What is the total?	REASONING
7	In a survey of 60, 34 use brand X and 27 use brand Y; 15 use both. How many use neither?	PROBLEM SOLVING
8	45 students; 28 study French, 22 study German; every student studies at least one. How many study both?	PROBLEM SOLVING
9	In a group of 50, 30 own a bike, 26 own a skateboard and 12 own neither. How many own both?	EXTENSION
10	Explain why the number in the overlap cannot exceed the smaller of the two set sizes.	EXTENSION

8.4 | Classification and fair decisions

C · III · D · III, V

To classify is to divide a group into categories using a rule. A classification is mathematically sound when every member lands in exactly one category and the rule gives the same answer every time.



A classification tree for whole numbers. Each branch divides a group into parts that do not overlap, so every number ends in exactly one final category.

DEFINITION

Classification

A **classification** assigns every member of a set to exactly one of several categories. The categories are **mutually exclusive** (no member belongs to two) and **exhaustive** (no member is left out). A classification tree makes the decisions visible: each branch applies one rule, and the branches at each stage cover every possibility.

KEY IDEA

The two tests for a fair classification

- **Mutually exclusive:** check that no object satisfies two categories. If “even” and “prime” were categories, 2 would belong to both.
- **Exhaustive:** check that every object fits somewhere. “Multiples of 4” and “not multiples of 4” is exhaustive; “multiples of 4” and “odd numbers” is not, because 6 fits neither.

Categories	Mutually exclusive?	Exhaustive?	Verdict
Even, odd	yes	yes	sound
Even, prime	no (2 is both)	no (9 fits neither)	unsound
Multiple of 4, not a multiple of 4	yes	yes	sound
Negative, zero, positive	yes	yes for integers	sound
Pass, fail, absent	yes if defined once	yes	sound if the rules are stated

Classifications tested

METHOD

Building a classification

1. Decide the universal set: the group being classified.
2. Choose a rule that every member either satisfies or does not.
3. Split first by one rule, then by another within each branch.
4. Test for overlaps and gaps with awkward cases, especially boundary values.
5. State the rule precisely enough that two people would decide the same way.

COMMON ERROR

Categories that overlap in practice

"Students who take the bus" and "students who walk" look exclusive until a student does both on different days of the week. The categories were defined by mode of travel without a time frame. Precision in the rule, not confidence in the answer, is what prevents this.

WORKED EXAMPLE 8.4.1

Testing a classification

GIVEN A class is divided into "has a pet" and "has a brother".

ASKED Decide whether the classification is sound.

METHOD Test for overlap and for gaps.

1. Overlap: a student can have both a pet and a brother, so the categories are not mutually exclusive.
2. Gap: a student with neither belongs to no category, so the categories are not exhaustive.

ANSWER Unsound; the categories overlap and leave a gap.

CHECK Sketching the two circles shows a two-region overlap and an outside region — the classification is not a division into separate parts.

WORKED EXAMPLE 8.4.2

A sound classification

GIVEN Whole numbers from 1 to 30.

ASKED Classify them so that each number falls in exactly one category, using factors.

METHOD Choose rules that are complementary at each stage.

1. First split: prime or composite — but 1 is neither, so the rule must be "prime", "composite", "neither".
2. Test boundary cases: 1 (neither), 2 (prime), 4 (composite).
3. Count the categories: 10 primes, 19 composites, 1 neither; total 30.

ANSWER Prime, composite and neither is an exhaustive, mutually exclusive classification for whole numbers above 0.

CHECK $10 + 19 + 1 = 30$.

WORKED EXAMPLE 8.4.3

Classification and fairness in a decision

GIVEN A school gives a prize to "the best all-round student", judged on mathematics, sport and service.

ASKED Suggest how to make the decision fair using classification and counting.

METHOD Replace a vague category with stated criteria and measurable evidence.

1. Define each criterion precisely, for example "service: at least 10 hours recorded".
2. Decide whether a student must meet all criteria or a stated number of them.
3. Count how many students meet each combination, and check that no one is excluded by an unstated rule.

ANSWER A fair scheme states the criteria, applies them to everyone, and shows the counts for each combination of criteria.

CHECK If the criteria leave a student unclassified, the scheme is not exhaustive and must be revised.

WATCH OUT A criterion that depends on opinion cannot be applied consistently. Mathematics can organise a decision, but it cannot make a vague rule precise on its own.

PRACTICE 8.4 · Classification and fair decisions

1	Are "multiple of 2" and "multiple of 3" mutually exclusive?	FLUENCY
2	Are "less than 10" and "at least 10" exhaustive for whole numbers?	FLUENCY
3	Classify the numbers 1 to 10 as prime or composite. Which number fits neither?	FLUENCY
4	Are "red" and "blue" exhaustive for the colours of cars in a car park?	FLUENCY
5	Explain why a classification must be both mutually exclusive and exhaustive.	REASONING
6	A survey asks "Do you walk or take the bus?" and offers only those two answers. What is wrong with the question?	REASONING
7	Classify the numbers 1 to 20 by "multiple of 5" and "not a multiple of 5". How many in each?	PROBLEM SOLVING
8	A club divides members by age: under 12, 12 to 16, over 16. Is this sound?	PROBLEM SOLVING
9	Design a two-stage classification tree for the numbers 1 to 30. State the categories and counts.	EXTENSION
10	Explain, with an example, why a classification that works for whole numbers may fail for decimals.	EXTENSION

F **INVESTIGATE AND APPLY**
A criterion B investigation and a criterion D application

8.5 | Number sets and the real line

A · I, II · C · II, III

Numbers come in families that fit inside one another like nested sets. Writing those families down, and testing a number to see where it belongs, is a good use of everything this chapter has built.

DEFINITION
The number sets

$\mathbb{N}$ — the **natural numbers** 1, 2, 3, … (sometimes including 0). $\mathbb{Z}$ — the **integers**: all whole numbers, positive, negative and zero. $\mathbb{Q}$ — the **rational numbers**: every number that can be written as $\frac{a}{b}$ with integers a and b and $b \neq 0$. This includes fractions, terminating decimals and recurring decimals. **Irrational numbers** cannot be written as a fraction of two integers: $\sqrt{2}$, π and 0.1010010001… are examples. $\mathbb{R}$ — the **real numbers**: every rational and every irrational number, which is to say every point on the number line.

KEY IDEA
The sets fit inside one another

Every natural number is an integer, every integer is rational, and every rational number is real:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

So the number -3 is in $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ but not in $\mathbb{N}$. A number that is not rational is irrational, and it is still real.

Number	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	Reason
7	yes	yes	yes	yes	a whole number
-4	no	yes	yes	yes	negative
$\frac{2}{3}$	no	no	yes	yes	a fraction of integers
0.66…	no	no	yes	yes	recurring decimals are fractions: $\frac{2}{3}$
$\sqrt{9}$	yes	yes	yes	yes	equals 3, so it is an integer
$\sqrt{2}$	no	no	no	yes	cannot be written as a fraction
π	no	no	no	yes	irrational

Testing where numbers belong

DEFINITION
Decimals on the number line

Every terminating or recurring decimal has an exact position on the number line. Between any two rational numbers there is another rational number — take their average — so the rationals are **dense**. The irrationals fill the remaining points, and together with the rationals they cover the whole line.

COMMON ERROR
“A square root is always irrational”

$\sqrt{9} = 3$ and $\sqrt{0.25} = 0.5$ are rational. A square root is irrational only when the number under it is not a perfect square of a rational. Simplify first, then classify.

WORKED EXAMPLE 8.5.1

Classifying numbers

GIVEN -7, $\frac{5}{8}$, $\sqrt{16}$, $0.4111\ldots$, $\sqrt{5}$.

ASKED Place each in the smallest set it belongs to.

METHOD Test in order: natural, integer, rational, otherwise irrational.

1. -7 is an integer but not natural: $\mathbb{Z}$.
2. $\frac{5}{8}$ is a fraction of integers: $\mathbb{Q}$.
3. $\sqrt{16} = 4$, a natural number: $\mathbb{N}$.
4. $0.4111\ldots$ recurs, and $0.4111\ldots = \frac{41}{99}$: $\mathbb{Q}$.
5. $\sqrt{5} \approx 2.236$ does not terminate or repeat: irrational.

ANSWER $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{N}$, $\mathbb{Q}$ and irrational respectively.

CHECK Each answer is the smallest set containing the number, and every one of them is also real.

WORKED EXAMPLE 8.5.2

Placing irrationals on the line

GIVEN $\sqrt{2}$ and π.

ASKED Give each to two decimal places and place it between two consecutive integers.

METHOD Estimate with perfect squares, then refine.

1. $1^2 = 1$ and $2^2 = 4$, so $1 < \sqrt{2} < 2$.
2. $1.4^2 = 1.96$ and $1.5^2 = 2.25$, so $1.4 < \sqrt{2} < 1.5$, giving $\sqrt{2} \approx 1.41$.
3. $3 < \pi < 4$, and $\pi \approx 3.14$.

ANSWER $\sqrt{2} \approx 1.41$ between 1 and 2; $\pi \approx 3.14$ between 3 and 4.

CHECK Squaring the approximation 1.41 gives 1.9881, close to 2.

WORKED EXAMPLE 8.5.3

Sets within a universal set

GIVEN U = whole numbers from 1 to 20, A = multiples of 3, B = perfect squares.

ASKED List A, B, $A \cap B$ and $A \cup B$, and state $n(A')$.

METHOD List first, then combine.

1. $A = 3, 6, 9, 12, 15, 18$ and $B = 1, 4, 9, 16$.
2. $A \cap B = 9$ and $A \cup B = 1, 3, 4, 6, 9, 12, 15, 16, 18$.
3. $n(A \cup B) = 9$, so $n(A') = 20 - 6 = 14$.

ANSWER as listed; $n(A') = 14$.

CHECK $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 6 + 4 - 1 = 9$.

PRACTICE 8.5 · Number sets and the real line

1	List the smallest set each number belongs to: (a) 12 (b) -5 (c) $\frac{3}{4}$ (d) $\sqrt{7}$	FLUENCY
2	True or false: (a) every integer is rational (b) every rational is an integer (c) π is real	FLUENCY
3	Write $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ and mark where 0 and $-\frac{1}{2}$ belong.	FLUENCY
4	Give the value of $\sqrt{25}$, $\sqrt{0.04}$ and $\sqrt{2}$ to two decimal places.	FLUENCY
5	Explain why $0.33\ldots$ is rational.	REASONING

6	A student claims $\sqrt{2} = 1.414$ exactly. What is wrong with the claim?	REASONING
7	Between which two consecutive integers does $\sqrt{40}$ lie? Give it to one decimal place.	PROBLEM SOLVING
8	U = integers from -5 to 5. List the elements of $\mathbb{N}$ and of $\mathbb{Z}$ *setminus* $\mathbb{N}$ within U.	PROBLEM SOLVING
9	Show that the average of $\frac{1}{3}$ and $\frac{1}{2}$ is rational and lies between them.	EXTENSION
10	Explain why there is no smallest positive rational number.	EXTENSION

INVESTIGATION 8 | How much overlap is possible?

B · I, II, III

Extend the Explore activity into a general result about the possible sizes of an overlap and the possible number outside both sets.

INVESTIGATION

Investigation brief

Question. Given two set sizes chosen from a group of n, what are the largest and smallest possible overlaps, and how many can belong to neither set? **What you need.** Your class data, a table of hypothetical cases, and the union formula. **What to produce.** A general rule for each extreme, a justification, and a test on a case where the sets must overlap.

PART 1 · TABULATE

$n(A)$	$n(B)$	Smallest overlap	Largest overlap	Smallest number outside both
18	14			
30	24			
35	8			
20	20			
40	40			

Copy and complete (group of 40)

PART 2 · GENERALISE

1. State the largest possible overlap in terms of $n(A)$ and $n(B)$.
2. State the smallest possible overlap when $n(A) + n(B) > n$.
3. Write the value of the smallest number who can be outside both sets.
4. What happens to your answers in the last row, where both sets are the whole group?

PART 3 · JUSTIFY

1. Explain why the overlap cannot exceed the smaller set.
2. Explain, using the union formula, why the overlap must be at least $n(A) + n(B) - n$.
3. A survey reports 26 like tea, 31 like coffee, in a group of 40, with 0 liking neither. Is the report possible? Justify.

CRITERION TASK
Criterion B task · Requirements

Complete table; general rules stated with variables; a justification using the union formula; a test of a reported figure for consistency.

APPLICATION 8 | The library survey

D · I, II, III, IV

A school library surveys students about two services: borrowing books and using the study room. The librarian must decide how to use the results and what to report.

THE BRIEF

Information	Value
Students surveyed	240
Borrow books	158
Use the study room	96
Use both	61
Students who answered the survey in the first week	150

Survey results

- The report must state how many use at least one service and how many use neither.
- The librarian also wants the figures as percentages of those surveyed.
- The survey was completed by all 240 students over two weeks.

TASK 1 · COMPLETE THE DIAGRAM

1. Calculate the number in each of the four regions of a two-set diagram.
2. State how many students use exactly one service.
3. Check that the four regions total 240.

TASK 2 · REPORT THE FIGURES

1. Express the number who use neither as a percentage of the total, to one decimal place.
2. Express each region as a percentage, and check that the four percentages total 100%.
3. Explain why the percentages must be calculated from 240 rather than 150.

TASK 3 · RECOMMEND AN ACTION

1. Identify which service has the lower take-up and by how much.
2. Propose one action for the library and state the evidence from your figures that supports it.
3. State one limitation of the survey that the report should acknowledge.

CRITERION TASK
Criterion D task · Requirements

All four regions correct and checked; percentages accurate and totalled; a recommendation supported by the data; a limitation identified; clear reporting language.

TASK C | Explaining a misleading headline

C · I, III, IV

CRITERION TASK

The task

A newspaper reports: **“Two thirds of students do not walk to school.”** The survey asked 90 students whether they walked or cycled. Write a response explaining what the headline does and does not establish.

1. Construct a possible set of survey counts consistent with the headline.
2. Identify what the survey did not ask, and how that affects the claim.
3. Show two different sets of counts that produce the same headline.
4. Rewrite the headline so that it states exactly what the data supports.

Criterion C strand	Evidence
Notation (i)	Correct set notation and a labelled Venn diagram.
Reasoning (iii)	The analysis distinguishes what the data show from what the headline claims.
Organisation (iv)	The response is ordered so a reader can follow the argument.

Success criteria

G REFLECT AND CONSOLIDATE

Summary, vocabulary, misconceptions and review

REVIEW | Consolidating Chapter 8

A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- A set is a well-defined collection; membership must be decidable for every object.
- Sets can be described by listing or by a rule, and the two descriptions can be equivalent.
- Union, intersection and complement have symbols and meanings that mirror the Venn diagram regions.
- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ corrects for double counting.
- Survey problems are solved by filling the overlap first, then subtracting.
- A sound classification is mutually exclusive and exhaustive, and its rules are stated precisely.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Set	A well-defined collection of objects.
Element	A member of a set.
Universal set	The set of everything under discussion.
Empty set	A set with no elements.
Subset	A set all of whose elements belong to another set.
Union	Everything in either set or both.

Term | Meaning in this course
Intersection | The elements common to both sets.
Complement | Everything in the universal set but not in the set.
Mutually exclusive | Categories with no member in common.
Exhaustive | Categories that cover every member.

Vocabulary for Chapter 8

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence
“Union means add the two sizes.”	Subtract the overlap.	$6 + 4 - 1 = 9$
“$\varnothing$ and 0 are the same.”	One has no elements.	$n(\varnothing) = 0$, $n(0) = 1$
“Everything outside A is empty.”	The complement may be large.	half the universal set
“Any two categories classify.”	They must not overlap or leave gaps.	“even” and “prime”
“The overlap can be as large as we like.”	It cannot exceed the smaller set.	$n(A \cap B) \le n(A)$

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
list and describe sets using both notation styles.			
find unions, intersections and complements.			
solve survey problems using a Venn diagram.			
use the union formula and check it against the diagram.			
test whether a classification is sound.			
explain a set result in the language of a real context.			

How confident are you?

REFLECTION

1. Which region of the Venn diagram caused you most difficulty, and why?
2. Describe a survey or report you have seen that could be analysed with sets.
3. Write one sentence explaining why the overlap is subtracted in the union formula.
4. In the library task, which figure was most useful to the librarian? Why?
5. How does stating a rule precisely change the fairness of a decision?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ8 · Applying the chapter

1	List the set of even numbers between 10 and 20.	FLUENCY
2	$n(a, e, i, o, u) = ?$	FLUENCY
3	$A = 1, 2, 3$, $B = 3, 4$. Find $A \cup B$.	FLUENCY
4	$n(A) = 15$, $n(B) = 11$, $n(A \cap B) = 6$. Find $n(A \cup B)$.	FLUENCY
5	In a class of 30, 18 like sport, 12 like music, 6 like both. How many like neither?	FLUENCY
6	Write the complement of 2, 4, 6 in $U = 1, \ldots, 8$.	FLUENCY
7	Explain why the four regions of a two-set diagram must add to the total.	REASONING
8	A survey reports 20 like A, 18 like B, in a group of 30, with 5 liking neither. Is this possible?	REASONING
9	Explain the difference between “exactly one” and “at least one”.	REASONING
10	In a group of 60, 35 own a phone, 28 own a tablet and 10 own both. How many own neither?	PROBLEM SOLVING
11	A Venn diagram shows 14 in A only, 9 in both, 12 in B only and 15 outside. Find the total.	PROBLEM SOLVING
12	Classify the numbers 1 to 24 as multiples of 3 or not. How many in each?	PROBLEM SOLVING
13	Explain why $n(A \cap B)$ cannot be larger than $n(A)$.	COMMUNICATING
14	Explain how a Venn diagram shows that the union formula is correct.	COMMUNICATING
15	In a survey of 100, 62 like tea, 48 like coffee and 15 like neither. Find the number who like both.	EXTENSION
16	Two sets have 12 and 20 elements and their union has 20. What does that tell you?	EXTENSION
17	Explain why a three-set Venn diagram has eight regions.	EXTENSION
18	A school has 180 students; 96 study French, 84 study Spanish, 30 study neither. How many study both?	PROBLEM SOLVING

MIXED 8 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX8 · End-of-chapter mixed set

1	$A = 2, 4, 6$, $B = 4, 6, 8$. Find $A \cap B$.	FLUENCY
2	State $n(\varnothing)$.	FLUENCY
3	Is 9 in $x : x \text{ is prime}$?	FLUENCY
4	$n(A) = 22$, $n(B) = 19$, $n(A \cap B) = 8$. Find $n(A \cup B)$.	FLUENCY

5	In a group of 25, 10 like tea, 12 like coffee and 4 like both. How many like neither?	FLUENCY
6	Explain why 1, 2, 3 and 3, 2, 1 are the same set.	REASONING
7	A classification has categories “under 10” and “over 10”. What is missing?	REASONING
8	In a class of 32, 20 have a pet, 14 have a bike, and 8 have neither. How many have both?	PROBLEM SOLVING
9	In a survey of 50, 30 read fiction, 22 read non-fiction and 12 read neither. How many read both?	PROBLEM SOLVING
10	A Venn diagram has 18 in A only, 7 in both and 11 in B only. Find $n(A)$ and $n(B)$.	PROBLEM SOLVING
11	Explain why the overlap must be subtracted exactly once in the union formula.	COMMUNICATING
12	Explain why a classification of “prime” and “composite” needs a third category for whole numbers.	COMMUNICATING
13	Two sets have 30 and 24 elements in a universal set of 40. Find the smallest possible overlap.	EXTENSION
14	In a group of 60, 40 like A, 35 like B and everyone likes at least one. How many like both?	EXTENSION
15	A survey of 200 finds 120 use app A, 90 use app B and 30 use neither. How many use exactly one app?	EXTENSION
16	A school must decide a prize using three criteria. Explain, with one example, how a classification tree can make the decision consistent.	PROBLEM SOLVING

CUMULATIVE PRACTICE 8 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP8 · Cumulative practice

1	$A = 1, 3, 5, 7$, $B = 3, 5, 9$. Find $A \cap B$.	FLUENCY
2	$n(A) = 17$, $n(B) = 12$, $n(A \cap B) = 6$. Find $n(A \cup B)$.	FLUENCY
3	Explain why a classification must be exhaustive.	REASONING
4	In a group of 45, 26 like tea, 21 like coffee and 8 like neither. How many like both?	PROBLEM SOLVING
5	Classify the numbers 1 to 30 by prime, composite and neither.	PROBLEM SOLVING
6	Explain why 0 is not the empty set.	COMMUNICATING
7	A survey of 150 finds 90 use A, 70 use B and 20 use neither. How many use both?	EXTENSION
8	Explain why the overlap cannot exceed the smaller set.	EXTENSION

PART IV · SHAPE, STRUCTURE AND MEASURE

CHAPTER 09

Perimeter, area, surface area and volume

A boundary has length, a surface has area and a solid encloses space. Each quantity needs its own formula, its own unit and its own check. Measuring well is therefore as much about choosing the right question as about calculating accurately.

KEY CONCEPT	**Form**
RELATED	**Measurement**
RELATED	**Space**
GLOBAL CONTEXT	**Globalization and sustainability**

09
CHAPTER

INQUIRY QUESTION

A community centre is redesigning its courtyard. It must buy turf for the lawn, paving for a semicircular stage, fencing for the boundary, soil for three raised beds and paint for their outsides. Which measurements does each purchase require, and how can the centre avoid paying for material it does not need?

A courtyard, a carton and a water tank raise the same question: how much material is needed for the surface, and how much space is available inside? Accurate measurement reduces waste; the wrong quantity wastes money as well.

IN THIS CHAPTER

9.1 Perimeter and area of rectangles and triangles
9.2 Area of parallelograms and compound shapes
9.3 Circles: circumference and area
9.4 Surface area of cuboids and prisms
9.5 Volume of cuboids and prisms
9.6 Using measurement to make decisions

Investigation 9 The packaging problem

Application 9 The water tank decision
Task C Writing the measurement guide
Review Consolidating Chapter 9
Review questions Practice across the whole chapter
Mixed 9 Fluency, reasoning and problem solving
Cumulative practice 9 Questions that mix this chapter with earlier ones

APPROXIMATELY 16 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

A boundary is one-dimensional, a surface is two-dimensional and a solid is three-dimensional. That single sentence explains why fencing is bought in metres, turf in square metres and soil in cubic metres — and why using the wrong unit is not a small slip but a different question.

KEY CONCEPT · FORM

Form. The shape of an object determines which measurements are needed and how they combine. A rectangle, a prism and a composite courtyard each have a different form.

RELATED CONCEPT · MEASUREMENT

Measurement. Length, area and volume are different quantities; the unit records which one has been measured.

RELATED CONCEPT · SPACE

Space. Decomposing a shape into familiar parts turns an unfamiliar space into a calculation that can be checked.

GLOBAL CONTEXT

Globalization and sustainability

EXPLORATION

How measurement supports responsible decisions about materials, packaging, land and water.

FACTUAL

What is the difference between perimeter, area, surface area and volume? Which units are appropriate for each?

CONCEPTUAL

Why does doubling every dimension multiply volume by eight but surface area by only four? How does decomposition simplify an unfamiliar shape?

DEBATABLE

Should packaging use the least material even when the result is less attractive? Who pays for unnecessary packaging — buyer, seller or environment?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Select and use measurement formulas; convert units correctly; classify measurements by dimension.
B · Investigating patterns	i, ii	Compare the surface areas of cuboids with the same volume, record the pattern and justify a conjecture.
C · Communicating	i, ii, iii	Label diagrams with dimensions and units, show the decomposition of composite shapes, and explain each step of a calculation.
D · Applying mathematics	i, ii, iii, iv	Cost a courtyard project, decide quantities of turf, soil and paint, and evaluate a storage-crate design.

ATL SKILL · RESEARCH SKILLS

In order for a student to compare two paint tins fairly, students must find the coverage, the number of coats, the price per litre and the area to be covered, then justify the comparison with the quantities rather than the price alone.

STATEMENT OF INQUIRY

Form determines how length, area and volume are measured, and measuring accurately supports responsible use of materials and space.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand why perimeter, area, surface area and volume are different quantities with different units.
- Calculate the perimeter and area of rectangles, triangles, parallelograms, trapeziums and circles.
- Break a composite shape into familiar parts and combine the results.
- Draw and interpret nets, and calculate the surface area of cuboids and triangular prisms.
- Calculate the volume of prisms and convert between cubic units, millilitres and litres.
- Investigate how the shape of a cuboid affects its surface area when the volume is fixed.

SUCCESS CRITERIA

- ✓ I can name the quantity being measured and state its unit.
- ✓ I can choose the correct formula and substitute the perpendicular dimensions.
- ✓ I can decompose a composite shape and account for every boundary edge.
- ✓ I can calculate surface area from a net or from a formula.
- ✓ I can calculate volume and convert capacity units accurately.
- ✓ I can evaluate a measurement-based design against a budget and a constraint.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Calculate 14×6.
2. Convert 2.4 m to centimetres.
3. Calculate $0.5 \times 10 \times 7$.
4. Calculate 3.14×4.
5. Calculate $3^2 + 4^2$.
6. Calculate $1000 \div 8$.

Answers are in the Chapter 9 answer section. If more than two are incorrect, review multiplication of decimals, metric length conversions and the meaning of square units before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Perimeter and area of rectangles and squares; naming common 2D and 3D shapes; simple metric conversions.	Area of triangles, parallelograms, trapeziums and circles; composite shapes; nets and surface area of prisms; volume and capacity.	Arc length and sector area, cylinders, cones and spheres, Pythagoras' theorem, similarity and scaling, and optimisation.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Gardens, packaging and water tanks are sized before they are built or bought.	Which measurement does each decision require, and which unit makes it meaningful?	Form, measurement and space.	Which cuboid uses the least material for a fixed volume?	Labelled nets, dimensioned diagrams and calculations that carry units.	When is an estimate sufficient, and when must a measurement be exact?

A ENGAGE
The box that holds more than it should

Two cereal boxes stand side by side. Box A is 20 cm by 12 cm by 6 cm. Box B is 18 cm by 14 cm by 7 cm. The manufacturer claims Box B holds more but uses less cardboard.

Both claims need checking, and they are different measurements. Volume tells us how much fits inside: Box A holds $20 \times 12 \times 6 = 1{,}440$ cm³ and Box B holds $18 \times 14 \times 7 = 1{,}764$ cm³. So the first claim is true. Surface area tells us how much cardboard is needed: Box A needs 1,008 cm² and Box B needs 952 cm². The second claim is also true.

The reason is that a shape can be more compact — closer to a cube — and hold more for the same amount of material. This is not a trick of the numbers; it is a geometric principle that packaging designers use, and it is why milk cartons are close to cuboids rather than long thin boxes.

Box	Dimensions	Volume	Surface area
A	20 × 12 × 6 cm	1,440 cm³	1,008 cm²
B	18 × 14 × 7 cm	1,764 cm³	952 cm²
Cube of side 12 cm	12 × 12 × 12 cm	1,728 cm³	864 cm²

Figure 9.1 Two boxes compared

NOTICE AND WONDER

- Which box holds more for the same cardboard?
- Why does a cube use less material than a long thin box of the same volume?
- Which measurement — length, area or volume — matches “how much cardboard”?
- What would a box shaped like a very flat envelope do to the surface area?

Keep this question open: how does the form of a shape decide how much material it takes to enclose a given space?

B EXPLORE
The packaging problem

Work in groups of three. You need 24 identical centimetre cubes, paper, scissors and a ruler.

YOUR INVESTIGATION

Step 1. Build every cuboid you can from exactly 24 cubes. Record length, width and height for each.

Step 2. Calculate the surface area of each cuboid by counting faces or by using a formula.

Step 3. Record your results in a table, ordering the shapes from smallest to largest surface area.

Step 4. Decide which shape a manufacturer should choose if cardboard cost matters, and which if shelf space matters.

Length	Width	Height	Volume	Surface area
24	1	1	24	
12	2	1	24	

Length	Width	Height	Volume	Surface area
8	3	1	24	
6	4	1	24	
6	2	2	24	
4	3	2	24	

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

1. Which cuboid uses the least material? How close is it to a cube?
2. Which uses the most? How many times more material does it need than the best one?
3. Does every cuboid with the same volume have the same surface area? Explain.
4. If the cardboard cost is 0.02 units per cm^2, how much does the cheapest box save compared with the most expensive?
5. Does the answer change if a lid is needed, so that the top face is counted twice?

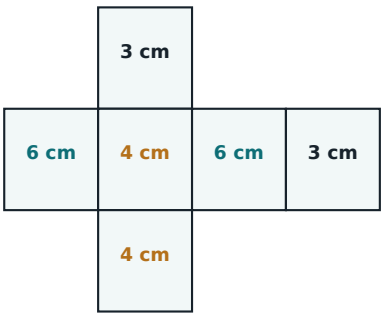
MAKING A CONJECTURE

> “For a fixed volume, the surface area is smallest when … because … .”

WHAT YOU SHOULD HAVE NOTICED

- The volume is the same for all six cuboids, but the surface area varies a great deal.
- The most compact cuboid, closest to a cube, uses the least material.
- The long thin 24 × 1 × 1 box uses the most, because it has the greatest amount of exposed surface per unit of volume.
- A lid changes the totals but not the ranking.

surface area = 2(lw + lh + wh)



The net of a cuboid. The surface area is the total area of all six rectangles, and opposite faces are equal.

INVESTIGATION

Explore further

Investigate the same question with 36 cubes. Which set of dimensions gives the smallest surface area, and how close to a cube is it? Then try 30 cubes, where no perfect cube is possible, and describe the best shape you can find.

9.1 | Perimeter and area of rectangles and triangles

A · I, II

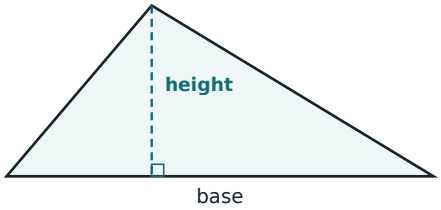
Perimeter measures a boundary in units of length; area measures a surface in square units. Confusing them is the commonest error in this chapter, and the units are the quickest way to detect it.

DEFINITION

Perimeter and area

The **perimeter** of a shape is the total length of its boundary. The **area** is the amount of surface it covers, measured in square units. For a rectangle of length l and width w: perimeter $= 2(l + w)$ and area $= lw$. For a triangle with base b and perpendicular height h: area $= \frac{1}{2}bh$.

$$\text{area} = \tfrac{1}{2} \times \text{base} \times \text{height}$$



The height of a triangle is measured perpendicular to the base, not along a side. The area is half the area of the rectangle with the same base and height.

KEY IDEA

Base and height must be perpendicular

The height of a triangle is measured at right angles to the chosen base, which may lie outside the triangle for an obtuse triangle. Using a sloping side instead of the perpendicular height gives the wrong area — sometimes a large overestimate.

Shape	Perimeter	Area
Rectangle	$2(l + w)$	lw
Square	$4s$	s^2
Triangle	sum of three sides	$\frac{1}{2}bh$
Parallelogram	$2(a + b)$	bh (perpendicular height)

The formulas of this section

METHOD

Choosing and applying a formula

1. Identify the shape and the quantity asked for: boundary (perimeter) or surface (area).
2. Write the formula before substituting, so the units are clear.
3. Substitute with units and calculate.
4. State the answer in the correct units: cm for perimeter, cm^2 for area.
5. Check plausibility: area grows with the square of the length, so doubling the sides quadruples the area.

COMMON ERROR

Mixing units of length and area

A rectangle 6 cm by 4 cm has area 24 cm², not 24 cm. Writing the unit as part of the calculation — 6 cm × 4 cm = 24 cm^2 — makes the difference visible and prevents the error at the end.

WORKED EXAMPLE 9.1.1

Rectangle and square

GIVEN A rectangle 12 cm by 7 cm, and a square of side 9 cm.

ASKED Find the perimeter and area of each.

METHOD Substitute into the formulas.

1. Rectangle: perimeter = 2(12 + 7) = 38 cm; area = 12 × 7 = 84 cm².
2. Square: perimeter = 36 cm; area = 81 cm².

ANSWER 38 cm and 84 cm²; 36 cm and 81 cm².

CHECK The square's perimeter is smaller than the rectangle's although its area is nearly the same — a reminder that the two measures are independent.

WORKED EXAMPLE 9.1.2

Area of a triangle

GIVEN A triangle with base 14 cm and perpendicular height 9 cm.

ASKED Find its area.

METHOD Use area = $\frac{1}{2}bh$.

1. Area = $\frac{1}{2}$ × 14 × 9.
2. = 63.

ANSWER 63 cm².

CHECK The triangle is half the rectangle 14 by 9, whose area is 126 cm².

WORKED EXAMPLE 9.1.3

Working backwards

GIVEN A rectangle has area 96 cm² and width 8 cm.

ASKED Find its length and perimeter.

METHOD Rearrange the area formula.

1. $l \times 8 = 96$, so $l = 12$ cm.
2. Perimeter = 2(12 + 8) = 40 cm.

ANSWER 12 cm; 40 cm.

CHECK 12 × 8 = 96.

WATCH OUT Dividing the area by the width gives a length, not an area. The unit check protects the answer.

PRACTICE 9.1 · Perimeter and area of rectangles and triangles

1	Find the area of a rectangle 15 cm by 6 cm.	FLUENCY
2	Find the perimeter of a square of side 7.5 cm.	FLUENCY
3	Find the area of a triangle with base 12 m and height 5 m.	FLUENCY

4	A rectangle has area 72 cm² and length 9 cm. Find its width.	FLUENCY
5	Explain why area is measured in square units.	REASONING
6	A student gives a perimeter as 24 cm². What is wrong?	REASONING
7	A triangle has area 48 cm² and base 12 cm. Find its height.	PROBLEM SOLVING
8	A rectangular garden is 18 m by 12 m. Find the cost of fencing at 4 units per metre.	PROBLEM SOLVING
9	A square and a rectangle have the same perimeter of 36 cm; the rectangle is 12 cm long. Compare their areas.	EXTENSION
10	Explain how doubling every side of a rectangle affects its perimeter and its area.	EXTENSION

9.2 | Area of parallelograms and compound shapes

A · II · C · II

New shapes are handled by cutting and rearranging them into shapes whose area is already known. This is the method behind every area formula in the chapter.

DEFINITION

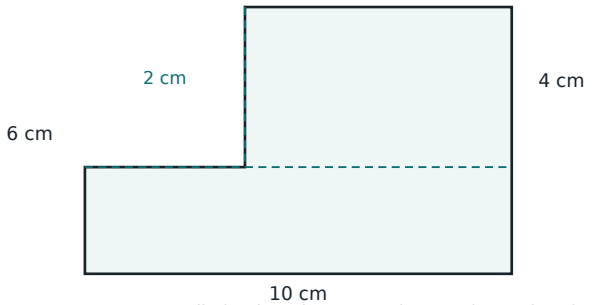
Parallelogram area

For a parallelogram with base b and perpendicular height h, area = bh. The height is the perpendicular distance between the parallel sides, not the length of a sloping side.

KEY IDEA

Why the parallelogram formula works

Cut a right-angled triangle from one end of the parallelogram and move it to the other end. The result is a rectangle with the same base and height. The area is unchanged because nothing was added or removed, only moved: bh.



split the shape into rectangles, or subtract the missing part

A compound shape can be split into rectangles, or treated as a large rectangle with a piece removed. Both methods give the same area.

METHOD

Area of a compound shape

1. Split the shape into rectangles, triangles or parallelograms, using lines you can measure.
2. Calculate the area of each part.
3. Add the parts, or subtract the parts removed from a larger shape.
4. Check that the parts do not overlap and that nothing is counted twice.

Method	Parts	Total
Add the parts	$10 \times 4 = 40$ and $4 \times 6 = 24$	64 cm²
Subtract from a rectangle	$10 \times 10 = 100$ minus $6 \times 6 = 36$	64 cm²

Two methods for one shape

COMMON ERROR

Using a sloping side as the height

A parallelogram has sides 8 cm and 5 cm with a perpendicular height of 4 cm. Its area is $8 \times 4 = 32$ cm², not $8 \times 5 = 40$ cm². The sloping side is a length of the boundary, not the distance between the parallel sides.

WORKED EXAMPLE 9.2.1

Area of a parallelogram

GIVEN A parallelogram with base 11 cm and perpendicular height 7 cm.

ASKED Find its area.

METHOD Multiply base by perpendicular height.

1. Area $= 11 \times 7$.
2. $= 77$.

ANSWER 77 cm².

CHECK The rectangle with the same base and height has the same area.

WORKED EXAMPLE 9.2.2

A compound shape by adding

GIVEN An L-shaped room made from a 9 m by 5 m rectangle and a 4 m by 3 m rectangle joined along a side.

ASKED Find the area and the perimeter.

METHOD Add the rectangles; add the outer boundary only.

1. Area: $45 + 12 = 57$ m².
2. Perimeter: the outer boundary measures $9 + 5 + 5 + 3 + 4 + 2 = 28$ m.

ANSWER 57 m² and 28 m.

CHECK The perimeter is not the sum of both rectangles' perimeters, because the joining edge is inside the shape.

WORKED EXAMPLE 9.2.3

A compound shape by subtracting

GIVEN A square of side 12 cm with a square of side 5 cm cut from one corner.

ASKED Find the area and the perimeter.

METHOD Subtract the removed area; the perimeter is unchanged by a corner cut.

1. Area: $144 - 25 = 119$ cm^2.
2. Perimeter: the two removed edges replace two edges of equal length, so it is still $4 \times 12 = 48$ cm.

ANSWER 119 cm^2 and 48 cm.

CHECK Cutting a corner from the boundary removes two segments and adds two of the same lengths.

WATCH OUT A corner cut leaves the perimeter the same; a cut from the middle of an edge increases it.

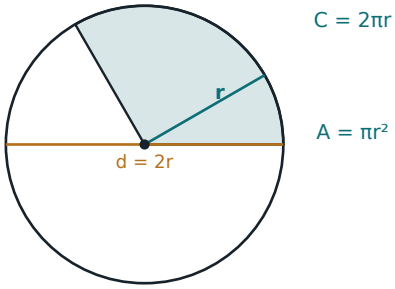
PRACTICE 9.2 · Area of parallelograms and compound shapes

1	Find the area of a parallelogram with base 9 cm and height 4 cm.	FLUENCY
2	A compound shape is made of a 6 by 4 rectangle and a 3 by 2 rectangle. Find the total area.	FLUENCY
3	Find the area of a square of side 11 m.	FLUENCY
4	A parallelogram has area 54 cm^2 and base 9 cm. Find its height.	FLUENCY
5	Explain why the sloping side of a parallelogram is not its height.	REASONING
6	A student finds the area of an L-shape by adding the areas of two rectangles and then adding both perimeters. What is wrong with adding the perimeters?	REASONING
7	A room 8 m by 6 m has a 2 m by 2 m fireplace removed from one wall. Find the floor area.	PROBLEM SOLVING
8	A parallelogram has sides 12 cm and 7 cm with height 5 cm. Find its area and explain why 12×7 is wrong.	PROBLEM SOLVING
9	A cross-shaped figure is made from five squares of side 3 cm. Find its area and perimeter.	EXTENSION
10	Explain how cutting and moving a triangle shows that the parallelogram formula is correct.	EXTENSION

9.3 | Circles: circumference and area

A · I, II · C · II

A circle has no straight edges, so its perimeter and area are not measured by counting squares. Both come from one length, the radius, and one constant, π.



The parts of a circle: radius r, diameter d = 2r, circumference C = 2πr and area A = πr². The shaded sector is a fraction of the whole circle.

DEFINITION

Circumference and area of a circle

The **circumference** is the distance around a circle: $C = 2\pi r$, or $C = \pi d$ because $d = 2r$. The **area** is the space inside: $A = \pi r^2$. Here r is the radius, d the diameter, and $\pi \approx 3.14159$, often used as 3.14 or $^{22}/_{7}$.

KEY IDEA

What π is

π is the same number for every circle: the circumference divided by the diameter. Measure round any circular lid and across it, and the first divided by the second is close to 3.14 each time. Because π never ends or repeats, any decimal answer is an approximation, so state the value of π you used.

METHOD

Choosing the right formula

1. Decide what is being measured: a length around the edge (circumference) or space inside (area).
2. Identify the radius. If the diameter is given, halve it.
3. Substitute into $C = 2\pi r$ or $A = \pi r^2$.
4. Give units: cm for circumference, cm² for area.
5. Round sensibly — two decimal places is usual unless the question says otherwise.

Radius	Diameter	Circumference $2\pi r$	Area πr^2
5 cm	10 cm	31.4 cm	78.5 cm²
7 cm	14 cm	44.0 cm	153.9 cm²
2.5 m	5 m	15.7 m	19.6 m²

The same circle, three questions

COMMON ERROR

Mixing up radius and diameter

$A = \pi r^2$ uses the radius. Using the diameter in the area formula multiplies the area by 4, not by 2: a circle of diameter 10 cm has area $25\pi \approx 78.5$ cm², not 100π. A second common slip is squaring the wrong quantity: $(2r)^2$ is not $2r^2$.

WORKED EXAMPLE 9.3.1

Circumference from the diameter

GIVEN A circular tabletop of diameter 1.2 m.

ASKED Find the length of edging needed to go right round it.

METHOD Use $C = \pi d$.

1. $C = \pi \times 1.2$.
2. $C \approx 3.14159 \times 1.2 = 3.77$ m.

ANSWER About 3.77 m of edging.

CHECK The circumference is a little more than three diameters: $3 \times 1.2 = 3.6$, and 3.77 is a little more.

WORKED EXAMPLE 9.3.2

Area from the radius

GIVEN A circular flower bed of radius 3 m.

ASKED Find the area to be covered with soil.

METHOD Use $A = \pi r^2$.

1. $r^2 = 9$.
2. $A = \pi \times 9 \approx 28.27$.

ANSWER About 28.3 m².

CHECK A square of side 6 m around it has area 36 m², and the circle fills a little less than that.

WORKED EXAMPLE 9.3.3

A composite shape with a semicircle

GIVEN A running track end: a rectangle 20 m by 12 m with a semicircle of diameter 12 m attached to one short side.

ASKED Find the total area and the perimeter of the outside edge.

METHOD Split the shape, then add.

1. Rectangle area $= 20 \times 12 = 240$ m².
2. Semicircle: radius 6 m, area $= \frac{1}{2}\pi \times 36 \approx 56.5$ m².
3. Total area ≈ 296.5 m².
4. Perimeter: two long sides (40 m) plus one short side (12 m) plus the semicircular arc $\frac{1}{2} \times \pi \times 12 \approx 18.8$ m.
5. Perimeter ≈ 70.8 m.

ANSWER Area about 296.5 m², perimeter about 70.8 m.

WATCH OUT The straight side of the semicircle is not part of the outside edge, so it is not counted in the perimeter. This is the most common error in composite circle questions.

PRACTICE 9.3 · Circles: circumference and area

1	Find the circumference, using $\pi \approx 3.14$: (a) $r = 4$ cm (b) $d = 10$ cm (c) $r = 2.5$ m	FLUENCY
2	Find the area, using $\pi \approx 3.14$: (a) $r = 3$ cm (b) $r = 10$ cm (c) $d = 8$ m	FLUENCY
3	A circle has diameter 14 cm. Find its circumference and area, using $\pi \approx \frac{22}{7}$.	FLUENCY
4	Find the area of a semicircle of radius 6 cm.	FLUENCY

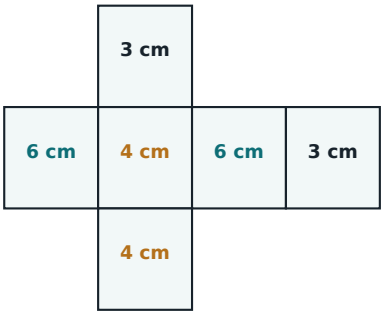
5	Explain why the area of a circle grows four times when the radius doubles.	REASONING
6	A student calculates the area of a circle of diameter 9 cm as $\pi \times 81$. What went wrong?	REASONING
7	A circular pond of diameter 6 m is surrounded by a path 1 m wide. Find the area of the path.	PROBLEM SOLVING
8	A wheel of diameter 60 cm rolls once. How far does it travel, and how many turns for 100 m?	PROBLEM SOLVING
9	The circumference of a circle is 62.8 cm. Find its radius to one decimal place.	EXTENSION
10	A quarter circle of radius 8 cm is cut from a square of side 8 cm. Find the area left.	EXTENSION

9.4 | Surface area of cuboids and prisms

A · II · C · II

Surface area is the total area of the faces of a solid. A net makes it visible: unfolding the solid into a flat shape turns a three-dimensional question into an area question.

surface area = 2(lw + lh + wh)



The net of a cuboid. The surface area is the total area of all six rectangles, and opposite faces are equal.

DEFINITION
Surface area

The **surface area** of a solid is the sum of the areas of all its faces, measured in square units. For a cuboid with dimensions l, w and h: surface area $= 2(lw + lh + wh)$.

KEY IDEA
Why the formula has a factor of 2

A cuboid has three pairs of opposite faces, and each pair is equal: top and bottom (lw), front and back (lh), left and right (wh). Adding one of each and doubling gives the total. The factor of 2 is not a rule to memorise but a consequence of the pairs.

METHOD

Surface area of a cuboid

1. Identify the three different dimensions.
2. Calculate the three different face areas.
3. Add them and double the total.
4. Check the total against the net: six faces, in three equal pairs.

Face	Area	How many
6×4	$24\ \text{cm}^2$	2
6×3	$18\ \text{cm}^2$	2
4×3	$12\ \text{cm}^2$	2
Total	$2(24 + 18 + 12)$	$108\ \text{cm}^2$

Surface area of a $6 \times 4 \times 3$ cm cuboid

COMMON ERROR

Forgetting the hidden faces

Counting only the faces you can see in a drawing gives half the surface area. A cuboid has six faces, and the back, bottom and one side are hidden in any single view. The net is the reliable picture because nothing is hidden.

WORKED EXAMPLE 9.4.1

A cuboid

GIVEN A cuboid 10 cm by 6 cm by 4 cm.

ASKED Find its surface area.

METHOD Use $2(lw + lh + wh)$.

1. $lw = 60$, $lh = 40$, $wh = 24$.
2. Sum $= 124$.
3. $2 \times 124 = 248$.

ANSWER $248\ \text{cm}^2$.

CHECK The net has two $60\ \text{cm}^2$, two $40\ \text{cm}^2$ and two $24\ \text{cm}^2$ faces: $120 + 80 + 48 = 248$.

WORKED EXAMPLE 9.4.2

A cube

GIVEN A cube of side 7 cm.

ASKED Find its volume and surface area.

METHOD Use the special cases of the cuboid formulas.

1. Volume $= 7^3 = 343\ \text{cm}^3$.
2. Surface area $= 6 \times 49 = 294\ \text{cm}^2$.

ANSWER $343\ \text{cm}^3$ and $294\ \text{cm}^2$.

CHECK Six equal faces of $49\ \text{cm}^2$.

WORKED EXAMPLE 9.4.3

A box without a lid

GIVEN An open-topped box 20 cm by 15 cm by 10 cm.

ASKED Find the area of card needed.

METHOD Count five faces instead of six.

1. Base: $20 \times 15 = 300$ cm².
2. Two sides of $20 \times 10 = 200$ each: 400 cm².
3. Two ends of $15 \times 10 = 150$ each: 300 cm².
4. Total = 300 + 400 + 300 = 1,000 cm².

ANSWER 1,000 cm².

CHECK The closed box would need $2(300 + 200 + 150) = 1{,}300$ cm², so removing the lid saves 300 cm².

WATCH OUT "Open box" changes the count of faces. Read whether a lid is included before using the formula.

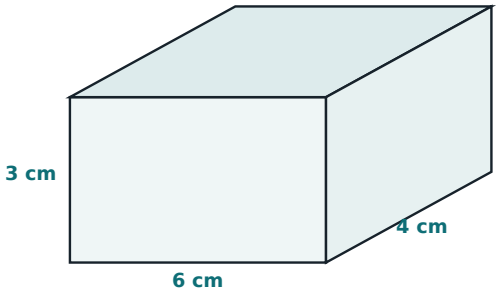
PRACTICE 9.4 · Surface area of cuboids and prisms

1	Find the surface area of a cuboid 5 cm by 3 cm by 2 cm.	FLUENCY
2	Find the surface area of a cube of side 4 cm.	FLUENCY
3	A cuboid is 8 cm by 5 cm by 5 cm. Find its surface area.	FLUENCY
4	How many faces does a cuboid have?	FLUENCY
5	Explain why the surface-area formula doubles the sum of three face areas.	REASONING
6	A student calculates the surface area of a $6 \times 4 \times 3$ cuboid as 72 cm². What has gone wrong?	REASONING
7	A gift box 24 cm by 16 cm by 9 cm is to be covered in paper. How much paper is needed, ignoring overlaps?	PROBLEM SOLVING
8	An open box is 30 cm by 20 cm by 12 cm. Find the card needed.	PROBLEM SOLVING
9	A cube has surface area 150 cm². Find its side length and volume.	EXTENSION
10	Explain why a cube uses less material than a long cuboid of the same volume.	EXTENSION

9.5 | Volume of cuboids and prisms

A · II, III

Volume measures the space a solid occupies, in cubic units. For a cuboid it is the product of three lengths, and for any prism it is the area of the cross-section multiplied by the length.



A cuboid with its three dimensions marked. Volume is length × width × height, and capacity is measured in litres.

DEFINITION

Volume and capacity

The **volume** of a cuboid is $l \times w \times$ h, measured in cubic units. The volume of any **prism** is the area of its cross-section multiplied by its length. **Capacity** is the volume of liquid or gas a container can hold, usually measured in litres: 1 *litre* = 1,000 cm^3 and 1 m^3 = 1,000 *litres*.

length	1 cm = 10 mm	× 10
area	1 cm² = 100 mm²	× 100
volume	1 cm³ = 1,000 mm³	× 1,000

Converting units for length, area and volume: area uses the square of the length factor, and volume its cube.

KEY IDEA

Units follow the dimensions

Length uses one factor of 10; area uses two, so 1 cm^2 = 100 mm^2; volume uses three, so 1 cm^3 = 1,000 mm^3. This is why a litre — 1,000 cm³ — is a cube about 10 cm on each side.

METHOD

Volume and capacity problems

1. Identify the solid and the dimensions, converting to the same unit first.
2. Apply the formula: *lwh* for a cuboid, cross-section × length for a prism.
3. State the volume in cubic units.
4. Convert to litres if the question asks about capacity: divide cm³ by 1,000, or m³ by 0.001.

Solid	Dimensions	Volume	Capacity
Cuboid	20 × 15 × 10 cm	3,000 cm³	3 litres
Cuboid	1 m × 50 cm × 40 cm	200,000 cm³	200 litres
Cube	10 cm side	1,000 cm³	1 litre
Tank	2 m × 1 m × 0.5 m	1 m³	1,000 litres

Volume and capacity

COMMON ERROR

Mixing units inside one formula

$2\ m \times 50\ \text{cm} \times 40\ \text{cm}$ must be calculated as $200 \times 50 \times 40\ \text{cm}^3$, not $2 \times 50 \times 40$. Converting all lengths to the same unit before multiplying prevents an error of a factor of 100.

WORKED EXAMPLE 9.5.1

Volume of a cuboid

GIVEN A box 25 cm by 18 cm by 12 cm.

ASKED Find its volume and its capacity in litres.

METHOD Multiply the dimensions, then convert.

1. $25 \times 18 = 450$; $450 \times 12 = 5{,}400\ \text{cm}^3$.
2. $5{,}400 \div 1{,}000 = 5.4$ litres.

ANSWER $5{,}400\ \text{cm}^3$, or 5.4 litres.

CHECK A litre is $1{,}000\ \text{cm}^3$, so $5{,}400\ \text{cm}^3$ is a little over five litres.

WORKED EXAMPLE 9.5.2

Volume of a prism

GIVEN A triangular prism whose cross-section is a triangle with base 8 cm and height 5 cm; the prism is 20 cm long.

ASKED Find its volume.

METHOD Find the cross-sectional area, then multiply by the length.

1. Triangle area $= \frac{1}{2} \times 8 \times 5 = 20\ \text{cm}^2$.
2. Volume $= 20 \times 20 = 400\ \text{cm}^3$.

ANSWER $400\ \text{cm}^3$.

CHECK The prism is equivalent to a cuboid with the same cross-section area and length.

WORKED EXAMPLE 9.5.3

A problem with mixed units

GIVEN A water tank is 1.2 m long, 80 cm wide and 60 cm deep.

ASKED Find its capacity in litres.

METHOD Convert all lengths to centimetres, then convert the volume.

1. $120 \times 80 \times 60 = 576{,}000\ \text{cm}^3$.
2. $576{,}000 \div 1{,}000 = 576$ litres.

ANSWER 576 litres.

CHECK $1.2 \times 0.8 \times 0.6 = 0.576\ \text{m}^3$, and $0.576 \times 1{,}000 = 576$ litres.

WATCH OUT A depth of 60 cm is 0.6 m, not 6 m. Estimating the size of a tank in litres before calculating catches factor-of-ten errors.

PRACTICE 9.5 · Volume of cuboids and prisms

1	Find the volume of a cuboid 6 cm by 4 cm by 3 cm.	FLUENCY
2	Find the volume of a cube of side 5 cm.	FLUENCY
3	Convert $2{,}500\ \text{cm}^3$ to litres.	FLUENCY

4	A cuboid has volume 120 cm³ and a base of 20 cm². Find its height.	FLUENCY
5	Explain why volume is measured in cubic units.	REASONING
6	A student calculates the volume of a 2 m by 50 cm by 40 cm box as 4,000 cm³. What has gone wrong?	REASONING
7	A fish tank is 60 cm by 30 cm by 40 cm. How many litres does it hold when full?	PROBLEM SOLVING
8	A prism has a cross-section of 15 cm² and a length of 24 cm. Find its volume.	PROBLEM SOLVING
9	A cuboid has volume 480 cm³ and dimensions 10 cm by 8 cm by h. Find h and its surface area.	EXTENSION
10	Explain why doubling all three dimensions of a cuboid multiplies its volume by 8.	EXTENSION

9.6 | Using measurement to make decisions

D · I, II, III, IV

Measurement answers questions about cost, capacity and material. The mathematics is in choosing the right measure, keeping the units consistent and stating what the answer does not cover.

METHOD

A decision routine

1. Decide which quantity matters: length for fencing, area for paint or card, volume for capacity.
2. Convert all measurements to one unit before calculating.
3. Calculate, then add a stated allowance for waste or overlap if the question requires it.
4. Compare cost per unit of the relevant quantity, not per object.
5. State an assumption and a limitation with the recommendation.

Job	Measure	Unit
Fencing a garden	perimeter	metres
Painting a wall	area	square metres
Filling a tank	volume	litres
Covering a box	surface area	square centimetres
Buying ribbon for a parcel	length	metres

Which measure for which job

KEY IDEA

Cost per unit is the fair comparison

Two boxes may cost the same but hold different volumes, or cost different amounts for the same capacity. Comparing cost per litre, or per square metre, makes the comparison meaningful — the same reasoning as comparing fractions by a common denominator.

COMMON ERROR

Comparing totals instead of rates

"Pack A costs 6 for 2 litres and pack B costs 8 for 3 litres, so A is cheaper" compares totals, not value. Pack A costs 3 per litre and pack B about 2.67 per litre, so B is the better buy.

WORKED EXAMPLE 9.6.1

Paint for a room

GIVEN A room has four walls, each 4 m by 2.5 m, ignoring doors and windows. One litre of paint covers 10 m² and paint is sold in 5-litre tins.

ASKED Find the number of tins needed.

METHOD Find the area, then the paint, then the tins, rounding up.

1. Area of one wall = 10 m²; four walls = 40 m².
2. Paint needed = 40 ÷ 10 = 4 litres.
3. One 5-litre tin covers 50 m², so one tin is enough.

ANSWER One tin.

CHECK 5 litres covers 50 m², which exceeds 40 m².

WORKED EXAMPLE 9.6.2

Comparing two cartons

GIVEN Carton A: 12 cm by 10 cm by 8 cm at 1.20 units. Carton B: 15 cm by 10 cm by 6 cm at 1.35 units.

ASKED Decide which carton is better value.

METHOD Find each volume and the cost per 100 cm³.

1. A: 12 × 10 × 8 = 960 cm³.
2. B: 15 × 10 × 6 = 900 cm³.
3. A costs 1.20 ÷ 960 × 100 = 0.125 per 100 cm³; B costs 1.35 ÷ 900 × 100 = 0.15 per 100 cm³.

ANSWER Carton A is better value, despite its lower apparent capacity being irrelevant: it holds more and costs less.

CHECK A holds 960 cm³ for 1.20 units and B holds 900 cm³ for 1.35 units.

WORKED EXAMPLE 9.6.3

Designing a box

GIVEN A manufacturer wants a box of volume 1,000 cm³ with a square base, using as little card as possible.

ASKED Choose the dimensions and justify the choice.

METHOD Test three square-based boxes and compare surface areas.

1. Base 10 × 10, height 10: surface area 6 × 100 = 600 cm².
2. Base 5 × 5, height 40: surface area 2(25) + 4(200) = 850 cm².
3. Base 20 × 20, height 2.5: surface area 2(400) + 4(50) = 1,000 cm².

ANSWER The cube of side 10 cm, using 600 cm² — the least card of the three.

CHECK All three have volume 1,000 cm³, and the cube has the smallest surface area.

WATCH OUT A very flat or very tall box of the same volume always uses more card. The compact shape wins.

PRACTICE 9.6 · Using measurement to make decisions

1	A wall is 5 m by 3 m. How many square metres must be painted?	FLUENCY
2	Ribbon goes once round a 20 cm by 15 cm by 10 cm box in each direction, plus 20 cm for a bow. Find the total length needed.	FLUENCY
3	A 1.5-litre bottle costs 0.90. Find the cost per litre.	FLUENCY
4	A tank holds 300 litres. How many 20-litre buckets fill it?	FLUENCY
5	Explain why cost per litre is a fairer comparison than total cost.	REASONING
6	A student compares a 2-litre pack at 6 units with a 3-litre pack at 8 units and says the first is cheaper. Explain the error.	REASONING
7	Paint covers 12 m^2 per litre and is sold in 4-litre tins. How many tins for 90 m^2?	PROBLEM SOLVING
8	A box 40 cm by 30 cm by 20 cm is wrapped. Find the card needed, ignoring overlaps.	PROBLEM SOLVING
9	Two boxes have volumes 720 cm^3 and 750 cm^3 and cost 2.40 and 2.60. Which is better value per 100 cm^3?	EXTENSION
10	Explain why a very tall thin box uses more card than a cube of the same volume.	EXTENSION

F INVESTIGATE AND APPLY

A criterion B investigation and a criterion D application

INVESTIGATION 9 | The packaging problem

B · I, II, III

Extend the Explore activity to a general claim about compactness, and test it where no perfect cube is possible.

INVESTIGATION
Investigation brief

Question. For a fixed volume, how does the shape of a cuboid affect the material needed to make it? **What you need.** Your table of cuboids, a calculator, and the surface-area formula. **What to produce.** A comparison table, a conjecture about the most compact shape, a justification, and a test with a volume that has no cubic factorisation.

PART 1 · COMPARE

Length	Width	Height	Volume	Surface area	Ratio area ÷ volume
24	1	1			
12	2	1			
6	4	1			
4	3	2			

Copy and complete

PART 2 · GENERALISE

1. Which shape gives the smallest ratio of surface area to volume? What is that ratio?
2. Which gives the largest? How many times larger is it?
3. State a conjecture about the shape that minimises material for a fixed volume.
4. Why does the ratio, rather than the surface area alone, give a fair comparison between boxes of different volumes?

PART 3 · TEST AND EVALUATE

1. Test your conjecture with volume 36, where a cube is impossible. Which cuboid comes closest?
2. Test with volume 64, where a cube is possible. Does the conjecture hold?
3. Explain, in two sentences, what a packaging designer gains from this result.
4. State one practical constraint that could make the best mathematical shape a poor choice.

CRITERION TASK
Criterion B task · Requirements

Complete table with correct values; a conjecture about compactness; a justification using the ratio; tests on two new volumes; an evaluation that names a practical constraint.

APPLICATION 9 | The water tank decision

D · I, II, III, IV, V

A school must choose a water tank. Three tanks are available, and the choice depends on capacity, floor space and cost.

THE BRIEF

Tank	Dimensions	Cost	Floor space
A	1.5 m × 1 m × 1.2 m	240 units	1.5 m²
B	2 m × 0.8 m × 1.2 m	270 units	1.6 m²
C	cylindrical, diameter 1.2 m, height 1.5 m	300 units	about 1.1 m²

The three tanks

- The school needs at least 1,400 litres.
- The available floor space is 2 m².
- The budget is 300 units, and the tank must fit through a door 1 m wide.
- For the cylinder, use volume $= \pi r^2$ h with $\pi \approx 3.14$.

TASK 1 · CALCULATE CAPACITIES

1. Find the volume of each tank in cubic metres, then in litres.
2. Decide which tanks meet the capacity requirement.
3. Check the floor space and door-width constraints for each candidate.

TASK 2 · COMPARE VALUE

1. Calculate the cost per 100 litres for each candidate.
2. Rank the candidates by value.

3. State which constraint eliminates the cheapest option, if any.

TASK 3 · RECOMMEND

1. Choose a tank and justify the choice against all four constraints.
2. State one assumption in your calculation and one way the answer could change.
3. Write a short recommendation for the school office, using your figures.

CRITERION TASK
Criterion D task · Requirements

All volumes correct with units; constraints checked explicitly; cost per 100 litres calculated; a recommendation that survives every constraint; assumptions and limitations stated.

TASK C | Writing the measurement guide

C · I, II, IV

CRITERION TASK
The task

Write a one-page guide for younger students: **"Perimeter, area, surface area and volume — which one is it?"**

1. A table of the four measurements with their units and one example each.
2. A labelled sketch for each, showing what is being measured.
3. Two worked examples where choosing the wrong measurement would give a silly answer, with the wrong answer shown.
4. A short test of understanding: four short questions with answers at the end.

Criterion C strand	Evidence
Notation (i)	Correct units and correct use of squared and cubed notation.
Representation (ii)	Sketches that show what each measurement refers to.
Organisation (iv)	A reader can choose the right measure without help.

Success criteria

G REFLECT AND CONSOLIDATE

Summary, vocabulary, misconceptions and review

REVIEW | Consolidating Chapter 9

A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- Perimeter is a length; area is measured in square units; volume in cubic units.
- Rectangle area lw; triangle area $\frac{1}{2}bh$; parallelogram area bh, using perpendicular height.
- Compound shapes are handled by adding or subtracting simple shapes.
- Surface area is the sum of all faces: $2(lw + lh + wh)$ for a cuboid, $6s^2$ for a cube.
- Volume of a cuboid is lwh; of a prism, cross-section area × length.

- 1 *litre* = 1,000 cm^3; unit conversions follow the number of dimensions.
- For a fixed volume, the most compact shape uses the least material.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Perimeter	The total length of a boundary.
Area	The surface covered, in square units.
Perpendicular height	The distance at right angles to a base.
Compound shape	A shape made from two or more simple shapes.
Surface area	The total area of all faces of a solid.
Volume	The space occupied, in cubic units.
Capacity	The volume a container can hold, often in litres.
Prism	A solid with the same cross-section along its length.
Net	A flat shape that folds into a solid.

Vocabulary for Chapter 9

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence
“Area and perimeter are the same.”	One is surface, one is boundary.	6 by 4: 24 cm^2, 20 cm
“The sloping side is the height.”	Height is perpendicular.	parallelogram 8 × 4
“A cuboid has five faces in a drawing.”	It has six.	the net
“1 m^2 = 100 cm^2.”	It is 10,000 cm^2.	100 × 100
“Doubling the sides doubles the volume.”	It multiplies by 8.	$2^3 = 8$

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
calculate perimeter and area of rectangles, triangles and parallelograms.			
find the area of a compound shape by adding or subtracting.			
calculate the surface area of a cuboid, including an open box.			
calculate volume and convert between cm^3, m^3 and litres.			
choose the right measure for a practical decision.			
compare value using cost per unit.			

How confident are you?

REFLECTION

1. Which measurement do you still confuse with another, and what will you check in future?
2. Describe a container you have seen that is shaped to use less material.
3. Write one sentence explaining why a cube is an efficient shape.
4. In the tank task, which constraint eliminated the most tempting option?
5. How did the unit of the answer help you check your work this chapter?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ9 · Applying the chapter

1	Find the area of a rectangle 14 cm by 5 cm.	FLUENCY
2	Find the perimeter of a rectangle 14 cm by 5 cm.	FLUENCY
3	Find the area of a triangle with base 16 m and height 7 m.	FLUENCY
4	Find the volume of a cuboid 9 cm by 5 cm by 4 cm.	FLUENCY
5	Find the surface area of a cube of side 6 cm.	FLUENCY
6	Convert 4,500 cm^3 to litres.	FLUENCY
7	Explain why the surface area of a cuboid is doubled in the formula.	REASONING
8	A square and a rectangle have equal areas. Must their perimeters be equal?	REASONING
9	Explain why a litre is 1,000 cm^3 and not 100 cm^3.	REASONING
10	A room is 6 m by 4.5 m. Carpet costs 12 units per m^2. Find the cost.	PROBLEM SOLVING
11	A tank is 1.5 m by 0.8 m by 0.7 m. Find its capacity in litres.	PROBLEM SOLVING
12	An open box is 25 cm by 20 cm by 15 cm. Find the card needed.	PROBLEM SOLVING
13	Explain why the area of a triangle is half that of a rectangle with the same base and height.	COMMUNICATING
14	Explain why doubling all dimensions multiplies the surface area by 4 and the volume by 8.	COMMUNICATING
15	A cuboid has volume 240 cm^3 and a square base of side 4 cm. Find its height and surface area.	EXTENSION
16	A cylinder has radius 5 cm and height 10 cm. Find its volume ($\pi \approx 3.14$).	EXTENSION
17	Explain why a sphere is more efficient than a cube for holding a given volume.	EXTENSION
18	A tank 1 m × 1 m × 1 m is filled at 5 litres per minute. How long to fill it?	PROBLEM SOLVING
19	A shop sells soap in a 6 × 4 × 3 cm box and in a 12 × 4 × 3 cm box. Compare surface areas and volumes, and say which is the better use of card per unit of soap.	COMMUNICATING

MIXED 9 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX9 · End-of-chapter mixed set

1	Find the area of a parallelogram with base 12 cm and height 5 cm.	FLUENCY
2	Find the perimeter of a square of side 8 m.	FLUENCY
3	Find the volume of a cube of side 9 cm.	FLUENCY
4	Convert 0.75 m^3 to litres.	FLUENCY
5	Find the surface area of a cuboid 7 cm by 5 cm by 4 cm.	FLUENCY
6	Explain why perimeter has units of cm while area has units of cm^2.	REASONING
7	A student says a box $10 \times 10 \times 10$ holds twice as much as one $5 \times 5 \times 5$. Refute this.	REASONING
8	A garden 15 m by 9 m is fenced and turfed. Find the fencing and turf costs at 3 and 8 units per unit respectively.	PROBLEM SOLVING
9	A carton 20 cm by 12 cm by 8 cm is filled with 1 cm cubes. How many fit?	PROBLEM SOLVING
10	A prism has cross-section area 24 cm^2 and length 15 cm. Find its volume.	PROBLEM SOLVING
11	Explain when cutting a corner from a rectangle changes the perimeter.	COMMUNICATING
12	Explain how to decide whether a problem needs area or volume.	COMMUNICATING
13	A cuboid has dimensions in the ratio 1 : 2 : 3 and volume 384 cm^3. Find the dimensions.	EXTENSION
14	A cube and a cuboid have equal volumes of 216 cm^3; the cuboid is 9 cm by 6 cm by 4 cm. Which has the smaller surface area?	EXTENSION
15	A tank 80 cm by 50 cm by 40 cm is half full. How many litres must be added to fill it?	EXTENSION
16	A cylindrical can has diameter 8 cm and height 12 cm. Find its volume and how many litres it holds.	PROBLEM SOLVING
17	Find the circumference of a circle of radius 7 cm, using $\pi \approx {}^{22}/_{7}$.	FLUENCY
18	Explain why doubling the radius of a circle multiplies its area by 4 but its circumference by only 2.	REASONING
19	A circular table top has diameter 1.4 m. How long is the strip that goes right round the edge?	PROBLEM SOLVING
20	A semicircular window has diameter 60 cm. Find its perimeter.	EXTENSION

CUMULATIVE PRACTICE 9 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP9 · Cumulative practice

1	Find the area of a triangle with base 21 cm and height 10 cm.	FLUENCY

2	Find the volume of a cuboid 8 cm by 5 cm by 5 cm.	**FLUENCY**
3	Convert 7,500 cm^3 to litres.	**FLUENCY**
4	Explain why the height of a triangle is not always a side.	**REASONING**
5	An open box is 40 cm by 30 cm by 20 cm. Find the card needed.	**PROBLEM SOLVING**
6	A tank 90 cm by 60 cm by 50 cm is $\frac{3}{4}$ full. How many litres does it hold?	**PROBLEM SOLVING**
7	Explain why a cube is an efficient container.	**COMMUNICATING**
8	A cube has volume 512 cm^3. Find its surface area.	**EXTENSION**

PART V

DATA AND DECISIONS

V

10 Data, tables and graphs

PART V · DATA AND DECISIONS

CHAPTER 10

Data, tables and graphs

A table can organise a crowd of responses into a pattern. A graph can make that pattern visible at a glance. But every display is a set of choices, and those choices decide what a reader notices, what they overlook and what they are willing to believe.

KEY CONCEPT	**Relationships**
RELATED	**Representation**
RELATED	**Patterns**
GLOBAL CONTEXT	**Fairness and development**

10
CHAPTER

INQUIRY QUESTION

A school surveys 200 students about where they eat lunch. The newsletter reports that "most students buy school lunch", while a second article claims that "off-site eating is out of control". Both use the same survey. Which tables and graphs would let you check each claim, and what could a careless display hide?

The same survey can support two opposite headlines. One number is not a finding until the question, the group and the display are known. This chapter teaches you to read the choices behind the chart.

IN THIS CHAPTER

10.1 Collecting and organising data
10.2 Displaying data: bar charts, dot plots and pictograms
10.3 Averages and range
10.4 Grouped data
10.5 Reading data critically
Investigation 10 How should data be grouped?
Application 10 The canteen survey
Task C Writing the graph-checking guide
Review Consolidating Chapter 10
Review questions Practice across the whole chapter
Mixed 10 Fluency, reasoning and problem solving
Cumulative practice 10 Questions that mix this chapter with earlier ones

APPROXIMATELY 16 HOURS including investigation, application and review

UNIT PLANNER

FRAMING THE INQUIRY

A frequency table is a decision about categories; a graph is a decision about scale; an average is a decision about summary. None of these decisions is neutral, and each can be checked. Statistical literacy begins with asking who chose and why.

KEY CONCEPT · RELATIONSHIPS

Relationships. A data set records how individuals vary. Tables, graphs and averages are ways of showing relationships between a variable and its frequency, or between two variables.

RELATED CONCEPT · REPRESENTATION

Representation. Every table or graph models the data and can reveal or distort a pattern through its scale and labels.

RELATED CONCEPT · PATTERNS

Patterns. Trends, clusters and outliers are patterns; describing them needs evidence, not first impressions.

GLOBAL CONTEXT

Fairness and development

EXPLORATION

How data displays inform decisions about shared resources, and how a misleading display can make a claim seem stronger than the evidence.

FACTUAL

What are the main types of data? How is a frequency table built, and how is a sector angle calculated?

CONCEPTUAL

How does the choice of representation reveal or conceal a pattern? When does one average represent a group fairly?

DEBATABLE

Is a graph that is technically accurate but visually exaggerated misleading? Who is responsible for the interpretation — the creator or the reader?

CRITERION	STRANDS	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Classify data, construct frequency and two-way tables, calculate averages and sector angles accurately.
B · Investigating patterns	i, ii	Compare how different class intervals change a histogram, describe the pattern and justify a choice.
C · Communicating	i, ii, iii	Label axes and units, choose an appropriate display, and explain a trend or a misleading feature precisely.
D · Applying mathematics	i, ii, iii, iv	Analyse survey data, cost a canteen decision, and evaluate a public claim against the evidence behind it.

ATL SKILL · CRITICAL THINKING

In order for a student to question the claim that a new phone has “twice the battery life”, students must identify the test conditions, the axis scale and the baseline, then state which feature of the graph could exaggerate the difference.

STATEMENT OF INQUIRY

Data becomes evidence when it is represented accurately; the choice of table, graph or average shapes the conclusion a reader can draw.

LEARNING INTENTIONS AND SUCCESS CRITERIA

YOU ARE LEARNING TO

- Understand the difference between categorical and numerical data, and between discrete and continuous measurements.
- Organise data in frequency tables and two-way tables.
- Construct and interpret bar charts, pie charts, dot plots, histograms and time-series graphs.
- Calculate and compare the mean, median, mode and range.
- Describe a trend and calculate a percentage change.
- Investigate how the choice of class interval changes the picture a histogram gives.

SUCCESS CRITERIA

- ✓ I can classify a variable and justify the classification.
- ✓ I can build a table with correct totals and relative frequencies.
- ✓ I can calculate sector angles and draw a labelled pie chart.
- ✓ I can choose an average and explain why it suits the data.
- ✓ I can describe a trend using values from a graph.
- ✓ I can identify how an axis, interval or missing context misleads.

NOTE

Prerequisite check · 10 minutes, no calculator

1. Calculate 25% of 80.
2. Write $\frac{3}{8}$ as a decimal and a percentage.
3. Calculate 360 ÷ 8.
4. Find the mean of 4, 7 and 13.
5. List these numbers in order: 12, 5, 9, 5, 14.
6. Calculate the difference between 186 and 104.

Answers are in the Chapter 10 answer section. If more than two are incorrect, review fractions, percentages and the idea of an average before continuing.

BEFORE YEAR 1	YEAR 1 · THIS CHAPTER	LATER
Tally charts, simple bar charts, reading tables, and finding the mean of a small set of numbers.	Data types; frequency and two-way tables; bar charts, pie charts, dot plots and histograms; averages and range; trends and misleading graphs.	Scatter graphs and correlation, sampling and bias, probability from data, cumulative frequency and statistical inference.

INQUIRY PATHWAY FOR THIS CHAPTER

CONTEXT	QUESTION	CONCEPT	INVESTIGATION	COMMUNICATION	REFLECTION
Schools and news outlets use surveys to make claims about groups of people.	Which display communicates each claim honestly, and which display exaggerates it?	Relationships, representation and pattern.	How does the width of a class interval change the shape of a histogram?	Labelled axes, stated totals and a caption that names the pattern.	When is an average a fair summary, and when does it hide the people behind the number?

A ENGAGE
Two charts, two impressions

Two newspapers report the same survey of 100 students. The first shows a bar chart with the vertical axis starting at zero. The second shows the same counts with the axis starting at 35. In the second, one category looks three times taller than another.

Both charts are drawn from accurate numbers and both are misleading, because the second exaggerates the differences by cutting off the bottom of the axis. A reader who glances rather than reads is being persuaded by the drawing, not by the data.

Data is not self-explanatory. A table has to be organised, a graph has to be chosen, an average has to be selected, and each of those decisions changes what a reader sees. This chapter is about making those decisions honestly and reading them critically.

Representation	What it shows well	What it can hide
Frequency table	Exact counts	The overall shape of the data
Bar chart from zero	Comparison between categories	Nothing, if the axis is honest
Bar chart from 35	A dramatic visual difference	The size of the difference relative to the whole

Figure 10.1 The same survey, reported three ways

NOTICE AND WONDER

- Why does the height of a bar matter only if the axis is honest?
- Which representation would you trust for a decision, and which for a first impression?
- What would happen to the picture if one extra student were added to the smallest group?
- Who benefits when a graph exaggerates a difference?

> Keep this question open: when does a representation reveal the story in the data, and when does it invent one?

B EXPLORE
How should data be grouped?

Work in groups of three. You need a stopwatch, a corridor or playground, and a tally sheet.

YOUR INVESTIGATION

Step 1. Time 30 students walking a fixed 10-metre course. Record each time to the nearest second.

Step 2. Display the raw times as a dot plot.

Step 3. Group the times into intervals — first 5-second intervals, then 2-second intervals — and draw a bar chart for each grouping.

Step 4. Compare the three displays. Which shows the shape of the data most clearly? Which loses the most information?

Interval	Tally	Frequency
6–10 s		
11–15 s		
16–20 s		

Interval	Tally	Frequency
21–25 s		
26–30 s		

Recording table (copy and complete)

QUESTIONS TO GUIDE YOUR OBSERVATION

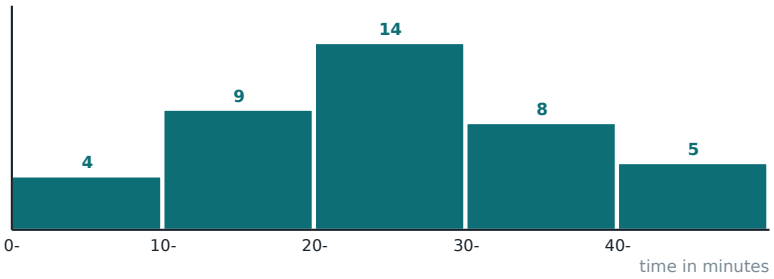
1. With 5-second intervals, can you tell whether most students took 12 seconds or 14 seconds? Why not?
2. What happens to the picture when you use 2-second intervals?
3. Does the number of groups change the total number of students? Should it?
4. What interval width would you choose to report the results to a teacher? Explain.
5. Could a different choice of interval make the class look faster than it is?

MAKING A CONJECTURE

> “The choice of interval changes … because … .”

WHAT YOU SHOULD HAVE NOTICED

- Grouping loses detail but reveals shape; the raw values are lost once an interval is chosen.
- Narrower intervals show more detail and more, smaller bars.
- The total frequency must stay the same however the data are grouped.
- The choice of interval is a decision, and it should be stated when results are reported.



A histogram of grouped data. The bars touch because the groups are intervals on a continuous scale.

INVESTIGATION
Explore further

Group the same 30 times in three different ways — 1-second, 3-second and 10-second intervals. For each grouping, find the mean of the grouped data. Compare the three means with the true mean of the raw times, and explain any differences.

10.1 | Collecting and organising data

A · I, II · D · I

Before any graph is drawn, data must be collected honestly and organised so that it can be checked. A frequency table is the working document of statistics.

DEFINITION
Data and frequency

Data are observations, measurements or counts. Data are **qualitative** when they describe a category (favourite sport) and **quantitative** when they are numbers (height in cm). Quantitative data are **discrete** if they can be counted (number of siblings) and **continuous** if they can be measured on a scale (time, mass, length). The **frequency** of a value is the number of times it occurs. A **frequency table** lists each value with its frequency.

number of pets	frequency
0	4
1	9
2	12
3	7
4	3

total 35

A frequency table records how often each value occurs; the frequencies add to the number of students surveyed.

KEY IDEA
A good table can be checked

Three checks catch almost every error in a frequency table: the frequencies must add to the number of observations; every observation must appear exactly once; and the categories must be mutually exclusive and exhaustive, so that no observation is lost or counted twice — the same requirement as a sound classification.

METHOD
Building a frequency table

1. Decide whether the data are qualitative or quantitative.
2. For qualitative data, list the categories and use tally marks.
3. For discrete quantitative data, list the possible values in order.
4. For continuous data, choose intervals of equal width and state whether boundaries are included.
5. Total the frequencies and check against the number of observations.

Number of pets	Tally	Frequency
0	𝍸	4
1	𝍸 𝍸	9
2	𝍸 𝍸 ||	12
3	𝍸 ||	7
4	|||	3
Total		35

Tally and frequency

COMMON ERROR

Intervals that overlap or leave gaps

Intervals "0–10, 10–20, 20–30" are ambiguous: a value of exactly 10 belongs to two groups. Write "$0 \le x < 10$, $10 \le x < 20$" or state the convention once. Unequal interval widths also distort a bar chart, because a wider interval contains more data for the same height.

WORKED EXAMPLE 10.1.1

Classifying data

GIVEN Four sets of data: shoe size, favourite colour, time to run 100 m, number of books read.

ASKED Classify each as qualitative or quantitative, and quantitative as discrete or continuous.

METHOD Ask whether the data are categories or numbers; then whether the numbers are counted or measured.

1. Shoe size: quantitative, discrete.
2. Favourite colour: qualitative.
3. Time to run 100 m: quantitative, continuous.
4. Number of books: quantitative, discrete.

ANSWER discrete, qualitative, continuous, discrete.

CHECK Shoe sizes are counted in half units and can be listed; times can take any value in a range.

WORKED EXAMPLE 10.1.2

Completing a frequency table

GIVEN A table of 40 observations with frequencies 7, 12, 9, x and 5 for five categories.

ASKED Find x.

METHOD Use the total.

1. $7 + 12 + 9 + 5 = 33$.
2. $x = 40 - 33 = 7$.

ANSWER $x = 7$.

CHECK $7 + 12 + 9 + 7 + 5 = 40$.

WORKED EXAMPLE 10.1.3

Choosing intervals

GIVEN 60 masses in kilograms ranging from 41.2 to 68.9.

ASKED Choose suitable intervals and explain the choice.

METHOD Aim for five to eight intervals of equal width covering the range.

1. Range: $68.9 - 41.2 = 27.7$ kg.
2. Five intervals of 6 kg from 40 give 40–, 46–, 52–, 58–, 64–, which covers the data.
3. State that each interval includes its lower bound and excludes its upper bound.

ANSWER Five intervals of width 6 kg starting at 40.

CHECK The highest value 68.9 falls in the interval 64–70, so no observation is left out.

WATCH OUT Rounding the range down can leave the largest value without an interval. Always check the extremes.

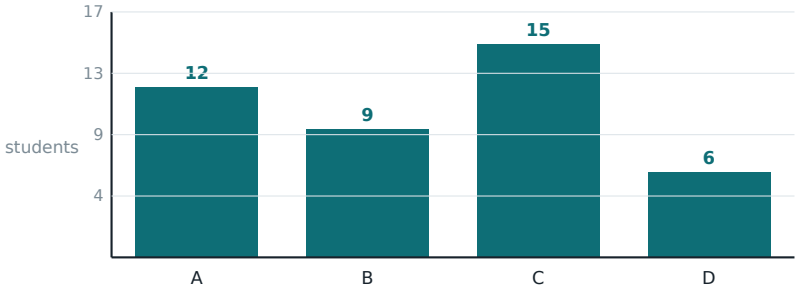
PRACTICE 10.1 · Collecting and organising data

1	Classify: number of goals scored in a match.	FLUENCY
2	Classify: type of pet.	FLUENCY
3	A frequency table has frequencies 5, 8, 6 and 4. How many observations in total?	FLUENCY
4	What interval width would suit data from 12 to 47 in five groups?	FLUENCY
5	Explain why intervals must not overlap.	REASONING
6	A table of 50 observations has frequencies summing to 47. What does that suggest?	REASONING
7	Design a frequency table for the number of letters in students' first names in your class. State the categories.	PROBLEM SOLVING
8	200 students were asked their preferred sport. 45 chose football, 60 basketball, 35 swimming and the rest chose other. Draw up the table and find the missing frequency.	PROBLEM SOLVING
9	Explain why grouping continuous data always loses information.	EXTENSION
10	A report groups ages as 0–20, 20–40, 40–60, 60–100. Give two problems with the grouping.	EXTENSION

10.2 | Displaying data: bar charts, dot plots and pictograms

C · I, II, IV

A good graph answers the question the data were collected to answer. The wrong choice of graph hides the pattern, or invents one.



A bar chart of four categories. The bars are separate because the categories are labels, not numbers on a continuous scale.



A dot plot of 13 values. Each dot is one student's result, and repeated values stack vertically, so the shape of the data is visible.

Display	Best for	Avoid when
Frequency table	exact values and totals	the shape matters
Bar chart	comparing categories	the data are continuous measurements
Dot plot	small sets of discrete values	there are hundreds of values

Display	Best for	Avoid when
Histogram	grouped continuous data	the categories are labels
Pictogram	a general audience, simple comparisons	precise comparison is needed
Pie chart	parts of a single whole	more than one whole is compared

Choosing a display

KEY IDEA
What every graph must have

- A title that says what is being shown.
- Labelled axes with units, or a key for a pictogram.
- A vertical axis that starts at zero for bar charts and histograms.
- Bars of equal width, separated for categories and touching for continuous intervals.

METHOD
Drawing a bar chart

1. Choose a scale for the vertical axis that starts at zero and fits the largest value.
2. Draw and label the axes, including units.
3. Draw bars of equal width with equal gaps.
4. Add a title and, where useful, the value on each bar.
5. Check that the tallest bar matches the greatest frequency.

COMMON ERROR
Bars that touch in a bar chart

In a bar chart the categories are separate, so the bars must not touch. Touching bars signal a histogram, where the horizontal axis is a continuous scale. Mixing the two conventions confuses the reader about whether the data are categories or intervals.

WORKED EXAMPLE 10.2.1
Reading a bar chart

GIVEN A bar chart of favourite sports for 120 students: football 40, basketball 30, swimming 20, athletics 15, other 15.

ASKED Find the fraction who chose football or basketball, and the angle each would occupy in a pie chart.

METHOD Combine the counts, then convert to fractions and to angles of 360°.

1. Football or basketball: $40 + 30 = 70$, which is $\frac{70}{120} = \frac{7}{12}$.
2. Football angle: $\frac{40}{120} \times 360° = 120°$.
3. Basketball: $\frac{30}{120} \times 360° = 90°$.

ANSWER $\frac{7}{12}$; 120° and 90°.

CHECK The five angles total $120 + 90 + 60 + 45 + 45 = 360°$.

WORKED EXAMPLE 10.2.2

Reading a dot plot

GIVEN A dot plot of the number of books read by 13 students: 1 (one student), 2 (two), 3 (three), 4 (two), 5 (four), 6 (one).

ASKED State the most common number, the range and the number of students who read more than 4 books.

METHOD Read the heights of the stacks.

1. The tallest stack is at 5, with four students.
2. Range: 6 - 1 = 5.
3. More than 4: the 5 and 6 stacks, 4 + 1 = 5 students.

ANSWER 5 books; range 5; 5 students.

CHECK Frequencies sum to 1 + 2 + 3 + 2 + 4 + 1 = 13.

WORKED EXAMPLE 10.2.3

Criticising a display

GIVEN A bar chart with no vertical scale, three-dimensional bars and an axis starting at 80.

ASKED Explain three ways it could mislead.

METHOD Check each convention in turn.

1. No scale: the reader cannot recover the values.
2. Axis from 80: differences are exaggerated.
3. Three-dimensional bars: the front face is proportional, but the volume is not, so the eye overestimates large bars.

ANSWER Missing scale, truncated axis and misleading perspective.

CHECK A correct version would show the axis from zero, a labelled scale and flat bars.

WATCH OUT A graph can be accurate in every number and still mislead through its design. Checking the axes is the reader's responsibility.

PRACTICE 10.2 · Displaying data

1	Which display suits comparing the favourite subjects of 30 students?	FLUENCY
2	Which display suits showing the shape of 200 grouped heights?	FLUENCY
3	In a pie chart of 180 students, how many degrees represent 45 students?	FLUENCY
4	A bar chart shows counts 8, 12 and 4. What is the total?	FLUENCY
5	Explain why bar-chart axes should start at zero.	REASONING
6	Why do histogram bars touch while bar-chart bars do not?	REASONING
7	240 students chose four sports. Football 90, basketball 60, swimming 50, other 40. Find the pie-chart angle for football.	PROBLEM SOLVING
8	A dot plot has stacks of height 3, 5, 4 and 2. How many values in total?	PROBLEM SOLVING
9	Explain how a pictogram can mislead even when the numbers are right.	EXTENSION
10	A bar chart shows category A at 20 and B at 22 with an axis from 18 to 24. Describe the visual impression and the true difference.	EXTENSION

10.3 | Averages and range

A · II, III · C · III

An average is a single number used to summarise a set of data. There are three common averages, and they answer slightly different questions.

DEFINITION
Mean, median, mode and range

The **mean** is the total divided by the number of values. The **median** is the middle value when the data are in order. The **mode** is the most frequent value. The **range** is the difference between the largest and smallest values and measures spread, not average.

KEY IDEA
Which average to use

- The mean uses every value, so it reflects the whole data set — and is pulled by extreme values.
- The median ignores the size of extreme values and is more representative when the data are skewed.
- The mode is the only average that works for qualitative data, such as the most popular colour.
- The range is easy to compute but depends on only two values.

Data: 3, 5, 5, 6, 8, 9, 14	Value	Reason
Mean	7.14 (to 3 s.f.)	50 ÷ 7
Median	6	fourth of seven ordered values
Mode	5	occurs twice
Range	11	14 - 3

Three averages of the same data

METHOD
Calculating each average

1. Order the data before finding the median.
2. For an odd number of values, the median is the middle one; for an even number, it is the mean of the two middle values.
3. For the mean, add all values and divide by how many there are.
4. For the mode, count frequencies; a data set may have no mode or several.

COMMON ERROR
Forgetting to order the data

Finding the median of 7, 2, 9, 4, 5 by taking the middle of the written list gives 9, which is the largest value. In order, 2, 4, 5, 7, 9, the median is 5. Ordering is not a preliminary step that can be skipped.

WORKED EXAMPLE 10.3.1

Finding all three averages

GIVEN The data 12, 9, 14, 9, 11, 15, 9, 10.

ASKED Find the mean, median, mode and range.

METHOD Order the data, then apply each definition.

1. Ordered: 9, 9, 9, 10, 11, 12, 14, 15.
2. Mean: 89 ÷ 8 = 11.125.
3. Median: mean of the fourth and fifth values, 10 and 11, so 10.5.
4. Mode: 9. Range: 15 - 9 = 6.

ANSWER mean 11.125, median 10.5, mode 9, range 6.

CHECK The mean lies between the median and the largest values, as expected with a slight upper tail.

WORKED EXAMPLE 10.3.2

Working backwards from the mean

GIVEN Five test scores have a mean of 14. Four of them are 12, 15, 13 and 16.

ASKED Find the fifth score.

METHOD Use the total implied by the mean.

1. Total = 5 × 14 = 70.
2. Known total = 12 + 15 + 13 + 16 = 56.
3. Fifth score = 70 - 56 = 14.

ANSWER 14.

CHECK (12 + 15 + 13 + 16 + 14) ÷ 5 = 70 ÷ 5 = 14.

WORKED EXAMPLE 10.3.3

Choosing the better average

GIVEN Salaries in a small company: 18,000; 19,000; 19,500; 20,000; 21,000; 22,000; 180,000.

ASKED Decide whether the mean or the median better represents a typical salary.

METHOD Compare the two and consider the effect of the extreme value.

1. Mean: total = 299,500, divided by 7 is about 42,786.
2. Median: 20,000.
3. The mean is more than twice the median because one salary is far larger.

ANSWER The median, 20,000, represents a typical salary better; the mean is distorted by the single extreme value.

CHECK Six of the seven salaries are below the mean.

WATCH OUT An average that most of the data lie below is usually the wrong choice to describe “typical”.

PRACTICE 10.3 · Averages and range

1	Find the mean of 4, 7, 9, 10.	FLUENCY
2	Find the median of 3, 8, 5, 6, 9.	FLUENCY
3	Find the mode of 2, 3, 3, 5, 6, 3.	FLUENCY
4	Find the range of 12, 5, 9, 20.	FLUENCY
5	Explain why the range is not an average.	REASONING

6	A data set has two modes. Is that a problem?	REASONING
7	The mean of six numbers is 12. Five are 10, 11, 13, 14 and 15. Find the sixth.	PROBLEM SOLVING
8	Find the median of 7, 2, 9, 4, 11, 6.	PROBLEM SOLVING
9	A data set of 9 values has median 20. What is the fifth value in order?	EXTENSION
10	Adding one value changes the mean from 10 to 11. What can you say about the added value?	EXTENSION

10.4 | Grouped data

A · III · B · II

When data are grouped, the individual values are lost, so averages can only be estimated. A histogram shows the shape, and the midpoint of each interval stands in for the values inside it.

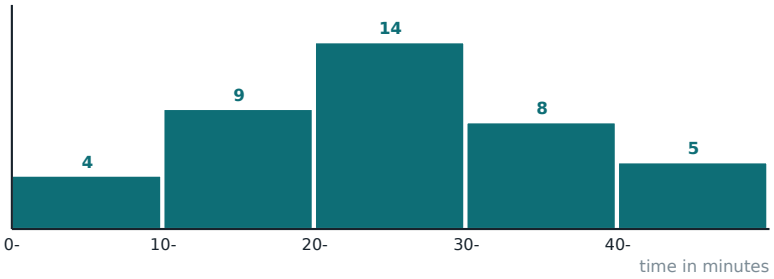
DEFINITION

Grouped data and the estimated mean

For grouped data, the **midpoint** of each interval represents its values, and the estimated mean is

$$\text{mean} \approx \frac{\Sigma\,(\text{midpoint} \times \text{frequency})}{\Sigma\,\text{frequency}}$$

The **modal class** is the interval with the greatest frequency, and the **median class** is the interval containing the middle value.



A histogram of grouped data. The bars touch because the groups are intervals on a continuous scale.

METHOD

Estimating the mean from a grouped table

1. Find the midpoint of each interval.
2. Multiply each midpoint by its frequency.
3. Add these products, and divide by the total frequency.
4. State that the answer is an estimate, and say why.

Interval	Midpoint	Frequency	Midpoint × frequency
6–10	8	4	32
11–15	13	9	117
16–20	18	14	252

Interval	Midpoint	Frequency	Midpoint × frequency
21–25	23	8	184
26–30	28	5	140
Total		40	725

Estimated mean of grouped times

Estimated mean = 725 ÷ 40 = 18.125 seconds. The true mean would need the individual times; using midpoints assumes the values in each interval are evenly spread, which is an assumption, not a fact.

COMMON ERROR

Using the upper boundary as the midpoint

For the interval 11–15 the midpoint is 13, not 15. Using the boundary overstates every value and inflates the estimated mean. The midpoint is the average of the two boundaries.

KEY IDEA

What grouping hides

Two data sets with the same grouped table can have different true means, medians and ranges. Once the raw values are replaced by intervals, the information is lost for ever: this is why the raw data should be kept, and why a grouped mean is always reported as an estimate.

WORKED EXAMPLE 10.4.1

Estimating the mean

GIVEN The grouped table above.

ASKED Estimate the mean and state the modal class.

METHOD Use midpoints and frequencies.

1. Products: 8 × 4 = 32, 13 × 9 = 117, 18 × 14 = 252, 23 × 8 = 184, 28 × 5 = 140.
2. Total = 725; frequency total = 40.
3. Estimated mean = 18.125.
4. The greatest frequency is 14, in the interval 16–20.

ANSWER about 18.1 seconds; modal class 16–20.

CHECK The mean lies inside the modal class, which is usual for a single-peaked distribution.

WORKED EXAMPLE 10.4.2

Finding the median class

GIVEN Frequencies 4, 9, 14, 8, 5 for five intervals, with a total of 40.

ASKED State the median class.

METHOD Find where the 20th and 21st values fall in the cumulative totals.

1. Cumulative: 4, 13, 27, 35, 40.
2. The 20th and 21st values lie in the third interval, since the count reaches 27 there.

ANSWER The median class is the third interval, 16–20.

CHECK The interval contains values 14 to 27 in order, which includes both middle positions.

WORKED EXAMPLE 10.4.3

Comparing two classes

GIVEN Class A: mean 62, range 8. Class B: mean 62, range 30.

ASKED Compare the two classes.

METHOD Compare both centre and spread.

1. The means are equal, so the classes performed similarly on average.
2. The ranges differ greatly: Class B's results are much more spread out.

ANSWER Same average, very different consistency; Class A is more consistent.

CHECK Two data sets can share an average and differ entirely in spread — which is why range or another measure of spread should be reported with any average.

WATCH OUT Reporting only the mean conceals differences that matter. A mean without a measure of spread is an incomplete summary.

PRACTICE 10.4 · Grouped data

1	Find the midpoint of the interval 20–30.	FLUENCY
2	Frequencies 3, 7, 5. Which is the modal class?	FLUENCY
3	Estimate the mean: intervals with midpoints 5, 15, 25 and frequencies 2, 6, 2.	FLUENCY
4	Total frequency of 4, 9, 14, 8, 5.	FLUENCY
5	Explain why a grouped mean is an estimate.	REASONING
6	Two classes have the same mean but different ranges. What does that tell you?	REASONING
7	Estimate the mean of: 0–10 (f 5), 10–20 (f 12), 20–30 (f 8).	PROBLEM SOLVING
8	Find the median class for frequencies 6, 10, 9 with a total of 25.	PROBLEM SOLVING
9	Explain why a grouped mean can differ from the true mean even when the grouping is accurate.	EXTENSION
10	A grouped table gives an estimated mean of 18.1 and the true mean is 17.4. Explain a possible reason.	EXTENSION

10.5 | Reading data critically

C · III · D · III, V

A statistic is a claim about the world, and claims can be checked. Reading critically means asking what was measured, how it was summarised and what the presentation leaves out.

the same three counts, drawn on two different vertical scales



The same counts drawn on two different scales. The second chart makes small differences look dramatic, because its axis does not start at zero.

METHOD

Five questions for any data claim

1. **What exactly was measured, and on whom?** A sample of volunteers is not a random sample.
2. **How many?** A difference of 10% in a sample of 20 may be one person.
3. **Which average was used, and why?** The mean and median can tell different stories.
4. **What is the spread?** An average without a range hides how consistent the data are.
5. **What does the graph do?** Check the axis, the scale, the units and whether the picture is proportional.

KEY IDEA

Correlation is not cause

Two quantities can rise together without one causing the other. Ice-cream sales and drowning incidents both rise in hot weather; the cause is the weather, not the ice cream. A data set can show an association; it cannot, by itself, show that one thing caused another.

Claim	Question to ask	Likely problem
“9 out of 10 prefer our brand”	How many were asked, and who?	Small or self-selected sample
“Average waiting time is 4 minutes”	Mean or median?	A few very short waits may pull the mean down
“Sales up 200%”	From what base?	A small base makes a large percentage
“Results improved”	Compared with what, and by how much?	No comparison group or effect size

Claims and their weaknesses

COMMON ERROR

Reporting a percentage without its base

“60% more students passed” means one thing if 10 passed last year and quite another if 200 did. The base belongs with the percentage — the same principle as in Chapter 5, where the base determined everything.

WORKED EXAMPLE 10.5.1

Interrogating a survey

GIVEN A school newsletter reports: “90% of students say the canteen is good value. 100 students were surveyed during one lunchtime.”

ASKED Identify three weaknesses and suggest an improvement.

METHOD Apply the five questions.

1. Sample: students present at one lunchtime may not represent those who never use the canteen.
2. Wording: “good value” is not defined, so responses may not be comparable.
3. Summary: no information on how many students exist overall, so the size of the sample relative to the school is unknown.

ANSWER Sampling bias, undefined wording and missing context; a random sample across year groups with a defined question would improve it.

CHECK A survey answered only by canteen users cannot measure satisfaction among non-users.

WORKED EXAMPLE 10.5.2

Comparing two summaries

GIVEN Two classes take a test. Class A: mean 65, median 66, range 12. Class B: mean 65, median 58, range 40.

ASKED Explain what each figure reveals and recommend which summary to report.

METHOD Compare centre and spread.

1. The means are equal, so the headline figure suggests no difference.
2. The medians differ by 8, showing Class B has more low scores; its range of 40 confirms a wide spread.
3. Reporting the mean alone hides the difference.

ANSWER Report the median and range as well as the mean; Class A is more consistent and higher in the middle.

CHECK A class with several very high scores can have a high mean and a low median.

WORKED EXAMPLE 10.5.3

Detecting a misleading graph

GIVEN A bar chart where bars have equal widths but the vertical axis starts at 40, and the shortest bar is cut off entirely.

ASKED State what the chart implies and what the data actually show.

METHOD Read the axes, not the picture.

1. Starting at 40 exaggerates the ratio between bars.
2. A cut-off bar makes a category look like zero when it may be 39.
3. The true differences must be read from the labels, not the heights.

ANSWER The chart implies large differences; the data may show small ones.

CHECK Redrawing with an axis from zero gives a proportional picture.

WATCH OUT A graph with a truncated axis is not necessarily dishonest, but it is misleading unless the break is clearly marked.

PRACTICE 10.5 · Reading data critically

1 A survey of 20 people reports 60% agree. How many people is that? FLUENCY

2	Which average can be distorted by one very large value?	FLUENCY
3	What is missing from a graph with no vertical scale?	FLUENCY
4	Why should a sample be random?	FLUENCY
5	Explain why a percentage should be reported with its base.	REASONING
6	Two variables rise together. Does that prove one causes the other?	REASONING
7	A shop reports sales up 50% from 20 units to 30. How many extra units would a 50% rise be from 200?	PROBLEM SOLVING
8	A class of 30 has a mean of 12 and a median of 14. What does this suggest about the data?	PROBLEM SOLVING
9	Explain how choosing the interval width can change the impression given by a histogram.	EXTENSION
10	Design a fair way to survey 200 students about homework time, and state one weakness of your design.	EXTENSION

F INVESTIGATE AND APPLY

A criterion B investigation and a criterion D application

INVESTIGATION 10 | How should data be grouped?

B · I, II, III

Extend the Explore activity into a general investigation of how the choice of interval changes what a reader can see.

INVESTIGATION

Investigation brief

Question. How does the choice of interval width change the appearance and the summary of a set of continuous data? **What you need.** The 30 times from the Explore activity, graph paper, and a calculator. **What to produce.** Three displays of the same data, a table of statistics for each grouping, and a recommendation with a justification.

PART 1 · DISPLAY

1. Draw a dot plot of all 30 raw times.
2. Draw a histogram with 5-second intervals.
3. Draw a histogram with 2-second intervals.
4. Describe one thing each display shows that the others do not.

PART 2 · SUMMARISE

Grouping	Modal class	Estimated mean	True mean
Raw data	not applicable		
5-second intervals			
2-second intervals			

Copy and complete

PART 3 · EVALUATE

1. Which grouping gives an estimated mean closest to the true mean? Why?
2. Can a grouping change the modal class? Explain with your data.
3. Write a recommendation for how the class should report its results, with two reasons.
4. Describe a situation in which wide intervals would be preferable to narrow ones.

CRITERION TASK
Criterion B task · Requirements

Three accurate displays; a completed table with estimates; a comparison with the true mean; a justified recommendation; an evaluation of when each grouping is appropriate.

APPLICATION 10 | The canteen survey

D · I, II, III, IV, V

The student council must survey the school canteen and report to the head teacher. Decisions about sampling, presentation and summary will shape what the head teacher concludes.

THE BRIEF

Item	Detail
Students in the school	600
Proposed sample	60 students at lunch on Monday
Questions	“Do you like the canteen?” and “How long do you wait?”
Waiting times collected	6, 8, 8, 9, 10, 10, 10, 11, 12, 12, 13, 15, 15, 16, 18, 20, 22, 25, 28, 30 minutes

The planned survey

- The head teacher wants a figure for average waiting time and a statement about satisfaction.
- The council has one week and no budget.
- The report must be honest about its limitations.

TASK 1 · ANALYSE THE DATA

1. Find the mean, median, mode and range of the 20 waiting times.
2. Group the times into intervals of your choice and find the estimated mean.
3. Compare the mean and median, and explain which better represents a typical wait.

TASK 2 · DESIGN A BETTER SURVEY

1. Propose a sampling method that reduces bias, and explain why it is better than the Monday lunchtime sample.
2. Suggest one improved question about satisfaction, and explain what was wrong with the original.
3. State what additional data the council should collect to support its recommendation.

TASK 3 · REPORT

1. Write three sentences reporting the findings, using at least two statistics.
2. Include one graph or table that presents the waiting times honestly, and state its scale.

3. State two limitations of the survey and one recommendation, each justified by your analysis.

CRITERION TASK
Criterion D task · Requirements

All statistics correct and reported with units; a suitable grouping and estimate; a sampling method justified; a presentation that follows the conventions from §10.2; limitations acknowledged.

TASK C | Writing the graph-checking guide

C · I, III, IV

CRITERION TASK
The task

Write a one-page guide called **“Five questions to ask about any graph”**, to be displayed in the mathematics room.

1. Five questions, each with a short explanation of why it matters.
2. Two examples of a graph that misleads, redrawn correctly, with a note on what changed.
3. A checklist a student can use in ten seconds.

Criterion C strand	Evidence
Notation (i)	Correct use of axis, scale, unit and average.
Reasoning (iii)	Each question explains the effect of getting it wrong.
Organisation (iv)	The guide can be used without further explanation.

Success criteria

G REFLECT AND CONSOLIDATE

Summary, vocabulary, misconceptions and review

REVIEW | Consolidating Chapter 10

A · I, II, III

WHAT THIS CHAPTER ESTABLISHED

- Data are qualitative or quantitative; quantitative data are discrete or continuous.
- A frequency table records counts and can be checked by its total.
- Bar charts suit categories, dot plots suit small discrete sets, histograms suit grouped continuous data.
- The mean, median and mode summarise the centre in different ways; the range measures spread.
- Grouped data can only give an estimated mean, using interval midpoints.
- Graphs and statistics can mislead through sampling, choice of average, truncated axes and missing context.

ESSENTIAL VOCABULARY

Term	Meaning in this course
Qualitative data	Data describing a category.
Discrete data	Countable values, usually whole numbers.
Continuous data	Measurements that can take any value in a range.
Frequency	The number of times a value occurs.
Mean	Total divided by the number of values.
Median	The middle value of ordered data.
Mode	The most frequent value.
Range	The difference between the largest and smallest values.
Histogram	A chart of grouped continuous data with touching bars.
Modal class	The interval with the greatest frequency.

Vocabulary for Chapter 10

COMMON MISCONCEPTIONS

Misconception	Correction	Evidence
“The median is the middle of the list as written.”	Order the data first.	9 in 7, 2, 9, 4, 5
“The range is an average.”	It measures spread.	15 is not a centre
“The mean is always the best summary.”	One extreme value can distort it.	7 salaries
“Touching bars are always wrong.”	Histograms need them.	continuous intervals
“A bigger bar means more if the axis is cut.”	Read the scale.	axis from 40

Misconception, correction and evidence

SELF-ASSESSMENT

I can …	Not yet	Getting there	Confident
classify data and build a frequency table.			
choose and draw a suitable display for a data set.			
calculate and interpret the mean, median, mode and range.			
estimate the mean of grouped data and name the modal class.			
criticise a graph or a statistical claim using specific evidence.			
report data honestly, including limitations.			

How confident are you?

REFLECTION

1. Which average do you now use first, and why?

2. Describe a graph or headline you have seen that this chapter would help you question.
3. Write one sentence explaining why a grouped mean is an estimate.
4. In the canteen task, which decision had the largest effect on the conclusion?
5. What will you check first the next time you see a bar chart?

REVIEW QUESTIONS | Practice across the whole chapter

A · I, II, III

PRACTICE RQ10 · Applying the chapter

1	Classify: time to complete a race.	FLUENCY
2	Find the mean of 6, 9, 12, 13.	FLUENCY
3	Find the median of 5, 8, 8, 11, 14.	FLUENCY
4	Find the mode of 4, 6, 6, 7, 9, 6.	FLUENCY
5	Find the range of 3, 11, 7, 15.	FLUENCY
6	Find the midpoint of the interval 30–40.	FLUENCY
7	Explain why the median can be a better summary than the mean.	REASONING
8	A histogram has bars of unequal width. Why is that a problem?	REASONING
9	Explain why the mode is the only average available for colours.	REASONING
10	The mean of 8 values is 15. Find their total.	PROBLEM SOLVING
11	Estimate the mean: 0–20 (f 4), 20–40 (f 10), 40–60 (f 6).	PROBLEM SOLVING
12	A pie chart of 300 students gives football 120°. How many chose football?	PROBLEM SOLVING
13	Explain how to check a frequency table.	COMMUNICATING
14	Explain why a 50% increase from 20 to 30 is the same percentage as from 200 to 300.	COMMUNICATING
15	A data set of 5 values has mean 10 and median 9. Find a possible set.	EXTENSION
16	Two data sets have the same range but different means. Illustrate.	EXTENSION
17	Explain how the choice of sample can change a survey's conclusion.	EXTENSION
18	40 students' times are grouped: 10–14 (4), 15–19 (9), 20–24 (14), 25–29 (8), 30–34 (5). Estimate the mean and name the modal class.	PROBLEM SOLVING
19	Explain why an estimated mean from grouped data may differ from the true mean.	COMMUNICATING

MIXED 10 | Fluency, reasoning and problem solving

A · I, II, III

PRACTICE MX10 · End-of-chapter mixed set

1	Find the mean of 12, 15, 18, 19.	FLUENCY
2	Find the median of 2, 7, 9, 10.	FLUENCY

3	Find the range of 14, 6, 9, 21.	FLUENCY
4	Total frequency of 5, 8, 12, 3.	FLUENCY
5	How many degrees in a pie chart represent 25% of the data?	FLUENCY
6	Explain why “average” is ambiguous in a newspaper report.	REASONING
7	A graph shows a rise from 40 to 44 with an axis starting at 39. What impression does it create?	REASONING
8	A student scores 60, 72, 68, 80, 60. Find the mean, median and mode.	PROBLEM SOLVING
9	A class of 25 has a mean height of 160 cm. A new student of 185 cm joins. Find the new mean.	PROBLEM SOLVING
10	A bar chart of 90 students shows 30 choosing A. What is the pie-chart angle for A?	PROBLEM SOLVING
11	Explain how to decide whether to use a bar chart or a histogram.	COMMUNICATING
12	Explain why a report should state its sample size.	COMMUNICATING
13	A data set has mean 20 and range 10. Suggest what the effect of adding a value of 60 would be.	EXTENSION
14	Eight values have a sum of 96. Two of them are removed and the mean of the remaining six is 11. Find the sum of the two removed.	EXTENSION
15	Explain why two data sets can have identical means and medians but different shapes.	EXTENSION
16	A survey of 200 students finds 120 walk, 50 cycle and 30 take the bus. Draw a table, find the pie-chart angles and state one conclusion.	PROBLEM SOLVING

CUMULATIVE PRACTICE 10 | Questions that mix this chapter with earlier ones

A · I, II

PRACTICE CP10 · Cumulative practice

1	Find the median of 4, 9, 11, 13, 15.	FLUENCY
2	Find the mean of 7, 9, 12, 16.	FLUENCY
3	Find the range of 6, 14, 9, 20.	FLUENCY
4	Explain why a histogram has no gaps between bars.	REASONING
5	A pie chart of 180 students shows 60 choosing sport. Find the angle.	PROBLEM SOLVING
6	Estimate the mean of grouped data: 0–10 (f 3), 10–20 (f 9), 20–30 (f 8).	PROBLEM SOLVING
7	Explain why a report should give the range as well as the mean.	COMMUNICATING
8	A data set of 6 values has mean 12. Five values are 8, 10, 11, 14, 15. Find the sixth.	EXTENSION

EFERENCE

CUMULATIVE REVISION

Retrieval practice across parts

The five papers in this section revisit earlier chapters deliberately. Each paper mixes topics, because the ability to choose a method — not only to apply one — is what revision develops.

Attempt each paper without notes first, then mark it and return only to the questions you could not finish. A paper that is completed quickly but inaccurately is less useful than one that is attempted slowly and checked.

Paper	Chapters	Main focus
Cumulative revision 1	1–3	Place value, powers, factors
Cumulative revision 2	4–5	Fractions, decimals, percentages
Cumulative revision 3	6–7	Algebra and geometry
Cumulative revision 4	8–9	Sets and measurement
Cumulative revision 5	1–10	End-of-year mixed paper

What each paper covers

CUMULATIVE REVISION 1 | Chapters 1–3

A · I, II · D · II

PRACTICE CR1 · Cumulative practice 1

1	Write 4,706 in expanded form.	FLUENCY
2	Round 38,472 to the nearest hundred.	FLUENCY
3	Evaluate $2^5 + 3^2$.	FLUENCY
4	Evaluate $48 \div 4 \times 2$.	FLUENCY
5	List the factors of 42.	FLUENCY
6	Write 84 as a product of primes.	FLUENCY
7	Explain why 1 is neither prime nor composite.	REASONING
8	A student writes $3^2 = 6$. Explain the error.	REASONING
9	Explain why the digit sum test works for 3.	REASONING
10	A museum reports 11,846 visitors. Write this rounded to the nearest thousand and say why a report might do that.	PROBLEM SOLVING
11	Find the HCF and LCM of 24 and 36.	PROBLEM SOLVING
12	72 students and 60 parents are split into identical groups containing both. Find the largest number of groups.	PROBLEM SOLVING
13	Explain why 91 is not prime.	EXTENSION
14	Find the smallest number by which 108 must be multiplied to make a square.	EXTENSION
15	Without calculating, decide whether 2^{10} is greater than 1,000.	EXTENSION
16	Explain why a number with an odd number of factors must be a square.	COMMUNICATING

CUMULATIVE REVISION 2 | Chapters 4–5

A · I, II · D · III

PRACTICE CR2 · Fractions and percentages

1	Simplify $^{42}/_{56}$.	FLUENCY
2	Write $^{7}/_{8}$ as a decimal.	FLUENCY
3	Calculate $^{3}/_{4} + {}^{2}/_{5}$.	FLUENCY
4	Calculate $^{5}/_{6} \times {}^{9}/_{10}$.	FLUENCY
5	Find 15% of 240.	FLUENCY
6	Write 0.45 as a percentage.	FLUENCY
7	Explain why $^{2}/_{3}$ is greater than $^{3}/_{5}$.	REASONING
8	Explain why adding fractions needs a common denominator but multiplying does not.	REASONING
9	Explain why a 20% discount followed by 10% is not 30%.	REASONING
10	A jacket costs 84 after a 30% reduction. Find the original price.	PROBLEM SOLVING
11	A recipe for 8 uses $^{3}/_{4}$ kg of flour. How much for 20?	PROBLEM SOLVING
12	A price rises 15% then falls 10%. Find the overall change.	PROBLEM SOLVING
13	Find a fraction between $^{3}/_{7}$ and $^{4}/_{7}$.	EXTENSION
14	Write 0.4545… as a fraction.	EXTENSION
15	Two successive increases of 25% equal what single increase?	EXTENSION
16	Explain how to compare $^{7}/_{12}$ and $^{9}/_{16}$ reliably.	COMMUNICATING

CUMULATIVE REVISION 3 | Chapters 6–7

A · I, II, III

PRACTICE CR3 · Algebra and geometry

1	Find the rule for 6, 10, 14, 18, …	FLUENCY
2	Evaluate $3x^2 - 5$ when $x = 4$.	FLUENCY
3	Simplify $5a + 3 - 2a + 7$.	FLUENCY
4	Solve $4x - 7 = 21$.	FLUENCY
5	Two angles of a triangle are 64° and 58°. Find the third.	FLUENCY
6	Two angles on a straight line: one is 115°. Find the other.	FLUENCY
7	Explain why $3(x + 2)$ is not $3x + 2$.	REASONING
8	Explain why a triangle cannot have angles 90°, 80° and 20°.	REASONING
9	Explain why a sequence with constant difference 5 has a rule containing $5n$.	REASONING

10	A taxi charges 4 plus 2 per km. Find the cost of 9 km and the distance for 30 units.	PROBLEM SOLVING
11	Solve $5(x - 2) = 3x + 4$.	PROBLEM SOLVING
12	A rectangle is 5 cm longer than it is wide and its perimeter is 46 cm. Find its area.	PROBLEM SOLVING
13	The angles of a triangle are $2x$, $3x$ and $4x$. Find the largest.	EXTENSION
14	Solve $\frac{x}{4} + 5 = 2x - 1$.	EXTENSION
15	Explain why the angles of an equilateral triangle are 60°.	EXTENSION
16	Explain how a balance model justifies solving an equation by inverse operations.	COMMUNICATING

CUMULATIVE REVISION 4 | Chapters 8–9

A · I, II · D · I

PRACTICE CR4 · Sets and measurement

1	$A = 2, 4, 6$, $B = 4, 6, 8$. Find $A \cup B$.	FLUENCY
2	$n(A) = 14$, $n(B) = 9$, $n(A \cap B) = 5$. Find $n(A \cup B)$.	FLUENCY
3	Find the area of a triangle with base 15 cm and height 8 cm.	FLUENCY
4	Find the volume of a cuboid 7 cm by 5 cm by 4 cm.	FLUENCY
5	Convert 3,250 cm^3 to litres.	FLUENCY
6	Find the surface area of a cube of side 5 cm.	FLUENCY
7	Explain why the overlap is subtracted in the union formula.	REASONING
8	Explain why area uses square units.	REASONING
9	Explain why a litre is 1,000 cm^3.	REASONING
10	In a class of 34, 20 study French, 16 study Spanish and 6 study both. How many study neither?	PROBLEM SOLVING
11	A room is 5 m by 4 m and a 2 m by 1 m rug covers part of it. What area of floor is uncovered?	PROBLEM SOLVING
12	A tank 60 cm by 40 cm by 30 cm is $\frac{2}{3}$ full. How many litres does it hold?	PROBLEM SOLVING
13	Two sets have 25 and 18 elements in a universal set of 30. Find the smallest possible overlap.	EXTENSION
14	A cuboid has volume 360 cm^3, length 10 cm and width 6 cm. Find its surface area.	EXTENSION
15	Explain why a cube uses less material than a long box of the same volume.	EXTENSION
16	Explain how a Venn diagram shows that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.	COMMUNICATING

CUMULATIVE REVISION 5 | Chapters 1–10

A · I, II, III · C · III

PRACTICE CR5 · End-of-year mixed paper

1	Round 5.076 to two decimal places.	FLUENCY
2	Evaluate $5 + 3 \times 4^2$.	FLUENCY
3	Write $^{18}/_{24}$ in simplest form.	FLUENCY
4	Increase 180 by 35%.	FLUENCY
5	Solve $7x + 5 = 40$.	FLUENCY
6	Find the mean of 8, 11, 14, 15.	FLUENCY
7	Find the area of a parallelogram with base 13 cm and height 6 cm.	FLUENCY
8	Explain why 0.3 and $^1/_3$ are not equal.	REASONING
9	Explain why the mean can misrepresent a data set.	REASONING
10	Explain why successive percentage changes multiply.	REASONING
11	A shop sells a 3-litre pack for 7.50 and a 5-litre pack for 11.80. Which is better value?	PROBLEM SOLVING
12	A survey of 120 students: 70 walk, 45 cycle, 25 do both. How many do neither?	PROBLEM SOLVING
13	A cube has surface area 294 cm². Find its volume.	PROBLEM SOLVING
14	Explain how you would decide whether to use the mean or the median for a set of house prices.	COMMUNICATING
15	Explain why a rule found from four terms is a conjecture rather than a proof.	COMMUNICATING
16	A rectangle's length is 3 cm more than twice its width and its perimeter is 48 cm. Find its area.	EXTENSION
17	A number is increased by 20% and then decreased by 20%; the result is 96. Find the original.	EXTENSION
18	Two angles of an isosceles triangle are 40° and 70°. Find the third and classify the triangle.	EXTENSION

PART REVIEW · PROBLEM SOLVING ACROSS EACH PART

Reference

Each part review is longer than the cumulative papers and is designed for the end of a term. The questions are deliberately unfamiliar: they require a choice of method rather than a recall of one.

Work through a part review in one sitting of about 45 minutes, without notes. Mark your work against the answers, then write down the two questions that caused most difficulty and the mathematical idea behind each. That short note is more useful for revision than a second attempt at the same paper.

PART REVIEW I | The grammar of number (Chapters 1–3)

A · I, III · D · II

PRACTICE PR1 · Part I problem solving

1	Round 4,096 to the nearest hundred and to the nearest thousand.	FLUENCY
2	Evaluate $6 + 2^3 \times 3$.	FLUENCY
3	Find the HCF and LCM of 40 and 60.	FLUENCY
4	A student claims that multiplying a number by 10 always adds a zero. Test the claim on 4.5 and on 0.3.	REASONING
5	Explain why the sum of two multiples of 6 is always a multiple of 6.	REASONING
6	A museum has 11,846 objects and reports “about 12,000”. Write the smallest and largest counts that could honestly be reported in the same way.	PROBLEM SOLVING
7	84 Year 7 students, 60 Year 8 and 48 Year 9 students are split into identical teams. Find the largest possible team size and the number of teams.	PROBLEM SOLVING
8	Explain how the place-value system makes the digit-sum test for 9 possible.	COMMUNICATING
9	Explain why the product of two consecutive whole numbers is always even.	COMMUNICATING
10	A number between 200 and 300 has exactly six factors and is divisible by 9. Find it.	EXTENSION
11	Find the smallest whole number with exactly 12 factors.	EXTENSION
12	Explain why a claim that “a number is divisible by 8 if its last two digits are divisible by 8” is false, and correct it.	COMMUNICATING
13	Two bells ring every 15 and 18 minutes. They ring together at 9:00. When do they next ring together, and how many times in 6 hours?	EXTENSION
14	A rectangular floor is 4.5 m by 3.6 m. Square tiles of whole-centimetre side cover it exactly. Find the largest possible tile side.	REASONING

PART REVIEW II | Parts and proportions (Chapters 4–5)

A · II · C · III

PRACTICE PR2 · Part II problem solving

1	Calculate $\frac{3}{4} - \frac{2}{5}$.	FLUENCY
2	Calculate $\frac{5}{6} \times 1\frac{1}{5}$.	FLUENCY
3	Increase 340 by 12.5%.	FLUENCY
4	Explain why $\frac{3}{8}$ is smaller than $\frac{2}{5}$ although $3 > 2$.	REASONING
5	A shop increases a price by 10% and later reduces it by 10%. Explain why the final price is not the original.	REASONING
6	A recipe for 12 uses $\frac{5}{8}$ kg of sugar. How much for 30, and how many 1 kg bags are needed?	PROBLEM SOLVING

7	A jacket is reduced by 20% to 96. A second jacket is reduced by 30% to 98. Which was more expensive originally?	PROBLEM SOLVING
8	Explain why dividing by a fraction less than 1 gives an answer larger than the original number.	COMMUNICATING
9	Explain why 0.375 and $^{3}/_{8}$ are the same number, using place value.	COMMUNICATING
10	A price rises 25%, then falls 20%. Find the overall change.	EXTENSION
11	Find a fraction between $^{5}/_{8}$ and $^{2}/_{3}$.	EXTENSION
12	Explain how to compare $^{7}/_{9}$ and $^{8}/_{11}$ reliably and state which is larger.	COMMUNICATING
13	A student scores 45% then 54%. Describe the change in the score as a percentage of the first score.	EXTENSION

PART REVIEW III | Algebra and shape (Chapters 6–7)

A · III · B · III

PRACTICE PR3 · Part III problem solving

1	Find the rule for 11, 18, 25, 32, ...	FLUENCY
2	Solve $6x - 5 = 2x + 11$.	FLUENCY
3	Two parallel lines; one corresponding angle is 116°. Find the co-interior angle.	FLUENCY
4	Explain why a sequence with constant difference 4 cannot have rule $4n + 3$ if its first term is 9.	REASONING
5	Explain why drawing an accurate diagram does not prove an angle property.	REASONING
6	A pattern of squares uses 4 matches in stage 1 and 7 in stage 2. How many matches in stage 20, and which stage uses 100?	PROBLEM SOLVING
7	A triangle has angles $2x$, $3x + 10$ and $4x - 10$. Find each.	PROBLEM SOLVING
8	Explain how the balance model justifies each step of solving $3(x - 2) = 15$.	COMMUNICATING
9	Explain why the exterior angle of a triangle equals the sum of the two opposite interior angles.	COMMUNICATING
10	A rectangle's perimeter is 54 cm and its length is 3 cm more than twice its width. Find its area.	EXTENSION
11	Construct, in writing, a triangle with sides 5 cm, 6 cm and 7 cm; explain why exactly one triangle is possible.	EXTENSION
12	Explain why three lengths 2 cm, 3 cm and 5 cm do not form a triangle.	COMMUNICATING
13	An isosceles triangle has a base angle of 3x and a vertex angle of 4x. Find each.	EXTENSION
14	A student writes the rule for 5, 9, 13 as $5n$. Explain the error using the difference.	REASONING

PART REVIEW IV | Sets and measurement (Chapters 8–9)

A · II · D · III

PRACTICE PR4 · Part IV problem solving

1	$A = 2, 3, 5, 7$, $B = 1, 2, 3$. Find $A \cap B$ and $A \cup B$.	FLUENCY
2	Find the surface area of a cuboid 9 cm by 6 cm by 5 cm.	FLUENCY
3	Convert 0.8 m^3 to litres.	FLUENCY
4	Explain why “pass” and “fail” is a sound classification but “good” and “bad” is not.	REASONING
5	Explain why doubling all three dimensions of a cuboid multiplies the volume by 8 but the surface area by 4.	REASONING
6	In a group of 80, 45 own a bicycle, 38 own a helmet and 12 own neither. How many own both?	PROBLEM SOLVING
7	A tank is 1.2 m by 0.9 m by 0.7 m. Find its capacity in litres and the cost of filling it at 0.03 per litre.	PROBLEM SOLVING
8	Explain how a classification tree prevents a student being counted twice.	COMMUNICATING
9	Explain why the surface area of an open box is less than that of the closed box.	COMMUNICATING
10	A cuboid has a square base of side 6 cm and volume 216 cm^3. Find its surface area.	EXTENSION
11	In a survey of 120, 70 like A, 55 like B and 15 like neither. Find the number who like exactly one.	EXTENSION
12	Explain why a litre is not 100 cm^3 even though 1 m = 100 cm.	COMMUNICATING
13	A cylindrical tank has radius 0.6 m and height 1.5 m. Find its capacity in litres ($\pi \approx 3.14$).	EXTENSION
14	A shop sells boxes 20 × 10 × 5 cm and 25 × 8 × 5 cm at the same price. Which is better value, and does the answer change if card cost matters?	REASONING

PART REVIEW V | Data and decisions (Chapter 10, with Chapters 1–9)

C · III · D · V

PRACTICE PR5 · Part V problem solving

1	Find the mean, median and range of 12, 15, 15, 18, 20.	FLUENCY
2	A pie chart of 240 students gives 90° to football. How many chose football?	FLUENCY
3	Estimate the mean of: 0–10 (f 4), 10–20 (f 10), 20–30 (f 6).	FLUENCY
4	Explain why the median is preferred for house prices.	REASONING
5	Explain why a bar chart with an axis from 50 can mislead even with correct numbers.	REASONING
6	A survey of 250 finds 150 walk, 90 cycle and 40 do both. How many do neither, and what percentage is that?	PROBLEM SOLVING

7	A class of 28 has a mean score of 62. One student's score is corrected from 40 to 58. Find the new mean.	PROBLEM SOLVING
8	Explain how to report a survey honestly in three sentences.	COMMUNICATING
9	Explain why a percentage without a base is uninformative.	COMMUNICATING
10	A data set of 9 values has median 30. Two values are removed and the median stays 30. What can you say about the removed values?	EXTENSION
11	The mean of 10 values is 24. One value of 44 is removed. Find the new mean.	EXTENSION
12	Explain why grouping data into intervals always changes the mean.	COMMUNICATING
13	Two classes have means 60 and 64 and ranges 22 and 8. Compare them fully.	EXTENSION
14	A report says “results improved by 12%” without a comparison group. Explain what is missing.	REASONING

EXTENDED PROJECT | Designing a mathematics trail

PROJECT

This project brings the whole course together. It is designed to be completed over two or three lessons, in groups, and it produces a piece of work that can be assessed against all four criteria.

THE BRIEF

Your group will design a **mathematics trail** for visitors to your school or neighbourhood: a route of six stations, each with a mathematical task that uses a measurement, a calculation or a data set from that place. The trail must be usable by a student in the year below without a teacher's help.

STAGE 1 · CHOOSE AND MEASURE

1. Walk the route and choose six stations: for example a staircase, a noticeboard, a sports court, a tree or a window.
2. At each station, take the measurements needed to write a task. Record them with units and note how accurate each measurement is.
3. Photograph or sketch each station, and note one safety point.

STAGE 2 · WRITE THE TASKS

1. Write one task per station. Across the six tasks, cover at least four of: place value and rounding; powers; factors; fractions or percentages; algebra; angle or shape; perimeter, area or volume; sets; data.
2. Give each task a clear instruction, the data needed, and a space for the answer.
3. Write the answers on a separate sheet, with the method for each.

STAGE 3 · TEST AND REVISE

1. Ask another group to attempt two of your tasks. Record what confused them.
2. Revise the wording of any task that was misunderstood, and say what you changed.
3. Check every answer again, including the units.

STAGE 4 · PRESENT

1. Produce a trail leaflet: a map, the six stations, the tasks and a safety note.
2. Write a short introduction explaining what a visitor will practise.

Criterion	Evidence in the project
A · Knowing and understanding	Correct mathematics in the tasks and answers, with consistent units.
B · Investigating patterns	At least one station where the visitor must find a general rule from measurements.
C · Communicating	The leaflet: clear instructions, labelled diagrams, accurate notation.
D · Applying mathematics	Real measurements, real places, a usable product, limitations stated.

How the project is assessed

> Keep this question for the end: which station taught the visitor the most mathematics, and why?

PRACTICE PAPER A | Chapters 1–5

45 MARKS · 60 MINUTES

A paper written to look like a school examination. Work through it without notes, then mark it with the answers at the back. The mark allocation tells you how much working is expected: one mark for a correct answer with no working, two or three where a method must be shown.

INSTRUCTIONS

1. Answer all questions. Write your working, not only your answer.
2. Calculators are not allowed in Section 1; they are allowed in Section 2.
3. The paper is worth 45 marks. Allow roughly one minute per mark, with ten minutes to check.
4. Where a question says “explain”, a calculation alone does not earn the mark.

PRACTICE PA · Practice paper A

1	Write 3,407,052 in words.	FLUENCY
2	Round 4,836 to the nearest hundred and to the nearest ten.	FLUENCY
3	Write $2^4 \times 5^2$ as a single number.	FLUENCY
4	Work out $48 \div (6 - 2) + 3^2$.	FLUENCY
5	List the factors of 84.	FLUENCY
6	Find the HCF and LCM of 36 and 48.	FLUENCY
7	Write $\frac{7}{8}$ as a decimal and as a percentage.	FLUENCY
8	Work out $\frac{5}{6} - \frac{3}{8}$, giving the answer in simplest form.	FLUENCY
9	Work out 0.6×0.45 and explain where the decimal point goes.	FLUENCY
10	Explain why 1 is not a prime number.	REASONING
11	A student writes $\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$. What has gone wrong, and what is the correct answer?	REASONING

12	A recipe for 6 people uses 450 g of rice. How much is needed for 15 people?	PROBLEM SOLVING
13	A jacket costs 240 units and is reduced by 15%. Find the sale price and the saving.	PROBLEM SOLVING
14	A bill of 354 units includes 18% VAT. Find the price before tax.	PROBLEM SOLVING
15	Show that a 20% increase followed by a 20% decrease does not return to the starting value.	EXTENSION

PRACTICE PA2 · Practice paper A

1	Work out 3.7×1.45, giving the answer to two decimal places.	FLUENCY
2	Write $^{5}/_{12}$ as a recurring decimal.	FLUENCY
3	Increase 640 by 12.5%.	FLUENCY
4	Find 15% of 80 and explain how a 10% and a 5% could be combined instead.	FLUENCY
5	Explain why multiplying by 0.9 is the same as reducing by 10%.	REASONING
6	Two shops advertise “20% off” and “one third off”. Which is the larger reduction?	REASONING
7	A car travels 132 km on 11 litres. Find the distance per litre and the fuel needed for 300 km.	PROBLEM SOLVING
8	A class of 32 has 12 students who walk to school. What percentage walk? Give the answer to one decimal place.	PROBLEM SOLVING
9	A phone plan charges 8 units a month plus 0.15 units per minute. Find the cost for 90 minutes.	PROBLEM SOLVING
10	A shop marks a price up by 25% and later discounts it by 25%. Is the final price above, below or equal to the original?	PROBLEM SOLVING
11	A price is reduced by 10% and then by a further 10%. Find the single percentage reduction.	EXTENSION
12	The value of a machine falls by 15% each year. Find its value after two years if it starts at 4,000 units.	EXTENSION
13	A student claims “a 50% increase and a 50% decrease cancel out”. Explain why the claim is false.	EXTENSION
14	Write 0.363636… as a fraction in simplest form, showing the method.	EXTENSION
15	A tank is 40% full. Adding 60 litres makes it 70% full. Find the capacity.	EXTENSION

PRACTICE PAPER B | Chapters 6–10

45 MARKS · 60 MINUTES

The second paper covers the algebra, shape, sets, measurement and data chapters. As in Paper A, the mark allocation shows how much working is expected, and the answers are in the back of the book.

INSTRUCTIONS

1. Answer all questions. All working should be shown.
2. A calculator may be used throughout. Use $\pi \approx 3.14$ unless a question says otherwise.
3. The paper is worth 45 marks, in two sections of 15 marks each.

PRACTICE PB · Practice paper B

1	Simplify $5x + 3y - 2x + y$.	FLUENCY
2	Expand $4(2a - 3)$.	FLUENCY
3	Solve $3x + 7 = 28$.	FLUENCY
4	Solve $2(x - 3) = 14$.	FLUENCY
5	Find the value of $3n^2$ when $n = 4$.	FLUENCY
6	Work out the size of each angle in an isosceles triangle with vertex angle 46°.	FLUENCY
7	Find the interior angle sum of a nonagon (9 sides).	FLUENCY
8	Find each exterior angle of a regular octagon.	FLUENCY
9	List the elements of $A \cup B$ when $A = 2, 4, 6, 8$ and $B = 3, 6, 9$.	FLUENCY
10	Explain why a triangle cannot have angles 100°, 90° and 10°.	REASONING
11	Explain why $\sqrt{2}$ is not a rational number.	REASONING
12	A parallelogram has base 14 cm and height 9 cm. Find its area.	PROBLEM SOLVING
13	A cuboid measures 8 cm by 5 cm by 4 cm. Find its volume and surface area.	PROBLEM SOLVING
14	A circle has radius 6 cm. Find its circumference and area, using $\pi \approx 3.14$.	PROBLEM SOLVING
15	The angles of a quadrilateral are $2x$, $3x$, $4x$ and $6x$. Find each angle.	EXTENSION

PRACTICE PB2 · Practice paper B

1	A survey of 40 students: 22 like football, 18 like basketball, 8 like both. How many like neither?	FLUENCY
2	Find the mean, median and mode of 4, 7, 7, 9, 13.	FLUENCY
3	Find the range of 12, 18, 9, 25, 14.	FLUENCY
4	Explain why a pie chart cannot show how many people gave each answer.	REASONING
5	A graph's vertical axis starts at 90 instead of 0. Explain how this can mislead.	REASONING
6	A water tank is a cuboid 1.2 m by 0.8 m by 0.5 m. Find its capacity in litres.	PROBLEM SOLVING
7	A rectangle 12 m by 5 m has a circular pond of diameter 4 m inside it. Find the area of the grass.	PROBLEM SOLVING
8	A set of 12 test scores has mean 15. One score of 27 is removed. Find the new mean.	PROBLEM SOLVING
9	The probability-free approach: a bag has 5 red and 7 blue counters. What fraction are red?	PROBLEM SOLVING
10	A cuboid has surface area 214 cm² and dimensions 9 cm by 5 cm by h. Find h.	EXTENSION
11	A regular polygon has interior angles of 150°. How many sides has it?	EXTENSION

12	In a survey, 60% of 250 people chose option A and 25% chose option B. How many chose neither?	EXTENSION
13	A histogram shows frequencies 4, 9, 14, 8 in classes of width 5. Estimate the mean using midpoints.	EXTENSION
14	Explain how you would check whether a reported “average” is a mean or a median.	EXTENSION
15	A box is 30 cm by 20 cm by 15 cm. A second box doubles each dimension. How many times larger is its volume?	EXTENSION

PRACTICE PAPER C | Mixed paper across all ten chapters

50 MARKS · 75 MINUTES

A longer mixed paper for the end of the year. Questions move between chapters deliberately: a question about a tank may need conversion, rounding and percentage work in the same answer. The mark allocation again shows where working must be shown.

INSTRUCTIONS

1. Answer all questions. A calculator may be used throughout, except in questions 1–6.
2. Give units with every measured answer, and money to two decimal places.
3. The paper is worth 50 marks. Answers are in the back of the book.

PRACTICE PC · Practice paper C

1	Write 5.06×10^3 as an ordinary number.	FLUENCY
2	Round 0.0467 to two significant figures.	FLUENCY
3	Work out $2.5 + 0.75 \times 4$.	FLUENCY
4	Simplify $7a - 3b - 2a + 5b$.	FLUENCY
5	Find 35% of 240.	FLUENCY
6	Write $\frac{9}{20}$ as a percentage.	FLUENCY
7	Explain why multiplying two numbers less than 1 can give an answer smaller than either.	REASONING
8	Explain the difference between a factor and a multiple, with one example of each.	REASONING
9	A rectangular field is 45 m by 32 m. Find its area and the length of fence needed to go round it.	PROBLEM SOLVING
10	A circular flower bed has radius 2.5 m. Find the area to be paved around it if the surrounding square is 6 m by 6 m.	PROBLEM SOLVING
11	A shop buys an item for 42 units and sells it for 58.80 units. Find the percentage profit.	PROBLEM SOLVING
12	A recipe for 8 people uses 600 g of flour. How much for 14 people?	PROBLEM SOLVING
13	A cube has volume 343 cm³. Find its side length and surface area.	PROBLEM SOLVING

14	The mean of five numbers is 12. Four of them are 9, 11, 14 and 15. Find the fifth.	PROBLEM SOLVING
15	A price rises by 12% and the new price is 336 units. Find the original price.	EXTENSION
16	Show that the sum of any three consecutive whole numbers is a multiple of 3.	EXTENSION

PRACTICE PC2 · Practice paper C

1	Solve $4x - 9 = 23$.	FLUENCY
2	Find the size of each angle in a regular pentagon.	FLUENCY
3	$U = \{1, 2, \ldots, 12\}$; A = multiples of 4. List A'.	FLUENCY
4	Explain why a square is a special rectangle but a rectangle is not always a square.	REASONING
5	Explain how a bar chart with a broken axis can mislead a reader.	REASONING
6	A tank 1.5 m by 1.2 m by 0.8 m is three-quarters full. How many litres are in it?	PROBLEM SOLVING
7	A train travels 240 km in 2 hours 30 minutes. Find its average speed in km/h.	PROBLEM SOLVING
8	A laptop costs 720 units; VAT at 18% is added. Find the total and the VAT.	PROBLEM SOLVING
9	A survey shows 45% of 240 students walk to school. How many is that?	PROBLEM SOLVING
10	A 6 m ladder leans against a wall with its foot 2 m from the wall. Explain why the right angle matters here, without calculating the height.	PROBLEM SOLVING
11	A cuboid 12 cm by 9 cm by 6 cm is painted and then cut into 3 cm cubes. How many small cubes have paint on exactly two faces?	EXTENSION
12	The mean of 20 numbers is 15. Each number is increased by 3 and then doubled. Find the new mean.	EXTENSION
13	A shop offers 30% off, then a further 10% off the reduced price. Find the single percentage saving.	EXTENSION
14	Explain why the median can be a better summary than the mean when one value is very large.	EXTENSION
15	A cylinder has radius 5 cm and height 10 cm. Estimate its volume and say whether a 1 litre bottle is large enough.	EXTENSION

WORKED SOLUTIONS | Twenty-five problems solved in full

METHOD, REASONING AND CHECK

An answer tells you whether you were right; a solution tells you how to think. The twenty-five problems below are chosen because each one is a place where Year 1 students commonly lose marks, and each solution is written in the same five steps: read, plan, work, answer, check.

NUMBER AND PLACE VALUE

1 · Rounding with a decision

Question. Round 4,836 to the nearest hundred. **Read.** The hundreds digit is 8; the digit that decides is the 3 in the tens place. **Plan.** Look only at the deciding digit, then choose the nearer hundred. **Work.** The number lies between 4,800 and 4,900. 4,836 is 36 above 4,800 and 64 below 4,900, so it is nearer 4,800. **Answer.** 4,800. **Check.** Rounding to the nearest ten gives 4,840, so the hundreds answer must be a multiple of 100 near it. 4,800 is consistent.

2 · The order of operations with a bracket

Question. Work out $48 \div (6 - 2) + 3^2$. **Plan.** Bracket first, then the power, then division, then addition. **Work.** $6 - 2 = 4$; $3^2 = 9$; $48 \div 4 = 12$; $12 + 9 = 21$. **Answer.** 21. **Check.** A quick estimate: $50 \div 4 \approx 12.5$, plus 9 gives about 21.5, and 21 is close to that. Changing the order would give $48 \div (4 + 9) \approx 3.7$, so the brackets are doing real work.

3 · HCF and LCM chosen by context

Question. Tiles 24 cm by 18 cm are to cover a square exactly. Find the smallest possible square. **Plan.** The side of the square must be a common multiple of both tile lengths; the smallest is the LCM. **Work.** $24 = 2^3 \times 3$ and $18 = 2 \times 3^2$, so LCM $= 2^3 \times 3^2 = 72$. **Answer.** A 72 cm square, using $3 \times 4 = 12$ tiles. **Check.** $72 \div 24 = 3$ and $72 \div 18 = 4$, both whole numbers, and no smaller number works because 24 and 18 share the factor 6.

4 · Prime factorisation in index form

Question. Write 360 as a product of primes. **Work.** $360 = 36 \times 10 = (2^2 \times 3^2) \times (2 \times 5) = 2^3 \times 3^2 \times 5$. **Check.** $8 \times 9 \times 5 = 360$. Only primes appear, and the factorisation is unique apart from order.

FRACTIONS, DECIMALS AND PERCENTAGES

5 · Adding fractions with unlike denominators

Question. Work out $\frac{5}{6} - \frac{3}{8}$. **Plan.** The LCM of 6 and 8 is 24, so both fractions can be written in twenty-fourths. **Work.** $\frac{5}{6} = \frac{20}{24}$ and $\frac{3}{8} = \frac{9}{24}$; the difference is $\frac{11}{24}$. **Answer.** $\frac{11}{24}$, which is already in simplest form because 11 is prime and does not divide 24. **Check.** $\frac{5}{6} \approx 0.833$ and $\frac{3}{8} = 0.375$, so the difference is about 0.458, and $\frac{11}{24} \approx 0.458$. The decimal check agrees with the fraction.

6 · A recurring decimal as a fraction

Question. Write 0.363636… as a fraction in simplest form. **Plan.** Let x be the number and multiply by a power of ten that shifts one whole repeating block. **Work.** $x = 0.363636\ldots$, so $100x = 36.363636\ldots$. Subtracting, $99x = 36$, hence $x = \frac{36}{99} = \frac{4}{11}$. **Check.** $4 \div 11 = 0.363636\ldots$, and the block 36 repeats.

7 · Multiplying decimals by counting places

Question. Work out 0.6×0.45. **Plan.** Multiply as whole numbers, then restore the total number of decimal places. **Work.** $6 \times 45 = 270$. There is one decimal place in 0.6 and two in 0.45, so three in total: 0.270. **Answer.** 0.27. **Check.** 0.6×0.45 is a little more than half of 0.45, so about 0.25; 0.27 is sensible.

8 · Percentage change in two steps

Question. A price of 240 is reduced by 15%, then the new price is reduced by 10%. Find the single equivalent reduction. **Work.** Multipliers: $0.85 \times 0.90 = 0.765$. The final price is 76.5% of the original, so the reduction is 23.5%. **Check.** A single 25% reduction would leave 180; two successive reductions leave $240 \times 0.765 = 183.6$, which is slightly more, as expected.

9 · Removing a tax that is already included

Question. A receipt shows 354 units including 18% VAT. Find the price before tax. **Plan.** The total is 118% of the price, so divide by 1.18. **Work.** 354 ÷ 1.18 = 300. The VAT is 54 units. **Check.** 300 × 1.18 = 354. Taking 18% of 354 would give 63.72, which is too large — a common error.

10 · Working backwards to an original amount

Question. After a 30% reduction a coat costs 84 units. Find the original price. **Work.** The sale price is 70% of the original, so the original is 84 ÷ 0.7 = 120. **Check.** 30% of 120 is 36, and 120 - 36 = 84.

ALGEBRA AND EQUATIONS

11 · Collecting like terms before solving

Question. Solve $5x + 3 - 2x = 18$. **Work.** $3x + 3 = 18$, so $3x = 15$ and $x = 5$. **Check.** $5(5) + 3 - 2(5) = 25 + 3 - 10 = 18$.

12 · Expanding then solving

Question. Solve $2(x - 3) = 14$. **Plan.** Either divide both sides by 2 first, or expand. Dividing is quicker here. **Work.** $x - 3 = 7$, so $x = 10$. **Check.** $2(10 - 3) = 14$. Expanding first gives $2x - 6 = 14$ and the same answer, which is a useful second method.

13 · Forming an equation from a context

Question. A taxi charges 3 units plus 2 units per kilometre. A journey costs 27 units. How far was the journey? **Plan.** Let d be the distance in kilometres; the cost is $3 + 2d$. **Work.** $3 + 2d = 27$, so $2d = 24$ and $d = 12$. **Check.** $3 + 2(12) = 27$, and the answer is a distance in kilometres, not an amount of money.

14 · Substitution with a power

Question. Find the value of $3n^2$ when $n = -4$. **Plan.** The power applies only to n, and squaring a negative gives a positive. **Work.** $(-4)^2 = 16$, and $3 \times 16 = 48$. **Check.** $3n^2$ is not $(3n)^2$, which would be 144. The position of the bracket changes the answer.

SHAPE AND MEASUREMENT

15 · Angles in a triangle with algebra

Question. The angles of a triangle are x, $2x + 10$ and $3x - 10$. Find each angle. **Work.** $x + 2x + 10 + 3x - 10 = 180$, so $6x = 180$ and $x = 30$. The angles are 30°, 70° and 80°. **Check.** They total 180° and all are less than 90°, so the triangle is acute-angled.

16 · Parallel lines and one known angle

Question. Two parallel lines are cut by a transversal. One angle is 115°. Find the alternate angle and the co-interior angle. **Work.** The alternate angle equals 115°; the co-interior angle is 180° - 115° = 65°. **Check.** Co-interior angles always total 180° between parallel lines, and 115 + 65 = 180.

17 · Interior angles of a regular polygon

Question. Find each interior angle of a regular 15-sided polygon. **Work.** Interior sum = (15 - 2) × 180° = 2,340°; each angle = 2,340° ÷ 15 = 156°. **Check.** Each exterior angle is 360° ÷ 15 = 24°, and 156 + 24 = 180, as it must at every vertex.

18 · Area of a compound shape

Question. A rectangle 12 m by 8 m has a semicircle of diameter 8 m attached to one short side. Find the total area. **Work.** Rectangle = 96 m²; semicircle radius 4 m = $\frac{1}{2}\pi(16) \approx 25.1$ m²; total ≈ 121.1 m². **Check.** The semicircle fits inside a 4 m by 8 m rectangle of area 32 m², and

25.1 is less than 32.

19 · Surface area of a cuboid

Question. Find the surface area of a cuboid 8 cm by 5 cm by 4 cm. **Work.** $2(8 \times 5 + 8 \times 4 + 5 \times 4) = 2(40 + 32 + 20) = 184$ cm². **Check.** The net has six rectangles in three equal pairs; 80 + 64 + 40 = 184.

20 · Capacity and volume units

Question. A tank is 1.2 m by 0.8 m by 0.5 m. Find its capacity in litres. **Work.** Volume = 0.48 m³. Since 1 m³ = 1,000 litres, the capacity is 480 litres. **Check.** $1.2 \times 0.8 = 0.96$ m² of base, and $0.96 \times 0.5 = 0.48$ m³; each cubic metre holds 1,000 litres.

SETS AND DATA

21 · A two-set problem with an overlap

Question. Of 40 students, 22 like football, 18 like basketball and 8 like both. How many like neither? **Work.** $n(F \cup B) = 22 + 18 - 8 = 32$, so 40 - 32 = 8 like neither. **Check.** Only football is 14, only basketball is 10, both are 8: 14 + 10 + 8 = 32, leaving 8.

22 · Mean, median and mode together

Question. Find the mean, median and mode of 4, 7, 7, 9, 13. **Work.** Sum 40 over 5 values gives mean 8; the middle value is 7, so median 7; 7 occurs twice, so mode 7. **Check.** The median equals the mode here but not in general; the mean is pulled up by the 13.

23 · Estimating a mean from grouped data

Question. Times are grouped as 10–14 (4), 15–19 (9), 20–24 (14). Estimate the mean. **Work.** Midpoints 12, 17, 22: 48 + 153 + 308 = 509 over 27 values, giving about 18.9. **Check.** The estimate of 18.9 sits inside the class 15–19, which holds the most values, so it is plausible; a grouped estimate always treats each value as if it were at the midpoint.

24 · Reading a misleading graph

Question. A bar chart starts its vertical axis at 90. Two bars show 92 and 95. Explain the distortion. **Work.** On a full scale from 0 the bars would differ by 3 in 100, about 3%; with the axis starting at 90 the visual difference is 3 in 5, or 60%. **Check.** The data are unchanged; only the axis has changed, which is why the axis must be read before the picture.

25 · Choosing between area and volume

Question. A paint tin covers 12 m² per litre. A wall is 9 m by 3 m. How many litres are needed for two coats? **Work.** Wall area = 27 m²; two coats cover 54 m²; 54 ÷ 12 = 4.5 litres. **Check.** One coat needs 2.25 litres, so two coats need 4.5 — and paint is sold in whole litres, so buy 5.

ANNOTATED STUDENT WORK

Four responses with examiner commentary

WHAT THE CRITERIA LOOK LIKE IN PRACTICE

Marks are not awarded for length. The four responses below are short, and each one is annotated to show what earns credit under the MYP mathematics criteria and what holds a response back. Reading them is faster than reading a rubric, and more useful.

RESPONSE 1 · KNOWING AND UNDERSTANDING (CRITERION A)

The task. "Explain why $\frac{3}{8}$ is a terminating decimal."

The response. "$8 = 2^3$, and $\frac{3}{8} = \frac{3 \times 125}{8 \times 125} = \frac{375}{1000} = 0.375$. The denominator is built only from 2s, so the fraction can always be written over a power of ten, and the decimal stops."

Commentary. Full credit. The student does more than state a fact: they name the condition, demonstrate it on the example, and give the reason the condition works. A response that says only "because the calculator shows 0.375" earns nothing, because a calculator result is not an explanation.

RESPONSE 2 · INVESTIGATING PATTERNS (CRITERION B)

The task. "How many matches are needed for stage n of a pattern that starts with 4 matches and adds 3 each time?"

The response. "Stage 1: 4. Stage 2: 7. Stage 3: 10. The differences are constant at 3, so the rule is $3n + 1$. Checking $n = 4$ gives 13, which matches the diagram. For stage 20, $3(20) + 1 = 61$."

Commentary. Full credit. Four things are present: data collected, a pattern described, a general rule expressed algebraically, and the rule verified against a case not used to find it. Without that verification the generalisation is a guess; with it, it is a justified conclusion.

RESPONSE 3 · COMMUNICATING (CRITERION C)

The task. "Explain how to decide whether a problem needs area or volume."

The response. "Area is for something that covers — paint for a wall, turf for a garden, tiles for a floor. Volume is for something that fills — water in a tank, sand in a box. If the answer needs square units it is area; if it needs cubic units it is volume. A paint problem can still need volume at the end, because paint is sold by the litre, so the surface area is divided by the coverage."

Commentary. Full credit at the top band. The response defines the terms through examples, gives a decisive test, and then handles the awkward case where a real problem needs both. The last sentence is what lifts it above a correct but thin answer.

RESPONSE 4 · APPLYING MATHEMATICS IN CONTEXT (CRITERION D)

The task. "A school needs to buy water for a sports day. Which option is better value?"

The response. "Option A: 24 bottles of 500 ml for 6 units, which is 12 litres for 6 units, so 0.50 units per litre. Option B: one 5 litre container for 2.40 units, so 0.48 units per litre. Option B is cheaper per litre, but it needs cups and a way to keep it cold, and the bottles can be handed out and resealed. For 200 students needing about 0.5 litres each we need 100 litres: A costs 50 units, B costs 48 units. The difference is 2 units, which is smaller than the cost of cups, so A is the better decision."

Commentary. Full credit. The student computes both unit rates, models the real quantity needed, and then makes a justified decision that includes a factor the mathematics alone cannot supply. A response that stops at "0.48 is less than 0.50, so B" earns partial credit only, because it has not applied the result to the situation.

WHAT THE FOUR RESPONSES HAVE IN COMMON

- Each states a method before using it, so the reader can follow the reasoning.
- Each answers the question that was asked, in the units that were asked for.
- Each checks or applies the result, rather than stopping at the calculation.
- None is long. Precision, not length, is what earns credit.

REVISION PLANNER

Twelve weeks to the assessment

A WORKING PLAN

A plan is useful only if it fits the time available. The twelve-week plan below assumes four short sessions a week of about twenty-five minutes, plus one longer session at the weekend. Adjust the number of weeks, not the order.

THE PLAN

Week	Focus	What to do	Evidence of progress	
1	Place value and rounding	Re-read 1.1–1.3	do the fluency items in 1.1 and 1.2	Explain rounding to a partner in one minute
2	Powers and roots	2.1–2.3	memorise squares to 15^2	Write 64 as a power in three ways
3	Factors, primes, HCF and LCM	3.3–3.5	ten mixed HCF/LCM items	Explain why 91 is not prime
4	Fractions	4.1–4.4	twenty fraction operations	Order five fractions without a calculator
5	Decimals and percentages	4.5–4.7, 5.1–5.3	Convert ten values between forms	
6	Percentage change	5.4–5.5	three real receipts or bills	Work backwards from a discounted price
7	Algebra	6.1–6.5	twenty equations with checks	Form an equation from a word problem
8	Angles and polygons	7.1–7.5	ten angle problems with reasons	Give the reason for every step
9	Measurement	9.1–9.6	five compound-shape and five volume items	Estimate before calculating
10	Sets and data	8.1–8.5, 10.1–10.5	Complete a survey and a two-way table	
11	Mixed practice	Paper A and Paper B under time	Mark honestly and list every lost mark	
12	Final revision	Re-do every question you got wrong in weeks 1–11	Explain your three weakest topics aloud	

Twelve weeks, four sessions a week

HOW TO MARK YOUR OWN WORK

1. Mark in a different colour from the one you worked in, so mistakes stay visible.
2. For each lost mark, write one line: “I lost this because …”. Group the reasons — method, arithmetic, units, misreading.
3. Re-do the question two days later without looking at the solution. If it works the second time, the gap has closed.
4. Keep a single sheet of the four errors you keep repeating; check it before every practice paper.

THE NIGHT BEFORE

Sleep matters more than one more paper. Read your error sheet, read the formula bank, and read the four annotated responses above. Do not start a new topic. In the assessment, read every question twice, write the units, and if a question is worth three marks, show the three steps those marks are for.

REFERENCE NOTES | REFERENCE

Using a calculator well

A calculator performs the operations you enter, in the order you enter them. Most errors are errors of entry or of order, not of arithmetic.

Habit	What it prevents
Estimate first	Accepting an absurd answer
Use brackets for whole numerators and denominators	7+3 ÷ 2 instead of (7+3) ÷ 2
Keep the full value in the calculator until the end	Rounding errors accumulating
Write down intermediate values	Losing the thread of a multi-step calculation
Check the units of the answer	Reporting an area as a length

Five habits that prevent calculator errors

A calculator is a tool for arithmetic, not for reasoning. It will compute 15% of 240 correctly, but it will not decide whether 15% is the right percentage to use; it will divide by 0.7 when asked, but it will not know that dividing is the correct operation for finding an original price. The decisions remain yours, and they are what this course assesses.

COMMON ERROR

The three commonest calculator errors

Entering 3 + 4 × 5 as (3 + 4) × 5; rounding a recurring decimal too early and then multiplying; and reading a display such as 1.2E3 as 1.23 rather than 1,200.

Answers and worked solutions

Answers are given so that you can check your reasoning, not replace it. Where a question has more than one sensible answer, a sample answer is shown. Where a question asks you to explain, the answer shows the kind of reasoning expected.

CHAPTER 01 · NUMBER SYSTEMS AND PLACE VALUE

Practice 1.1 · Number families and order

1. (a) natural, whole, integer, rational (b) whole, integer, rational (c) rational (d) integer, rational (e) rational

2. -4, -3, -2, -1, 0, 1, 2

3. (a) < (b) > (c) <

4. -7, -2, 0, 3, 5 °C

5. No. The natural numbers are the counting numbers 1, 2, 3, …, so zero is a whole number but not a natural number. "Counting zero items" describes an empty collection, not a counting number.

6. $\frac{7}{4}$ = 1.75 (or $\frac{5}{5}$ = 1), which is not less than 1.

7. First movement: 3 floors down to -2. Second: 5 floors up to 3. The final floor is 2 floors above the starting floor.

8. Marks at the correct positions; explanation: 2.5 is halfway between 2 and 3 because it equals 2 + $\frac{1}{2}$.

9. -120 < -18 < 45; distance = 45 - (-120) = 165 m.

Practice 1.2 · Place value and the role of zero

1. (a) 4,000 (b) 400 (c) 0.4 (d) 0.04

2. (a) 20 (b) no 7 (c) 0.07 (d) 0.7

3. 8,016

4. (a) 362 (b) 548 (c) 92 (d) 0.073

5. The zero holds the tenths place. Removing it gives 2.6, ten times larger in the tenths position: 2.06 means 2 units and 6 hundredths, while 2.6 means 2 units and 6 tenths.

6. 3.5, because 3.5 = 3.50 and 50 hundredths is greater than 5 hundredths.

7. 1,250 cents, which is the same amount expressed in a smaller unit.

8. 3.4 (dividing by 10 overall)

9. He has written the digits in the wrong places: the 3 is worth 3,000 and the 4 is worth 40, so the number is 3,045. Zero must hold the hundreds place.

Practice 1.3 · Expanded form and equivalent representations

1. (a) 4 × 1,000 + 7 × 100 + 0 × 10 + 6 × 1 (b) 9 × 10,000 + 0 × 1,000 + 2 × 100 + 1 × 10 + 5 × 1 (c) 6 × 1 + 0 × 0.1 + 8 × 0.01

2. (a) 8,305 (b) 27.4 (c) 0.053

3. (a) three thousand and sixty (b) twelve and five tenths (c) nine hundredths

4. (a) 7,040 (b) 0.206 (c) 90.9

5. She has put the 5 in the hundredths place. It is in the tenths place, so the correct expansion is 4 × 1 + 5 × 0.1.

6. They are equal: 0.70 = 7 × 0.1 + 0 × 0.01 and 0.7 = 7 × 0.1. The extra zero contributes nothing.

7. The record shows 5 hundredths instead of 5 tenths, so 0.45 has been lost: 12.05 is 45 hundredths less than 12.50.

8. Smallest 2,047; largest 7,420.

9. The smallest value is when $a = 1$, $b = 0$, $c = 0$, giving 100; the largest is $a = 9$, $b = 9$, $c = 9$, giving 999.

10. The display becomes 3,000: the tens place changed from 5 tens to 0 tens, so the value fell by 50.

Practice 1.4 · Comparing, ordering and rounding

1. (a) > (b) < (c) = (d) <

2. (a) 7,460 (b) 7,500 (c) 7,000

3. (a) 5.28 (b) 5.3 (c) 5

4. -3 < -0.9 < 0.45 < 2.04 < 2.4

5. Compare tenths: 0.305 has 3 tenths and 0.35 has 3 tenths; compare hundredths: 0 vs 5, so 0.305 < 0.35.

6. Any value from 6.75 up to but not including 6.85, such as 6.76, 6.8 and 6.849.

7. 12.5 s; the hundredths digit (0.06 s) is lost, so two runners could appear tied.

8. Rounded prices 4 + 3 + 2 = 9; exact total 7.25; the customer pays 1.75 more.

9. 450 and 549.

10. It lies between 2.95 and 3.05, so it could be 2.98 or 3.02. You cannot tell its exact value, and you cannot tell whether it is above or below 3.

11. Rounding twice can move a value across a boundary: 3,450 to the nearest hundred is 3,500, which then rounds to 4,000 to the nearest thousand, but 3,450 rounds directly to 3,000.

12. (a) 24,000: a sign needs scale, not precision. (b) 23,847: a funding formula depends on the exact count.

Practice 1.5 · Estimation and sensible accuracy

1. (a) about 900 (b) about 900

2. (a) about 800 (b) about 20

3. (a) about 8,000 (b) about 800 (c) about 200

4. Greater: 49 > 45, so 49 × 21 > 45 × 21 = 945 is not enough; use 49 > 50 - 1, giving 49 × 21 = 1,050 - 21 = 1,029. A quicker check: 49 × 21 > 49 × 20 = 980 and 50 × 21 = 1,050, so the product lies between 980 and 1,050.

5. 20 × 24 = 480; exact = 444; the estimate is above because the price was rounded up.

6. 12 × 400 = 4,800 m = 4.8 km, so "about 5 km" is reasonable.

7. 40 × 50 = 2,000, so 191 is far too small; correct value 39 × 49 = 1,911.

8. 2,000 ÷ 7 ≈ 285, so "about 300" overstates each share by about 15 units; with 7 clubs that is 105 units overspent.

9. e.g. 19 × 19: rounding to 20 × 20 = 400 against an exact 361 is 10.8% high, because both factors were small and each was increased by more than 5%.

10. 2,000 overstates by 386, about 24%, so "nearly 2,000" is a significant overstatement; "about 1,600" would be accurate.

Practice RQ1 · Applying the chapter

1. 9,070

2. 600

3. 8,700

4. -3.4 < -0.8 < 0 < 2.05 < 2.5

5. 470

6. 3 × 0.1 + 6 × 0.01

7. In 4,050 the 5 is in the tens place (value 50); in 4,005 it is in the ones place (value 5).

8. The number lies between 650 and 749; it rounds to 1,000 only if it is 650 or more.

9. 3,000 is correct: the tens digit is 5, so round up. 2,900 would come from looking at the wrong digit.

10. About 20 × 20 = 400; exact 432.

11. About 1,200 + 1,000 = 2,200; exact 2,233, so the estimate is below.

12. -1 °C

13. Sample: In 305 the zero holds the tens place. Without it the digits would move along and give 35, which is a much smaller number. The zero keeps the 3 worth three hundred

and the 5 worth five.

14. Six thousand and eighty-four; 6 × 1,000 + 0 × 100 + 8 × 10 + 4 × 1.

15. Any number such as 3.45: 3.40 < 3.45 < 3.50.

16. Ten numbers: 55 to 64.

17. 4,000, 400, 40, 4; sum 4,444.

18. For whole numbers an extra digit means a higher place value. For decimals, extra digits are to the right of the point and represent smaller parts, so 0.305 < 0.35.

19. About 11,800 - 4,100 = 7,700; exact 7,750, so the estimate is below.

Practice MX1 · End-of-chapter mixed set

1. 10,050

2. (a) 62 (b) 3.4 (c) 70

3. 12.48 and 12.5

4. -2.5 < -2.05 < -0.25 < 1.05 < 1.5

5. 3,405.62

6. Multiplying by 100 moves digits two places left: 240. Writing 2.400 changes nothing in value.

7. Neither: they are equal, because 0.7 L = 700 ml.

8. Floor 0.

9. About 667 units each; 2,000 ÷ 3 = 666 remainder 2, so each receives 666 units and 2 units remain.

10. About 2,000 ÷ 70 ≈ 28 minutes; exact ≈ 29.4 minutes.

11. Expanded form shows the value of every digit, so a digit written in the wrong place produces a sum that does not match the intended number.

12. Example: recording a race time of 12.46 s as 10 s makes two different times look equal and loses the result.

13. 5.55 and 5.649… (any value below 5.65).

14. e.g. 451 and 541 (both round to 500).

15. Every digit moves one place to the right, so the number is divided by 10: 0.5 becomes 0.05.

16. About 240 m, roughly 13.6% too large; the sign overstates the distance noticeably and should read “about 1.8 km”.

Practice CP1 · Cumulative practice

1. 7,000

2. 6.85

3. 2.5 = 2.50, and 50 hundredths is greater than 5 hundredths.

4. Rounded values are not the original values, so differences can shift.

5. 0.5 units

6. Three thousand and forty-eight

7. The zero holds the tens place, so the 3 means 300, not 3 tens.

8. Any values from 55 to 64, such as 57 and 63.

Prerequisite check

1. 4,060

2. 0.5, because 0.50 > 0.45

3. 56

4. 3,050

5. 0.7

6. -2 < 0 < 3

CHAPTER 02 · POWERS, EXPANDED FORM AND ORDER OF OPERATIONS

Practice 2.1 · From repeated multiplication to powers

1. (a) 3^4 (b) 7^2 (c) 10^5

2. (a) 5 × 5 × 5 (b) 2 × 2 × 2 × 2 × 2 (c) 8 × 8

3. (a) 36 (b) 16 (c) 1,000 (d) 5

4. (a) 81 (b) 343 (c) 128

5. The exponent counts factors, not multiplications: $3^5 = 3 \times 3 \times 3 \times 3 \times 3 = 243$.

6. $2^5 = 32$ and $5^2 = 25$, so 2^5 is larger; the base matters more than the exponent here.

7. $5^3 = 125$ cm³

8. 2^6 and 4^3 (also 8^2).

9. $2^{11} = 2 \times 1{,}024 = 2{,}048$, so it is greater; doubling $2^{10} = 1{,}024$ gives the value.

10. 3^4, 9^2 and 81^1.

Practice 2.2 · Powers of ten and expanded form

1. (a) $3 \times 10^3 + 5 \times 10^2$ (b) $6 \times 10^1 + 1 \times 10^0$ (c) $8 \times 10^3 + 4 \times 10^0$

2. (a) 1 (b) 10 (c) 1,000,000

3. (a) 2,500 (b) 0.6 (c) 40

4. 40,703

5. Each power is ten times the next: $10^2 = 100$, $10^1 = 10$, so $10^0 = 10 \div 10 = 1$.

6. $6^3 = 216$, which is not 6,000. The correct form is 6×10^3: the base is the place-value base, 10.

7. 90,009 — five digits.

8. $4 \times 10^5 = 400{,}000$ items.

9. 10^6, 100^3, $1{,}000^2$.

10. *a* can be 1 to 9, because a six-digit number has a non-zero digit in the hundred-thousands place.

Practice 2.3 · Squares, cubes and roots

1. (a) 49 (b) 81 (c) 144 (d) 125

2. (a) 8 (b) 10 (c) 13

3. 36, 49, 64

4. There is no whole number whose square is 30: $5^2 = 25$ and $6^2 = 36$.

5. No: 5 + 3 = 8, while √34 ≈ 5.83. Roots cannot be added inside a single root.

6. Side 14 m; perimeter 56 m.

7. 6 cm, because $6^3 = 216$.

8. Between 8 and 9; √75 ≈ 8.7.

9. $x^2 > 100$ requires $x \geq 11$, while $x^3 < 1{,}000$ requires $x \leq 9$. No whole number satisfies both, and $x = 10$ gives exactly 100 and exactly 1,000, so it fails both strict inequalities.

10. The gaps are 3, 5, 7, 9 — consecutive odd numbers; each new square adds the next odd number.

Practice 2.4 · Order of operations

1. (a) 11 (b) 2 (c) 7

2. (a) 21 (b) 2 (c) 13

3. (a) 12 (b) 12 (c) 25

4. 16

5. 15 is correct (division first). 6 comes from subtracting first: 12 ÷ 2.

6. (3 + 4) × 2 = 14

7. 3 + 2 × 7 = 17 units

8. 75

9. 24 ÷ (6 + 2) × 3 = 9 and (24 ÷ 6 + 2) × 3 = 18 (also 24 ÷ (6 + 2 × 3) = 2).

10. Without brackets multiplication is performed first: 2 + 12 = 14. Brackets force the sum first: 5 × 4 = 20.

11. It evaluated left to right, ignoring the agreed order. A calculator applies whatever order it was programmed with, so the person must write the expression unambiguously.

12. $12 + 5p = 112$, so $5p = 100$ and $p = 20$ pages.

Practice 2.5 · Substitution and expressions with powers

1. 48

2. 43

3. 100

4. The first squares *x* only (2 × 9 = 18); the second squares $2x$ ($6^2 = 36$).

5. 9 and 25

6. 15.625 cm³

7. Brackets decide what the power applies to: without them the power covers only the last symbol, with them it covers the whole bracket.
8. $x = 2$ (also $x = 0$).
9. 25 and 13; the first squares the sum, the second squares each part separately.
10. Multiplying the whole bracket by 2 multiplies each term inside it by 2.

Practice RQ2 · Applying the chapter

1. (a) 16 (b) 32 (c) 10,000
2. 7^3
3. 11
4. 10
5. 49
6. 340
7. $5^2 = 25$ and $2^5 = 32$: the base and exponent play different roles.
8. Greater: 64 against 36.
9. The base: $8^3 = 512$. The correct form is 8×10^3.
10. 35
11. 10
12. Side 13 m; perimeter 52 m.
13. 27 m³ = 27,000 L
14. Two people could otherwise read one expression in two ways and obtain different values; an agreement makes the written expression unambiguous.
15. $((2 + 3) \times 4)^2 = 400$
16. $2^5 + 2^6 = 32 + 64 = 96$
17. In the first, the base is −2; in the second the index applies only to 2, and the negative sign is applied afterwards.
18. Need $2^n > 10{,}000$; $2^{13} = 8{,}192$ and $2^{14} = 16{,}384$, so 14 folds.
19. Dividing 100 by 10 gives 10, and dividing again gives 1; each division reduces the exponent by 1, so the exponent 0 must give 1.
20. About 7%, caused by counting in powers of two rather than powers of ten.

Practice MX2 · End-of-chapter mixed set

1. (a) 27 (b) 36 (c) 256
2. (a) 15 (b) 240
3. $6 \times 10^4 + 0 \times 10^3 + 3 \times 10^2 + 0 \times 10^1 + 4 \times 10^0$
4. (a) 700 (b) 2.5
5. 14
6. 81, because $9^2 = 81$ and $10^2 = 100$ has three digits.
7. $2^{n+1} = 2 \times 2^n$; adding 1 to n changes n^2 by $2n + 1$, which is not a constant multiple.
8. 10^3 pages (1,000 pages).
9. 75
10. Each tile is 20 cm × 20 cm = 400 cm²; the rectangle is 80,000 cm², so 200 tiles.
11. Expanded form needs a value for the ones place; the pattern of dividing by 10 from 10^1 gives 1, so the ones place is 10^0.
12. $3n^2 + 5$; value 53.
13. $n = 9$
14. 2^{12}; the exponent counts the folds, each of which doubles the layers.
15. 2^{13} is larger: multiplying powers of the same base adds the exponents.
16. $3 \times 2 = 6 = \sqrt{36}$; for addition $3 + 2 = 5$, but $\sqrt{13} \approx 3.6$, so the rule fails for addition.

Practice CP2 · Cumulative practice

1. 81
2. 12
3. 8
4. Three factors of 2 multiplied by two more gives five factors.
5. 9 cm
6. 5,304
7. Brackets, then the index, then multiplication, then addition, giving 36.
8. $n = 6$

Prerequisite check

1. 42 and 81
2. 4,536
3. 24
4. 27
5. 64 cm²
6. $(2 + 4) \times 3$

CHAPTER 03 · FACTORS, MULTIPLES AND PRIMES

Practice 3.1 · Factors, multiples and factor pairs

1. (a) 1, 2, 3, 6, 9, 18 (b) 1, 2, 4, 5, 8, 10, 20, 40 (c) 1, 5, 25
2. (a) 6, 12, 18, 24, 30 (b) 9, 18, 27, 36, 45 (c) 11, 22, 33, 44, 55
3. (1, 30), (2, 15), (3, 10), (5, 6)
4. Yes: 96 ÷ 8 = 12 with no remainder.
5. 15 is a factor of 45 because 45 ÷ 15 = 3; 45 is a multiple of 15 because 15 × 3 = 45.
6. A number with four factors, such as 8 (1×8, 2×4) or 10 (1×10, 2×5).
7. 4, 5, 6, 10, 12, 15 (each divides 60 and lies in the range).
8. 24 seconds.
9. 12, whose factors are 1, 2, 3, 4, 6, 12.
10. Every factor pairs with another except the square root, which pairs with itself, leaving one unpaired factor.

Practice 3.2 · Divisibility tests

1. 84, 120, 132
2. 315 (digit sum 9), 216 (digit sum 9); 420 has digit sum 6.
3. By 4 (36 ÷ 4 = 9); not by 8 (236 ÷ 8 = 29.5).
4. 0, because 5+0+4=9 is divisible by 3.
5. Powers of ten are one more than multiples of 9, so only the digit sum matters; divisibility by 4 depends on the last two digits, not on the other digits.
6. No: 48 ÷ 8 = 6 is exact here, but the reasoning is unsound; 12 is divisible by 2 and 4 but not by 8.
7. 45
8. 62 groups; 372 is divisible by 2 and 3 (digit sum 12), so divisible by 6.
9. 120, 240, 360, 480, 600, 720, 840, 960 — any multiple of 60 whose digit sum is not 9 or 18.
10. 6 = 2 × 3 includes a factor 2; 4 divides 12 but 8 does not, since 12 = 4 × 3 has only two factors of 2.

Practice 3.3 · Prime and composite numbers

1. 23, 29, 31, 37
2. composite, prime, composite, prime
3. 3 + 7 or 5 + 5
4. It has only one factor; a prime must have exactly two.
5. No: 121 = 11 × 11.
6. The second is right: the digit sum 15 is divisible by 3, so 87 = 3 × 29.
7. 46 = 2 × 23 and 49 = 7 × 7.
8. Every other even number has 2 as a factor as well as 1 and itself.
9. 101
10. Not always: 2 + 3 = 5 is odd, because 2 is the only even prime.

Practice 3.4 · Prime factorisation

1. (a) $2^3 \times 3$ (b) $3^2 \times 5$ (c) $2^2 \times 5^2$
2. $2 \times 3^2 \times 7$
3. 200

4. 6 and 10 are composite; the factors must be prime.
5. Yes; the second is the same product without index notation.
6. $252 = 2^2 \times 3^2 \times 7$, so yes.
7. $108 = 2^2 \times 3^3$, so multiply by 3 to obtain 324.
8. Examples: 4, 15, 25, 100 — any product of a selection of the primes.
9. Every composite is built by splitting into primes; the same set of primes must reappear however the splitting is done, so only the order can differ.
10. $324 = 2^2 \times 3^4$; possibilities include 18×18 (2×3^2 each) and 6×54.

Practice 3.5 · Highest common factor and lowest common multiple

1. (a) 4 (b) 9 (c) 8
2. (a) 12 (b) 18 (c) 24
3. HCF 5, LCM 30
4. They share no prime factors, so no smaller number can be a multiple of both.
5. The LCM.
6. HCF of 48 and 36 is 12.
7. LCM is 120 s, so at 8:02.
8. HCF is 12 cm.
9. 12 and 18 (also 6 and 36).
10. LCM = 24 minutes, so at 12:24.

Practice RQ3 · Applying the chapter

1. 1, 3, 5, 9, 15, 45
2. 12, 24, 36, 48
3. Yes (digit sum 6); yes ($32 \div 4 = 8$)
4. $2 \times 3 \times 5^2$
5. 18
6. 40
7. 2 has factors 1 and 2 only; 4 has 1, 2 and 4.
8. Yes to the first ($9 = 3 \times 3$); no to the second (6 is divisible by 3 but not 9).
9. 90 is a common multiple but not the lowest: $6 \times 5 = 30$ and $15 \times 2 = 30$.
10. 24 cm
11. 180 seconds (3 minutes)
12. $441 = 3^2 \times 7^2$, so yes: 21^2.
13. A composite can be split into factors; if a factor is composite it can be split again, and the process must stop at primes because the factors get smaller.
14. $84 = 2^2 \times 3 \times 7$; factors 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84.
15. 24 (factors 1, 2, 3, 4, 6, 8, 12, 24).
16. 4 and 48, or 12 and 16.
17. One of them must be even, and an even factor makes the product even.
18. HCF of 120 and 72 is 24 groups.
19. 5 divides 10, so only the last digit matters; 7 does not divide any power of ten, so every digit can affect the remainder.

Practice MX3 · End-of-chapter mixed set

1. 5, 10, 15, 20, 30, 60
2. 8
3. 30
4. By 5 yes; by 9 no (digit sum 14).
5. $2^5 \times 3$
6. It divides every number exactly, but it has only one factor, so it fails the definition of prime.
7. False. 22 has digit sum 4, which is even, but $22 \div 3$ leaves remainder 1. Divisibility by 3 needs a digit sum that is a multiple of 3, not merely even.
8. 6 cm
9. 75 minutes later, at 8:15.
10. 62 (1, 2, 31, 62) — also 65 (1, 5, 13, 65) and 69 (1, 3, 23, 69).
11. Every branch ends in a prime, so no factor can be split further.
12. The HCF is the largest group size that divides both numbers; the LCM is the first moment two repeating events coincide.
13. 6 and 30, or 12 and 15.
14. Each odd number is one more than a multiple of 2; the two extra ones combine to make another multiple of 2.
15. $(a+1)(b+1)$ factors: test $12 = 2^2 \times 3$ gives 6 factors.
16. 60 cm; $9 \times 7 = 63$ tiles.

Practice CP3 · Cumulative practice

1. 1, 2, 3, 4, 6, 9, 12, 18, 36
2. 6
3. 36
4. $51 = 3 \times 17$.
5. 24 minutes later
6. 12 cm
7. Every branch ends in a prime.
8. 6, whose factors are 1, 2, 3, 6.

Prerequisite check

1. 56 and 54
2. 12
3. 4, 8, 12, 16, 20
4. 10×10
5. $3^3 = 27$
6. Yes, $7 \times 8 = 56$

CHAPTER 04 · FRACTIONS AND DECIMALS

Practice 4.1 · What a fraction means

1. Four ninths; numerator 4, denominator 9.
2. (a) $\frac{7}{3}$ (b) $\frac{23}{4}$ (c) $\frac{15}{8}$
3. (a) $3\frac{2}{5}$ (b) $3\frac{2}{9}$ (c) 1
4. 30
5. The numerator is larger than the denominator, so there are more than four quarters.
6. A larger denominator means smaller parts, so eighths are smaller than thirds.
7. $\frac{5}{6} \times 240 = 200$; need 40 litres.
8. 24 pages remain ($\frac{4}{7} \times 56$).
9. $\frac{50}{60} = \frac{5}{6}$
10. Each has all its parts, and all its parts make the whole.

Practice 4.2 · Equivalent fractions and simplifying

1. (a) $\frac{2}{3}$ (b) $\frac{2}{5}$ (c) $\frac{3}{4}$
2. (a) 5 (b) 20
3. $\frac{6}{10}$ and $\frac{9}{15}$
4. $\frac{3}{4}$
5. It regroups the pieces; the number of pieces and their size change together so the total is unchanged.
6. No: both divide by 2, giving $\frac{2}{3}$.
7. $\frac{3}{8}$
8. $\frac{4}{5} = 80\%$
9. $\frac{40}{64}$
10. 49, because $63 \div 9 = 7$ and $7 \times 7 = 49$.

Practice 4.3 · Comparing and ordering fractions

1. (a) < (b) < (c) >
2. $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$
3. $\frac{21}{30}$ and $\frac{20}{30}$
4. $\frac{6}{7}$ ($0.857 > 0.833$)
5. The smaller denominator gives larger pieces, so that fraction is larger.
6. The conclusion is right but the reasoning is not: denominators must never be compared directly.

7. $\frac{1}{3} < \frac{3}{8} < \frac{2}{5}$ (0.333 < 0.375 < 0.4)
8. $\frac{13}{14}$ (difference $\frac{1}{14}$ against $\frac{1}{12}$)
9. Any of $\frac{9}{20}$, $\frac{19}{40}$, $\frac{5}{11}$: e.g. $\frac{9}{20} = 0.45$
10. Averaging two fractions gives a value halfway between, and the process can be repeated without limit.

Practice 4.4 · Fractions and decimals

1. (a) 0.25 (b) 0.6 (c) 0.45
2. (a) $\frac{7}{10}$ (b) $\frac{1}{4}$ (c) $\frac{3}{5}$
3. $\frac{1}{4}$ and $\frac{1}{5}$
4. 0.555…
5. One third never divides a power of ten; the denominator 4 has only a factor 2.
6. 0.3 is $\frac{3}{10}$, not one third; the difference matters when multiplying.
7. $\frac{3}{5} = 0.6 < \frac{5}{8} = 0.625 < 0.65$
8. $\frac{3}{8}$ kg
9. $\frac{2}{3}$
10. Its digits end, so it equals a whole number divided by a power of ten.

Practice 4.5 · Calculating with decimals

1. (a) 5.15 (b) 4.55 (c) 13.16
2. (a) 4.2 (b) 0.2 (c) 2
3. (a) 0.8 (b) 4 (c) 12
4. 24.05; 240.5; 2405
5. They added wrongly aligned digits; the correct answer is 6.02.
6. Two tenths of three tenths is six hundredths.
7. 2.1 litres; 0.4 litres more
8. 18.125 ≈ 18.13
9. Both equal 0.9; multiplying divisor and dividend by 10 keeps the quotient.
10. Smaller, because 0.9 of a number less than 1 is less than the number.

Practice 4.6 · Adding and subtracting fractions

1. (a) $\frac{3}{5}$ (b) $\frac{1}{2}$ (c) $\frac{7}{9}$
2. (a) $\frac{3}{4}$ (b) $\frac{1}{2}$
3. (a) $\frac{1}{2}$ (b) $\frac{3}{10}$
4. $3\frac{5}{6}$
5. The denominators are not parts of the same size; the sum must be greater than $\frac{1}{2}$.
6. The answer is right by coincidence; the method happens to work here because 2 + 3 = 5 and 2 × 3 = 6, but it fails in general.
7. $\frac{17}{12} = 1\frac{5}{12}$ cups
8. $\frac{3}{8}$
9. $\frac{29}{24} = 1\frac{5}{24}$
10. Convert both to improper fractions over 12, giving $\frac{39}{12}$ - $\frac{22}{12}$.

Practice 4.7 · Multiplying and dividing fractions

1. (a) $\frac{2}{5}$ (b) $\frac{2}{9}$
2. (a) 2 (b) 2
3. $\frac{3}{5}$
4. 1
5. Because four quarters fit in one whole, so each division by a quarter multiplies the count by 4.
6. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$, which is smaller.
7. $\frac{7}{3} \div \frac{1}{6} = 14$ bottles
8. 21 girls; 14 play a sport.
9. $\frac{3}{4} = \frac{12}{16}$, and $\frac{12}{16} \div \frac{9}{16} = 12 \div 9 = \frac{4}{3}$.
10. $\frac{5}{4}$, because $\frac{5}{6} \div \frac{2}{3} = \frac{5}{4}$.

Practice RQ4 · Applying the chapter

1. $\frac{2}{3}$
2. $\frac{19}{5}$
3. 0.45
4. 45
5. $\frac{9}{10}$
6. $\frac{2}{3}$
7. Both simplify to $\frac{3}{4}$; multiplying $\frac{3}{4}$ by $\frac{2}{2}$ and $\frac{3}{3}$ gives them.
8. $\frac{4}{7} = \frac{36}{63}$ and $\frac{5}{9} = \frac{35}{63}$, so $\frac{4}{7}$ is larger by $\frac{1}{63}$.
9. 0.333… repeats for ever and equals $\frac{1}{3}$; 0.3 terminates and equals $\frac{3}{10}$.
10. 1.2 litres
11. $\frac{1}{4}$
12. 6 pieces
13. A 3 by 4 grid has 12 cells; the overlap is 6 cells, which is $\frac{6}{12} = \frac{1}{2}$.
14. Each fraction is below 1, and multiplying by a number below 1 shrinks.
15. $\frac{2}{5}$ and $\frac{3}{8}$, among others.
16. $\frac{3}{11}$
17. Exactly one three quarters fits in three quarters.
18. $\frac{5}{3} = 1\frac{2}{3}$ cups
19. $\frac{1}{2} \div \frac{3}{4} = \frac{2}{3}$
20. Adding a positive quantity makes it larger than $\frac{1}{2}$; since $\frac{1}{3} < \frac{1}{2}$, the total is less than $\frac{1}{2} + \frac{1}{2} = 1$.

Practice MX4 · End-of-chapter mixed set

1. $\frac{3}{5} < \frac{2}{3} < \frac{7}{10}$
2. $\frac{5}{8}$
3. $\frac{3}{4}$
4. $\frac{17}{20}$
5. 12
6. $\frac{4}{5} = 0.8$; it is 0.05 from $\frac{3}{4}$ and 0.2 from 1, so closer to $\frac{3}{4}$.
7. Numerators and denominators were added rather than multiplied; the answer must be less than $\frac{2}{3}$, but $\frac{6}{7}$ is larger.
8. 2.1 kg
9. 27 chose football; 6 chose goalkeeper.
10. 3 hours
11. Each whole contains two halves, so the count doubles.
12. Sum is $\frac{15}{16}$, which is $\frac{1}{16}$ short of 1.
13. $\frac{16}{25}$, $\frac{11}{17}$, $\frac{21}{32}$ among others.
14. $\frac{18}{42}$, since $4k = 24$, $k = 6$.
15. Each numerator cancels the previous denominator, leaving $\frac{1}{5}$.
16. $\frac{15}{8} = 1\frac{7}{8}$ kg; one 2 kg bag suffices.

Practice CP4 · Cumulative practice

1. $\frac{3}{5} < \frac{2}{3} < \frac{7}{10}$
2. $\frac{7}{8}$
3. $\frac{3}{2} = 1\frac{1}{2}$
4. The numerator and denominator cancel completely.
5. 0.75 kg
6. 12 students
7. Its denominator has a factor of 3, and $\frac{1}{12} = 0.08333\ldots$.
8. Five fifths fit in one whole, so the count is five times as large.

Prerequisite check

1. 6 of the 8 parts
2. $\frac{1}{2}$ and $\frac{1}{4}$
3. 10
4. $\frac{3}{4}$
5. $\frac{1}{2}$
6. 0.7 = $\frac{7}{10}$

CHAPTER 05 · PERCENTAGES AND APPLICATIONS

Practice 5.1 · What a percentage is

1. (a) 35% (b) 70% (c) 125%
2. (a) 0.06 (b) 0.45 (c) 1.2
3. (a) $\frac{1}{5}$ (b) $\frac{3}{4}$ (c) $\frac{2}{25}$
4. 60%

5. 0.5% = 0.005, while 50% = 0.5.
6. 150% of 40 is 60; percentages compare with a base and may exceed one whole.
7. 96 units; a 20% increase.
8. 18 students
9. 60% ($^{15}/_{25}$)
10. $^{1}/_{3}$; $33\frac{1}{3}$ is a recurring decimal approximation of one third, and the fraction is exact.

Practice 5.2 · Finding a percentage of a quantity

1. (a) 8 (b) 25 (c) 6.4
2. (a) 15 (b) 11
3. 6
4. 25
5. 5% is half of 10%, and halving a quantity halves its percentage.
6. The number of groups of 20 in 50; the correct answer is 10.
7. Case 59.2; total 379.2 units
8. 204 people
9. 18 (10% = 24, 5% = 12, 2.5% = 6, so 24 - 6 = 18)
10. 90 units, since 63 ÷ 0.7 = 90

Practice 5.3 · One quantity as a percentage of another

1. 35%
2. 75%
3. 75%
4. 85%
5. The base differs; the first is $66\frac{2}{3}$% and the second 150%.
6. Yes; both are 50%, although the counts differ.
7. 20%
8. Walkers 70%, cyclists 45%; walkers.
9. 34 marks
10. 20%

Practice 5.4 · Percentage increase and decrease

1. 66
2. 60
3. 280
4. 20%
5. The decrease is calculated on the larger amount, so the result is 1.2 × 0.8 = 0.96 of the original.
6. 10 units was added instead of 10% of 40, which is 4.
7. 450 × 0.88 × 1.08 = 427.68
8. 15%
9. No change: 1.25 × 0.8 = 1.
10. 21%, since 1.1 × 1.1 = 1.21.

Practice 5.5 · Discounts, taxes and budgets

1. 50
2. 80
3. 204
4. 840
5. Rising 200% triples the amount; rising to 200% doubles it.
6. No; 1.1 × 0.9 = 0.99, so it is 1% lower.
7. 220 units; saved 44.
8. First 20%, second 20%; the same.
9. 21%
10. No; 0.8 × 1.2 = 0.96, so it is 4% below.
11. 566.40 units; 86.40 units
12. 500 units
13. 810 units
14. The discount is taken from the pre-tax price, which is smaller than the price shown.

Practice RQ5 · Applying the chapter

1. 72%
2. $^{7}/_{20}$
3. 17
4. 154
5. 63
6. 40%
7. Both equal 20, since multiplication is commutative.
8. No; 0.9 × 1.1 = 0.99, which is 1% lower.
9. The increase is 60, so the new value is 100.
10. 80 units
11. 396 girls
12. 80 units
13. It adds 15% of the amount to the amount itself: $x + 0.15x = 1.15x$.
14. Each change multiplies the current amount by a factor, and the factors are applied in succession.
15. +20% then +20% (1.2 × 1.2 = 1.44).
16. 4% decrease (0.8 × 1.2 = 0.96).
17. 20% rise.
18. 30%
19. The second discount applies to 70% of the original, removing 7% of it: 0.7 × 0.9 = 0.63.
20. Each fall is calculated on a smaller base: $0.95^2 = 0.9025$, a fall of 9.75%.

Practice MX5 · End-of-chapter mixed set

1. 18
2. 62.5%
3. 310
4. 45
5. 75%
6. No; the second is 20% less, because the base changes.
7. The bases differ, so the money saved differs.
8. 62.5%
9. 400 units
10. 4,960.8, so about 4,961
11. Find 10%, halve it for 5%, then add.
12. First 80; second 93.5; the first.
13. 25% decrease (1.5 × 0.5 = 0.75).
14. 96%
15. 200 units
16. 100%, so no profit and no loss.

Practice CP5 · Cumulative practice

1. 8%
2. 21
3. 204
4. 0.5% = 0.005 while 50% = 0.5.
5. 80
6. 37%, since 0.7 × 0.9 = 0.63.
7. Each applies to a different base.
8. 21%

Prerequisite check

1. 0.75
2. 8
3. 30
4. $^{9}/_{20}$
5. 60
6. 0.6

CHAPTER 06 · PATTERNS, EXPRESSIONS AND EQUATIONS

Practice 6.1 · Patterns and rules

1. (a) 15, 18 (b) 11, 8
2. add 7
3. $4n + 1$
4. 59
5. Each step adds one more group of 4, so the term grows in proportion to the position.
6. 6 × 2 = 12, not 10; the difference is 4, so the rule is $4n + 2$.

7. $3 \times 12 + 2 = 38$
8. the 9th term
9. $109 - 9n$
10. First: 7 against 5; tenth: 34 against 41.

Practice 6.2 · Expressions and substitution

1. 27
2. 3
3. (a) 49 (b) 144
4. 48
5. $2x = 12$ but $x^2 = 36$.
6. They calculated $2 \times (3 + 5)$; the multiplication applies only to x, giving 11.
7. 11
8. 17 units
9. 9.5
10. -7

Practice 6.3 · Simplifying expressions

1. (a) $8x$ (b) $5y$ (c) $4a$
2. $6p + 2q$
3. (a) $3x + 6$ (b) $10y - 5$
4. $5x + 11$
5. The letters are different, so the terms are not like terms.
6. $8x - 2$: the 5 and 3 are like terms, and the -2 stays.
7. $4a + 5$
8. $4s + 8$
9. $13 - 5x$
10. $6x + 2$ and $6x + 3$; difference 1.

Practice 6.4 · Solving linear equations

1. (a) 5 (b) 13 (c) 7
2. (a) 4 (b) 4
3. 24
4. 5
5. An equation balances two whole sides; an expression inside one side is not a separate balance.
6. Dividing by 3 must divide the whole side, including the 2, which was not divided.
7. $x = 3$
8. $x = 10$
9. $x = 4$
10. $x = 12$

Practice 6.5 · Forming and using equations

1. $x + 12 = 30$; $x = 18$
2. $4x = 52$; $x = 13$
3. $4s = 48$; 12 cm
4. 20 and 21
5. The phrase “less than” reverses the order: three times the number, minus 5.
6. The equation or the interpretation of the unknown is wrong, since a length cannot be negative.
7. 240 minutes
8. 8 cm by 12 cm
9. 26, 28, 30
10. son 12, father 36

Practice RQ6 · Applying the chapter

1. 21
2. $4n + 5$
3. 31
4. $4a + 5b$
5. $x = 5$
6. $x = 6$
7. The 4 multiplies both terms in the bracket.
8. Yes: $5n = 50$, $n = 10$.
9. An expression has no equals sign; an equation states two expressions are equal.
10. 23 units
11. $x = 5$
12. 20, 21, 22
13. It regroups the same quantities without changing how many there are.
14. Substitute the value into the original equation and confirm both sides are equal.
15. $x = 6$
16. $x = 24$
17. $4n - 1$
18. With width w: $2w + 4w = 48$, so $w = 8$ and the length is 16. Area = 128 cm².
19. Subtracting the smaller letter term from both sides leaves letters on one side only.
20. False: for $3n + 1$ the 100th term is 301, not 400.

Practice MX6 · End-of-chapter mixed set

1. 82
2. 36
3. $14 - 3x$
4. 45
5. $x = 5$
6. Yes, by expanding: $2 \times 3x = 6x$ and $2 \times 4 = 8$.
7. $3(-2) + 4 = -2$, not 10.
8. 63
9. 3 months
10. $x = 11$
11. Compare the letters and their powers; only identical terms combine.
12. An equation can be solved correctly from a wrong model, and a negative length is still impossible.
13. 31 and 33
14. $x = 9$
15. 12
16. 9 cm by 14 cm; 126 cm²

Practice CP6 · Cumulative practice

1. $5n + 3$
2. $4x + 10$
3. $x = 6$
4. It states no equality, so there is nothing to solve.
5. 50
6. $x = 7$
7. Substitute: $12 - 5 = 7$.
8. The first is $4n + 2$; the second is $4n + 8$.

Prerequisite check

1. 42
2. 8
3. 4×5
4. 14, 17
5. 14
6. $\square = 9$

CHAPTER 07 · LINES, ANGLES, TRIANGLES AND CONSTRUCTIONS

Practice 7.1 · Points, lines, angles and notation

1. acute, right, obtuse, reflex
2. 112°
3. 75°
4. $\angle ABC$
5. Their sum would exceed 180°.
6. The wrong protractor scale was read; the correct angle is 50°.
7. $9x = 360$, $x = 40$; angles 80°, 120°, 160°.
8. 120°
9. $5y + 30 = 180$, $y = 30$.

10. Each is 180° - the shared angle.

Practice 7.2 · Angle relationships

1. 133°
2. 110°
3. 125°, 55°, 125°
4. 55°
5. Each is 180° minus the same adjacent angle.
6. No; angles on a line sum to 180°.
7. $10x = 360$, $x = 36$; 36°, 72°, 108°, 144°.
8. $5x + 10 = 180$, $x = 34$.
9. 10 parts = 360, one part 36; largest 144°.
10. Acute angles are each below 90°, so four would sum to less than 360°.

Practice 7.3 · Parallel lines and transversals

1. 70°
2. 42°
3. 128°
4. 65°, 115°, 65°, 115°
5. Alternate angles are equal; co-interior angles add to 180°.
6. Measure or calculate both angles and compare them exactly.
7. $2x = 40$, $x = 20$; angles 80°.
8. $5x + 30 = 180$, $x = 30$.
9. $2y = 30$, $y = 15$, both 65°.
10. The co-interior angle and the alternate angle of the given angle lie on a straight line.

Practice 7.4 · Triangles

1. 60°
2. 58°
3. 60°
4. 80°
5. Their sum would be 180°, leaving nothing for the third angle.
6. Obtuse-angled.
7. 40°, 60°, 80°
8. 35° each
9. $9x + 45 = 180$, $x = 15$; the angles are 30°, 45° and 105°, so the largest is 105°.
10. If all three were less than 60°, the sum would be below 180°.

Practice 7.5 · Polygons and their angles

1. (a) 540° (b) 1080° (c) 1800°
2. (a) 120° (b) 135° (c) 140°
3. (a) 72° (b) 36° (c) 18°
4. 70°
5. A diagonal splits it into two triangles of 180°.
6. 12 sides, since the exterior angle is 30° and $360 \div 30 = 12$.
7. 15 sides; 156°
8. $540 - 400 = 140°$, so not all five angles are equal.
9. $12 - 8 = 4$ sides
10. A convex polygon has every interior angle below 180°; 200° would make the corner point inwards.

Practice 7.6 · Constructions with compasses

1. more than 5 cm
2. three: one at the vertex, two inside
3. arcs of radius 5 cm from both ends of the base
4. 90° each
5. They are the evidence that the construction is exact.
6. Otherwise the arcs from the two ends do not meet.
7. Isosceles.
8. Any point on the perpendicular bisector of the line joining them.
9. Construct a right angle, then bisect it.
10. Each point on a bisector is equidistant from two sides; the three conditions coincide at one point.

Practice RQ7 · Applying the chapter

1. 53°
2. 52°
3. 45°
4. 50°
5. No, $5 + 6 = 11$, so they lie flat.
6. 73°
7. The sum would be 180° only with a zero angle, which is not an angle of a triangle.
8. 360°, since two triangles contribute 180° each.
9. A construction uses only compasses and a straight edge and relies on equal distances.
10. 120°
11. 75° each
12. $5x + 20 = 180$, $x = 32$.
13. It equals the sum of the two interior opposite angles, since it and the adjacent interior angle sum to 180°.
14. Every point on it is equidistant from the two endpoints.
15. $5x - 20 = 180$, $x = 40$; 50°, 80°, 50°.
16. No, the two shorter sides sum exactly to the longest.
17. 40° each
18. Anywhere on the perpendicular bisector of the diagonal joining the corners.

Practice MX7 · End-of-chapter mixed set

1. 112°
2. reflex
3. 150°
4. 83°
5. Yes, $7 + 8 > 9$.
6. The third angle must be positive.
7. The sum is 190°, not 180°.
8. $6x + 30 = 180$, $x = 25$; the angles are 25°, 50° and 105°, so the largest is 105°.
9. 25°
10. vertex 70°, base angles 55° each
11. Equal angles summing to 180° must each be one third.
12. Check whether one angle is exactly 90°.
13. Each transversal gives its own equal pairs, but angles from the two transversals need not be equal.
14. $12x = 180$, $x = 15$; angles 45°, 60°, 75°, acute.
15. Draw a segment, then arcs of the same radius from both ends to locate an equilateral triangle.
16. Valid; perimeter 42 m; 21 panels.

Practice CP7 · Cumulative practice

1. 70°
2. 84°
3. Translation along the transversal maps one crossing to the other.
4. 68°
5. No: $6 + 7 = 13 < 14$.
6. They show that the construction is exact.
7. $4x + 20 = 180$, $x = 40$; the largest is 80°.
8. Reflecting the segment in it maps each endpoint to the other.

Prerequisite check

1. $x = 140$
2. 115
3. $x = 60$
4. Isosceles
5. 90°
6. 220

CHAPTER 08 · SETS AND CLASSIFICATION

Practice 8.1 · Sets and set notation

1. 1, 3, 5, 7, 9, 11
2. 4
3. 1, 2, 4, 8, 16
4. Yes.
5. "Interesting" is not defined precisely enough to decide membership.
6. The empty set has no elements; 0 has one element, the number 0.
7. x : *x is a multiple* of 4, $0 < x < 24$
8. 2, 3, 5, 7, 11; 5
9. 15, 30
10. A set is determined by membership, not by arrangement.

Practice 8.2 · Subsets, union, intersection and complement

1. 3, 5 and 1, 3, 5, 7
2. 1, 3, 5, 7, 9
3. Yes.
4. 17
5. The overlap is counted twice in the sum.
6. When $A \subseteq B$, for example $A = 1, 2$, $B = 1, 2, 3$.
7. 3
8. 6
9. $A \cap B = 12, 24$, so 2; 7 + 5 - 2 = 10.
10. The union is a subset of the universal set, so it cannot have more elements than it.

Practice 8.3 · Venn diagrams and survey problems

1. 6
2. 20
3. 15
4. 8
5. Every other region is found by subtracting it.
6. 35
7. 14
8. 5
9. 18
10. Every member of the overlap is in both sets, so the overlap is a subset of each.

Practice 8.4 · Classification and fair decisions

1. No: 6 is both.
2. Yes.
3. 1.
4. No; other colours exist.
5. Otherwise a member is counted twice or not at all.
6. Some students do neither, and some do both.
7. 4 and 16.
8. Yes, provided "12 to 16" includes its endpoints and the boundaries are stated.
9. For example, prime/composite/neither, then odd/even within composites: 10 primes, 9 even composites, 10 odd composites, 1 neither.
10. "Even" is undefined for 2.5, so the rule does not decide membership.

Practice 8.5 · Number sets and the real line

1. (a) $\mathbb{N}$ (b) $\mathbb{Z}$ (c) $\mathbb{Q}$ (d) irrational
2. (a) true (b) false (c) true
3. 0: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$; $-½$: $\mathbb{Q}$, $\mathbb{R}$
4. 5; 0.2; 1.41
5. It equals $\frac{1}{3}$, a fraction of two integers.
6. 1.414 is a rounded decimal; squaring it gives 1.999396, not 2.
7. 6 and 7; 6.3
8. $\mathbb{N} = \{1,2,3,4,5\}$; the rest = $\{-5,-4,-3,-2,-1,0\}$
9. $\frac{5}{12}$, and $\frac{1}{3} < \frac{5}{12} < \frac{1}{2}$
10. Halving any positive rational gives a smaller positive rational.

Practice RQ8 · Applying the chapter

1. 12, 14, 16, 18
2. 5
3. 1, 2, 3, 4
4. 20
5. 6
6. 1, 3, 5, 7, 8
7. They are non-overlapping parts of the whole group.
8. 20 + 18 - 5 = 33 > 30, so no: the overlap would be negative.
9. Exactly one excludes the overlap; at least one includes it.
10. 7
11. 50
12. 8 and 16
13. Every element of the intersection is also in A.
14. The overlap is counted in both circles and should be counted once.
15. 25
16. The smaller set is a subset of the larger.
17. Each set contributes a yes/no choice, and $2^3 = 8$.
18. 30

Practice MX8 · End-of-chapter mixed set

1. 4, 6
2. 0
3. No.
4. 33
5. 7
6. Sets are determined by membership, not order.
7. The value 10 itself, so the categories are not exhaustive.
8. 10
9. 14
10. 25 and 18
11. It is counted twice in the sum of the two set sizes and should appear once.
12. The number 1 is neither.
13. 14
14. 15
15. 140
16. The tree applies each criterion in a stated order so every student follows the same path.

Practice CP8 · Cumulative practice

1. 3, 5
2. 23
3. Otherwise a member belongs to no category.
4. 10
5. 10 prime, 19 composite, 1 neither.
6. It contains one element, the number 0.
7. 30
8. Every element of the overlap belongs to both sets.

Prerequisite check

1. 2, 4, 6, 8, 10
2. $\frac{3}{4}$
3. 26
4. −4
5. 3 and 6 (or 9)
6. 5

CHAPTER 09 · PERIMETER, AREA, SURFACE AREA AND VOLUME

Practice 9.1 · Perimeter and area of rectangles and triangles

1. 90 cm²

2. 30 cm
3. 30 m²
4. 8 cm
5. Area counts the number of unit squares that cover the shape.
6. Perimeter is a length, so the unit should be cm.
7. 8 cm
8. Perimeter 60 m; cost 240 units.
9. Square 81 cm²; rectangle 12 by 6 gives 72 cm².
10. Perimeter doubles; area quadruples.

Practice 9.2 · Area of parallelograms and compound shapes

1. 36 cm²
2. 30 cm²
3. 121 m²
4. 6 cm
5. The height must be perpendicular to the base.
6. The shared edge is counted twice; perimeter is the outer boundary.
7. 44 m²
8. 60 cm²; 7 cm is a sloping side, not the perpendicular height.
9. Area 45 cm²; perimeter 36 cm.
10. The moved triangle leaves a rectangle of the same base and height.

Practice 9.3 · Circles: circumference and area

1. (a) 25.12 cm (b) 31.4 cm (c) 15.7 m
2. (a) 28.26 cm² (b) 314 cm² (c) 50.24 m²
3. 44 cm; 154 cm²
4. $\frac{1}{2}\pi \times 36 \approx 56.5$ cm²
5. The area depends on r^2, and $(2r)^2 = 4r^2$.
6. They used the diameter instead of the radius; the correct area is $\pi \times 4.5^2 \approx 63.6$ cm².
7. $\pi(16 - 9) \approx 22.0$ m²
8. About 188.5 cm; about 53 turns
9. $r = 62.8 \div (2\pi) \approx 10.0$ cm
10. $64 - \frac{1}{4}\pi \times 64 \approx 13.7$ cm²

Practice 9.4 · Surface area of cuboids and prisms

1. 62 cm²
2. 96 cm²
3. 210 cm²
4. 6
5. The faces occur in three equal pairs.
6. Only three faces were counted, not six.
7. 2(384 + 216 + 144) = 1,488 cm²
8. 600 + 2(360) + 2(240) = 1,800 cm²
9. side 5 cm; volume 125 cm³
10. Its faces are closer to squares, so the sum of products is smaller for a fixed product.

Practice 9.5 · Volume of cuboids and prisms

1. 72 cm³
2. 125 cm³
3. 2.5 litres
4. 6 cm
5. Three lengths are multiplied, so the unit is cubed.
6. The 2 m was not converted to 200 cm; the correct volume is 400,000 cm³.
7. 72 litres
8. 360 cm³
9. h = 6 cm; surface area 376 cm².
10. Each length doubles, so the product gains three factors of 2.

Practice 9.6 · Using measurement to make decisions

1. 15 m²
2. 2(20+10) + 2(15+10) + 20 = 60 + 50 + 20 = 130 cm.
3. 0.60 per litre
4. 15
5. It compares value for the same quantity.
6. The totals differ; per litre, 3.00 against about 2.67.
7. 2 tins (8 litres covers 96 m²)
8. 2(1200 + 800 + 600) = 5,200 cm²
9. 0.333 against 0.347; the first.
10. Its surface-to-volume ratio is larger.

Practice RQ9 · Applying the chapter

1. 70 cm²
2. 38 cm
3. 56 m²
4. 180 cm³
5. 216 cm²
6. 4.5 litres
7. There are three pairs of equal faces.
8. No; a square has the smallest perimeter for a given area.
9. A litre is a cube of side 10 cm, and $10^3 = 1{,}000$.
10. 27 m²; 324 units.
11. 840 litres
12. 500 + 2(375) + 2(300) = 1,850 cm²
13. Two congruent triangles make the rectangle.
14. Area depends on two lengths and volume on three.
15. height 15 cm; surface area 2(16) + 4(60) = 272 cm².
16. 785 cm³
17. Its surface area per unit volume is smaller.
18. 1,000 litres; 200 minutes.
19. 108 cm² for 72 cm³ against 192 cm² for 144 cm³; the larger box is more efficient.

Practice MX9 · End-of-chapter mixed set

1. 60 cm²
2. 32 m
3. 729 cm³
4. 750 litres
5. 166 cm²
6. Perimeter is a length; area is a product of two lengths.
7. The volumes are 1,000 and 125 cm³, so eight times as much.
8. Fence 48 m costs 144; turf 135 m² costs 1,080.
9. 1,920 cubes
10. 360 cm³
11. It does not if the cut replaces two edges with equal lengths; it does if the cut is not at a corner.
12. Area covers a surface; volume fills a space.
13. $6k^3 = 384$, $k^3 = 64$, $k = 4$; 4, 8, 12 cm.
14. Cube 216 cm²; cuboid 228 cm²; the cube.
15. Capacity 160 litres, half full holds 80, so add 80 litres.
16. $r = 4$; volume ≈ 603 cm³, about 0.6 litres.
17. 44 cm
18. Area depends on r^2; circumference depends on r.
19. $\pi \times 1.4 \approx 4.4$ m
20. Arc ≈ 94.2 cm plus the 60 cm diameter, so about 154.2 cm

Practice CP9 · Cumulative practice

1. 105 cm²
2. 200 cm³
3. 7.5 litres
4. It must be measured perpendicular to the base.
5. 1,200 + 2(800) + 2(600) = 4,000 cm²
6. 202.5 litres
7. It has a small surface area for its volume.
8. side 8 cm; surface area 384 cm².

Prerequisite check

1. 84
2. 240 cm
3. 35
4. 12.56
5. 25
6. 125

CHAPTER 10 · DATA, TABLES AND GRAPHS

Practice 10.1 · Collecting and organising data

1. quantitative, discrete
2. qualitative
3. 23
4. 7 (12–, 19–, 26–, 33–, 40–)
5. A value would be counted twice.
6. Three observations are missing or mis-tallied.
7. Categories from 3 to 12 letters, with tallies and frequencies totalling the class size.
8. Other = 60.
9. Individual values are replaced by an interval.
10. Unequal widths and ambiguous boundaries.

Practice 10.2 · Displaying data

1. bar chart
2. histogram
3. 90°
4. 24
5. Otherwise differences are exaggerated.
6. Intervals on a continuous scale are adjacent; categories are separate.
7. 135°
8. 14
9. Symbols of different sizes or partial symbols invite wrong comparison.
10. B looks much larger although it is only 2 more.

Practice 10.3 · Averages and range

1. 7.5
2. 6
3. 3
4. 15
5. It measures spread, not a central value.
6. No; it is bimodal, and both values are reported.
7. 11
8. 6.5
9. 20
10. It is greater than 10; specifically, if there were n values, the new value is $10 + 11(n+1) - 10n = 11 + n$... explain that the added value exceeds the old mean.

Practice 10.4 · Grouped data

1. 25
2. the second
3. 15
4. 40
5. Midpoints stand in for unknown individual values.
6. The spread of results differs.
7. 15.2
8. the second interval (positions 13–23)
9. The true values within an interval are not all at the midpoint.
10. Many values lie near the lower bound of each interval, so midpoints overstate them.

Practice 10.5 · Reading data critically

1. 12
2. the mean
3. the values
4. To represent the whole group fairly.
5. The same percentage of different totals is a very different count.
6. No; a third factor may cause both.
7. 100 units
8. Some low values pull the mean below the median.
9. Wide intervals smooth the shape; narrow intervals show detail and can make small groups look important.
10. Random selection across year groups; weakness: self-reported times may be inaccurate.

Practice RQ10 · Applying the chapter

1. quantitative, continuous
2. 10
3. 8
4. 6
5. 12
6. 35
7. It is not distorted by extreme values.
8. Heights no longer represent frequency directly.
9. Colours cannot be added or ordered.
10. 120
11. 30
12. 100
13. Frequencies must sum to the number of observations, with no gaps or overlaps.
14. Both are half of the base.
15. For example 4, 7, 9, 11, 19.
16. 2, 5, 8 and 10, 13, 16 both have range 6.
17. A biased sample over-represents a group.
18. Midpoints 12, 17, 22, 27, 32 give 48 + 153 + 308 + 216 + 160 = 885; mean = 885 ÷ 40 = 22.125; modal class 20–24.
19. Values within an interval are not all at the midpoint.

Practice MX10 · End-of-chapter mixed set

1. 16
2. 8
3. 15
4. 28
5. 90°
6. It may mean mean, median or mode.
7. A dramatic rise although the increase is 10%.
8. mean 68, median 68, mode 60
9. 160.96 cm
10. 120°
11. Categories need a bar chart; continuous intervals need a histogram.
12. A percentage means little without knowing how many people it summarises.
13. The mean rises and the range increases greatly.
14. 96 − 66 = 30.
15. Centre does not describe spread or the position of clusters.
16. 216°, 90°, 54°; walking is the main mode of travel.

Practice CP10 · Cumulative practice

1. 11
2. 11
3. 14
4. The intervals are continuous.
5. 120°
6. $^{15 + 135 + 200}/_{20} = 17.5$
7. The mean alone does not show how spread out the data are.
8. 14

Prerequisite check

1. 20
2. 0.375 and 37.5%
3. 45

4. 8
5. 5, 5, 9, 12, 14
6. 82

CUMULATIVE REVISION

Practice CR1 · Cumulative practice 1

1. $4 \times 1{,}000 + 7 \times 100 + 0 \times 10 + 6 \times 1$
2. 38,500
3. $32 + 9 = 41$
4. 24
5. 1, 2, 3, 6, 7, 14, 21, 42
6. $2^2 \times 3 \times 7$
7. It has exactly one factor.
8. $3^2 = 3 \times 3 = 9$.
9. Every power of ten is one more than a multiple of 3, so only the digit sum affects the remainder.
10. 12,000; a rounded figure communicates size without false precision.
11. 12 and 72
12. 12 groups (HCF)
13. $91 = 7 \times 13$.
14. 3, giving 324.
15. Yes: $2^{10} = 1{,}024$.
16. Every factor pairs with another except the square root.

Practice CR2 · Fractions and percentages

1. $\frac{3}{4}$
2. 0.875
3. $\frac{23}{20} = 1\frac{3}{20}$
4. $\frac{3}{4}$
5. 36
6. 45%
7. 0.667 against 0.6, or $\frac{10}{15}$ against $\frac{9}{15}$.
8. Only like-sized parts can be added.
9. The second discount is calculated on the reduced price.
10. 120 units
11. $1\frac{7}{8}$ kg
12. $1.15 \times 0.9 = 1.035$, a rise of 3.5%.
13. $\frac{13}{28}$ (about 0.464), since $\frac{3}{7} = \frac{12}{28}$ and $\frac{4}{7} = \frac{16}{28}$; $\frac{1}{2}$ also lies between.
14. $\frac{5}{11}$
15. 56.25%
16. Use a common denominator of 48: $\frac{28}{48}$ against $\frac{27}{48}$.

Practice CR3 · Algebra and geometry

1. $4n + 2$
2. 43
3. $3a + 10$
4. $x = 7$
5. 58°
6. 65°
7. The 3 multiplies both terms.
8. They sum to 190°.
9. Each step adds another group of 5.
10. 22 units; 13 km.
11. $x = 7$
12. 9 cm by 14 cm; 126 cm².
13. $x = 20$; largest 80°.
14. Multiply by 4: $x + 20 = 8x - 4$, so $7x = 24$ and $x = \frac{24}{7}$.
15. Equal angles summing to 180°.
16. Both sides must change equally to keep the balance.

Practice CR4 · Sets and measurement

1. 2, 4, 6, 8
2. 18
3. 60 cm²
4. 140 cm³
5. 3.25 litres
6. 150 cm²
7. It is counted twice in the sum.
8. It counts unit squares.
9. It is a cube of side 10 cm.
10. 4
11. 18 m²
12. 48 litres
13. 13
14. height 6 cm; surface area 312 cm².
15. Its surface-to-volume ratio is smaller.
16. The overlap is counted in both circles and once in the union.

Practice CR5 · End-of-year mixed paper

1. 5.08
2. 53
3. $\frac{3}{4}$
4. 243
5. $x = 5$
6. 12
7. 78 cm²
8. $0.3 = \frac{3}{10}$, and $\frac{1}{3} = 0.333\ldots$.
9. An extreme value pulls it away from the typical case.
10. Each change applies to the current amount.
11. 2.50 against 2.36 per litre; the 5-litre pack.
12. 30
13. side 7 cm; volume 343 cm³.
14. The median, because a few very high prices distort the mean.
15. Other rules fit the same four terms but differ later.
16. width 7, length 17; area 119 cm².
17. 100
18. 70°; acute-angled isosceles.

Practice PR1 · Part I problem solving

1. 4,100 and 4,000
2. 30
3. 20 and 120
4. $4.5 \times 10 = 45$ and $0.3 \times 10 = 3$; no zero is added, but the digits shift one place.
5. Both are divisible by 6, so their sum is too.
6. 11,500 to 12,499 if rounding to the nearest thousand with one significant figure — the report is consistent with any count in that range.
7. 12 students per team; $7 + 5 + 4 = 16$ teams.
8. Every power of ten leaves remainder 1 on division by 9, so only the digits matter.
9. One of them is even.
10. 225 ($3^2 \times 5^2$) has nine factors; 252 has 18; 243 has six: 3^5. So 243.
11. 60
12. 100 is not a multiple of 8, so the hundreds digit matters; the test uses the last three digits.
13. 90 minutes later at 10:30; 4 times in 6 hours.
14. 90 cm: the HCF of 450 and 360 is 90, and both sides are multiples of 90.

Practice PR2 · Part II problem solving

1. $\frac{7}{20}$
2. 1
3. 382.5
4. The denominators differ; as decimals $0.375 < 0.4$.
5. $1.1 \times 0.9 = 0.99$.
6. $1\frac{9}{16}$ kg; 2 bags.
7. First 120; second 140; the second.
8. There are more than one of those fractions in each whole.
9. $0.375 = \frac{375}{1000}$, which simplifies to $\frac{3}{8}$.

10. No change: 1.25 × 0.8 = 1.
11. $^{21}/_{32}$ (0.656), among others.
12. Common denominator 99: $^{77}/_{99}$ against $^{72}/_{99}$, so $^{7}/_{9}$ is larger.
13. 20% rise.

Practice PR3 · Part III problem solving

1. $7n + 4$
2. $x = 4$
3. 64°
4. 4(1) + 3 = 7, not 9; the rule would be $4n + 5$.
5. A drawing shows one case; a proof must cover all cases.
6. 61 matches; stage 33.
7. $9x = 180$, $x = 20$; 40°, 70°, 70°.
8. Each operation is applied to both sides, preserving equality.
9. It and the adjacent interior angle sum to 180°, as do the three interior angles.
10. width 8, length 19; area 152 cm².
11. The two arcs meet at a single point above the base (and its mirror image).
12. The two shorter sides sum exactly to the longest, so the shape is flat.
13. $10x = 180$, $x = 18$; base angles 54°, vertex 72°.
14. The difference is 4, so the rule is $4n + 1$; $5n$ gives 5, 10, 15.

Practice PR4 · Part IV problem solving

1. 2, 3 and 1, 2, 3, 5, 7
2. 258 cm²
3. 800 litres
4. The first is defined by an objective boundary; the second is a judgement.
5. Volume uses three lengths, surface area two.
6. 15
7. 756 litres; 22.68 units.
8. Each student follows exactly one branch at every stage.
9. One face is missing.
10. height 6 cm; surface area 216 cm² (it is a cube).
11. 75
12. Volume scales with the cube of the length factor.
13. about 1,696 litres.
14. Volumes 1,000 and 1,000 cm³, so equal value; surface areas 700 and 730 cm², so the first is cheaper to make.

Practice PR5 · Part V problem solving

1. mean 16, median 15, range 8
2. 60
3. 16
4. A few very high prices distort the mean.
5. Differences are exaggerated visually.
6. 50 students, 20%.
7. 62.64 (to 2 d.p.)
8. State the sample and how it was chosen, the statistics found, and at least one limitation.
9. The same percentage of different totals is a very different count.
10. One was at or below 30 and one at or above, so the middle value is unchanged.
11. 220 ÷ 9 ≈ 24.4
12. Midpoints replace the true values.
13. The second is higher and much more consistent.
14. What the results are compared with, and how large the sample was.

Practice PA · Practice paper A

1. Three million, four hundred and seven thousand and fifty-two.
2. 4,800 and 4,840
3. 16 × 25 = 400
4. 12 + 9 = 21
5. 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84
6. HCF 12; LCM 144
7. 0.875; 87.5%
8. $^{11}/_{24}$
9. 0.27; two decimal places in total
10. A prime has exactly two factors; 1 has only one.
11. Denominators were added; the correct answer is $^{11}/_{12}$.
12. 1,125 g
13. 204 units; 36 units saved
14. 300 units
15. 1.2 × 0.8 = 0.96, a 4% fall

Practice PA2 · Practice paper A

1. 5.37
2. 0.41666…
3. 720
4. 12; half of 10% is 5%, and 8 + 4 = 12
5. 0.9 is 90% of the amount, so 10% has been removed.
6. One third, since 33.3% > 20%
7. 12 km per litre; 25 litres
8. 37.5%
9. 21.5 units
10. Below: 1.25 × 0.75 = 0.9375
11. 19%
12. $4000 \times 0.85^2 = 2{,}890$
13. 1.5 × 0.5 = 0.75, a 25% fall
14. $100x - x = 36$, so $x = {}^{36}/_{99} = {}^{4}/_{11}$
15. 30% is 60 litres, so capacity is 200 litres

Practice PB · Practice paper B

1. $3x + 4y$
2. $8a - 12$
3. $x = 7$
4. $x = 10$
5. 48
6. 67° and 67°
7. 1,260°
8. 45°
9. {2, 3, 4, 6, 8, 9}
10. They total 200°, not 180°.
11. It cannot be written as a fraction of two integers; its decimal never ends or repeats.
12. 126 cm²
13. 160 cm³; 184 cm²
14. 37.68 cm; 113.04 cm²
15. $15x = 360$, so $x = 24$; 48°, 72°, 96°, 144°

Practice PB2 · Practice paper B

1. 8
2. mean 8; median 7; mode 7
3. 16
4. It shows proportions, not totals.
5. Small differences look large.
6. 480 litres
7. 60 - 12.56 ≈ 47.4 m²
8. 153 ÷ 11 ≈ 13.9
9. $^{5}/_{12}$
10. $2(45 + 9h + 5h) = 214$, so $14h = 62$ and h ≈ 4.4 cm
11. 12
12. 37.5 → 38 people (with rounding explained)
13. Midpoints 12, 17, 22, 27 give 16.5
14. Ask whether the middle value or the total-shared value was used; compare with the data.
15. Eight times

Practice PC · Practice paper C

1. 5,060

2. 0.047
3. 5.5
4. $5a + 2b$
5. 84
6. 45%
7. A fraction of a fraction takes a part of a part.
8. 4 is a factor of 12; 12 is a multiple of 4.
9. 1,440 m²; 154 m
10. 36 - 19.6 ≈ 16.4 m²
11. 40%
12. 1,050 g
13. 7 cm; 294 cm²
14. 5 × 12 = 60; the fifth is 11
15. 300 units
16. $n + (n+1) + (n+2) = 3n + 3 = 3(n+1)$

Practice PC2 · Practice paper C

1. $x = 8$
2. 108°
3. {1, 2, 3, 5, 6, 7, 9, 10, 11}
4. A square has four right angles and all sides equal; a rectangle need not have equal sides.
5. Equal-looking intervals represent unequal amounts, so the picture exaggerates.
6. 1,080 litres
7. 96 km/h
8. 849.60 units; 129.60 units
9. 108
10. The wall, ground and ladder form a right-angled triangle, so the height can be found from the two known lengths.
11. 12
12. 36
13. 37%
14. The median is not affected by extreme values; the mean is pulled towards them.
15. $V = \pi(25)(10) \approx 785$ cm³, so yes.

Formula bank

Formulas are collected here for reference. A formula is a compressed relationship: read it slowly, name each symbol and its unit, and check that the result is sensible before using it in a problem.

NUMBER AND PLACE VALUE

Place value in a base-ten number	each place is 10 times the place on its right
Expanded form	$3{,}048 = 3\times1000 + 0\times100 + 4\times10 + 8\times1$

POWERS AND ROOTS

Power notation	$a^n = a \times a \times \ldots \times a$ (n factors)
Multiplying powers of the same base	$a^m \times a^n = a^{m+n}$
Dividing powers of the same base	$a^m \div a^n = a^{m-n}$ $(m>n)$
Power of a power	$(a^m)^n = a^{mn}$
Zero and first powers	$a^0 = 1$, $a^1 = a$ $(a \neq 0)$
Square root	$\sqrt{a}$ is the positive number whose square is a

FACTORS AND MULTIPLES

Highest common factor	HCF: the largest number dividing both exactly
Lowest common multiple	LCM: the smallest number divisible by both
Relationship	HCF × LCM = $a \times b$ for positive integers a, b

FRACTIONS AND DECIMALS

Equivalent fractions	$\frac{a}{b} = \frac{ak}{bk}$ $(k \neq 0)$
Adding and subtracting	$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$
Multiplying	$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$
Dividing	$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$
Fraction to decimal	divide the numerator by the denominator

PERCENTAGES

Percentage of a quantity	amount = $\frac{p}{100}$ × quantity
Increase by p%	multiply by $1 + \frac{p}{100}$
Decrease by p%	multiply by $1 - \frac{p}{100}$
Successive changes	multiply the multipliers: 1.2×0.9
Percentage change	$\frac{\text{change}}{\text{original}} \times 100\%$

ALGEBRA

Collecting like terms	$3x + 2x = 5x$
Expanding	$a(b + c) = ab + ac$
Solving a two-step equation	undo the addition, then undo the multiplication
Perimeter of a rectangle	$P = 2(l + w)$
Area of a rectangle	$A = lw$

GEOMETRY

Angles on a straight line	sum to 180°
Angles at a point	sum to 360°
Vertically opposite angles	equal

Alternate angles between parallel lines	equal
Corresponding angles between parallel lines	equal
Co-interior angles between parallel lines	sum to 180°
Angle sum of a triangle	180°
Exterior angle of a triangle	equals the sum of the two opposite interior angles
Angle sum of a polygon	$(n\text{-}2) \times 180°$
Exterior angle of a regular polygon	$^{360°}/_{n}$

MEASUREMENT

Perimeter	the total distance around a shape
Area of a triangle	$A = {}^{1}/_{2}bh$
Area of a parallelogram	$A = bh$
Area of a trapezium	$A = {}^{1}/_{2}(a+b)\text{h}$
Circumference of a circle	$C = 2\pi r = \pi d$
Area of a circle	$A = \pi r^2$
Volume of a prism	V = area of cross-section × length
Volume of a cuboid	$V = lwh$
Surface area of a cuboid	$A = 2(lw + lh + wh)$
Capacity	1 L = 1000 cm^3, 1 m^3 = 1000 L
Unit conversions	1 m = 100 cm, 1 km = 1000 m; 1 m^2 = 10,000 cm^2

SETS

Number of elements	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Complement	$n(A') = n(U) - n(A)$

DATA

Mean	sum of values ÷ number of values
Median	middle value of an ordered list
Mode	the most frequent value
Range	largest value - smallest value
Sector angle	$^{frequency}/_{total} \times 360°$
Frequency density idea	a histogram shows frequency by area

Glossary

Each entry gives the meaning used in this course. Where a term has a wider meaning in mathematics, the entry states the Year 1 boundary.

Acute angle	An angle smaller than 90°.
Alternate angles	A pair of equal angles on opposite sides of a transversal between parallel lines.
Angle	The amount of turn between two straight lines that meet at a point, measured in degrees.
Area	The amount of surface a shape covers, measured in square units.
Average	A single value used to summarise a set of values; in this course the mean, median and mode.
Bar chart	A display in which the height or length of each bar shows the frequency of a category.
Base	In a power, the number that is multiplied repeatedly, as in 2^5.
Bidmas	The agreed order of operations: brackets, indices, division and multiplication, addition and subtraction.
Budget	A plan for spending a fixed total, in which each item can be compared with the total as a percentage.
Capacity	The amount a container can hold, usually measured in litres or millilitres.
Cardinality	The number of elements in a set, written $n(A)$.
Categorical data	Data that records a category or label rather than a number.
Circumference	The perimeter of a circle, $C = 2\pi r$.
Coefficient	The number multiplying a variable in a term, as in $7x$.
Common factor	A number that divides two or more numbers exactly.
Complement	The elements of the universal set that are not in the set, written A'.
Composite number	A whole number greater than 1 with more than two factors.
Congruent	Exactly the same shape and size.
Constant	A quantity that does not change.
Continuous data	Numerical data that can take any value in a range, such as height.
Corresponding angles	Angles in the same position at each intersection when a transversal crosses parallel lines; they are equal.
Cross-section	The shape you see when a solid is sliced parallel to its base.
Cube number	A number of the form $n \times n \times n$, such as 27.
Cuboid	A solid with six rectangular faces.
Decimal	A number written using place value with tenths, hundredths and so on.
Denominator	The number below the line in a fraction; it names the size of each part.
Diameter	A straight line across a circle through its centre; $d = 2r$.
Difference	The result of subtracting one number from another.
Discount	An amount taken off a marked price, usually quoted as a percentage of that price.
Discrete data	Numerical data that can take only particular values, usually whole numbers.
Divisible	A whole number a is divisible by b if b divides a with no remainder.
Equation	A statement that two expressions are equal.
Equivalent fractions	Fractions with the same value, such as $\frac{1}{2}$ and $\frac{3}{6}$.
Estimate	An approximate answer obtained by rounding or by another deliberate simplification.
Expanded form	A number written as the sum of the values of its digits.
Exponent	The number in a power that records how many copies of the base are multiplied, as in 2^5.
Expression	A combination of numbers, variables and operations without an equals sign.
Exterior angle	The angle between one side of a polygon and the extension of the next side. The exterior angles of a convex polygon total 360°.
Factor	A whole number that divides another exactly, with no remainder.
Factor pair	Two factors that multiply to give a number, such as 4 and 9 for 36.
Factor tree	A diagram that breaks a number into prime factors step by step.
Formula	A rule written with symbols that relates quantities.

Frequency	How often a value or category occurs.
Greater than	The symbol > means is greater than; $5 > 3$.
HCF	Highest common factor: the largest number that divides two or more numbers exactly.
Histogram	A display of continuous data grouped into equal class intervals, where the area of each bar shows the frequency.
Hypotenuse	The side opposite the right angle in a right-angled triangle.
Improper fraction	A fraction whose numerator is greater than or equal to its denominator, such as $\frac{7}{4}$.
Index	Another word for exponent.
Integer	Any whole number, positive, negative or zero.
Interior angle	An angle inside a polygon, between two of its sides. The interior angles of an n-sided polygon total $(n\text{-}2) \times 180°$.
Intersection	The elements that belong to both sets, written $A \cap B$.
Inverse	An operation that undoes another, such as addition and subtraction.
Irrational number	A number that cannot be written as a fraction of two integers, such as $\sqrt{2}$ or π; its decimal never ends or repeats.
Irregular	Not regular: sides or angles are not all equal.
LCM	Lowest common multiple: the smallest number that two or more numbers divide into exactly.
Less than	The symbol < means is less than; $3 < 5$.
Like terms	Terms with exactly the same variable part, such as $3x$ and $5x$.
Mean	The sum of the values divided by the number of values.
Median	The middle value of an ordered data set.
Mixed number	A whole number together with a fraction, such as $2\frac{1}{3}$.
Mode	The value that occurs most often.
Multiple	The result of multiplying a whole number by another whole number.
Natural number	A counting number: 1, 2, 3, …
Negative number	A number less than zero.
Net	A flat shape that folds into a solid; the area of the net is the surface area of the solid.
Notation	A agreed system for writing symbols and values.
Numerator	The number above the line in a fraction; it counts the parts.
Obtuse angle	An angle between 90° and 180°.
Origin	The point where the x-axis and y-axis meet, (0, 0).
Outlier	A value that lies far from the rest of the data.
Parallel	Lines in the same plane that never meet; the symbol is ∥.
Parallelogram	A quadrilateral with two pairs of parallel sides.
Percentage	A proportion expressed out of 100.
Perimeter	The total distance around the outside of a shape.
Perpendicular	At right angles; the symbol is ⊥.
Pi (π)	The ratio of the circumference of a circle to its diameter, approximately 3.14159.
Place value	The value a digit has because of its position in a number.
Polygon	A closed shape made of straight sides.
Prime factorisation	A number written as a product of prime factors.
Prime number	A whole number greater than 1 with exactly two factors, 1 and itself.
Prism	A solid with the same cross-section all along its length.
Product	The result of a multiplication.
Proper fraction	A fraction whose numerator is smaller than its denominator.
Proportion	A relationship in which quantities keep the same ratio.
Quadrilateral	A polygon with four sides.
Quotient	The result of a division.
Range	The largest value minus the smallest value in a data set.
Ratio	A comparison of two quantities by division.

Rational number	A number that can be written as a fraction of two integers.
Real number	Any number that has a position on the number line: every rational number and every irrational number together.
Reciprocal	The number you multiply by to get 1; the reciprocal of $^{3}/_{4}$ is $^{4}/_{3}$.
Recurring decimal	A decimal in which a digit or block of digits repeats for ever, such as 0.333... or 0.3636...; every recurring decimal is rational.
Regular polygon	A polygon with all sides and all angles equal.
Remainder	What is left over after a division that is not exact.
Round	Write a number to a required degree of accuracy.
Scale drawing	A drawing in which every length is a fixed fraction of the real length.
Sector	The region of a circle between two radii.
Set	A well-defined collection of objects.
Simplifying	Rewriting an expression or fraction in a shorter equivalent form.
Solution	A value of the variable that makes an equation true.
Square number	A number of the form $n \times n$, such as 49.
Square root	A value that, multiplied by itself, gives the number; $\sqrt{49} = 7$.
Standard form of a number	A number written with digits rather than in expanded form.
Subset	A set whose elements all belong to another set, written $A \subset B$.
Substitute	Replace a variable with a number.
Surface area	The total area of all the faces of a solid.
Tax (VAT)	An amount added to a price by government, quoted as a percentage. To remove a tax that is already included, divide by $1 + {}^{rate}/_{100}$.
Term	A part of an expression separated by addition or subtraction.
Terminating decimal	A decimal with a finite number of digits after the point, such as 0.375; it equals a fraction whose denominator has no prime factors other than 2 and 5.
Transversal	A line that crosses two or more other lines.
Trapezium	A quadrilateral with exactly one pair of parallel sides.
Union	The elements that belong to either set, written $A \cup B$.
Universal set	The set of all elements being considered, written U.
Variable	A letter that stands for a number that can change.
Venn diagram	A diagram using overlapping circles to show sets and their relationships.
Vertex	A corner of a shape, or the point where two lines meet.
Volume	The amount of space a solid occupies, measured in cubic units.
Whole number	Zero together with the natural numbers: 0, 1, 2, 3, …

Index

Page numbers are the printed page numbers at the foot of each page; the first number is usually the section where the idea is introduced.

A

Acute angle 155–156, 163
Alternate angles 159–162
Angle 149–151, 154
Area 19, 30–31, 33
Average 192, 194, 228–229

B

Bar chart 230, 233–235
Base 2–3, 9, 23
Budget 19, 22, 96, 102

C

Capacity 29, 45–46, 202
Cardinality 179, 182, 184
Circumference 200, 209–211
Co-interior angles 159–161, 172
Coefficient 127, 133, 144
Common factor 51, 53, 65–66
Complement 108, 121, 174, 177
Composite number 53, 60–61, 63
Composite shape 202, 211
Constant 130–132, 144
Construction 127, 149–151
Continuous data 232, 234–235, 244
Corresponding angles 159–161, 173
Cross-section 214–217
Cube number 38, 47
Cuboid 202–204, 212

D

Decimal 2–3, 7, 9
Denominator 6, 8, 76, 78
Diameter 210–212, 220
Difference 3–4, 26, 29
Discount 102–105
Discrete data 247
Divisibility test 58, 71–72
Divisible 52–54, 58
Dot plot 230, 234, 236, 244

E

Equation 42, 115, 126–127
Equivalent fractions 75–77, 79
Estimate 3, 6, 11, 18
Estimation 1, 3, 18–19
Expanded form 1–3, 12
Expanding 40, 44, 58, 82
Exponent 29–32
Expression 28–31
Exterior angle 151, 162, 164–165

F

Factor 4, 19, 28, 32
Factor pair 54–57
Factor tree 53, 63–64, 73
Formula 39, 44, 118–119
Frequency 228–231
Frequency table 228, 230–232

G

Greater than 7, 19, 26, 40

H

HCF 52–53, 65–66
Histogram 228–229, 231, 235

I

Improper fraction 80–81, 98–99
Integer 3, 6–8
Interior angle 160–161, 164–165
Intersection 167, 169, 177–178
Inverse 30, 137, 144, 252
Irrational number 192
Irregular 61, 164, 166

L

LCM 52–53, 65–66
Less than 9, 15–16, 19
Like terms 127–128, 135–136

M

Mean 1, 10–11, 29
Median 229, 237–239
Misleading graph 243, 265
Mixed number 14, 80–81, 85
Mode 190, 229, 237–238
Multiple 51–53, 56

N

Natural number 6, 9, 24, 192
Negative number 7, 15, 134
Net 202, 204, 212–213
Numerator 76, 79–81

O

Order of operations 28–31

P

Parallel 149–151, 154
Parallelogram 205, 207–209
Percentage 27, 46, 49, 78
Percentage change 112–115
Perimeter 39, 136–137, 140
Perpendicular 149, 151, 153–154
Perpendicular bisector 151, 153, 167–168
Pie chart 229, 235–236, 248
Place value 1–4
Polygon 150–151, 164–165
Power 1, 12, 28–29
Prime factorisation 51–53, 61
Prime number 52, 60, 62, 71
Prism 201, 214–216
Product 19, 30, 32, 34
Proper fraction 80, 98
Proportion 77, 83, 102–103

Q

Quadrilateral 164–166, 170
Quotient 63, 80, 94

R

Range 171, 182, 227, 229
Ratio 77, 103–104, 111
Rational number 6, 24, 192, 194
Real number 192
Reciprocal 77, 93–94, 97
Recurring decimal 86–88, 98
Regular polygon 165–166, 260, 264
Remainder 26, 56–58
Round 2–3, 12, 15
Rounding 1–3, 14

S

Scale drawing 151, 171–172
Sector 202, 210, 228–229
Set 1–2, 4, 20
Simplifying 75, 82–83, 126
Solution 42, 127–128, 137
Solving an equation 129, 252
Square number 38–39, 47, 49
Square root 38–39, 47, 55
Subset 179, 184, 196
Substitute 24, 30, 43–44
Substitution 28–30, 42
Surface area 148, 200–202

T

Tax (VAT) 116, 118
Term 12–13, 24, 36
Terminating decimal 86–89
Transversal 159–161, 172
Trapezium 165
Two-way table 267

U

Union 177–179, 181
Universal set 178–179, 181–182

V

Variable 43, 47, 127, 228
VAT 116–118, 259
Venn diagram 179–181, 184
Vertex 154–156, 163
Vertically opposite 155–158
Volume 31, 33, 35, 39

W

Whole number 2–3, 6, 9

References and acknowledgements

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